

BS-480/BS-490 Chemistry Analyzer

Service Manual



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Mindray is responsible for safety, reliability and performance of this product only in the condition that:

- all installation operations, expansions, changes, modifications and repairs of this product are conducted by Mindray authorized personnel;
- the electrical installation of the relevant room complies with the applicable national and local requirements;
- the product is used in accordance with the instructions for use.

Upon request, Mindray may provide, with compensation, necessary circuit diagrams, calibration illustration list and other information to help qualified technician to maintain and repair some parts, which Mindray may define as user serviceable.



NOTE

This equipment is to be operated only by medical professionals trained and authorized by Mindray or Mindray-authorized distributors.



WARNING

It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or injury of human health.

Warranty

THIS WARRANTY IS EXCLUSIVE AND IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE.

Exemptions

Mindray's obligation or liability under this warranty does not include any transportation or other charges or liability for direct, indirect or consequential damages or delay resulting from the improper use or application of the product or the use of parts or accessories not approved by Mindray or repairs by people other than Mindray authorized personnel.

This warranty shall not extend to:

- any Mindray product which has been subjected to misuse, negligence or accident;
- any Mindray product from which Mindray's original serial number tag or product identification markings have been altered or removed;
- any product of any other manufacturer.

Return Policy

Return Procedure

In the event that it becomes necessary to return this product or part of this product to Mindray, the following procedure should be followed:

- 1** Obtain return authorization: Contact the Mindray Service Department and obtain a Customer Service Authorization (Mindray) number. The Mindray number must appear on the outside of the shipping container. Returned shipments will not be accepted if the Mindray number is not clearly visible. Please provide the model number, serial number, and a brief description of the reason for return.
- 2** Freight policy: The customer is responsible for freight charges when this product is shipped to Mindray for service (this includes customs charges).
- 3** Return address: Please send the part(s) or equipment to the address offered by Customer Service department

Company Contact

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Summary of Hazards

Introduction

Observe the following safety precautions when using the product. Ignoring any of these safety precautions may lead to personal injury or equipment damage.



WARNING

If the product is used in a manner not specified by our company, the protection provided by the product may be impaired.

Electric Shock Hazards

Observe the following instructions to prevent electric shock.



WARNING

- When the MAIN POWER is turned on, users other than the servicing personnel authorized by our company must not open the rear cover or side cover.
 - Spillage of reagent or sample on the product may cause equipment failure and even electric shock. Do not place sample and reagent on the product. In case of spillage, switch off the power immediately, remove the spillage and contact our Customer Service Department or your local distributor.
-

Moving Parts Hazards

Observe the following instructions to prevent personal injury caused by moving parts.



WARNING

- Do not touch such moving parts as sample probe, reagent probe, mixers , cuvette wash station, sample carousel, reagent carousel and reaction carousel,when the system is in operation.
 - Do not put your fingers or hands into any open part when the system is in operation.
-

Photometer Lamp Hazards

Observe the following instructions to prevent personal injury caused by photometer lamp.



WARNING

- Eye injury could occur from light emission from the photometer lamp. Do not stare into the lamp when the system is in operation.
 - If you want to replace the photometer lamp, first switch off the MAIN POWER and then wait at least 15 minutes for the lamp to cool down before touching it. Do not touch the lamp before it cools down, or you may get burned.
-

Laser Beam Hazards

Observe the following instructions to prevent personal injury caused by laser beam.



WARNING

Light emitted by the bar code reader may cause eye injury. Do not stare into the laser beam radiated from the bar code reader when the system is in operation.

Sample, Calibrator and Control Hazards

Observe the following instructions to protect against the biohazardous infection by samples, calibrators and control samples.



BIOHAZARD

- Inappropriately handling samples, controls and calibrators may lead to biohazardous infection. Do not touch samples, mixtures or waste with your bare hands. Wear gloves and lab coat and, if necessary, goggles.
 - In case your skin contacts the sample, control or calibrator, follow the standard laboratory safety procedure and consult a doctor.
-

Reagent and Wash Solution Hazards

Observe the following instructions to protect against the biohazardous infection by reagents and wash solution.



WARNING

Reagents and concentrated wash solution are corrosive to human skins. Exercise caution when using reagents and concentrated wash solution. In case your skin or clothes contact them, wash them off with soap and clean water. If reagents or wash solution spills into your eyes, rinse them with much water and consult an oculist.

Waste Hazards

Observe the following instructions to prevent environmental pollution and personal injury caused by waste.



BIOHAZARD

- Some substances contained in reagent, control, concentrated wash solution and waste are subject to regulations of contamination and disposal. Dispose of the waste in accordance with your local or national rule for biohazard waste disposal and consult the manufacturer or distributor of the reagents for details.
 - Wear gloves and lab coat and, if necessary, goggles.
-

System Disposal Hazards

Observe the following instructions to dispose of the waste analyzer.



WARNING

Materials of the analyzer are subject to contamination regulations. Dispose of the waste analyzer in accordance with your local or national rule for waste disposal.

Fire and Explosion Hazards

Observe the following instructions to prevent fire and explosion.



WARNING

Ethanol is flammable substance. Please exercise caution while using ethanol around the instrument in order to prevent fire and explosion.

Removal of Analyzer from Use for Repair or Disposal

To minimize or eliminate the hazards involved in repair, transportation, disposal process, please observe the following instruction.



WARNING

When the analyzer is not in use, for example, in repair, transportation or disposal process, please clean and sterilize the parts that may cause biohazards (sample probe, reagent probe and mixers, etc.) and remind the person who handles the device of the related hazards.

Precautions on Use

Introduction

To use the product safely and efficiently, pay attention to the following operating precautions.

Intended Use



WARNING

The instrument is an automated chemistry analyzer for in vitro diagnostic use in clinical laboratories and designed for in vitro quantitative determination of clinical chemistries in serum, plasma, urine and cerebrospinal fluid samples. Please consult us before you use the instrument for other purposes.

When drawing a clinical conclusion, please also refer to patients' clinical symptoms and other test results.

Environment Precautions



CAUTION

Evaluate the electromagnetic environment prior to operating the system.

Please install and operate the system in an environment specified by this manual. Installing and operating the system in other environment may lead to unreliable results and even equipment damage.

To relocate the system, please contact our Customer Service Department or your local distributor.

Installation Precautions



WARNING

The product is switched on and off via a switch or breaker. Before installing the system, ensure that the building in which the system will be located has been equipped with a switch or breaker that complies with IEC 61010-1:2001, is in close proximity to the equipment and within easy reach of you, and is marked as the disconnecting device for the equipment.

Electromagnetic Noise Precautions



CAUTION

Electromagnetic noise may interfere with operations of the system. Do not install devices generating excessive electromagnetic noise around the system. Do not use such devices as radio transmitters in the room housing the system. Do not use other CRT displays around the system.

Do not use other medical instruments around the system that may generate electromagnetic noise to interfere with their operations.

Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. mobile phones or radio transmitters), as these may interfere with the proper operation.

The electromagnetic environment should be evaluated prior to operation of the device.

This device has been designed and tested to CISPR 11 Class A, and in a domestic environment may cause radio interference, in which case, you may need to take measures to mitigate the interference.



NOTE

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.

It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that it will perform as intended.

Operating Precautions



CAUTION

- Take the clinical symptoms or other test results of the patient into considerations when making a diagnosis based on the measuring results produced by the system.
 - Operate the system strictly as instructed by this manual. Inappropriate use of the system may lead to unreliable test results or even equipment damage or personal injury.
 - When using the system for the first time, first run calibrations, and then QC tests to make sure the system is in proper state.
 - Be sure to run QC tests every time when you use the system, otherwise the result may be unreliable.
 - Do not uncover the reagent carousel when the system is in operation. Keep the reagent carousel cover closed.
 - The RS-232 port on the analyzing unit is used for connection with the operation unit only. Do not use it for other connections. Use the cables provided by our company or your local distributor for the connection.
 - The operation unit is a personal computer with the operating software installed. Installing other software or hardware on the computer may interfere with the system operation. Do not run other software when the system is working.
 - Computer virus may destroy the operating software or test data. Do not use the computer for other purposes or connect it to the Internet. If the computer is infected by virus, please install anti-virus software to check for and clear virus.
 - Do not touch the display, mouse or keyboard with wet hands or hands with chemicals.
 - Do not place the MAIN POWER to ON again within 10 seconds since placing it to OFF; otherwise the system may enter the protection status. If it does so, place the MAIN POWER to OFF and place it to ON again.
-

Maintenance and Servicing Precautions



CAUTION

- Maintain the system strictly as instructed by this manual. Inappropriate maintenance may lead to unreliable results, equipment damage or personal injury.
 - To wipe off dust from the system surface, use a soft, clean and moistened cloth soaked with soap water rather than organic solvents such as ethanol. After cleaning, wipe the surface dry with dry cloth.
 - Shut down and turn off analyzer and disconnect the power plug before cleaning. Take necessary measures to prevent water ingress, otherwise equipment damage or personal injury may be caused.
 - Replacement of major parts as photometer lamp, sample probe, reagent probe, mixers and syringe plunger assembly must be followed by re-calibration.
 - Replacement of the photometer lamp should be done when the system power has been switched off for at least 15 minutes.
 - If the system is failed and needs servicing, contact our Customer Service Department or your local distributor. The system may need to be stopped or transported during servicing, which will probably cause biohazards, electric shock hazards and moving part hazards. Exercise caution when prepare the system for servicing.
-



NOTE

Check the safe state of the equipment after repair. Make sure the equipment is safe and then offer it to the customer.

Chemistry Parameter Configuration Precautions



CAUTION

To define such parameters as sample volume, reagent volume and wavelength, follow the instructions in this manual and the instructions of reagents.

ISE Module Precautions



CAUTION

To prevent ISE electrodes from being damaged due to water scarcity, if the system, when equipped with an ISE module, will be powered off for a long time, perform the electrode storage maintenance.

Sample Precautions



CAUTION

- Use samples that are completely free of insoluble substances like fibrin or suspended matter; otherwise the sample probe may be clogged.
 - Medicines, anticoagulants or preservative in the samples may lead to unreliable results.
 - Hemolysis, icterus or lipemia in the samples may lead to unreliable test results; running a sample blank, therefore, is recommended.
 - Store the samples properly. Improper storage may change the compositions of samples and lead to unreliable results.
 - Sample volatilization may lead to unreliable results. Do not leave the sample open for a long period.
 - The system has a specific requirement on the sample volume. Refer to this manual for proper sample volume.
 - Load samples to correct positions on the sample carousel before the analysis begins; otherwise reliable results may not be obtained.
-

Reagent, Calibrator and Control Precautions



CAUTION

- Use proper reagents, calibrators and controls on the system.
 - Select appropriate reagents according to the performance characteristics of the system. Consult the reagent suppliers, our company or our authorized distributor for details, if you are not sure about your reagent choice.
 - Store and use the reagents, calibrators and controls strictly as instructed by the suppliers; otherwise, reliable results or best performance of the system may not be obtained. Improper storage of reagents, calibrators and controls may lead to unreliable results and bad performance of the system even in validity period.
 - Perform calibration after changing the reagents, otherwise reliable results may not be obtained.
 - Contamination caused by carryover among reagents may lead to unreliable test results. Consult the reagent suppliers for details.
-

Data Archiving Precautions



NOTE

The system automatically stores the data to the built-in hard disk. Data loss, however, is still possible due to mis-deletion or physical damage of the hard disk. You are recommended to regularly archive the data to such medium as CDs.

To avoid the data loss caused by unexpected power failure, UPS (uninterrupted power supply) is recommended.

External Equipment Precautions



WARNING

For operating instructions and precautions of the computer and printer, please refer to their operation manuals.

External equipment connected to the analogue and digital interfaces must be authorized and complied with relevant safety and EMC standards (e.g., IEC 60950 Safety of Information Technology Equipment Standard and CISPR 22 EMC of Information Technology Equipment Standard (CLASS B)). Any person, who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies with the safety and EMC requirements. If you have any questions, consult the technical services department of your local representative.

External Vacuum Pump Precautions



WARNING

Make sure the vacuum pump tubing is connected properly without any twists or sharp angles so that it can work normally.

Tubing and cables connected to the vacuum pump must be protected to prevent damage and breaks due to human or other causes.

Set the vacuum pump on a solid flat platform or ground.

Tube and Liquid Container Precautions



WARNING

When the tube or the part that contain liquid become aged or damaged, please stop its use immediately and contact our customer service department or your local distributor to check and replace it.

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1

System Description

1 System Description

1.1 Overview

The BS-480/BS-490 is a fully automated and computer-controlled chemistry analyzer designed for the in vitro determination of clinical chemistries in serum, plasma, urine and cerebrospinal fluid (CSF) samples. BS-480/BS-490 consists of the analyzing unit and operation unit.

Figure 1.1 BS-480 chemistry analyzer



Throughput of biochemistry: 400 tests/hour.

Throughput of the ISE module: 240 tests/hour (Na⁺, K⁺, Cl⁻) for serum and plasma. 171 tests/hour (Na⁺, K⁺, Cl⁻) for urine.

Maximum throughput with ISE included: 560 tests/hour.

1.2 Components of Analyzing Unit

The BS-480 Chemistry Analyzer consists of the analyzing unit (analyzer), operation unit (computer), and output unit (printer).

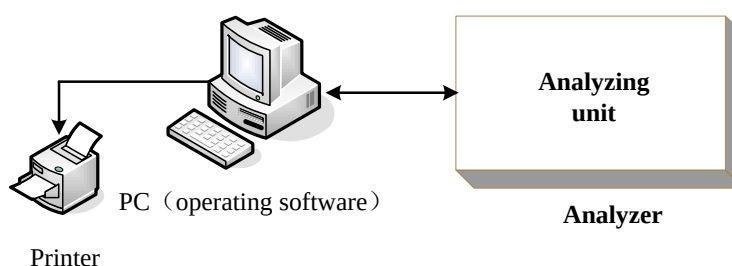
The analyzing unit performs all operations of sample analysis including dispensing sample and reagent, mixing, reaction and measurement, auto washing of cuvettes, ISE analysis, etc.

The operation unit is a computer installed with the operating software, which controls operation of the analyzer and processes the test data.

The printer is used to print test results.

The operation unit and the analyzing unit are independent structurally and communicate with each other through the serial port. The operation unit sends data and instructions to the analyzing unit and acquires test data and status information from the analyzing unit.

Figure 1.2 System structure



The BS-480/BS-490 has a throughput of 400 tests per hour for single- or double- reagent analysis, and its maximum throughput is 560 tests per hour with ISE module. The working period is 9 seconds. The instrument is composed of the following components: one reaction carousel, one sample carousel, one reagent carousel, one reagent probe, one sample probe, two mixers, one automatic cuvette wash station, one photometric system, one ISE module (optional). The photometric system, which is composed of gratings and diode array, performs photometric measurement to the reaction cuvettes that hold sample/reagent mixture. After testing, eight-phase automatic cuvette washing will be completed by the automatic cuvette wash station.

1 System Description

Figure 1.3 Layout of analyzer's operation panel

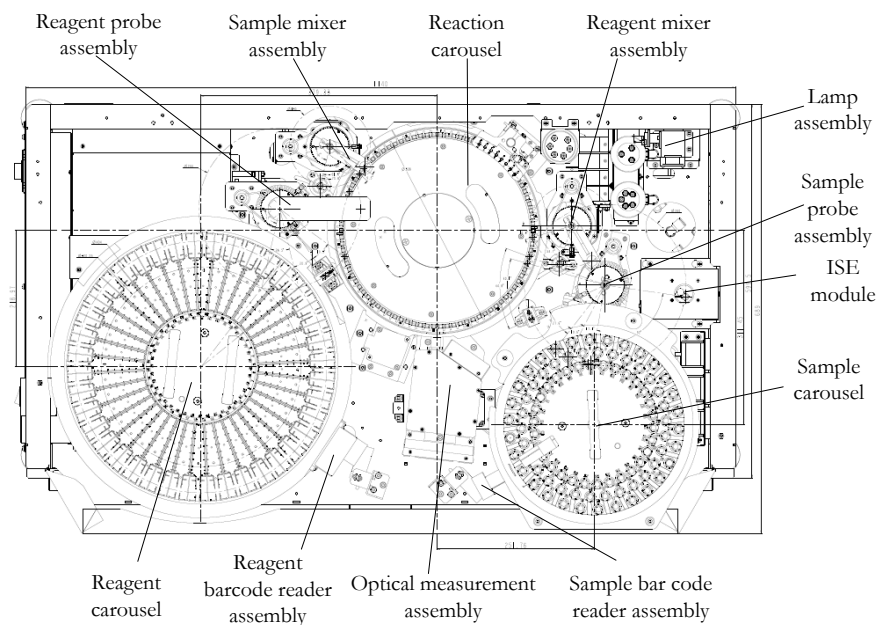
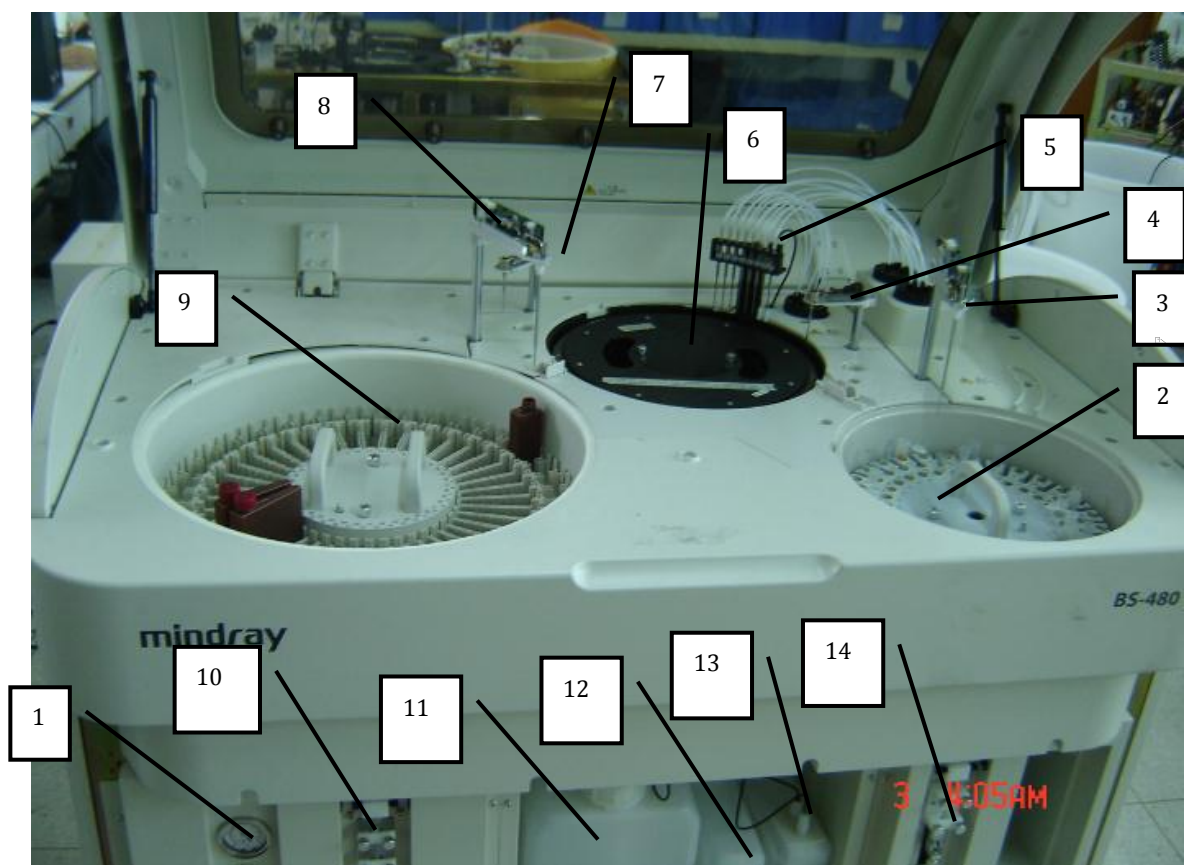


Figure 1.4 Overall structure of the analyzer



1-Pressure gauges for vacuum and deionized water

2-Sample carousel

3-Sample probe

4-Reagent mixer

5-Cuvette wash station

6-Reaction carousel

7-Sample mixer

8-Reagent probe

9-Reagent carousel

10-Reagent syringe

11-Water tank

12-Concentrated wash solution tank

13-Diluted wash solution tank

14-Sample syringe

1.3 Functions of Analyzing Unit

The following content describes the main procedures of the analyzer.

- 1 Resetting, initializing all the mechanical units and washing the exterior and interior of the two probes and the mixers;
- 2 Performing 8-phase automatic washing for each reaction cuvette, and running water blank test after phase 6;
- 3 Rotating the reagent carousel to the first specified reagent position, and lowering the reagent probe to aspirate the reagent;
- 4 Rotating reaction cuvette to the reagent dispense position after completing 8-phase washing; then rotating reagent probe with R1 to the reaction carousel to dispense R1 into the reaction cuvette;
- 5 Incubating R1 in the reaction cuvette for 12 periods;
- 6 Rotating the sample carousel to the specified sample dispense position, and lowering the sample probe into the sample carousel to aspirate a specified amount of sample;
- 7 Rotating the reaction cuvette with R1 to the sample dispense position, and rotating the sample probe with sample to the reaction carousel to dispense sample into the reaction cuvette;
- 8 Rotating the reaction cuvette with samples to the sample mixing position for mixing;
- 9 For double-reagent tests, rotating the reagent carousel to the specified R2 position after a fixed period, and lowering the reagent probe into the reagent carousel to aspirate reagent;
- 10 Rotating the reaction cuvette to the R2 dispense position, and then rotating reagent probe with R2 to the reaction carousel to dispense R2 into the reaction cuvette;
- 11 With R2 added, the reaction cuvette will be carried to the reagent mixing position for stirring;
- 12 During each period, the reaction cuvette receives photometric measurement (absorbance reading taking) when passing by the photometric unit;
- 13 Triple-/Quadruple-reagent analysis is similar to single-/double-reagent analysis stated above; (As for triple- or quadruple-reagent tests, the reaction cuvette with R2 dispensed will not be washed when passing by the wash unit.)
- 14 The reaction cuvettes in which reaction is finished will be washed automatically when they pass by the cuvette wash station.

Table 1.1 Main functions of each unit

Unit Name	Descriptions
Sample probe	Performs aspiration and dispensing for all biochemistry and ISE tests
Sample carousel unit	90 positions are available for loading samples. The outer and middle rings support barcode scanning.
Reagent probe unit	Performs aspiration and dispensing of R1, R2, R3 and R4 for all biochemistry tests

1 System Description

Unit Name	Descriptions
Reagent carousel unit	80 positions are available for placing reagents. All the positions support reagent bar code scanning. The outer ring supports 40ml and 20ml reagent bottles. The inner ring supports both 40ml and 62ml reagent bottles, Reagents can be placed on any position of the reagent carousel. The reagent carousel provides a refrigerating environment which is constant within 2°C-10°C for 24 hours each day.
Reaction carousel unit	90 positions provided for holding reaction cuvettes. Plastic cuvette is used.
Sample and reagent mixer unit	Two mixer units driven by step-in motor.
Photometric unit	Reversed optics; supporting 12 wavelengths: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm, 800nm
Cuvette wash station	8-phase automatic washing for each reaction cuvette
ISE module (optional)	Provides the function of ISE measurement

2

System Performance and Workflow

2.1 Technical Specifications

2.1.1 Common

- System

Random selection, multiple channels, multiple chemistries, pausing and adding new tests

- System Structure

Analyzing unit + computer + printer

- Sample type

Serum, urine, plasma

- Maximum concurrent tests

39 double-reagent tests/78 single-reagent tests

- Throughput

400 tests/hour; maximum: 560 tests/hour (with ISE module)

- Analytical method

Endpoint, Kinetic and Fixed-time measurements; supporting single-/double-/triple-/quadruple-reagent tests, and single-/double-wavelength tests

- Reaction time

The longest reaction time is 10 minutes for single-reagent measurement and 5 minutes for double-reagent measurement.

- Reaction temperature

37±0.3°C

- Test scope

Clinical biochemistries, Immunoassays, Therapeutic Drug Monitoring (TDM)

- Predilution

Dilution ratio: 3~175. Dilution is done in reaction cuvette.

- Operation mode

System and tests are configured via the operating software; test one by one; panels and calculation tests are allowed.

- Calibration math model

Single-point linear, two-point linear, multi-point linear; Logit-Log 4P; Logit-Log 5P; spline curve, exponential function; polynomial expression; parabola.

- Control

Westgard rules, Cumulative sum check, Twin-plot

- Data processing

Store and output various data and diagrams and calculate among different tests.

- Dimensions

Analyzing unit: l×b×h: 1180 mm×710 mm×1145 mm;

- Weight

Analyzing unit: 261.5 Kg;

- STAT samples

Emergent samples can be analyzed at any time with highest priority.

- Networking

Allowed to connect with LIS (Laboratory Information Management System).

2.1.2 Sample Specifications

- Sample loading

Samples are loaded via sample carousel.

- Sample tube type

Table 2.1 Sample tube volume and dead volume

Sample Container	Specification	Dead Volume
Sample container	Φ14×25mm, 0.5ml (Beckman microtube)	70μl
Sample container	Φ14×25mm, 2ml (Beckman microtube)	150μl
Sample container	Φ12×37mm, 2ml (Hitachi microtube)	100μl
Primary tube or plastic tube	Φ12×68.5 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ12×99 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ12.7×75 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ12.7×100 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ13 X 75 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ13 X 95 mm	8mm more over the unacceptable sample level height
Primary tube or plastic tube	Φ13 X 100 mm	8mm more over the unacceptable sample level height

- Sample inventory

Hitachi sample container (Φ12×37mm): sample volume ≥ 100μl; Beckman microtube (Φ14×25mm, 0.5ml): sample volume ≥ 70μl; Beckman microtube (Φ14×25mm, 2ml), sample volume ≥ 150μl.

For primary tube or plastic tube, sample should be kept 8mm higher than the unacceptable sample level height.

- Sample carousel

2 System Performance and Workflow

Coaxial sample carousel consists of inner, middle and outer rings. The outer and middle rings support barcode scanning.

- Sample positions on sample carousel

30 sample positions on each ring and 90 positions in total. The inner ring provides the routine, control, STAT and wash solution positions.

- STAT samples

Emergent samples can be analyzed at any time with highest priority.

- Sample volume

1.5µl ~ 45µl, with an increment of 0.1µl

ISE: Serum:70µl, Urine: 140µl

- Sample probe

One sample probe available, featuring level sense, horizontal/vertical obstruct detection, clog detection and level tracking.

- Sample probe wash

Wash the exterior and interior of the probe; carryover contamination rate $\leq 0.05\%$.

- Sample input mode (bar code)

Name	Value
Symbology	Codabar, ITF, code128, code39, UPC/EAN, Code93
Minimum bar code density	0.19mm~0.5mm
Data bits	3~27
Format and content	User-defined
Maximum width	55mm
Maximum height	10mm
Maximum inclination angle	$\pm 5^\circ$
Print quality	No less than Class C according to the ANSI MH10.8M Print Quality Specification.
Width and narrowness	2.5:1 to 3.0:1

2.1.3 Reagent Specifications

- Reagent loading

Reagents can be set in both the inner ring and outer ring.

- Reagent bar code

Name	Value
Symbology	Codabar, ITF, code128, code39, UPC/EAN, Code93
Minimum bar code density	0.25mm~0.5mm
Data bits	13~30
Format and content	User defined

Maximum width	55mm
Maximum height	10mm
Maximum inclination angle	±5°
Print quality	Class A (ANSI MH10.8M)
Width and narrowness	2.5:1

- Reagent refrigeration

Refrigeration temperature: 2~10°C

- Reagent addition approach

Reagent addition and check of fluid level and remaining volume are performed precisely by syringe.

- Supported reagent types

R1, R2, R3 and R4

- Reagent volume and Diluent volume

Reagent	Reagent volume	Diluent volume	Combined volume of Reagent and Diluent
Non-Concentrated Reagent			
R1	120µl~350µl, With 1µl increment	None	120µl~350µl
R2	10µl~350µl, With 1µl increment	None	10µl~350µl
R3	10µl~350µl, With 1µl increment	None	10µl~350µl
R4	10µl~350µl, With 1µl increment	None	10µl~350µl
Concentrated Reagent			
R1	10µl~340µl, With 1µl increment	10µl~340µl, With 1µl increment	120µl~350µl
R2	10µl~340µl, With 1µl increment	10µl~340µl, With 1µl increment	10µl~350µl
R3	10µl~340µl, With 1µl increment	10µl~340µl, With 1µl increment	10µl~350µl
R4	10µl~340µl, With 1µl increment	10µl~340µl, With 1µl increment	10µl~350µl

- Reagent carousel

The reagent carousel is coaxial with the inner and outer rings driven together; all the positions support built-in bar code scanning.

- Number and volume of reagent bottle

The reagent positions support Mindray reagent bottles: 40ml and 20ml for outer ring; 40ml and 62ml for inner ring.

- Reagent probe

2 System Performance and Workflow

One reagent probe is available, featuring level sense, horizontal/vertical obstruct detection and empty aspiration alarm.

- Reagent probe wash

Wash the exterior and interior of the probe;

- Reagent inventory

Maximum remaining volume of reagent bottles

- Outer ring 40ml reagent bottle: 3ml
- Outer ring 20ml reagent bottle: 2ml
- Inner ring 40ml reagent bottle: 3ml
- Inner ring 62ml reagent bottle: 3.5ml
- Prevention of cross contamination

User defined; wash the exterior and interior of the reagent probe intensively.

2.1.4 Reaction Specifications

- Optical pathlength of reaction cuvette

5mm

- Specification and material of reaction cuvette

5mm×5mm×29mm; made of plastic

- Number of reaction cuvettes

90

- Mixing procedure

There are two mixer assemblies for reagent and sample. The mixers start to work after sample, R2, R3 and R4 are respectively dispensed.

- Reactant volume

120~360μl

- Photometric system

Reversed optics of holographic concave flat-field gratings; Photodiode detects each wavelength.

- Range of wavelengths

12 wavelengths: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm, 800nm.

- Light source

12V/20W tungsten-halogen lamp; optical fiber incidence.

- Gratings type

Reversed optics of holographic concave flat-field gratings

- Wavelength accuracy

±2nm

- Photometric measurement method

Photodiode

- Concurrent wavelengths for each test

One or two wavelengths

- Absorbance range

0-3.3A, optical pathlength conversion: 10mm

- Resolution of photometer

0.00010D

2.1.5 Operation Specifications

- Monitor

17" LCD monitor with resolution 1280*1024

- Operating system

Windows 8 (64 bit)

- Communication interface

RS232; compatible with TCP/IP

- Printer

The system supports three types of printer, which include inkjet printer, laser (black and white) printer and stylus printer.

- Data input

Keyboard; mouse; 17-inch monitor; bar code reader; LIS: HL7, ASTM1394 (communicating through the TCP/IP or serial interface of static IP address)

- Data output

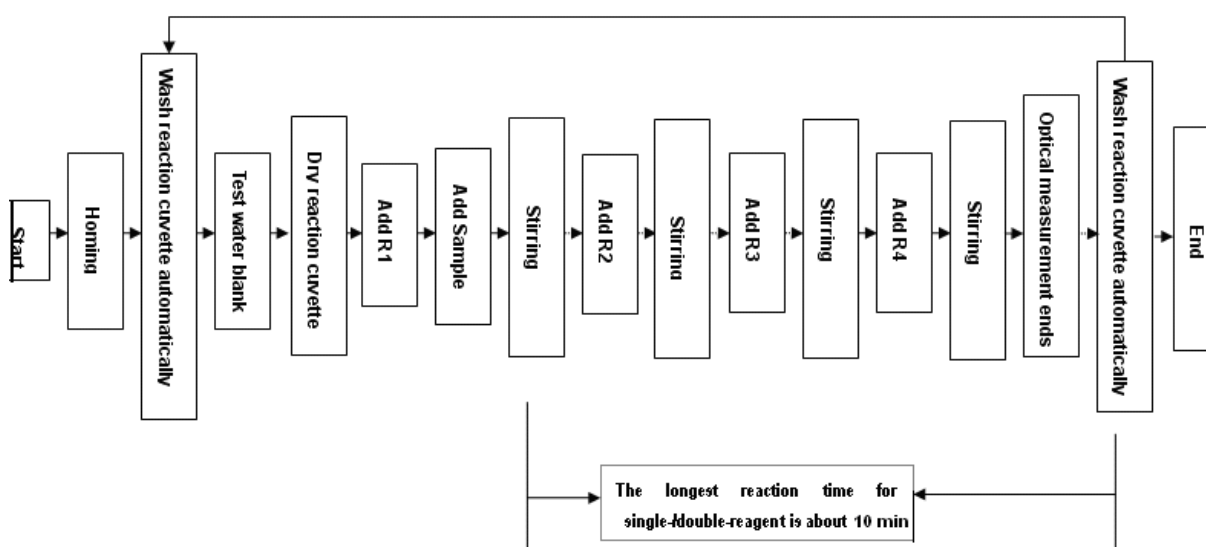
Monitor, printer, LIS

- Data storage

Hard disk; USB

2.1.6 Common Workflows

Figure 2.1 Analysis workflow



2.1.7 Startup/Shutdown Procedure

Startup procedure:

- 1 Power on the analyzer.
- 2 The operating software is run and checks the running environment.
- 3 The middle-low layer unit is checked.
- 4 The software checks the parameters and status of the analyzer, and inquires the unconfigured units, wrong parameters and status of the fans.
- 5 The sample carousel indicator flashes.
- 6 If sample barcode reader is configured, it is communicated.
- 7 If reagent barcode reader is configured, it is communicated.
- 8 Mechanical reset is performed.
- 9 The reaction carousel temperature control is turned on.
- 10 The lamp is turned off to inquire the dark current and maximum luminance value of the reaction carousel lamp, and then turned on. (It takes 10 minutes for the lamp to become steady. If the system finishes initialization in less than 10minutes, it will enter the Incubation status.)
- 11 The hydropneumatic unit is initialized.
- 12 The wash solution circulating pump P03 is switched on.
- 13 Preheating of cuvette cleaning water and wash solution is turned on.
- 14 The system is reset.
- 15 The system vacuum is released.
- 16 Wash solution circulating pump P03 is switched off.
- 17 Reagent bar code is scanned (when the reagent bar code reader is configured.)
- 18 The sample carousel indicator is off.
- 19 The ISE module, when configured, is initialized.
- 20 After confirming the standby conditions are satisfied, the system enters into the Standby status.

Shutdown procedure:

- 1 When the system is in Sleep or Standby status, the shutdown procedure can be performed.
- 2 The lamp power is turned off.
- 3 The reaction carousel temperature control and the preheating of the cuvette cleaning water and wash solution are turned off.
- 4 The hydropneumatic unit is turned off; pumps and valves are reset. The DI water pump P03 is switched off; the waste in the primary vacuum container is discharged.
- 5 The database is shutdown. If the operation fails, the system gives a warning, and the procedure will continue after confirmation.
- 6 The system is exited.

Sleep procedure:

- 1 The lamp is turned off.
- 2 The bar code reader motors are shutdown if the reagent/sample bar code readers are configured.

- 3 The hydropneumatic unit is turned off; pumps and valves are reset. The DI water pump P03 is switched off; the waste in the primary vacuum container is discharged.
- 4 The system enters into Sleep status.

Wake up procedure:

- 1 The lamp is turned on; and during the following 5min countdown, the following operations continue.
- 2 The bar code reader motors are turned on and laser is turned off if the reagent/sample bar code readers are configured.
- 3 Mechanical reset is performed.
- 4 Fluidic initialization is performed.
- 5 The system is reset.
- 6 The system vacuum is released.
- 7 The wash solution circulating pump P03 is switched off.
- 8 The hydropneumatic unit is initialized.

2.1.8 Workflow Descriptions

Sequential actions of sample probe in each period

Wash -> rotate to sample carousel -> lower into sample tube -> aspirate sample -> raise to "home" position vertically -> rotate to reaction carousel -> lower to reaction carousel -> dispense sample -> raise to the specified height for washing -> rotate to wash well -> go to next period

Sequential actions of reagent probe in each period

Wash -> Rotate to reagent carousel -> lower to reagent bottle -> aspirate R1 -> raise to "home" position vertically -> rotate to reaction carousel -> lower to reaction carousel -> dispense R1 -> raise to the specified height for washing -> rotate to wash well -> wash -> Rotate to reagent carousel -> lower to reagent bottle -> aspirate R2 -> raise to "home" position vertically -> rotate to reaction carousel -> lower into reaction carousel -> dispense R2 -> raise to the specified height for washing -> rotate to wash well -> go to the next period

Sequential actions of reagent and sample mixers in each period

Raise from wash well -> rotate to reaction carousel -> lower to reaction carousel -> mixing -> raise to above reaction carousel -> rotate to wash well -> lower to /above the wash well -> wash -> go to next period

Sequential actions of reaction carousel

The analyzer provides 90 positions for reaction cuvettes; 9 seconds for one cycle; one stop after each rotation. In the first rotation, 140 positions are rotated and photoelectric signal collection of all the cuvettes is done. During the first stop, R1 is dispensed and sample and reagent are mixed. In the second rotation, 41 positions are rotated. During the second stop, the sample and R2 are dispensed and 8-phase wash is performed.

2.1.9 Measuring Points

Figure 2.2 Measuring points for single-/double-reagent measurement

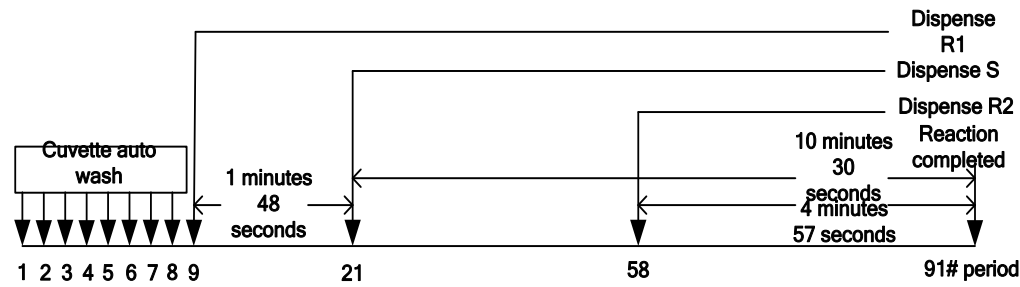
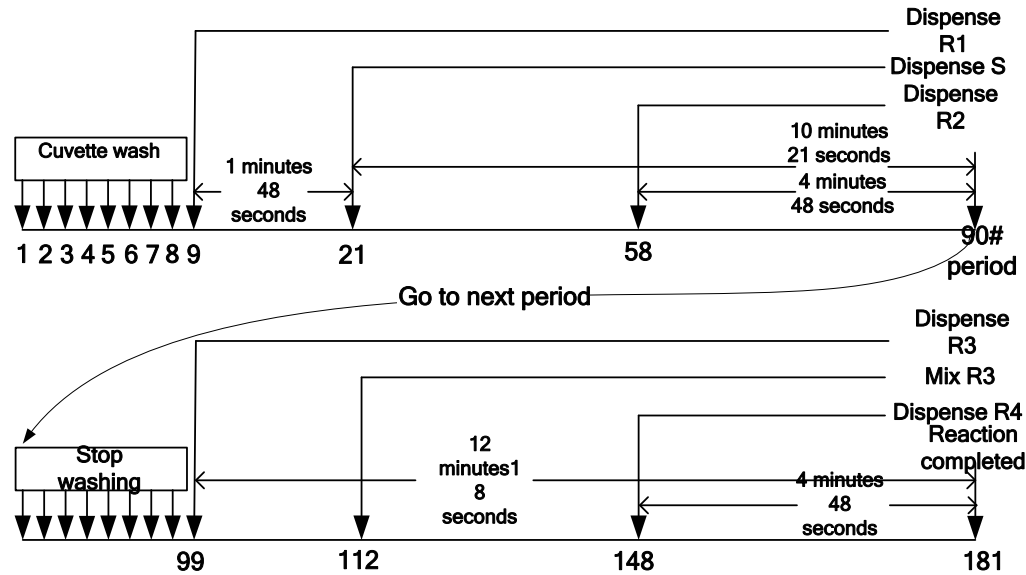


Figure 2.3 Measuring points for triple-/quadruple-reagents measurement



3

Installation Requirements and Procedure

3.1 Check before Installation

3.1.1 Installation Environment

- Altitude: -400- 4,000m.
- The system is for indoor use only.
- The bearing platform (or ground) should be level (with gradient less than 1/200).
- The bearing platform (or ground) should be able to support at least 450Kg weight.
- The installation site should be well ventilated.
- The installation site should be free of dust.
- The installation side should not be in direct sun.
- The installation site should be kept away from a heat or draft source.
- The installation site should be free of corrosive gas and flammable gas.
- The bearing platform (of ground) should be free of vibration.
- The installation site should be kept away from large noise and power supply interference.
- Keep the system away from brush-type motors and electrical contact device that is frequently switched on and off.
- Do not use such devices as mobile phones and radio transmitter near the system.
- Operating temperature: 15°C-30°C with fluctuation <2°C/H.
- Relative humidity: 35%-85%, without condensation.
- Provide air conditioning equipment if the room temperature does not meet the requirements.
- Install the equipment near a sewer for convenient discharging of waste liquid.
- Dimensions before packing: 1180mm(length)×710mm(depth)×1145mm(height)
- Dimensions after packing: 1530mm(length)×935mm(depth)×1390mm(height)

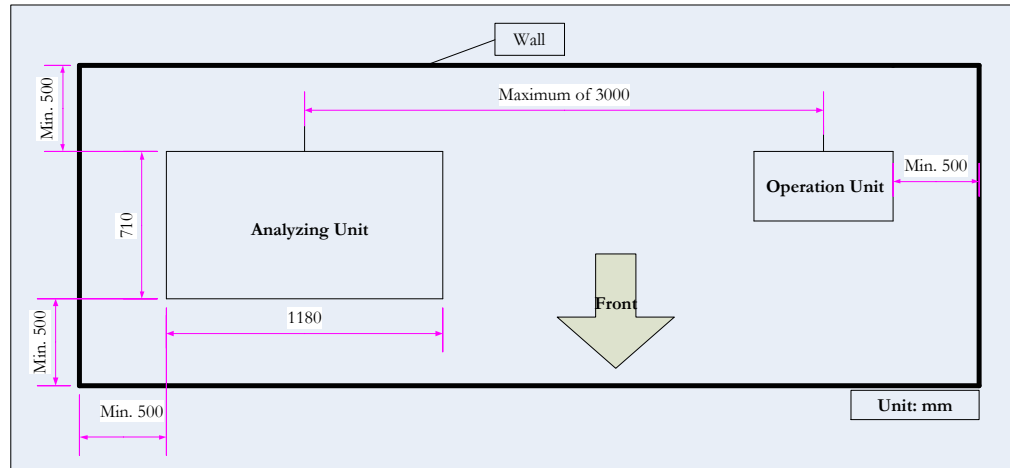
3.1.2 System Configuration

- BS-480/BS-490 chemistry analyzer
- Packing list of accessory kit
- Packing list of ISE accessory kit (for ISE module only)
- Optional external module (water supply module, drainage module)

3.2 Installation Requirements

3.2.1 Space and Accessibility Requirements

Figure 3.1 System Clearance



3.2.2 Power Supply and Noise

- Power supply: 220V: 220-240V~ 50Hz, 220/230V~ 60Hz; 110V: 110/115V~ 60Hz. Voltage fluctuation: +/-10% Line frequency: +/-3Hz. Three-wire power cord with good grounding performance (ground voltage $\leq 5V$).
- The instrument should be powered by a properly-grounded power supply no less than 1500VA, and the mains socket used to connect the instrument should have its load no less than 10A.
- Keep the instrument less than 2.5m away from the mains socket.
- If you are going to use a UPS to power the instrument, make sure that the UPS can provide power supply greater than or equal to 1875VA.



WARNING

Make sure the power socket is grounded correctly. Improper grounding may lead to electric shock or equipment damage.

Measurement of ground voltage:

Set the multimeter to the AC 250V scale, connect the black probe to the earth wire and the red probe to the live wire and neutral wire. If grounding is proper, the voltage between earth wire and neutral wire should be less than 5V, and the voltage between earth wire and live wire should be similar to that of live wire and neutral wire.

Check if the power socket outputs voltage meeting the specified requirements and has a proper fuse installed.

3.2.3 Water Supply and Drainage

- Water quality: meeting requirements of CLSI type II with specific resistance no less than 1 ($M\Omega \cdot cm @ 25^{\circ}C$) and silicate less than 0.1mg/L.
- Water supply pressure: 95-392kPa. Use a water supply module if the water unit does not meet the requirements of water supply pressure.

3 Installation Requirements and Procedure

- Flow: no less than 38L/H for continuous flow, and 2L/M for transient peak flow.
- Water temperature: 5-32°C.
- When the instrument uses water supplied by a water unit,
 - Make sure that the tube connecting the water unit outlet with the instrument inlet is no longer than 10m.
 - The water tube provided by the instrument is ID4mm OD6mm PU tube, which should be installed by the water unit supplier.
- When the instrument uses water supplied by a water supply module(optional),
 - Lay the water supply module between the deionized water tank and the instrument.
 - Make sure that the tube connecting the deionized water tank with the instrument inlet is no longer than 10m.
 - Choose a deionized water tank (with cap) with capacity of 80-120L.
 - Lay the water supply module on the ground.
- Make the instrument discharges low-concentration waste to a sewer,
 - Low-concentration waste tube is shorter than 5m.
 - Make sure that the low-concentration waste outlet is no less than 50mm wide.
 - Make sure that the low-concentration waste outlet is no higher than 100mm.
- The analyzer discharges low-concentration waste through the drainage module(optional).
 - The drainage module must be placed on the ground, otherwise low-concentration waste may overflow.
 - The tube connecting the low-concentration waste outlet and the drainage module inlet should be no longer than 5m.
 - the tube connecting the drainage module outlet and the waste outlet should be no longer than 10m.
 - Make sure that the low-concentration waste outlet is no less than 50mm wide.
 - Make sure that the low-concentration waste outlet connected to the drainage module is no higher than 1.2m.



BIOHAZARD

Dispose of waste liquid according to the local regulations.



CAUTION

1. The supplied water must meet the requirements of CLSI type II; otherwise insufficiently-purified water may result in misleading test results.
 2. The drainage module must be placed on the ground; otherwise low-concentration waste may overflow.
-

3.3 Computer Configuration (for International Customers)

3.3.1 CPU

At least Intel Dual Core, P4 2.6GHz

3.3.2 Memory

At least 2GB for each RAM

3.3.3 Network Adapter

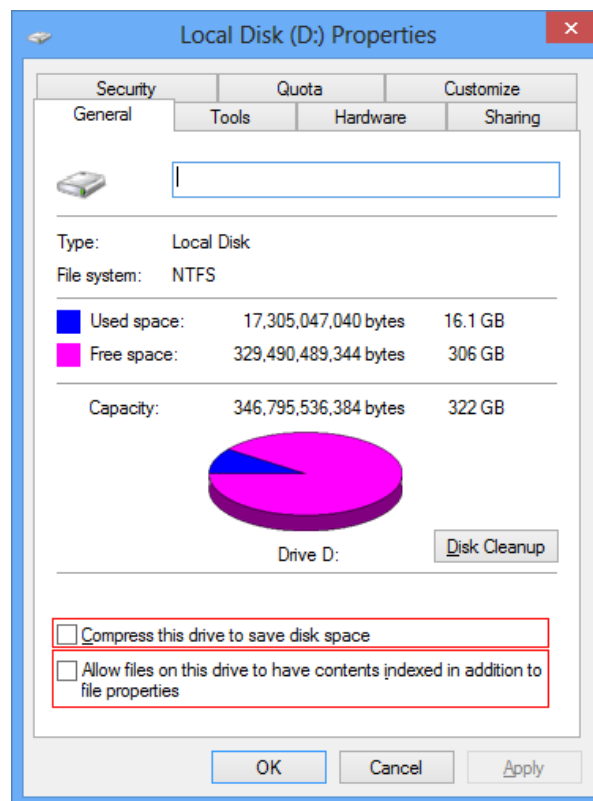
The computer is connected to the analyzer through a network adapter. If you are going to connect the computer with the LIS or Internet, you should prepare another network adapter (Intel gigabit network adapter)

3.3.4 Serial Port

The computer should provide an RS232 serial port, which is used to connect it with the analyzer.

3.3.5 Hard Disk Defragment

Install the operating system in the C drive and the operating software of the instrument in the D drive. Make sure that the C drive is over 30G and D drive over 100G, and the disk file system is of NTFS format. Deselect the two options at the bottom of the disk properties window: "Compress drive to save disk space" and "Allow Indexing Service to index this disk for fast file searching".



3 Installation Requirements and Procedure

Operating system

The operating software supports Microsoft Windows 8(64 bit).

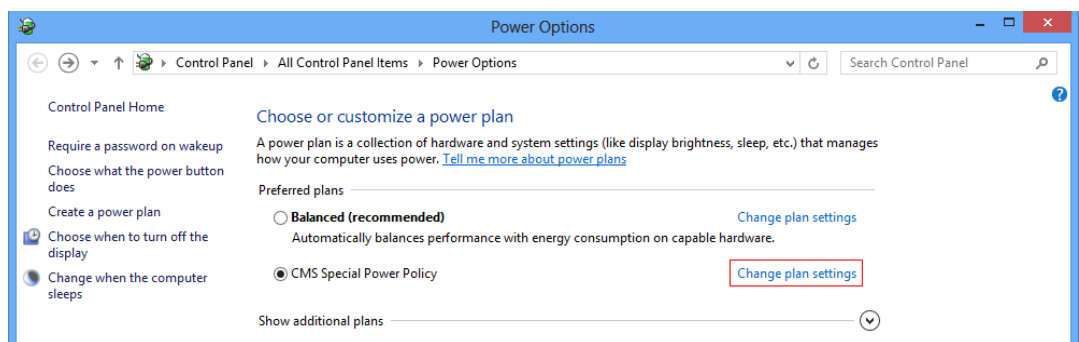
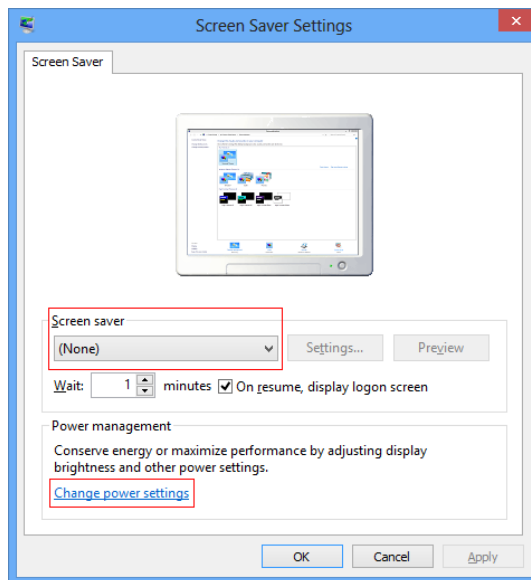
The default login user of the computer must belong to the Administrator group and has permissions for administrator.

Application Software

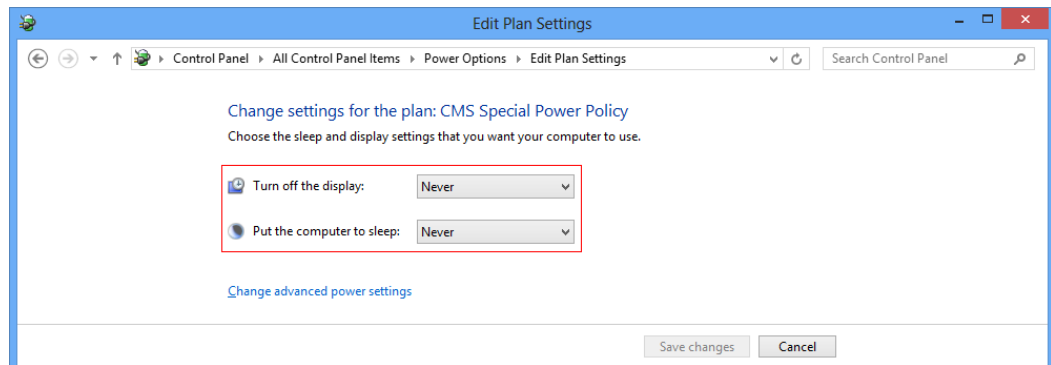
Except for the operating system, other application software must not be installed or reserved on the computer. If anti-virus application has been installed, remove the automatic scheduled scanning and add BS480 and BSLOG to the trust list.

3.3.6 Remove Screen Saver and System Standby

To remove the screen saver and system standby, click the right mouse button on the desktop, select Properties to display the Display Properties window. Select None in the Screen saver pull-down list box, and then select the Power button to go to the Power Options Properties window. Perform the settings as shown in the figures below.



Select the Hibernate page, and then deselect the Enable hibernation checkbox.

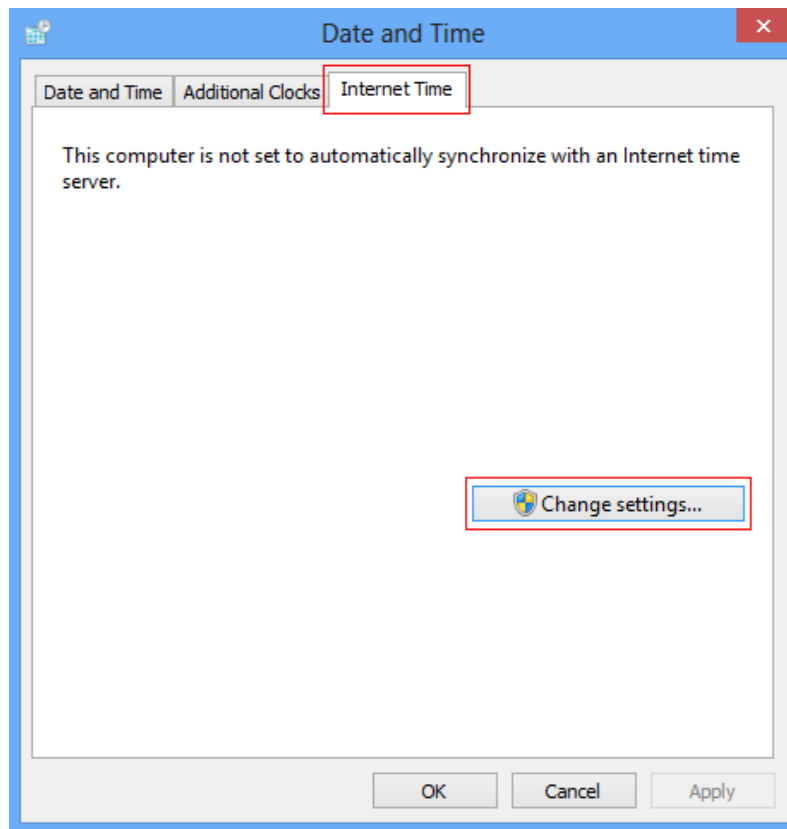


3.3.7 Screen Display Properties

Set the screen resolution as 1280*1024 pixels and color quality as Highest (32 bit).

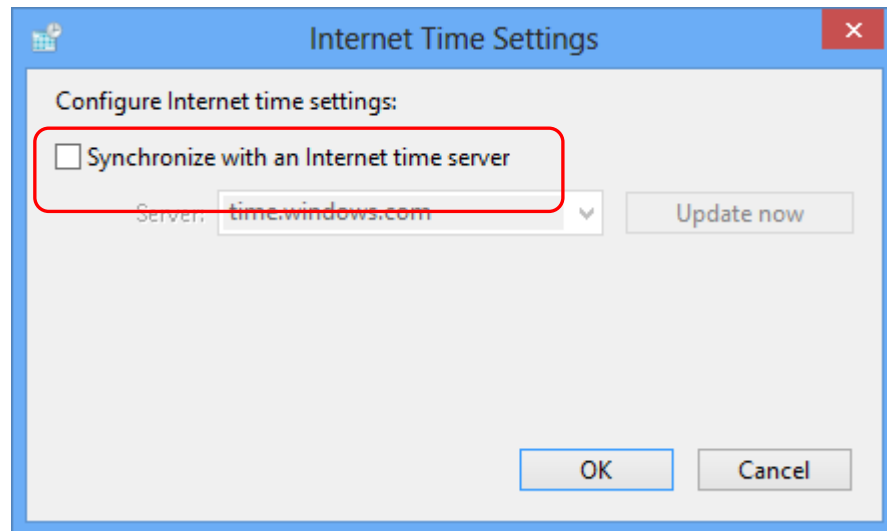
Disable Automatic Synchronization with Internet Time Server

Double click the time icon on the lower-right corner of the task bar. The Date and Time Properties window is displayed.



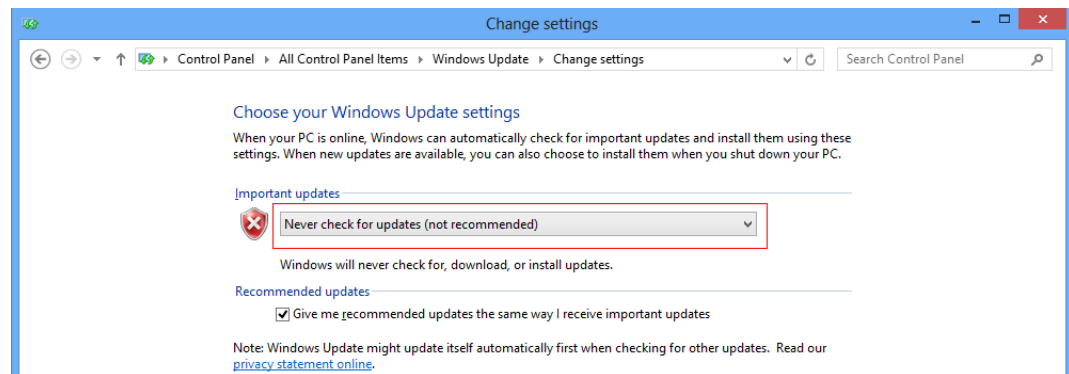
Select the Internet Time tab, and then deselect the checkbox in front of Automatically synchronize with an Internet time server.

3 Installation Requirements and Procedure



Turn off Automatic Updates

Turn off the automatic updates on the Automatic Updates window.



3.4 Installation Procedure

3.4.1 Unpacking

- 1 Before unpacking, check the packing list and see if the following items are included: chemistry analyzer, accessory kit, high-concentration waste tank, wash solution, ISE module (accessory kit, optional), water supply module (optional), drainage module (optional).
- 2 The chemistry analyzer is about 364kg. Unload it from the truck with a forklift.
- 3 Use an adjustable wrench to loosen the retaining screws on the upper cover and slope cover of the package, remove the two covers and keep 2.5m space in front of the slope cover (1.3m for the slope cover and 1.2m for the instrument).

Figure 3.2 Remove screws on the upper cover and slope cover of the package



Figure 3.3 Remove a side cover from the package



- 4 Unscrew the anchors, and ensure the four roller wheels are not contacting the mounting plates. Use an adjustable wrench to remove the four stationary barriers of the analyzer and all screws on the steel angles, and then remove the steel angles and screw up the anchors.

3 Installation Requirements and Procedure

Figure 3.4 Screw down the anchors and tighten them to support the analyzer



Figure 3.5 Remove the mounting plate, and then screw up the anchors carefully.



- 5** A slope cover is fixed on the side cover. Loosen the two screws on the slope cover to remove it.

Figure 3.6 Remove the slope cover



- 6** Fix the slope cover in proper direction on the bottom cover, and push the analyzer along the slope cover to the ground. Store all of the wooden boards and retaining screws properly so that they can be used when the instrument is transported.

Figure 3.7 Align the slope cover with the analyzer



- 7** After removing the analyzer from the package, screw down the four anchor screws, lift the roller wheels slightly from the ground and make sure all the four anchors steadily support the analyzer.

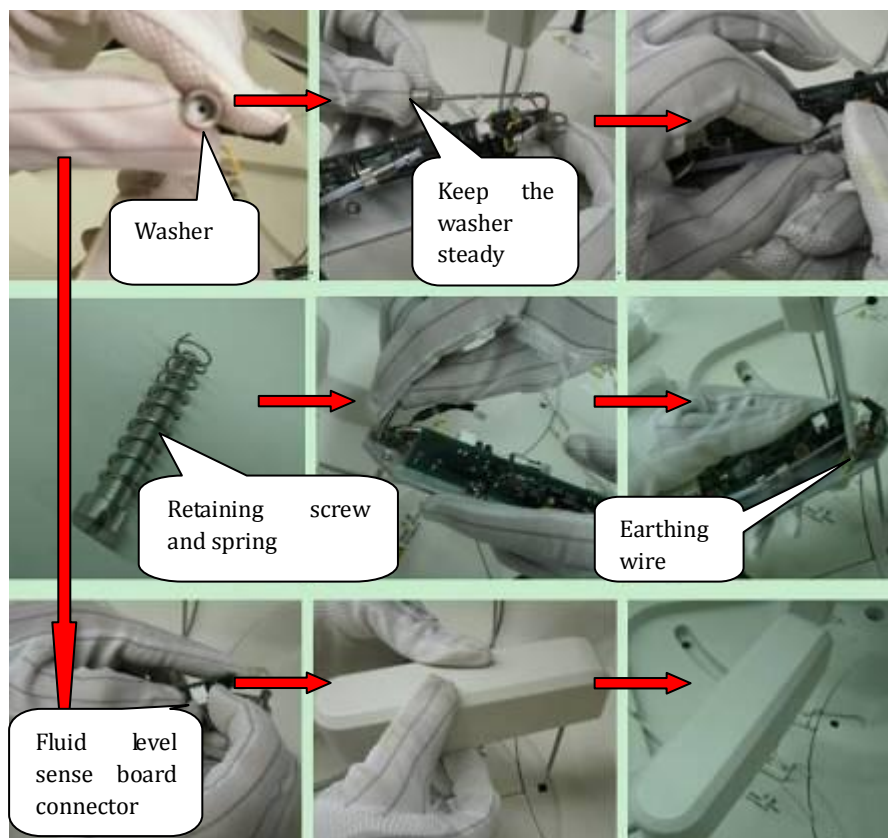
3.4.2 Installation

Remove the packaging material and plastics on the analyzer panels. Take out the sample probe and reagent probe from the accessory kit and insert them into the corresponding metal connectors. Note not to forget the white washers in the package.



Installation of Probes

Figure 3.8 Installation procedure of probes



- 1 Before installing the probe, remove the probe arm cover. Use a cross screwdriver to loosen the two screws on the arm cover, and remove the arm cover by lifting it up.
- 2 Insert the probe downwards into the hole on the probe arm while aligning the screw hole on the probe plate to the rod inside the arm.
- 3 Install the washer in the tube connector. To replace the washer, remove the old one from the tube connector and install the new one. Connect the tube connector to the probe and then tighten it.
- 4 Fix the earth wire of the probe to the earthing terminal inside the arm, and connect the probe connector to the fluid level detection board.
- 5 Sleeve the spring on the rod and tighten the retaining screw. Pay attention to the spring direction and make the thread opening face downwards.
- 6 Pinch the probe by the part near the probe arm, push the probe upwards and then release it to check if the spring works well. If it does, proceed to the next step. If not, check if the spring is clamped or fixed too tightly.
- 7 After installation, power on the analyzer, and check if the No.D2 LED indicator on the circuit board inside the probe arm is lit. If it is, it indicates that the level detection system works normally.
- 8 Install the arm cover, and use the cross screwdriver to tighten the two screws on it.
- 9 Pinch the probe by the part near the probe arm. Push the probe upwards and then release it to check if the spring works well. If it does, proceed to the next step. If not, it indicates that the arm cover is not installed correctly. Reinstall the arm cover and check the spring until it can move freely.

Installation of Mixers

- 1 Take out the mixer from the accessory kit and lift the mixer arm to the vertical extreme position.



- 2 Screw the mixer into the large hole of the retaining screw till it is level with the small hole end.

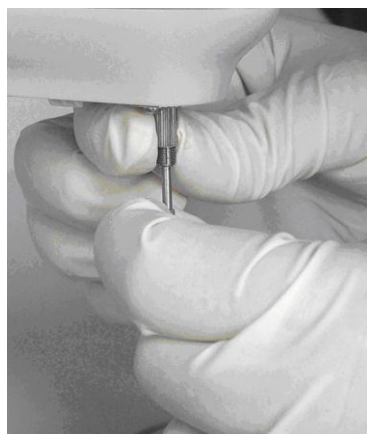


- 3 Insert the mixer motor shaft into the mounting hole and make sure they contact closely and there is no gap in the vertical direction.



- 4 Tighten the retaining screw of the mixer clockwise and check whether the mixer is perpendicular. If not, reinstall the mixer.

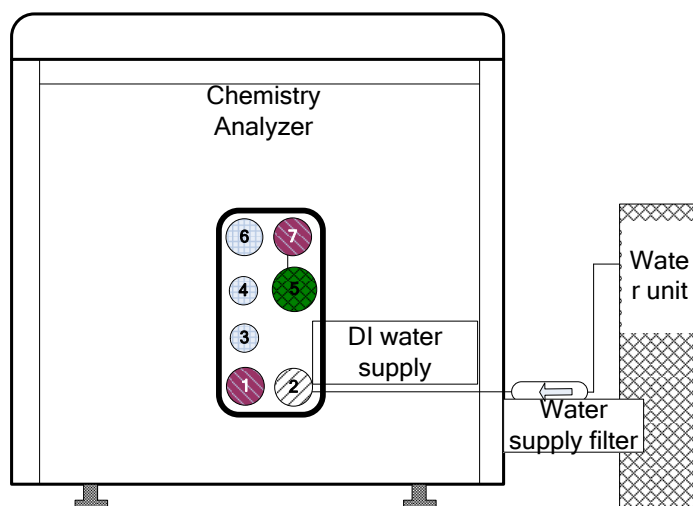
3 Installation Requirements and Procedure



3.4.3 Fluidic Connection

Connect BS-480 with Water Unit

Figure 3.9 Connect BS-480 with Water Unit



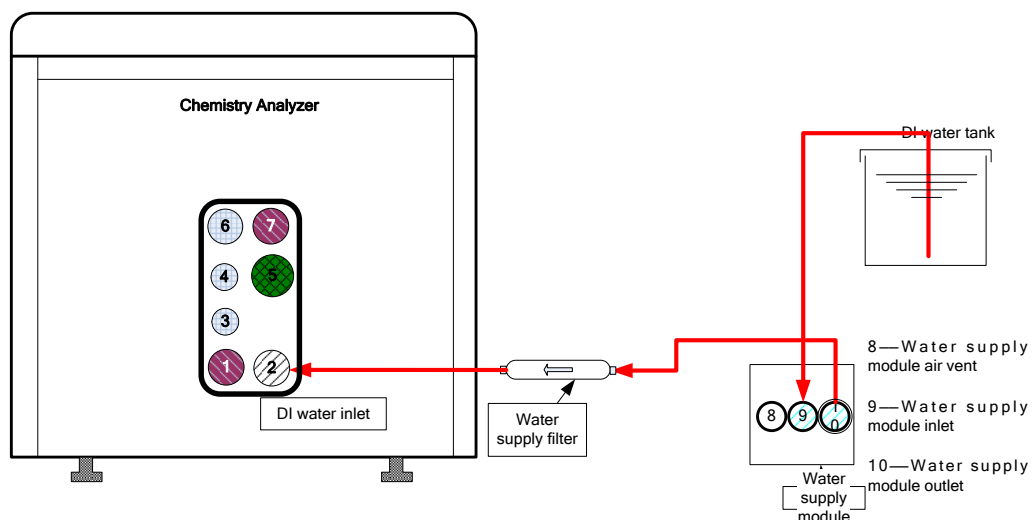
The water tube provided with the instrument is ID4mm OD6mm PU tube, which should be installed by the water unit supplier as shown in the above figure.

Requirements for connecting the analyzer and the water unit:

- 1 Use the PU tube in the accessory kit to connect the analyzer with the water unit. Install the water inlet filter between the water unit outlet and the analyzer inlet and make sure that the filter will not be influenced by external force and is convenient for replacement.
- 2 Make sure that the arrow on the filter shell that indicates the water flow points at the analyzer inlet.
- 3 Ensure that the water inlet tube between the water unit and the analyzer is less than 10m long. When installing the tube, prevent it from being bent, twisted, or pressed, and keep it away from any sharp-edged objects or others that may scratch the tube.
- 4 If the tube is not installed correctly, remove it, cut off the part that has contacted the connectors, and then reinstall the tube properly.

Connect Analyzer with Water Supply Module (optional)

Figure 3.10 Connect BS-480 with Water Supply Module

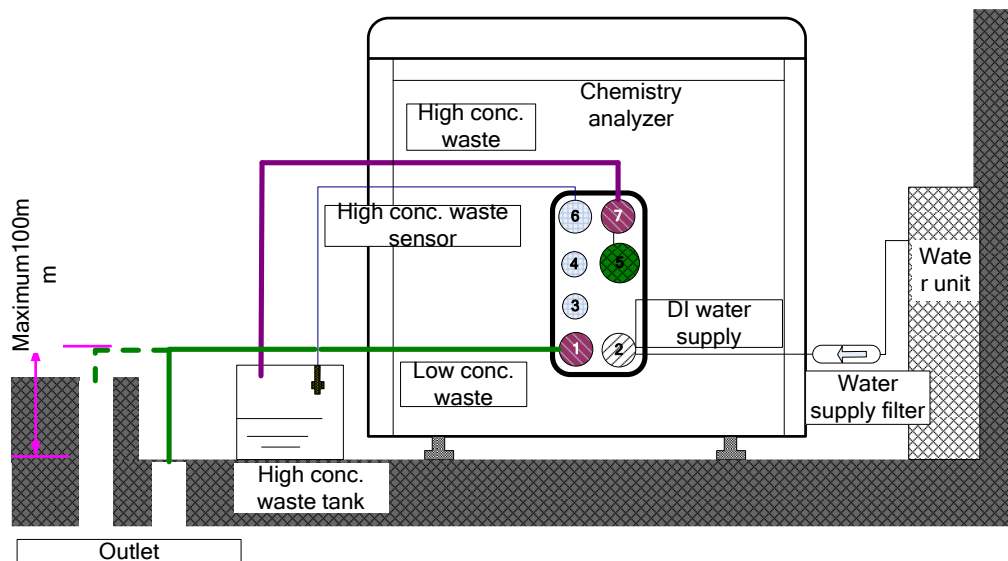


Requirements for connecting the analyzer and the water supply module:

- 1 Use the PU tube in the accessory kit to connect the analyzer with the water supply module. Install the water supply module between the deionized water tank and the analyzer inlet. Install the water inlet filter between the water supply module outlet and the analyzer inlet, and make sure that the filter will not be influenced by external force and is convenient for replacement.
- 2 Install the water supply module filter at the end of the inlet tube and then place the tube at the bottom of the deionized water tank.
- 3 Make sure that the arrow on the filter shell that indicates the water flow points at the analyzer inlet.
- 4 Ensure that the water inlet tube between the deionized water tank and the analyzer inlet is less than 10m long. When installing the tube, prevent it from being bent, twisted, or pressed, and keep it away from any sharp-edged objects or others that may scratch the tube.
- 5 Connect the air vent of the water supply module to the sewer and keep the tube less than 10m long.
- 6 If the tube is not installed correctly, remove it, cut off the part that has contacted the connectors, and then reinstall the tube properly.

Connect Analyzer with Waste Drainage Facilities

Figure 3.11 Connect BS-480 with Waste Drainage Facilities

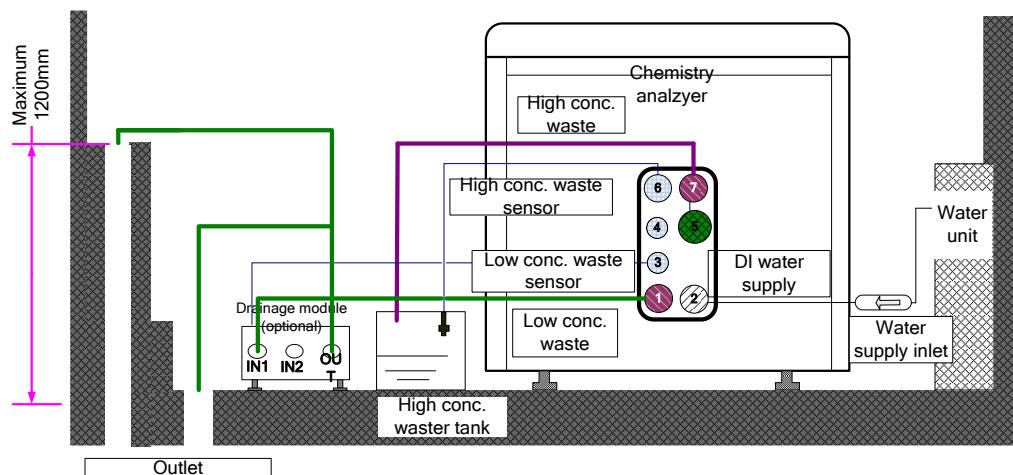


Requirements for connecting the waste drainage facilities:

- 1 Lay the high-/low-concentration waste tubes separately.
- 2 Connect the high-concentration waste tube and sensor (high-concentration waste tank cap assembly) to the high-concentration waste tank.
- 3 Insert the low-concentration waste tube into the sewer, and make sure that the tube is less than 5m long.
- 4 Make sure that the low-concentration waste outlet is no less than 50mm wide and less than 100mm high.
- 5 To prevent leaks, fix all tubes and connectors properly with the tube clamps in the accessory kit.
- 6 When installing the tubes, prevent them from being bent, twisted, or pressed, and keep them away from any sharp-edged objects or others that may scratch the tubes.
- 7 Ensure that all electric connectors are higher than and away from fluidic parts and prevent fluidic connectors from facing electric components.

Connect Analyzer with Drainage Module (optional)

Figure 3.12 Connect BS-480 with Drainage Module

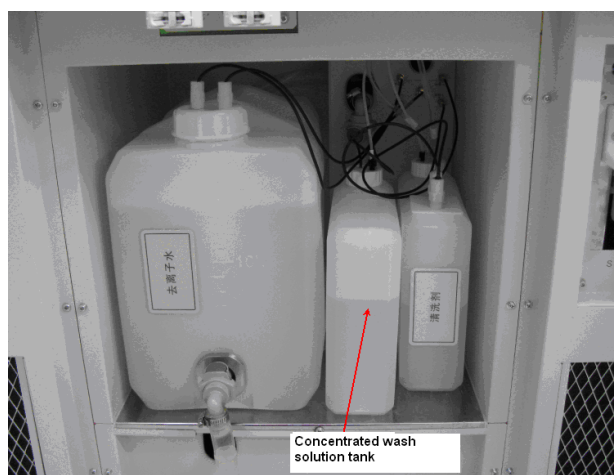


Requirements for connecting the BS-480 with the drainage module:

- 1 Lay the high-/low-concentration waste tubes separately.
- 2 Connect the high-concentration waste tube and sensor (high-concentration waste tank cap assembly) to the high-concentration waste tank.
- 3 Connect the low-concentration waste outlet to the inlet of the drainage module with a tube no more than 5m long; and then connect the low-concentration waste sensor to the drainage module.
- 4 Connect the drainage module outlet to the low-concentration waste outlet with a tube no more than 10m long.
- 5 Make sure that the low-concentration waste outlet is no less than 50mm wide and less than 1.2m high.
- 6 To prevent leaks, fix all tubes and connectors properly with the tube clamps in the accessory kit.
- 7 When installing the tubes, prevent them from being bent, twisted, or pressed, and keep them away from any sharp-edged objects or others that may scratch the tubes.
- 8 Ensure that all electric connectors are higher than and away from fluidic parts and prevent fluidic connectors from facing electric components.

3.4.4 Connection of Analyzer

- 1 Prepare new concentrated wash solution and place it in the front cabinet of the analyzer. Make sure it is capped properly.



- 2 If you are using a UPS, install it and make it properly grounded.
- 3 Connect the analyzer, computer, printer, monitor, water unit (or water supply module), power cord and data cable. Do not switch on the power before finishing all connections.
- 4 Connect the computer with the analyzer with the serial port cable, and choose the COM1 port for the computer.

3.4.5 Starting up the System

- 1 Turn on the power switch of the water unit or the water supply module (optional) and the water drainage module (optional).
- 2 Turn on the analyzing unit power switch and the main power switch. The water inlet valve in the back of the analyzer is turned on automatically to supply water for the water tank. In normal conditions, the water tank can be filled full in 20-30 minutes.

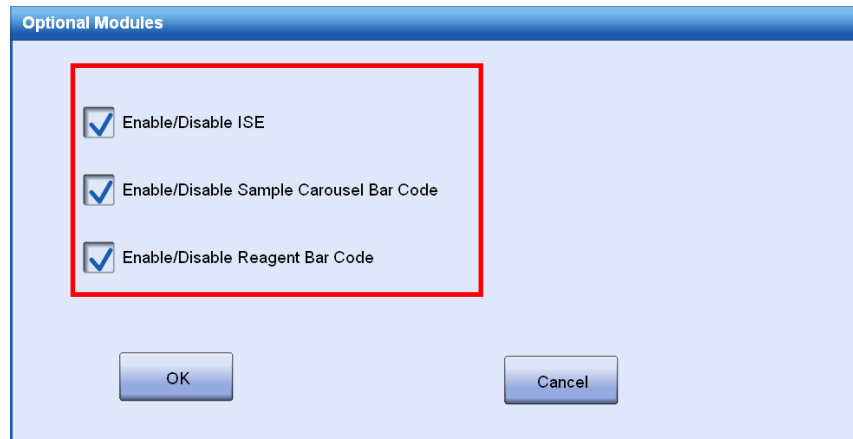
3 Installation Requirements and Procedure

- 3 Start the computer. The BS-480 operating software is run automatically when the Windows is started. You are also allowed to run the operating software by selecting the shortcut icon on the desktop or in the **Start** menu. See the figure below. The system skips the initialization procedure when detecting that the fluidic prime is not performed. Enter the username "ServiceUser" and password "#BS8A#SEU" in the login window.

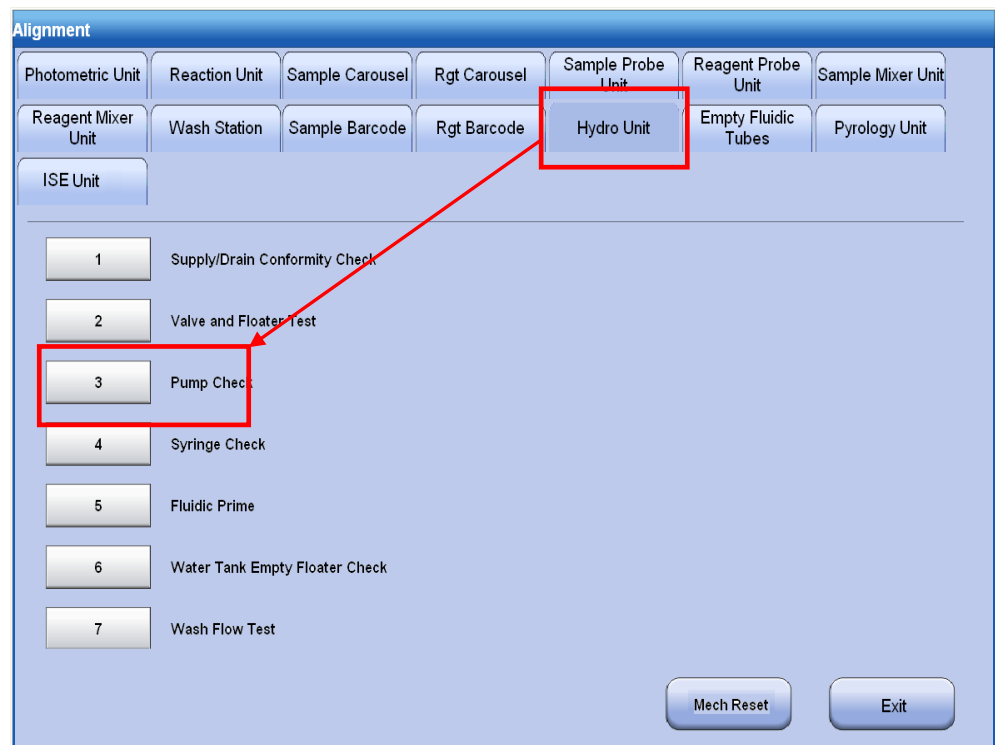


- 4 When the main screen is shown, select **Utility** -> **System Setup** -> **Factory F2**. Select **1 Optional Modules**, enable/disable ISE module, sample bar code and reagent bar code, and then select **OK**. See the figure below:

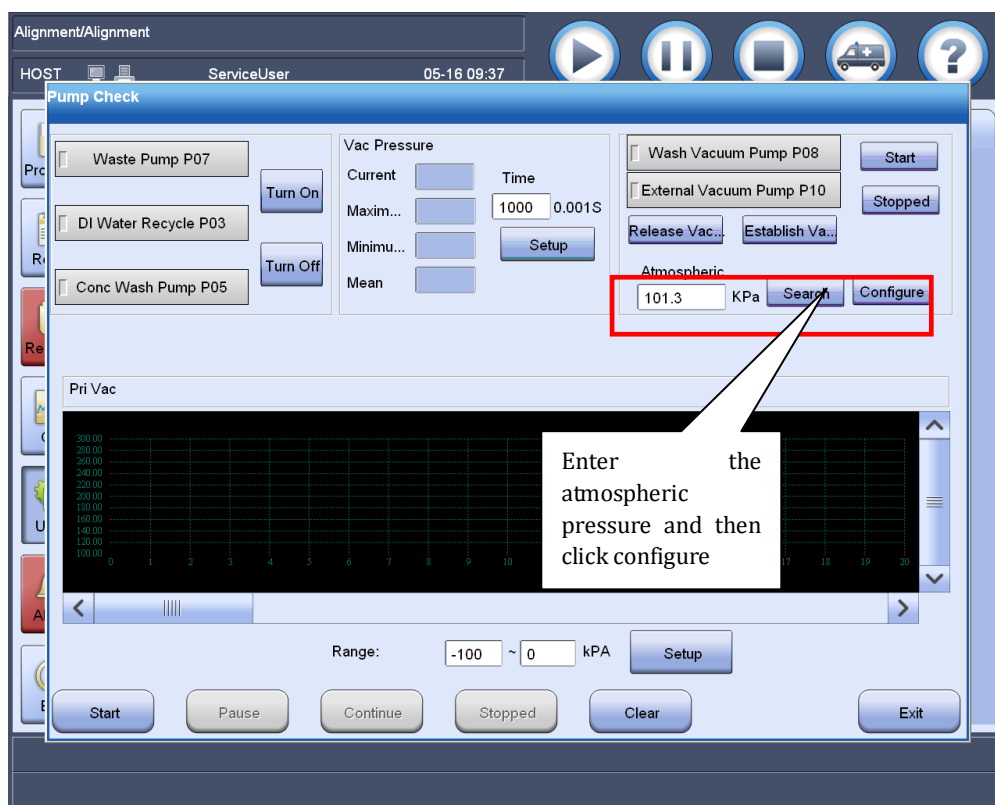
3 Installation Requirements and Procedure



- 5 To set the local atmospheric pressure, select **Utility -> Maintenance -> Alignment -> Hydro Unit**, select the **Pump Check** button, enter the atmospheric pressure as indicated in the figure below, and then select **Config** to save the input. Check the altitude of the installation site of the analyzer, and then find the corresponding atmospheric pressure.



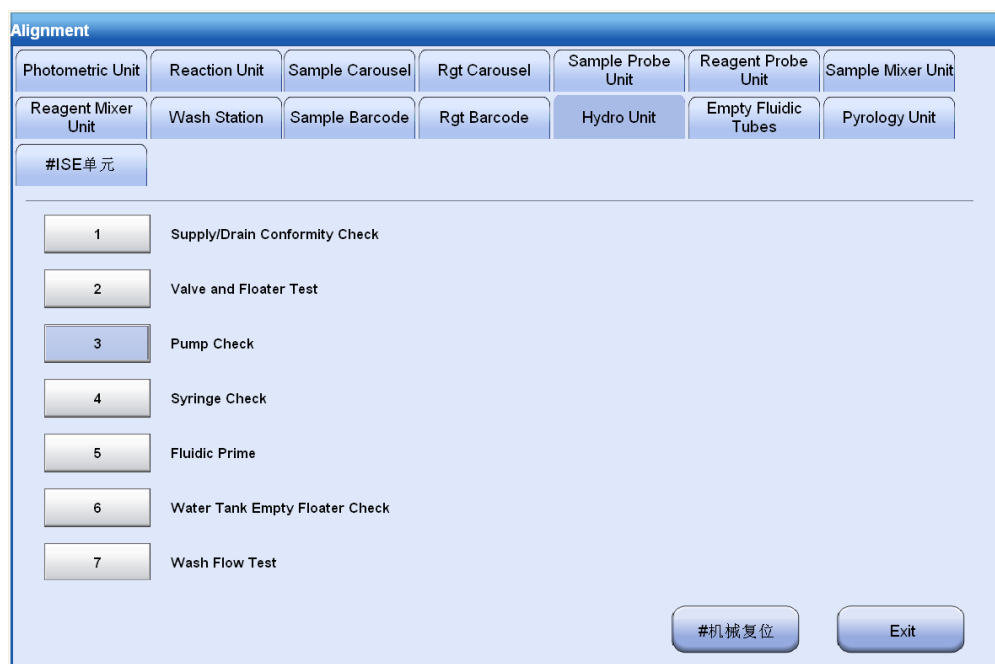
3 Installation Requirements and Procedure



- 6 Perform the Fluidic Prime procedure in 3.4.6 Fluidic Prime.

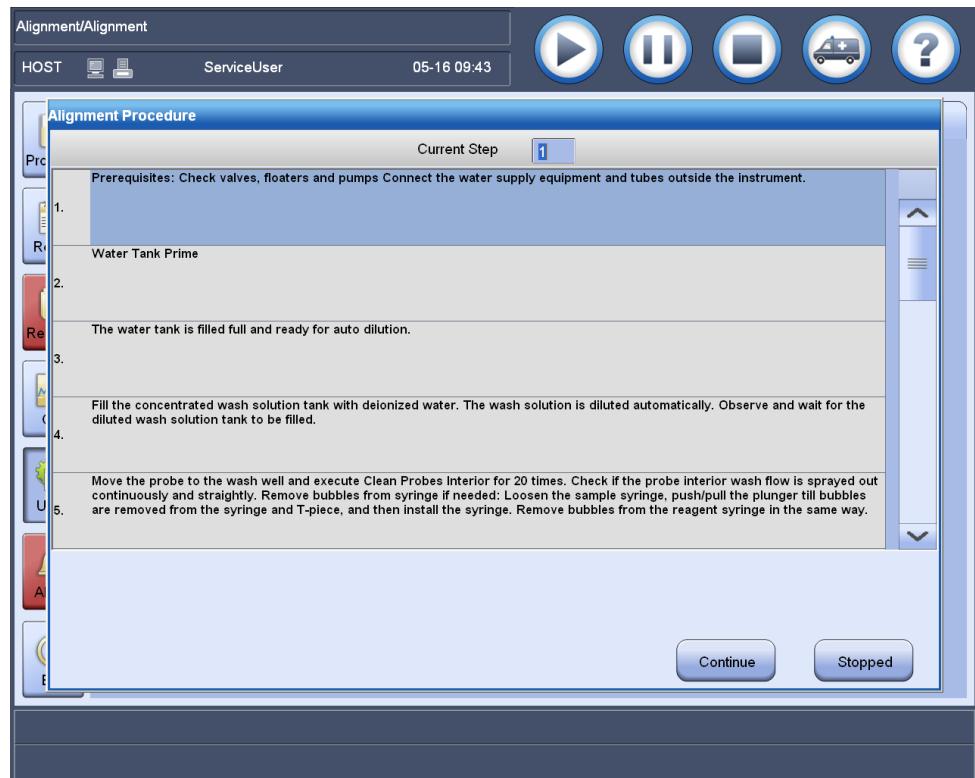
3.4.6 Fluidic Prime

- 1 Select **Utility -> Maintenance -> Alignment -> Hydro Unit**. The screen is displayed as follows.



- 2 Select **5 Fluidic Prime** to display the following window. The check of solenoid valves, fluid level floaters, pumps, and syringes has been finished by default. Select Continue to proceed to the next step.

3 Installation Requirements and Procedure

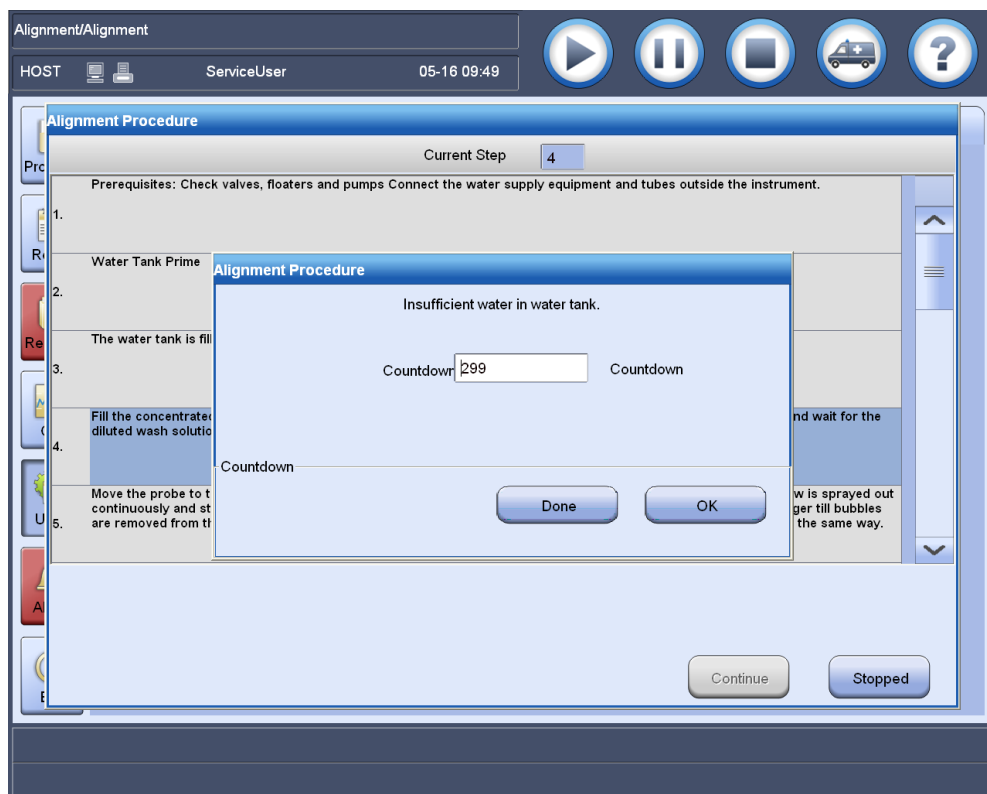


- 3 Select **Continue** and then select Start to prime the water tank. When the priming is finished, select OK.



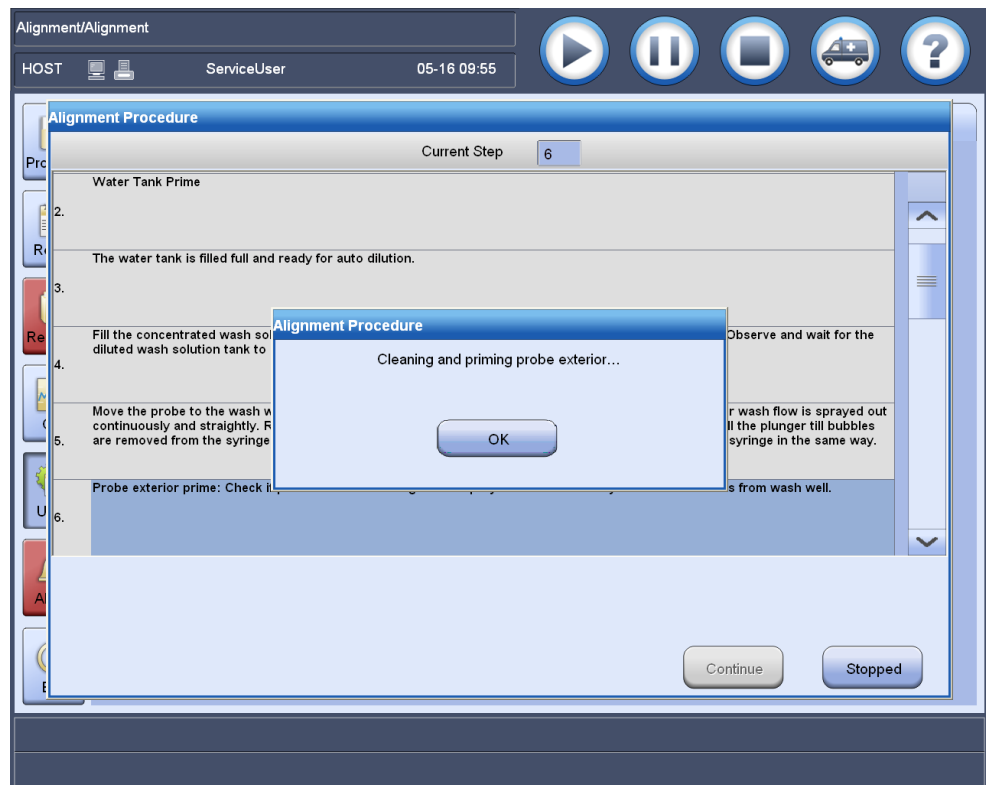
3 Installation Requirements and Procedure

- 4 Select **Continue** to prime the system with diluted wash solution. This step will take about 5 minutes as shown in the following figure. The DI water pump P03 is turned on during the process. Observe the water pressure gauge and make sure that the water supply pressure is within 34-40kPa. When the auto dilution is finished (The diluted wash solution container is full or the wash solution is sufficient for the test.), select **Done** and then select **Continue**.



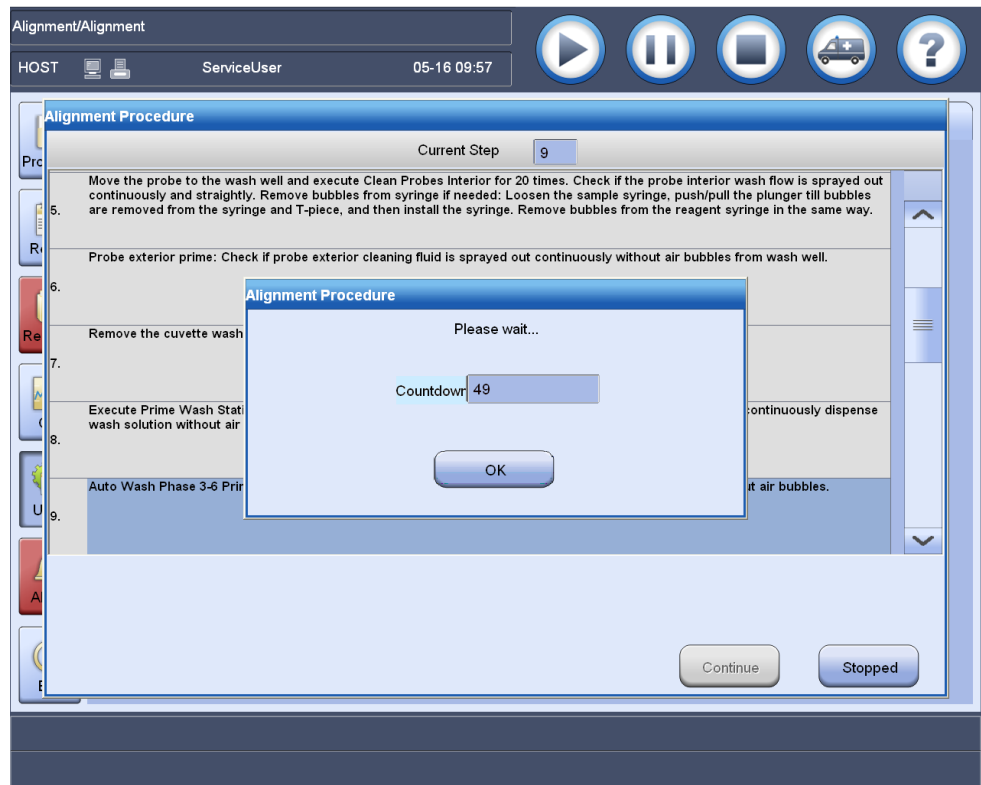
- 5 Select **Continue** and loosen the retaining screws and sample/reagent syringes. Select **Start** to prime the probe interior for 20 times. Check if the cleaning liquid is sprayed out straightly from the probes. Pull and push the syringe plunger until there is no air bubbles in the syringe. Install the syringes back and make the V-shape slot level to scale 11 of the sample syringe and scale 15 of the reagent syringe. Select **OK** button to exit.
- 6 Select **Continue** to prime the probe exterior and check if the cleaning liquid is sprayed out continuously without bubbles. Select **OK** and **Continue** to move to the next step.

3 Installation Requirements and Procedure

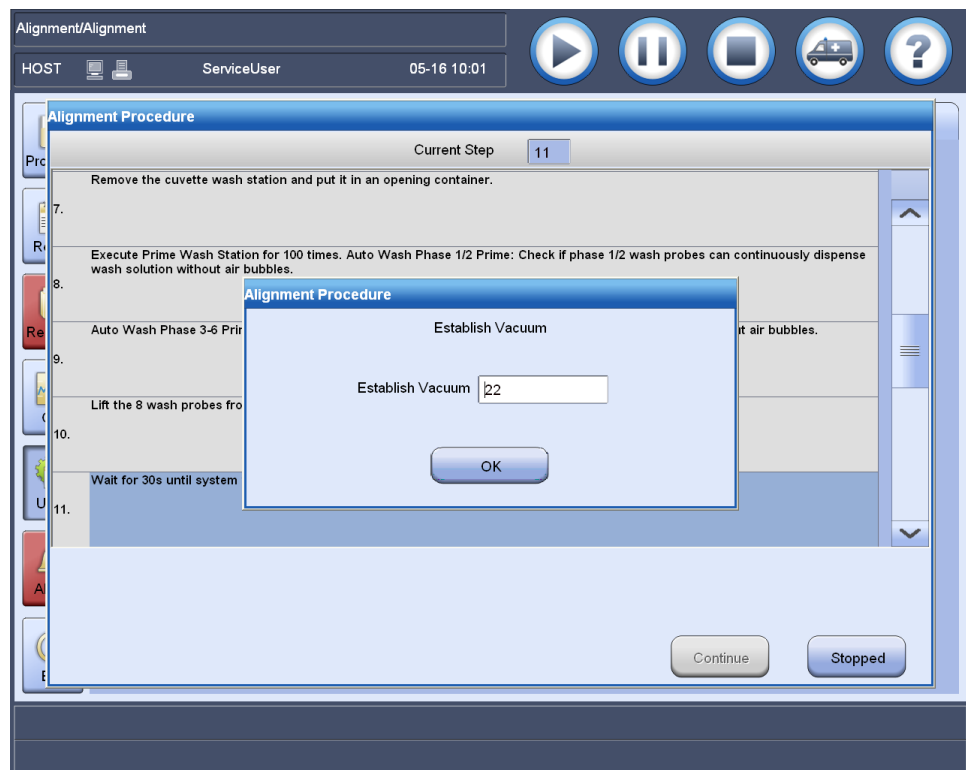


- 7 Follow the screen instructions, remove the cuvette wash station and put it in an opening container. (This step is important, otherwise the cuvettes may overflow.) Select **Continue** and Start to perform the phase 1-2 probe priming. When the wash probes for Phase 1-2 can spray out wash solution continuously without air bubbles, select **Stop** to exit.
- 8 Select **Continue** to perform the phase 3-6 probe priming. When the wash probes for Phase 3-6 can spray out wash solution continuously without air bubbles, select **Stop** to exit.

3 Installation Requirements and Procedure



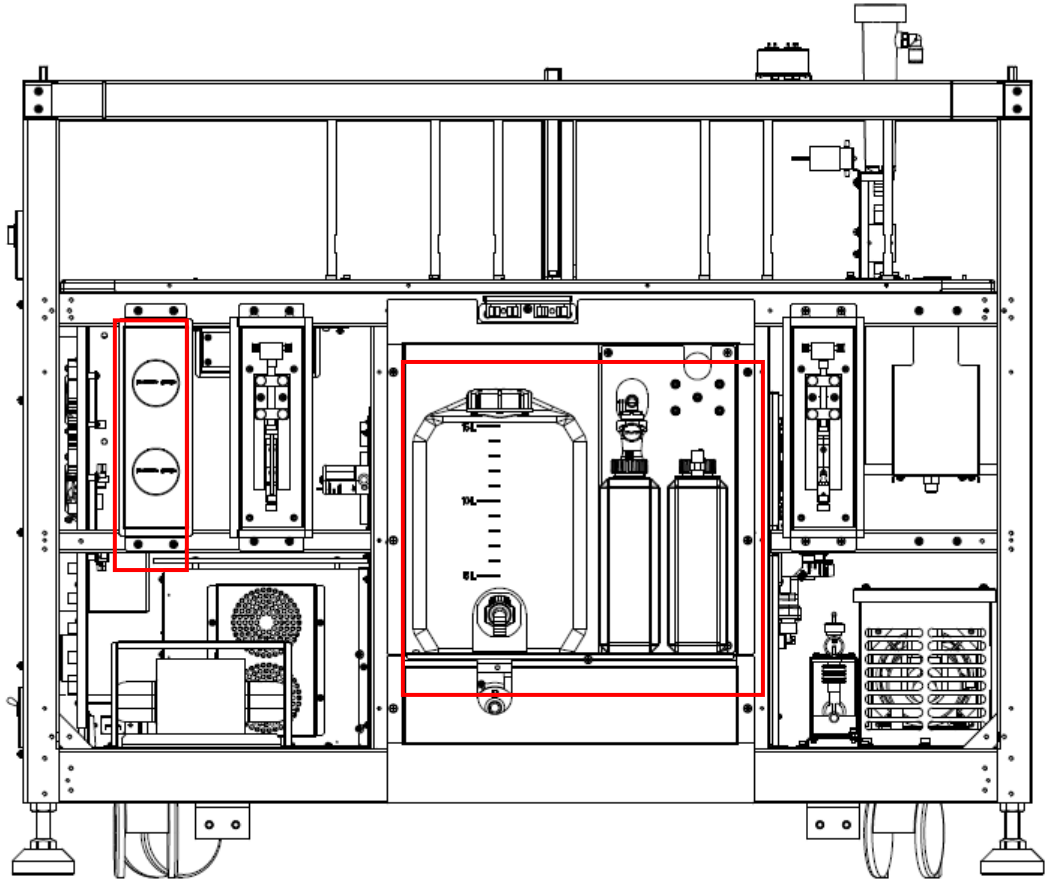
- 9 Follow the screen instructions to lift the wash probes from the fluid level and select **Continue**. After vacuum is created, check the primary vacuum pressure value. Select **Continue**, and touch the wash probes to feel suction force.



- 10 Reinstall the cuvette wash station in step 13.
- 11 The cuvettes are cleaned for 10 times in step 14. Check that there is no liquid overflowing the cuvettes and the fluid level in cuvettes is 5-10mm from the cuvette opening.

- 12 The system is homed for 3 times in Step 15. There should be no failure alarms, and the containers and pressure gauges in correct status as indicated in the following figure.
- 13 The fluidic prime is finished.

Figure 3.13 Container status and pressure gauges



3.4.7 Initial Maintenance Logs

Select **Utility** -> **Maintenance** -> **Maintenance** to display the maintenance window. Confirm all scheduled maintenance procedures and set up start time for them. The maintenance frequencies include Daily, Weekly, Two-Week, Monthly, Three-Month, Six-Month, and Other. Select the Select All button on each scheduled maintenance tab page, and then select **OK**. The current date and time appears in the Date Performed column of each maintenance procedure.

3 Installation Requirements and Procedure

Maintenance

Scheduled Maintenance | Biochemistry Maintenance | ISE Maintenance | Engineer

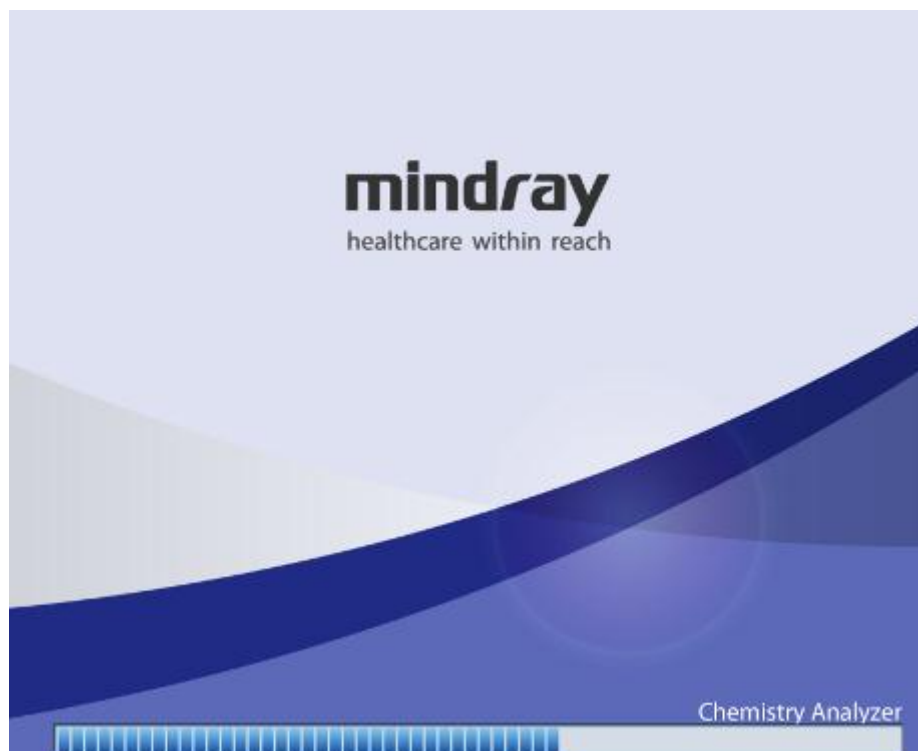
Daily | **Weekly** | Two-week | Monthly | Three-Month | Six-Month | Other

Procedure	Select	Property	Operator	Date Performed
Check DI Water Connection	☐	System		
Check Waste Tank Connection	☐	System		
Check Concentrated/Diluted Wash Solution	☐	System		
Check Sample/Reagent Syringes	☐	System		
Check Probes/Mixers/Wash Wells	☐	System		
Check Sample Probe Wash Solution	☐	System		
Clean Electrode Tubes	☐	System		
	☐			
	☐			
	☐			
	☐			

Select All | OK | Log | History | Customize | Delete | Exit

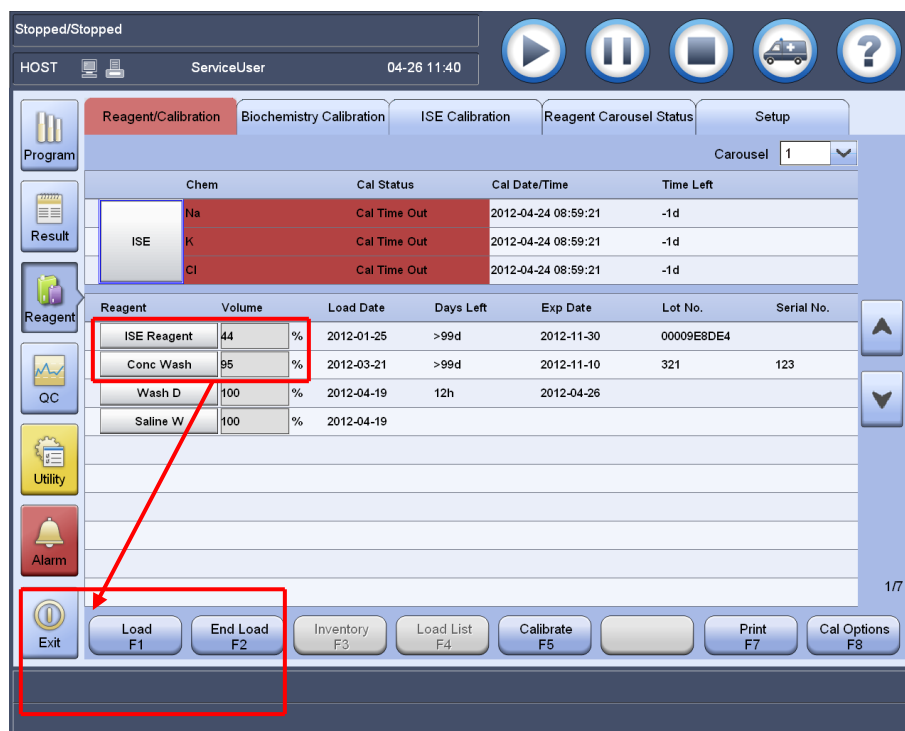
3.4.8 Startup Initialization

When the fluidic prime is finished, exit the operating software and reboot the computer. The operating software is automatically run and starts initializing. When the login window is displayed, enter the user name "ServiceUser" and password "#BS8A#SEU".





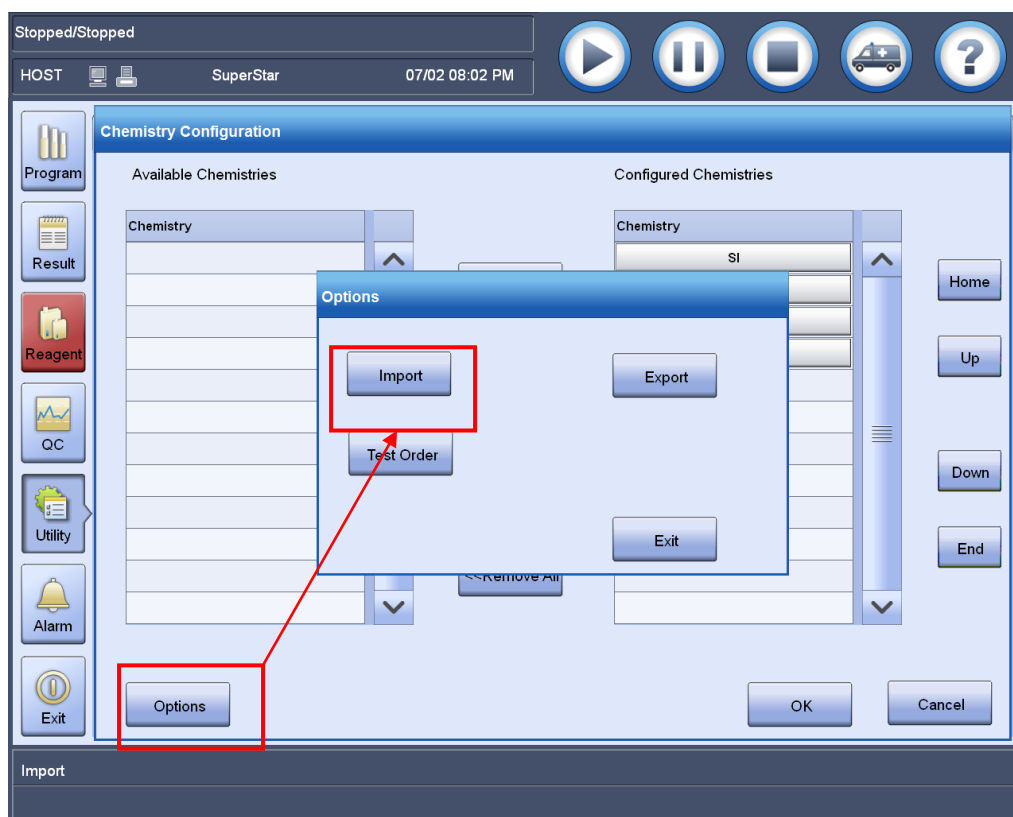
When the main screen is shown, select **Rgts/Cal -> Reagent/Calibration**, and then load concentrated wash solution and ISE reagent through the **Load** option.



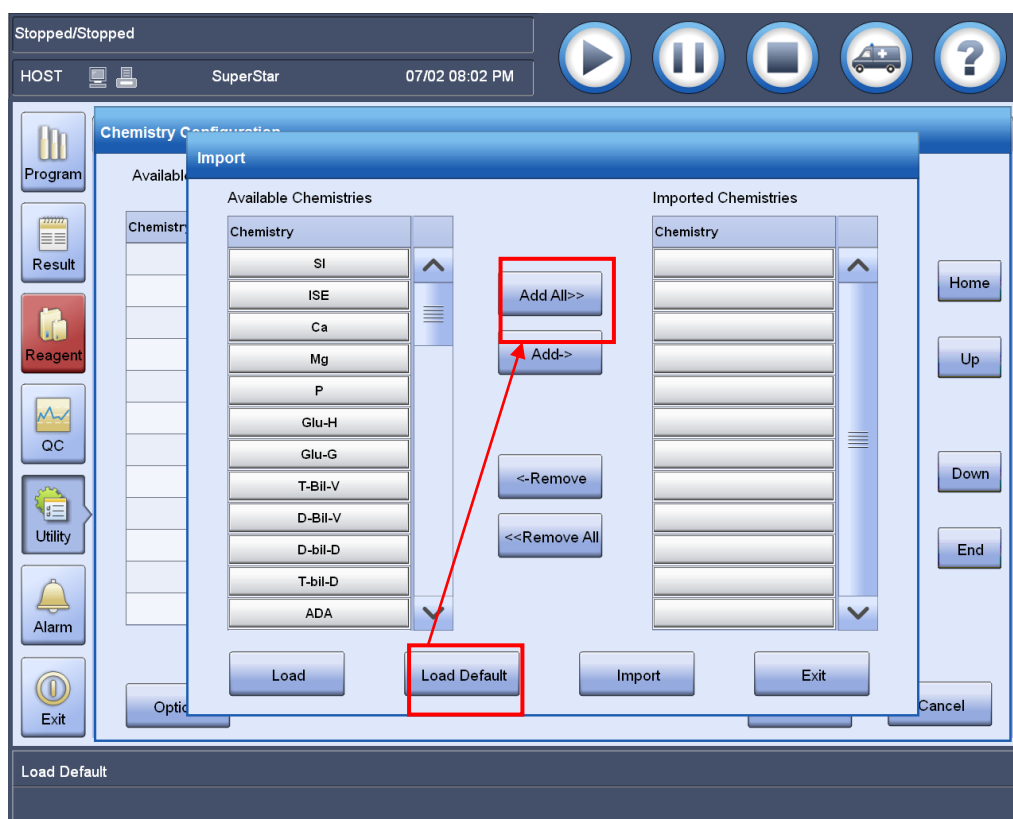
3.4.9 Importing and Configuring Chemistry Parameters

Select **Utility -> Chemistries -> Config**, and select **Options** to display the **Option** window. Select **Import** to display the **Import** window.

3 Installation Requirements and Procedure



Select **Load Default** to display all Mindray reagent chemistries in the left column. Select **Add All** to add all chemistries to the right column. Select **Exit** to close the window.



All Mindray reagent chemistries are displayed in the **Available Chemistries** column. You can delete chemistries that will not be used in your laboratory.



Select **OK** to complete the chemistry configuration.

3.4.10 Basic performance

Running test with DI water

Before test:

1. The mechanical parts of the analyzer are normal and no alarm is given (including photometric system, temperature control, mechanical system and analyzer communication).
2. The probes are normally washed and no liquid is adhered to the probe tip.
3. The wash station lowers to the cuvette bottom every time, it is pushed up.

Set up the chemistry:

1. Set up a double reagent chemistry "Water" with chemistry parameters: Kinetic method, increased, pri wave 340nm, secondary wave 412nm, unit U/L decimal 0.01, reaction time 80-82, sample volume 5, reagent volume R1 120 and R2 10.
2. Select calibration setup-calibration rule and select water and K factor. Save the settings and exit.
3. Load the reagent.

Preparations:

1. Place DI water in R1 and R2 (more than 15ml)
2. Place DI water in sample position (more than 2ml)
3. Place concentrated wash solution in reagent carousel D position and sample carousel D2 position.

Note: wear gloves when loading the reagents.

3 Installation Requirements and Procedure

Start analysis:

Request test for Waster chemistry and repeat 90 replicates. Start the test and observe the working status of the parts. If alarm is given, troubleshoot the failure.

3.5 Stop use

3.5.1 Stop use in Short term

When to do

Stop use for 3 days

How to do

- 1 Select utility-alignment- Empty Fluidic Tubes
- 2 Empty the fluidic tubes according to the software prompts.
- 3 If ISE is configured, enter Utility-Maintenance-ISE. Empty the ISE tubes according to the software prompts. And then remove the ISE electrodes and inject proper amount of Calibrator A into the Na,K and Cl electrodes with Pipette. Use the tape to seal the electrodes before storing them. For Ref electrode, you should use a soft tube to insert into it.
- 4 Exit the software and shut down the PC. Turn off the analyzer.
- 5 If the analyzer is stored in another place, disconnect the cables and power cords among the analyzing unit, PC, monitor, water unit or water supply module and follow the reverse order in 3.4.3 to disconnect the tube connection of the analyzer.

3.5.2 Stop use in long term

When to do

Stop use for above 3 days to several years

How to do

- 1 Select utility-alignment- Empty Fluidic Tubes
- 2 Empty the fluidic tubes according to the software prompts.
- 3 If ISE is configured, enter Utility-Maintenance-ISE. Empty the ISE tubes according to the software prompts. And then remove the ISE electrodes and inject proper amount of Calibrator A into the Na, K and Cl electrodes with Pipette. Use the tape to seal the electrodes before storing them. For Ref electrode, you should use a soft tube to insert into it.
- 4 Exit the software and shut down the PC. Turn off the analyzer.
- 5 Follow the reverse order in 3.4.4 Connection of Analyzer to disconnect the analyzer and follow the reverse order in 3.4.3 to disconnect the tube connection of the analyzer.
- 6 Follow the reverse order in 3.4.2 to dismantle the analyzer .and place the probes and related components into the accessory kit to store them.
- 7 Follow the reverse order in 3.4.1 to pack the analyzing unit and put the water supply and drainage module into the boxes to store them.

3 Installation Requirements and Procedure

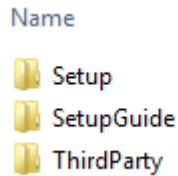
4

Software Introduction

4.1 Software installation

4.1.1 Installation package

There are three files in the installation package: Setup, SetupGuide, Thirdparty, as shown in the following figure:



- 1 Setup file: including Setup.exe and Kill blog used to terminate the Bslog program.

Introduction to Killbog: It is run when the operating software is started, used to record the software log. When the operating software is started, it will check if Killbog is started. If not, the software will give out an alarm and the startup will fail. After the operating software is successfully started, you can terminate the Killbog program and the software log will not be recorded.

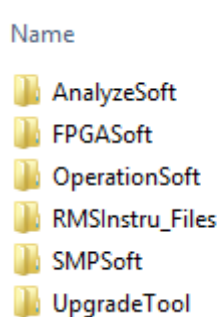
- 2 SetupGuide: including software installation guide and upgrade guide.

- 3 Thirdparty: including dotnetfx installation package and SQLExp installation package. The former includes the .net Framework program of Microsoft, which is the SQL running environment and the latter is the database program.

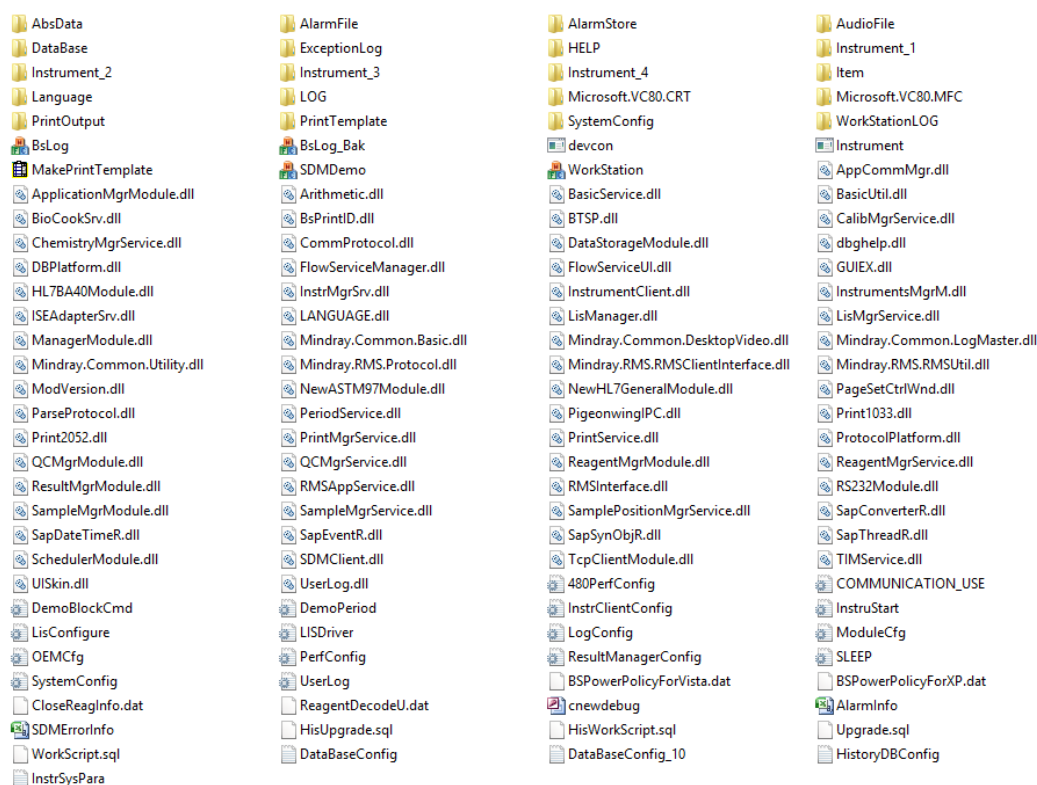
4.1.2 Folder structure

The default installation directory: D:\Mindray\Workstation.

After installation, the folder structure is as follows:



In which, the rather important folder is OperationSoft as displayed in the following figure:



The following files are displayed after the OperationSoft is unfolded.

AbsData: used to save the reaction curve data.

AlarmFile: used to save the alarm log.

Database: database for all parameters, results, calibration and QC data.

ExceptionLog: When the software breaks down, it records the abnormal address field to facilitate the troubleshooting (Two in total and one is under Instrument 1 folder.)

4 Software Introduction

Help: Help file in CHM format.

Instrument 1: The log file of the analyzing unit.

Item: Mindray parameter list file.

LOG: software upgrade information (upgrade record)

Logs: RMS (Remote Manager System) log.

Workstation. Exe: BS-480 execution program.

4.1.3 Log file

Bslog records in real time the information of communication and actions of the analyzer including but not limited to the following information:

PC keyboard input

Software alarm, operation log and maintenance log.

Instructions between the PC and analyzing unit.

Action instructions of the analyzer

The status of the floaters and the optical couplers (real time)

The power status of the pump and valve (real time)

Photoelectric data (real time)

...

Log file is generated daily and includes two parts:

Located under D:\Mindray\Workstation\OperationSoft\WorkstationLOG

Located under D:\Mindray\Workstation\OperationSoft\Instrument_1\InstrumentLOG

The folder structure is as follows:

Name
Arithmetic
ArithmeticData
AssertLog
BioCook
BioCookSrvIO
BioReag
CommPlatLog
CtrlSysLog
EventLog
FlowServiceGlobal
InstrService
IPC
ISEAdapterSrv
ISEMgsLog
MonitorTally
NotifyLog
Period
PeriodIO
RPC
ServiceLog
ThreadLog
TIMService
TIMServiceIO

InstrService: water blank data

Lislog: LIS communication data

RPC: basic control instructions

ISEAdapterSrv: ISE instructions and test data.

4.1.4 Running software

- 1 Check if Bslog program is started. If not, run Bslog. exe.
- 2 Run Sqlserver(If Sqlserver cannot be started, the alarm is given prompting database initialization failed)
- 3 Start the operating software
- 4 Input the user name and password
- 5 Handshake with the middle-lower layer units (If handshake failed, the system prompts the related unit is abnormal).

4.1.5 Upgrade software

Please refer to the Setup Guide in the installation package or guide under D:\Mindray\Workstation\UpgradeTool.

Note: after the software is upgraded, reboot the analyzer otherwise the refrigeration unit will not be refreshed.

4.2 Removing the software

4.2.1 Removing BS-480 software

- 1 Terminate Bslog program (You can use the KillBSlog tool in the installation package).
- 2 Select Control panel- Workstation and remove it.

4.2.2 Removing SQL database

Select Control panel- SQL Server and remove it.

5

Modules and Units

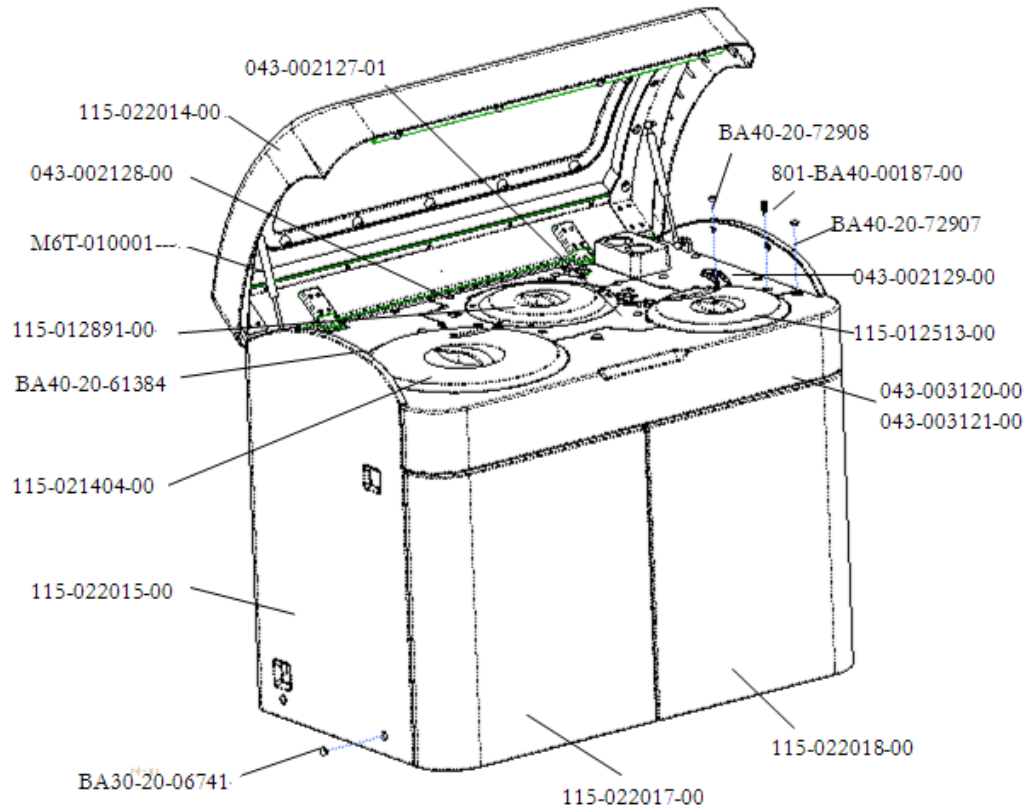
5.1 Shells Assembly

5.1.1 Module Functions

The shells assembly indicates the apparent structures of the whole analyzer and is designed for protecting the internal assemblies. It provides interfaces for each module, and is also a representation of industrial design.

5.1.2 Locations and FRU Details

The shells assembly consists of the shielding cover assembly, panel assembly, left and right plates, doors, rear panel, etc.



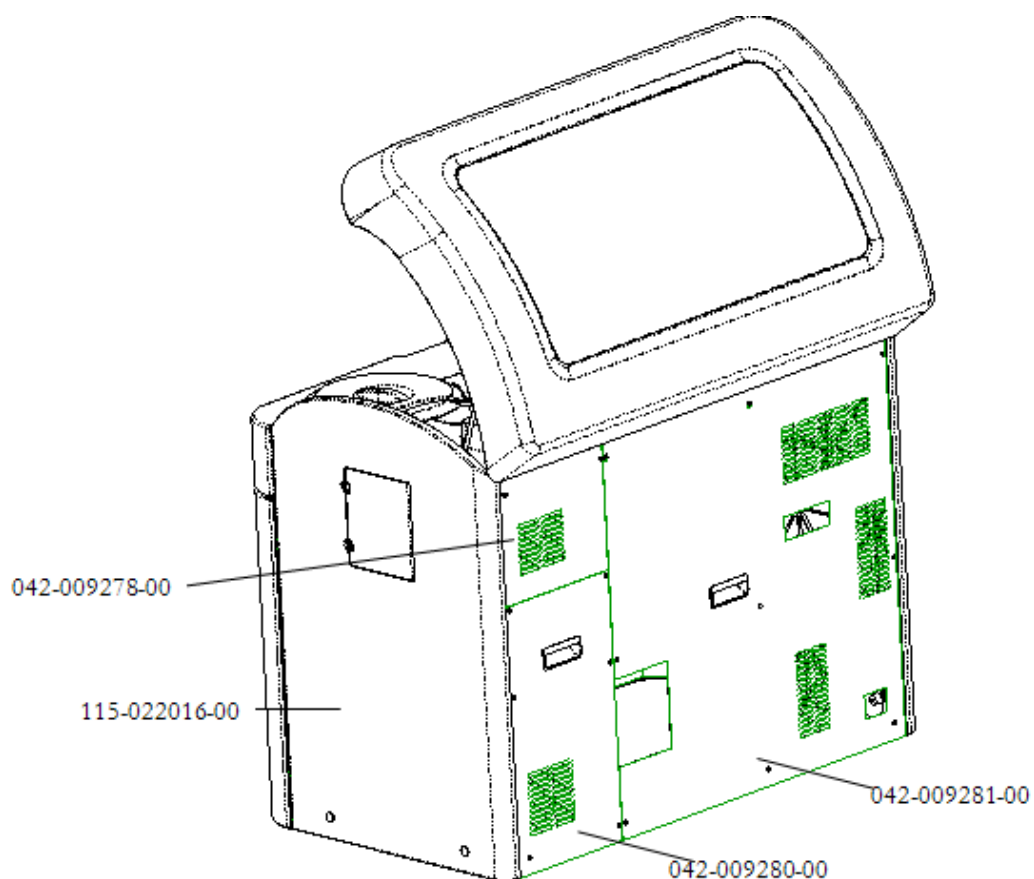


Table 5.1 List of materials

No.	FRU Code/PN	Material name	Remark
1	BA40-20-72908	ISE plug for panel (Mould MR72908)	
2	801-BA40-00187-00	Sample indicator cable	
3	BA40-20-72907	Screw plug for panel (Mould MR72907)	
4	043-002129-00	Wash station panel	
5	043-003120-00	BS-480 front panel	
6	043-003121-00	BS-490 front panel	
7	115-022018-00	Right door assembly	
8	115-022017-00	Left door assembly	
9	BA30-20-06741	Rubber cap (Mould 8692)	
10	115-022015-00	Left plate assembly	
11	BA40-20-61384	Panel 2 (Mould MR61384)	
12	043-002128-00	Reaction carousel panel (back)	
13	043-002127-01	Reaction carousel panel (front)	
14	042-009281-00	Rear panel 1	
15	042-009280-00	Rear panel 2	
16	042-009278-00	Light source assembly rear cover	
17	115-022016-00	Right plate assembly	

No.	FRU Code/PN	Material name	Remark
18	115-012891-00	Reaction carousel cover assembly	
19	115-012513-00	Sample carousel cover assembly	
20	115-021404-00	Reagent carousel cover assembly	
21	115-022014-00	Shielding cover assembly	
22	M6T-010001---	Air spring	

5.1.3 Removing and Reinstalling Shielding Cover

When to do

The shielding cover should be removed when needing placement due to damage.

Tools

Cross screwdriver, flathead screwdriver

Steps

- 1 Cut off the power of the analyzer to ensure all the assemblies are not in working status.
- 2 Open the shielding cover and push outwards the cir clip on the air spring head with a flathead screwdriver to the position that it cannot be dropped.
- 3 Hold the body of the shielding cover and pull the air spring to remove it
- 4 Remove M4X10 powder screws on the hinge of the shielding cover with a crosshead screwdriver, and remove the shielding cover.
- 5 To reinstall the shielding cover, follow the steps mentioned above in a reverse order.

Alignment and confirmation

Adjust the hinge of the shielding cover to ensure that the shielding cover is reinstalled in a proper position.

5.1.4 Removing and Reinstalling Side Plates

When to do

The side plates should be removed if the host PCB or the assemblies at left and right sides of the analyzer need to be maintained.

Tools

Cross screwdriver

Steps

- 1** Remove the rubber caps on the side plates.
- 2** Loosen the retaining screws on the side plates.
- 3** Hold the upper rim of the plate and lift it slightly to remove the side plates.
- 4** To reinstall the side plates, follow the steps mentioned above in a reverse order.

Alignment and confirmation

The two screw holes at the bottom of the side plates should be aligned.

5.1.5 Removing and Reinstalling Front Panel

When to do

The front panel should be removed if the components inside the analyzer need to be maintained.

Tools

Cross screwdriver

Steps

- 1** Unscrew rubber plugs of the three screws on the panel, and loosen the three screws to the position that they cannot be dropped.
- 2** Open the front door, unscrew the four retaining screws on the front panel, and then remove the panel.
- 3** To reinstall the front panel, follow the steps mentioned above in a reverse order.

Alignment and confirmation

You can adjust the gaps between the panels through the three screw holes on the front panel.

5.1.6 Replacement of Air Spring

When to do

The air spring should be replaced if it loses elasticity.

Tools

Flathead screwdriver

Steps

- 1** Open the shielding cover and push outward the cir clips on both heads of the air spring with a flathead screwdriver to the position that it cannot be dropped.
- 2** Hold the air spring and pull it out.
- 3** Install a new air spring.

Alignment and confirmation

N/A.

5.1.7 Remove the dust on the fans

When to do

Much dust is accumulated on the fans or this maintenance has not been performed for 1 year.

Tools

5 Modules and Units

Crosshead screwdriver, Hair brush and suction cleaner

How to do

- 1** Turn off the analyzer
- 2** If refrigeration fans need cleaning, remove back panel 1 (042-009281-00)
- 3** If air pump fan needs cleaning, remove back panel 2 (042-009280-00)
- 4** If light source fan needs cleaning, remove back plate of the lamp (042-009278-00)
- 5** Rotate the fan manually to remove the dust.
- 6** Install back the removed panel or plate.
- 7** Turn on the power of the analyzer.

Alignment and confirmation

Check if the fans work normally.

5.2 Frame assembly

5.2.1 Introduction

The frame assembly is a basic unit used for the installation and support of the components of the analyzer.

5.2.2 Locations and FRU Details

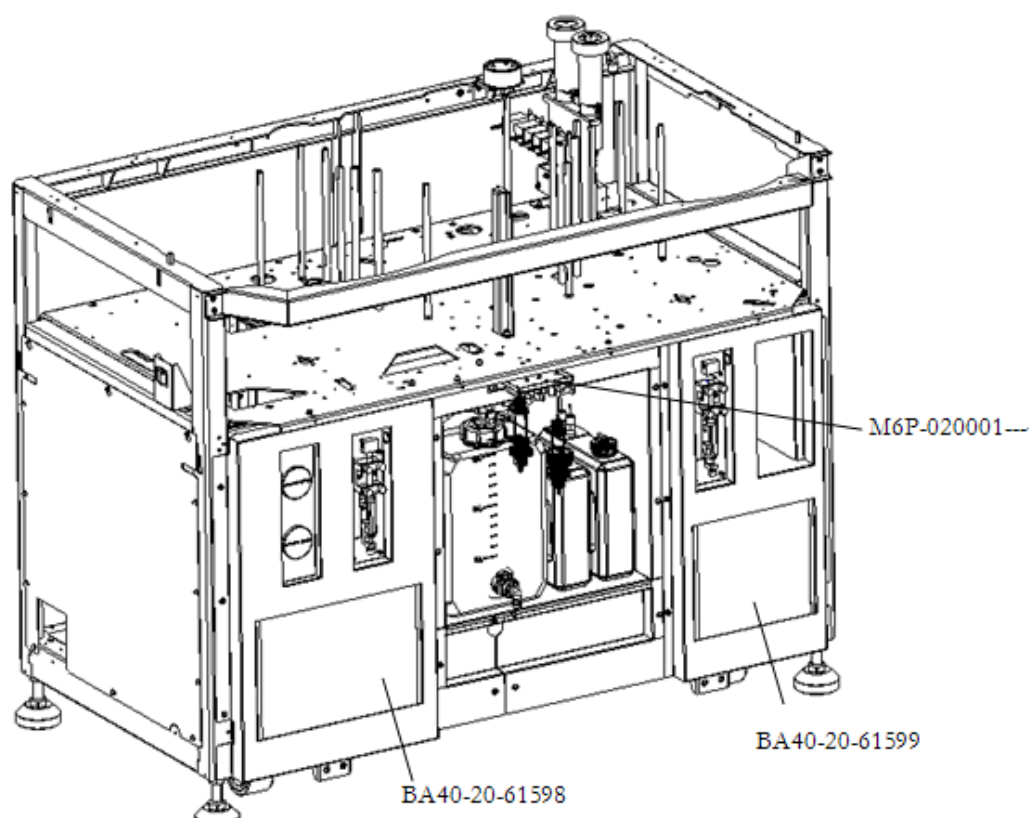


Table 5.2 FRU list

No.	FRU code/PN	Material	Remark
1	M6P-020001---	lock catch	FRU
2	BA40-20-61598	Left dust screen	FRU
3	BA40-20-61599	Right dust screen	FRU

5.2.3 Replacement of Lock Catch

When to do

The lock catch should be replaced if it is damaged.

Tools

Cross screwdriver

Steps

- 1** Unscrew the three M4X8 screws to remove the lock catch assembly.
- 2** Unscrew the retaining screws on the lock catch to remove it.
- 3** Install a new lock catch.
- 4** Place the new lock catch to the original position.

Alignment and confirmation

When installing the lock catch, ensure it is aligned completely with the door assembly, and the two doors must be level and not interfere with each other.

5.2.4 Replacement of Left/right dust screen

When to do

The left/right dust screen should be replaced if it is damaged.

Tools

None

Steps

- 1** Push upward the dust screen to remove it.
- 2** Replace it with a new one.

Alignment and confirmation

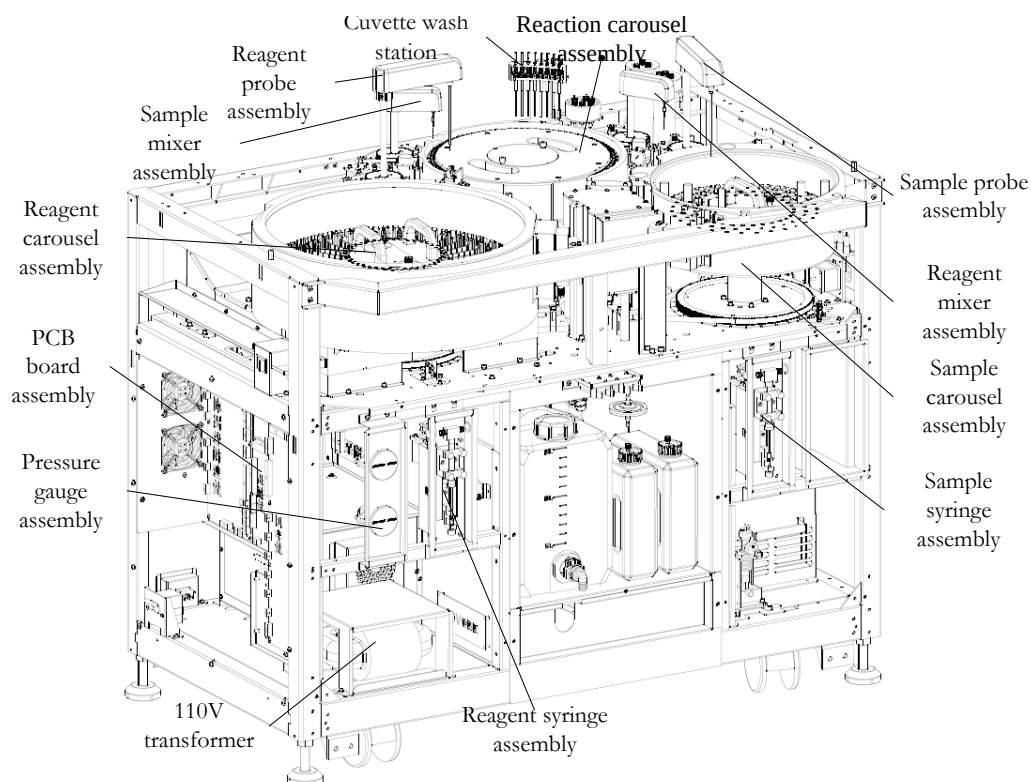
Make sure the dust screen can completely cover the air inlet.

5.3 Reaction Carousel Module

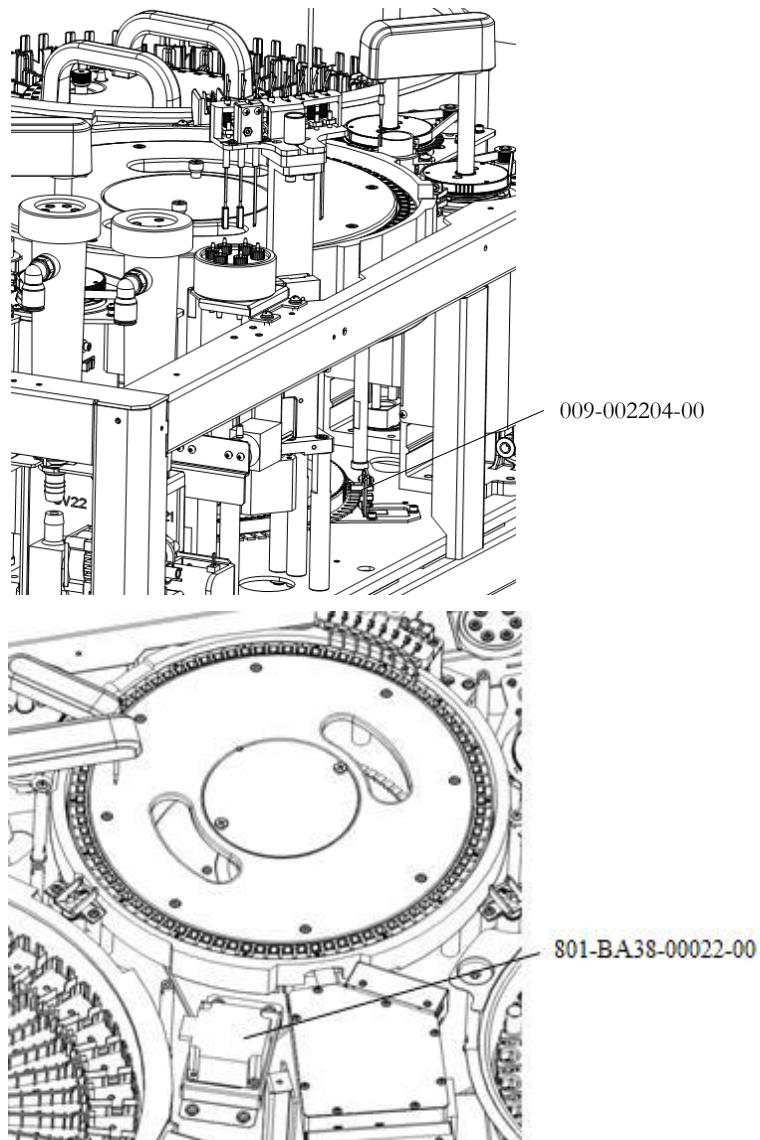
5.3.1 Module Functions

Reaction carousel assembly is situated at the middle and rear of the analyzer, including the reaction carousel body assembly, reaction carousel drive assembly, reaction carousel chamber, reaction carousel sensor, and the reaction carousel motor assembly. It is used to complete various actions, such as carrying and rotating the reaction cuvette to the specified position, adding sample/reagent to the cuvette, mixing and washing, together with the sample probe, reagent probe, mixer and the 8-phase wash probes. It, as well as, provides a constant temperature for the reaction cuvette for analysis.

Figure 5.1 Locations of components on the instrument



5.3.2 Locations and FRU Details



No.	FRU code and material code	Material name	Remark
1	115-013229-00	Coder sensor assembly	
2	115-013221-00	Reaction carousel motor assembly	
3	BA40-20-73118	Metal plate for cuvette	
4	009-002673-00	Cable of reaction carousel sensor 1	
5	009-002674-00	Cable of reaction carousel sensor 3	
6	BA40-21-61646	Cable of reaction carousel sensor 2	
7	009-00045-00	Temperature protection switch wire (for reaction carousel)	
8	043-000568-00	Cuvette	
9	009-002588-00	Heater cable	
10	009-002589-00	Slip ring cable	

No.	FRU code and material code	Material name	Remark
1	009-002204-00	Sensor	FRU
2	BA40-20-73118	Metal plate for cuvette	FRU
3	009-002673-00	Cable of reaction carousel temperature sensor 1	FRU
4	009-002674-00	Cable of reaction carousel temperature sensor 3	FRU
5	801-BA40-00173-00	Cable of reaction carousel temperature sensor 2	FRU
6	801-BA80-00060-00	Temperature protection switch wire (for reaction carousel)	FRU
7	115-004014-00	Plastic cuvette	FRU
8	009-002588-00	Heater cable	FRU
9	009-002589-00	Slip ring cable	FRU
10	801-BA80-00055-01	Temperature collection board	FRU
11	801-BA38-00022-00	SST59D5300 motor with pulley	FRU
12	045-000514-00	Reaction carousel chamber	FRU

Note: For locations of FRU 3~10, please refer to the related descriptions of replacement.

5.3.3 Replacement of Reaction Carousel Body Cables

When to do

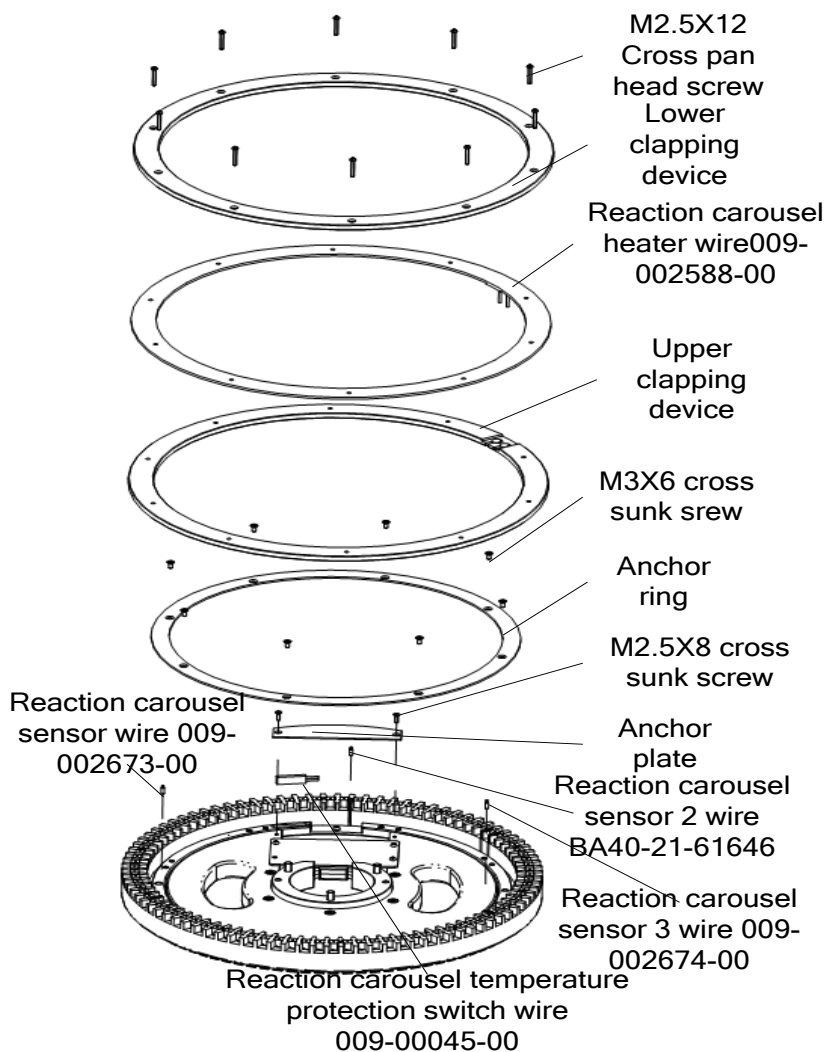
The reaction sensors, heater, protection switch or temperature collection board should be replaced if the reaction temperature is abnormal due to their failure.

Tools

Name	Code	Quantity
Cross head screwdriver	/	1
Hexagon wrench	/	1

Exploded view for installation

Figure 5.2 Exploded view for replacing reaction carousel body cables



NOTE

Please be careful when removing the reaction carousel body to protect the cuvettes and the top of the carousel's rotary shaft.

It is not suggested to remove the upper clamping device. If it is difficult to clean the stains in the cuvette slot, you should remove the parts and clean the interior of the cuvette slot.

Do not pull the cables during the installation; especially pay much attention to protect cables of the three sensors.

Further decomposition may invalidate the functions of the reaction carousel.

Steps

- 1 Switch off the power of the analyzer.
- 2 Open the front shielding cover and remove the auto wash station, and store them properly to protect the wash probes. Rotate the reagent probe to the reagent aspirate position and rotate the sample probe to the sample aspirate position to remove the reaction carousel cover.

- 3 Unscrew the two M3 cross head screws on the reaction carousel's skylight cover to remove it, and unplug the slip ring from the reaction carousel temperature collection board.
- 4 Unscrew the four M5X12 hexagon socket cap head screws with spring washer on the reaction carousel body to remove it. Please take the carousel carefully in order not to damage the reaction cuvettes.
- 5 Remove all the reaction cuvettes and store them properly.
- 6 Place the carousel body with the bottom surface upwards and unscrew the two M2.5X8 cross pan head screw to remove the anchor plate.
- 7 Unscrew the 8 M3*6 screws on the wire clamping ring to remove it.
- 8 Unscrew the 10 M2.5*16 cross sunk head screws on the lower clamping device to remove it.
- 9 Unplug the connector between the heater and the temperature collection board and remove the cable of the reaction carousel heater.
- 10 Remove the upper clamping device.
- 11 Unplug the connectors between the three temperature sensors and the temperature collection board and remove the temperature sensor cable 1, reaction carousel temperature sensor cable 2 and reaction carousel temperature sensor cable 3.
- 12 Daub some heat-conducting glue on the sensors of the new temperature sensor wire 1, 2 and 3 and the installation holes and place the three sensors into the holes on the body of the reaction carousel. Please ensure the positions of the three sensors are correct, the joints should not be higher than the slot surface of the reaction carousel. Plug the connectors of the sensor in the temperature collection board of the reaction carousel. Write down the serial numbers and the corresponding parameters of the three sensors which will be used to query the electric resistance value during the alignment.
- 13 Unplug the connector between the temperature protection switch and the temperature collection board and remove the cable of the temperature protection switch of the reaction carousel.
- 14 Daub proper amount of thermal paste on the new temperature protection switch and plug the connector of the temperature protection switch into the temperature collection board.
- 15 Install the upper clamping device.
- 16 Daub some heat-conducting glue on the root of the heater wire and place it on the upper clamping device, and the cable passes through the slot of the upper clamping device to the slot of the reaction carousel slot. Insert the cable connector on the temperature collection board.
- 17 Put the lower clamping device on the heater and tighten the lower clamping device with 10 M2.5*16. When tightening the screws, push the part of the heater which protrudes the outer ring of the reaction carousel to the center and fix the screws in diagonal pair to make the clamping device contact closely with the heater so that heat transmission is even.
- 18 Press cables of the heater, the temperature protection switch and the three sensors into the slot of the carousel body, and then install the wire clamping ring onto the carousel body with eight M3X6 cross countersunk head screws.
- 19 Use two 2.5X8 cross countersunk head screws to fix the anchor plate on the reaction carousel body.

- 20** For the replacement of the temperature collection board, first perform step 1-6 and disconnect the cable of the temperature collection board. Loosen the 4 M2.5*6 cross pan head screws used to fix the temperature collection board. Replace the board with new one and use the screws to fix it. Restore the assembly in the reverse order.

Alignment and confirmation

Re-configure the related parameters of the sensors after replacing the cables of the sensor, the heater and the temperature protection switch and test the performance of heat-insulating. Refer to 8.13.1 Reaction Carousel Temperature Control, 8.13.2 Sensor Configuration window, 8.13.3 Reaction Carousel Temperature Test, 8.13.4 Observe Temperature Curve.

5.3.4 Replacement of Metal Plate for Cuvette

When to do

If the metal plate has no elasticity and cannot retain a cuvette, then it should be replaced.

Tools

Tweezers and knife

Exploded view for installation

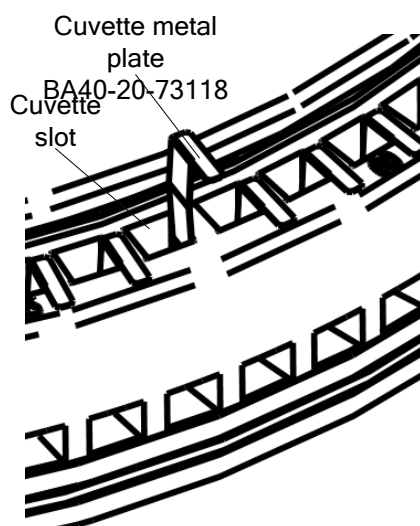


Figure 5.3 Schematic diagram of replacing metal plate for cuvette



NOTE

The spring plate should be pressed to the bottom.

Please do not press the spring plate with a sudden force to avoid bending the spring plate.

Edges of the spring plate may have some small burrs. Please be careful when replacing it.

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover and remove the auto wash station, and store them properly to protect the wash probes. Rotate the reagent probe to the reagent aspiration position and rotate the sample probe to the sample aspiration position to remove the reaction carousel cover

- 3 Unscrew the two M3 cross pan head combination screws on the reaction carousel's skylight cover to remove it.
- 4 Unscrew the four M5X12 hexagon socket cap head screws on the body of the reaction carousel to remove it. Please take the carousel carefully in order not to damage the reaction cuvettes.
- 5 Remove the cuvette of which the spring plate needs to be replaced as well as the 3 cuvettes on its left and right sides (2 on each side), and store them properly.
- 6 Extract the spring plate about 3~5mm with a blade and grip it out with a pair of sharp nose pliers.
- 7 Press a new spring plate into the cuvette slot in the correct direction.

Alignment and confirmation

N/A.

5.3.5 Replacement of Carousel Home Sensor and Coder Sensor Cable

When to do

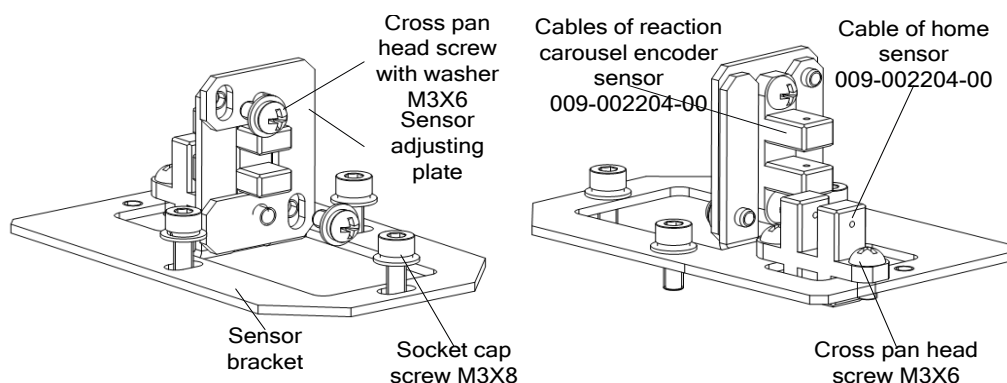
If reaction carousel sensor becomes invalid, please replace cables of the carousel home sensor and the coder sensor.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation

Figure 5.4 Schematic diagram of replacing sensor cables



NOTE

Please install the sensor according to the identifications of the cable.

Do not tighten the screws with a sudden force, in order to protect the sensor.

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Unscrew the 10 M4×8 cross pan head combination screws on the rear panel and remove the right rear panel 1.
- 3 Disconnect the home position sensor and coder sensor.
- 4 Unscrew the 3 M3×8 hexagon socket cap head screws with spring and flat washers on the reaction carousel sensor assembly, remove the sensor assembly and avoid dropping the screws into the analyzer.
- 5 Unscrew the two M3X6 cross pan head combination screws on the sensor's mounting plate and remove the coder sensor cable with the sensor.

- 6** Unscrew the two cross pan head screws on the coder sensor, and remove the coder sensor from the mounting plate.
- 7** Fix the new coder sensor on the mounting plate using 2 M3×6 cross pan head screws.
- 8** Use two M3X6 cross combination screws to fix the sensor on the mounting plate on the sensor bracket and do not tighten them.
- 9** Unscrew the two M3X6 cross pan head screws on the cables of carousel home sensor and remove the cable of the home sensor.
- 10** Fix the new cables of carousel home sensor onto the sensor bracket using two M3X6 cross pan head screws.
- 11** Fix the reaction carousel sensor assembly on the large bottom plate using three M3X8 hexagon socket cap head screws with spring and flat washers. Adjust the mounting plate and the sensor to ensure the coder is positioned in the middle of the coder sensor. Tighten the retaining screws of the mounting plate.
- 12** Connect the home position sensor and coder sensor.
- 13** After alignment and confirmation of the sensor assembly, fix the rear panel 1 to the framework using 10 M4×8 cross pan head combination screws.

Note:

To replace only the coder sensor cable, neglect step 9 and 10.

To replace only the home position sensor of the reaction carousel, neglect step 5 to 8.

Alignment and confirmation

After replacing the home sensor of the reaction carousel and the coder sensor cable, refer to 8.3.2 to perform the Signal Collecting Position Adjustment and refer to 8.4-8.10 to perform circumferential Position alignment of the reaction carousel, sample carousel and reagent carousel and the horizontal position alignment of the sample probe, reagent probe, sample mixer, reagent mixer and cuvette wash station.

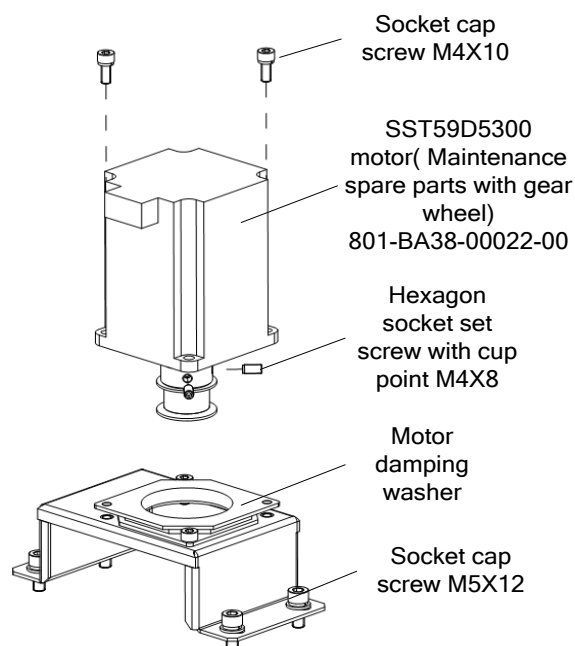
5.3.6 Replacement of reaction carousel motor and synchronous belt

When to do

The reaction carousel motor and synchronous belt should be replaced if they are damaged.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation**Figure 5.5** Schematic diagram of replacing reaction carousel motor assembly**WARNING**

During removing and installation, avoid dropping screws and washers into the analyzer.
Do not remove the two screws on the damping washer.

Removing/Reinstallation procedure**Replace the motor.**

- 1 Refer to 5.3.3 Replacement of Reaction Carousel Body Cables to remove the reaction carousel body assembly.
- 2 Loosen the three M4*20 hexagon socket screws used to fix the reaction carousel chamber and remove it.
- 3 Disconnect the motor cable, unscrew the 4 M5×12 hexagon socket head screws with spring washer on the reaction carousel motor assembly, remove the motor assembly and avoid dropping screws into the analyzer.
- 4 Unscrew the two M4X10 hexagon socket head screws with spring washer on the reaction carousel motor to remove the motor with the pulley.
- 5 Fix a new motor with the pulley on the damping board using two M4X10 hexagon socket head screws with spring washer, and interface direction of the motor should be away from the reaction carousel.
- 6 Fix the motor assembly to the large bottom plate using four M5X12 hexagon socket cap head screws with spring washer. Tighten the belt and screws and connect the cable of the motor.
- 7 Restore the reaction carousel chamber and make the chamber and the drive assembly align.
- 8 Restore the reaction carousel body.

Change the synchronous belt

- 1 Refer to 5.3.3 Replacement of Reaction Carousel Body Cables to remove the reaction carousel body assembly.

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- 2** Loosen the three M4*20 hexagon socket screws used to fix the reaction carousel chamber and remove it.
- 3** Unscrew the 4 M5×12 hexagon socket head screws on the reaction carousel motor assembly and remove the belt from the wheel.
- 4** Install a new synchronous belt.
- 5** Screw the four M5×12 hexagon socket head screws slightly and then tighten the synchronous belt and all the screws.
- 6** Restore the reaction carousel chamber and make the chamber and the drive assembly align.
- 7** Restore the reaction carousel body.

Alignment and confirmation

N/A.

5.3.7 Replacement of Slip Ring Cables

When to do

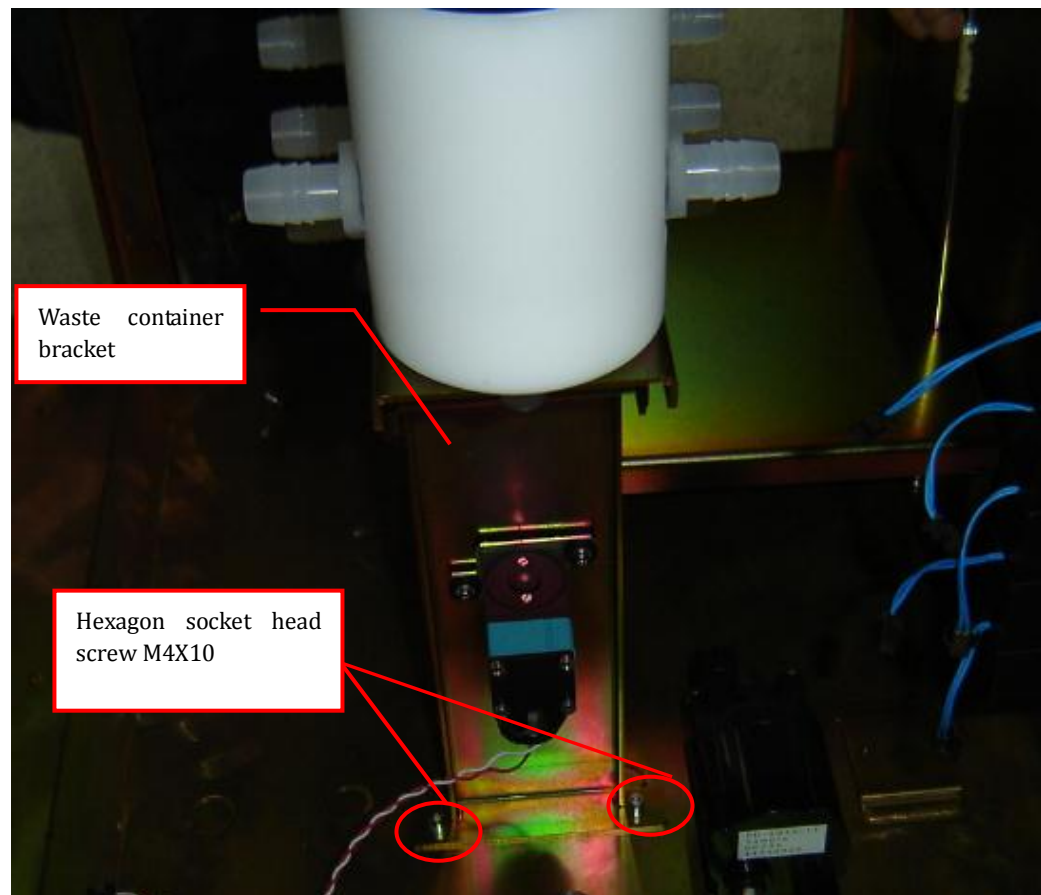
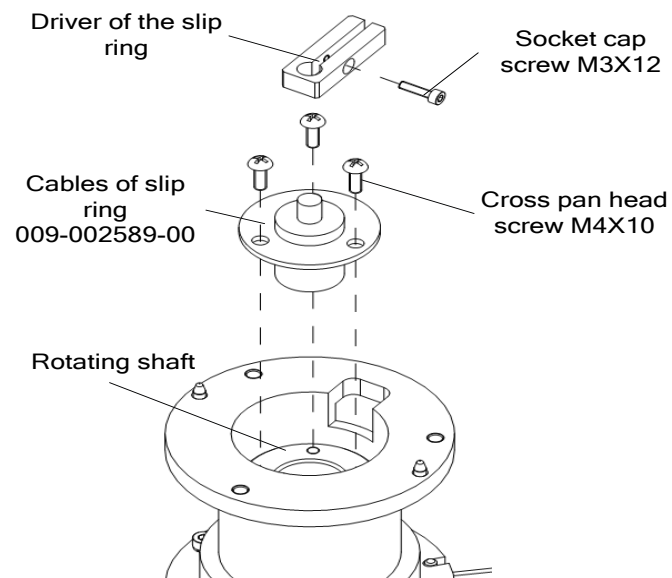
The cables of the slip ring should be replaced if they are damaged.

Tools

Cross head screwdriver, hexagon wrench

Exploded view for installation

Figure 5.6 Schematic diagram of replacing cables of slip ring





WARNING

Before removing or connecting the cable connectors, loosen the two M4X10 hexagon cap head screws on the cables of slip ring behind the waste container bracket, and then lay down the bracket.

The driver of the slip ring should not be installed at the bottom of the cables; it should be kept about 2mm distance from the cables;

Otherwise, the slip ring may be damaged.

Do not tighten screws forcefully to avoid distortion;

Otherwise, cables of the slip ring will be damaged.

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover and remove the auto wash station, and store them properly to protect the wash probes.
- 3 Rotate the reagent probe to the reagent aspirate position and the sample probe to the sample aspirate position, and remove the reaction carousel cover and all panels around the reaction carousel assembly and rear panel 1.
- 4 Unscrew the two M3 screws on the reaction carousel's skylight cover to remove it, and unplug the slip ring from the reaction carousel temperature collection board.

Unscrew the three M4X10 cross pan head screws used to fix the Slip ring cable and disconnect the terminal under the bottom plate of the analyzer to remove the cables of the slip ring.

- 6 Unscrew the one M3X12 hexagon socket cap head screw on the driver of the slip ring to remove it.
- 7 Install the driver of the slip ring onto the new cables of slip ring using one M3X12 hexagon socket cap head screw. Please lift the driver about 2mm from the cables and then tighten the screw.
- 8 Fix the cables of the slip ring to the rotating shaft using the three M4X10 cross pan head screws with spring washer, and connect the terminals under the bottom plate of the analyzer.
- 9 Install the skylight cover, reaction carousel cover assembly, and the rear panel 1 by steps mentioned above in a reverse order.

Alignment and confirmation

N/A.

5.3.8 Replacement of Reaction Cuvette

Refer to 5.9 replacement of cuvette.

5.4 Sample Carousel Assembly

5.4.1 Module Functions

The sample carousel assembly, situated in the right front of the analyzer, consists of the sample carousel body assembly, drive assembly, sample chamber assembly, sample carousel sensor assembly, sample carousel motor and bar code reader. It holds and carries sample tube to the specified position, and cooperates with the dispenser and the sample mixer for sample aspiration and dispensing and mixing. The functions of the sample carousel assembly include:

- 1 Carry samples: Sample containers (tube or microtube) with samples are located on the sample carousel, and the sampler will aspirate samples and dispense them to the cuvettes.
- 2 Sequential feeding: The drive assembly drives the sample carousel to carry certain sample in the given order to the aspirate position for aspiration.
- 3 Input sample data automatically: Bar code reader (optional).

5.4.2 Locations and FRU Details

Figure 5.7 Sample carousel assembly

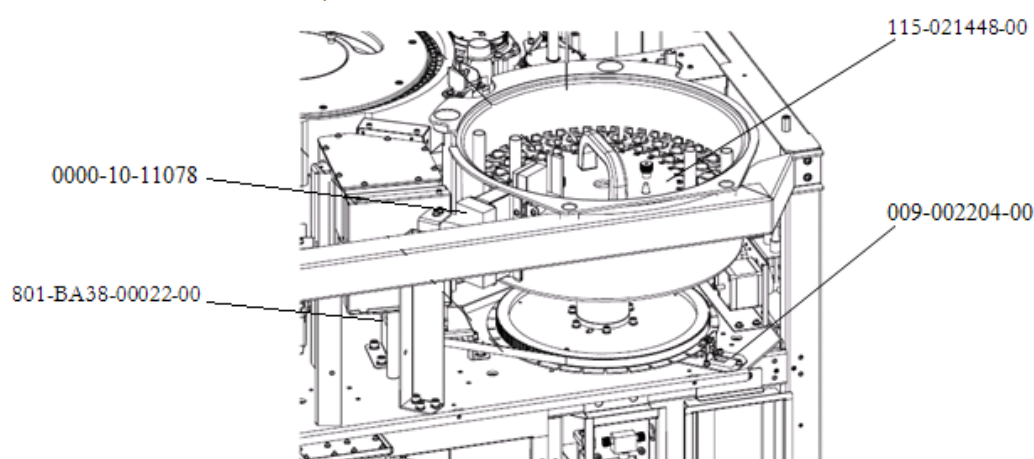


Table 5.3 List of materials

No.	FRU code and material code	Material Name	Remark
1	115-021448-00	Sample carousel assembly	FRU
2	009-002204-00	Sensor	FRU
3	801-BA38-00022-00	SST59D5300 motor (including belt pulley)	FRU
4	0000-10-11078	Built-in bar code reader, laser, with angle	FRU

5.4.3 Removing and Reinstalling Sample Carousel Body Assembly

When to do

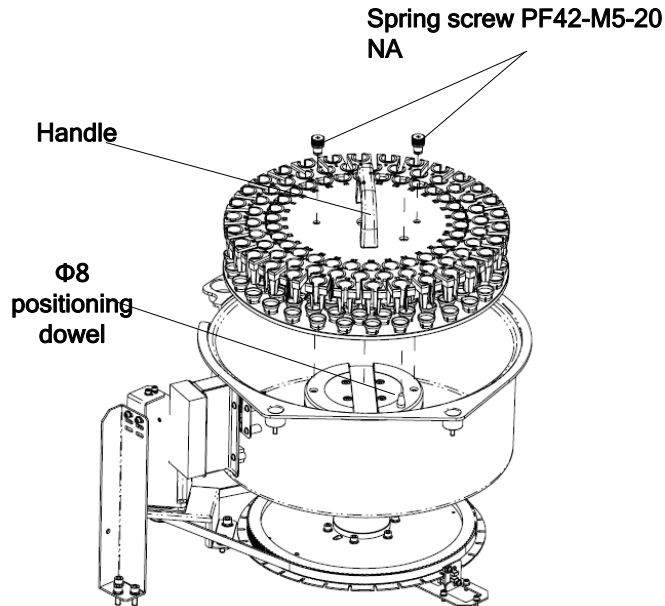
The sample carousel body assembly should be removed when it is damaged or replaced.

Tools

N/A.

Exploded view for installation

Figure 5.8 Removing and Reinstalling Sample Carousel Body Assembly



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Remove the sample carousel cover.
- 3 Loosen the two screws on the sample carousel body assembly.
- 4 Align the locating holes on the new sample carousel body with the stop bolt and place it on the bearing shaft of the sample carousel.
- 5 Tighten the two screws on the sample carousel body assembly.

Alignment and confirmation

Be sure to align with the stop bolt when installing the sample carousel body assembly.

5.4.4 Replacement of Sample Carousel Home Sensor and Coder Sensor Cables

When to do

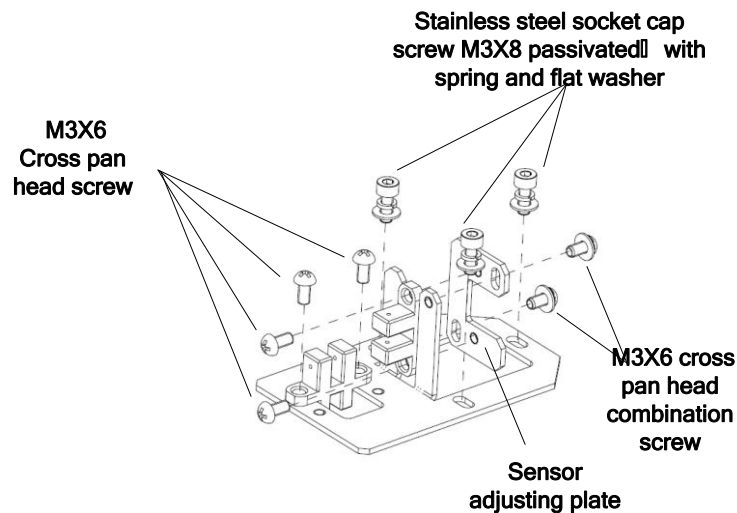
If the coder sensor or home position sensor of the sample carousel fails, replace them immediately.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation

Figure 5.9 Removing and Reinstalling sensor assembly



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Loosen the screws used to fix the front panel and remove the panel.
- 3 Disconnect the home position sensor and coder sensor.
- 4 Loosen the screws of the sample carousel sensor and remove the sensor assembly.
- 5 Unscrew the screws on the sensor's mounting plate and remove the coder sensor cable with the sensor's mounting plate.
- 6 Unscrew the screws on the coder sensor, and remove the coder sensor cable from the mounting plate.
- 7 Fix the new coder sensor on the mounting plate using 2 M3×6 cross pan head screws.
- 8 Install the sensor's mounting plate together with cables of the encoder sensor onto the bracket of the sensor using two M3X6 cross pan head screws. Do not tighten them.
- 9 Unscrew the screws on the home position sensor of the sample carousel, and remove the sensor from sensor bracket.
- 10 Fix the new cables of sample carousel home sensor onto the sensor bracket using two M3X6 cross pan head screws.
- 11 Fix the sample carousel sensor assembly on the large bottom plate using three M3X8 hexagon socket cap head screws. Adjust the mounting plate and the sensor to ensure the coder is positioned in the middle of the coder sensor. Tighten the retaining screws of the mounting plate.
- 12 Connect the home position sensor and coder sensor.
- 13 After the position of the sample carousel sensor assembly is aligned, fix the front panel with 3 M4*10 screws on top and 4 M4*8 cross pan head screws with washer in the front. Adjust the gap of the panels and then tighten the screws.

Note:

To replace only the coder sensor cable, neglect step 9 and 10.

To replace only the home position sensor, neglect step 5 to 8.

Alignment and confirmation

Refer to 8.5 Sample Carousel Unit and 7.7 sample probe unit to perform the alignment of Sample carousel circumferential position and sample probe rotary to sample carousel.

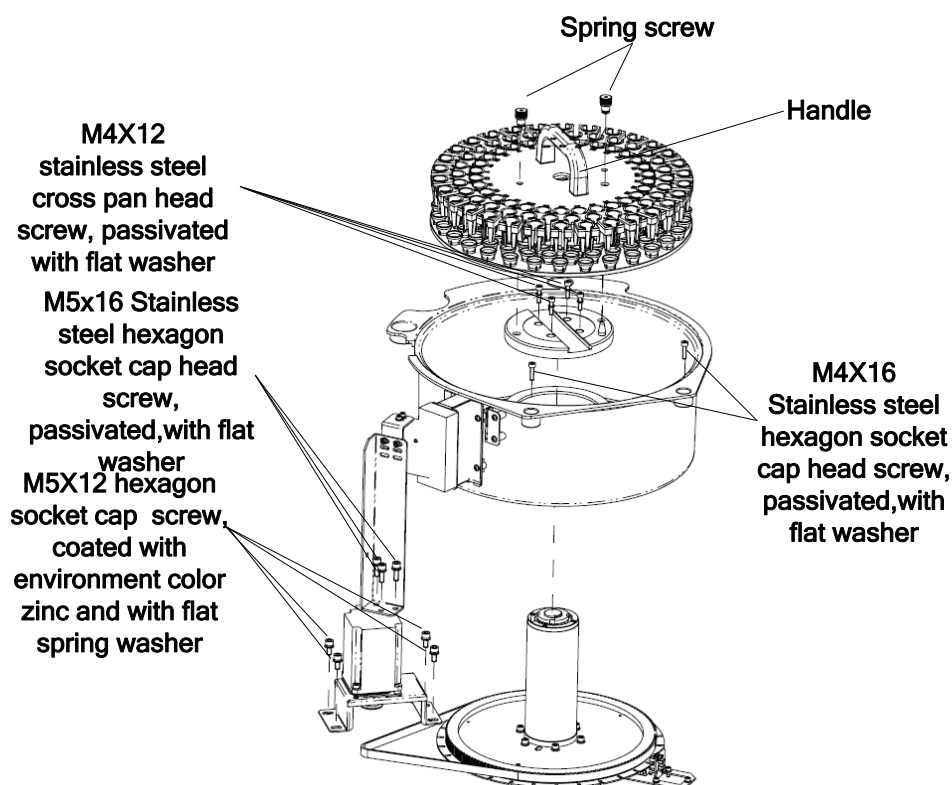
5.4.5 Replacement of sample carousel motor and synchronous belt

When to do

The sample carousel motor and synchronous belt should be replaced when damaged.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation**Figure 5.10** Replacement of sample carousel motor and synchronous belt**Removing/Reinstallation procedure****Replace the motor:**

- 1 Switch off the power of the analyzer.
- 2 Remove the front panel and unplug the motor cable.
- 3 Unscrew the 4 screws on the motor bracket and remove the belt from the wheel. Remove the motor assembly.
- 4 Loosen the two retaining screws on the motor and remove the motor with the belt.
- 5 Fix a new motor with the pulley on the damping board using two M4X10 hexagon socket head screws with spring washer, and interface direction of the motor should be away from the sample carousel.
- 6 Sleeve the belt on the wheel. Fix the motor assembly to the large bottom plate using four M5X12 hexagon socket cap head screws with spring washer. Tighten the belt and screws and connect the cable of the motor.
- 7 Restore the front panel.

Change the synchronous belt:

- 1 Switch off the power of the analyzer, and unplug the motor cable.
- 2 Remove the sample carousel cover and the front panel and the panel around the sample carousel.
- 3 Loosen the two screws on the sample carousel body assembly and remove the screws.
- 4 Unscrew the three screws on the bearing shaft of the sample carousel and remove the screws.

- 5** Loosen the 4 screws on the sample chamber and remove the sample chamber (If bar code reader is configured, unscrew the screws on the bar code reader.)
- 6** Unscrew the 4 screws on the motor bracket and remove the belt from the wheel.
- 7** Install a new synchronous belt.
- 8** Screw the four M5×12 hexagon socket head screws slightly and then tighten the synchronous belt and all the screws.
- 9** Use 4 M4*10 pan head screws with wash to fix the sample carousel chamber. Make the sample carousel chamber align with the drive assembly. If bar code reader is installed, check its position.
- 10** Use three M3*10 hexagon socket screws with spring wash to fix the sample bearing shaft.
- 11** Restore the sample carousel assembly and tighten the spring screws.
- 12** Restore the front panel and the panel around the sample carousel.

Alignment and confirmation

N/A.

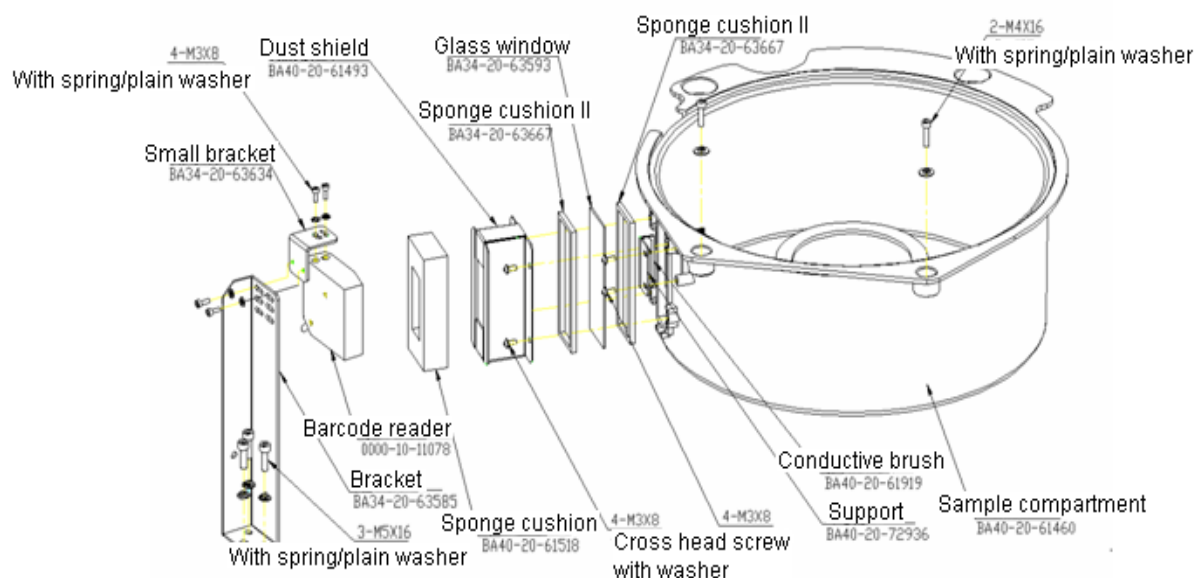
5.4.6 Removing the sample bar code reader

When to do

The sample bar code reader should be replaced when it does not work normally.

Tools

Hexagon wrench

Exploded view for installation**Figure 5.11** Sample bar code reader and sample chamber**Removing/Reinstallation procedure****Install the sample bar code assembly**

- 1 Apply sponge cushion 2 on two sides of the glass window, and note not to contaminate the window surface. The sponge cushion should align with the glass verge. Apply sponge cushion on the dust shield for bar code reader, and note to align its opening with that on the dust shield.
- 2 Place the glass window in the position of window of the sample carousel chamber. Use 4 M3*8 pan head screws to fix the bar code reader to the sample chamber.
- 3 Use two M3*8 hexagon socket screws with spring washer to fix the small bracket of the bar code reader to the square hole on the upper end of the bar code reader bracket.
- 4 Use two M3*10 hexagon socket screws with spring washer to fix the bar code reader to the small bracket.
- 5 Use two M5*12 hexagon socket screws with spring washer to fix the bar code reader bracket to the big bottom plate. Tighten the screws and connect the bar code reader cable. Use the strip to fix it to the bracket.

Replacing Bar Code Reader:

- 1 Remove the sample carousel cover and three the rubber plugs on the panel. Unscrew the three screws to the position that they cannot be dropped. Open the front door, unscrew the four retaining screws on the front panel, and then remove the panel.
- 2 Disconnect the bar code reader cable, loosen the wire clamp, unscrew the 2 M3 × 10 hexagon socket cap head screws on the bar code reader, and then remove the bar code reader.
- 3 Install the bar code reader on the small bracket with two M3X10 hexagon socket cap head screws with flat and spring washers. Connect the bar code reader cable. Note not to let the sponge cushion shield the emergent ray from bar code reader.
- 4 Restore the front panel and use three M4*10 pan head screws with rubber plug to fix the top panel and fix the front panel with four M4X8 pan head screw with wash. Close the sample carousel cover.

5 Modules and Units

Alignment and confirmation

Align the sample bar code reader according to Section 8.11.1 .

5.5 Reagent Carousel Assembly

5.5.1 Module Functions

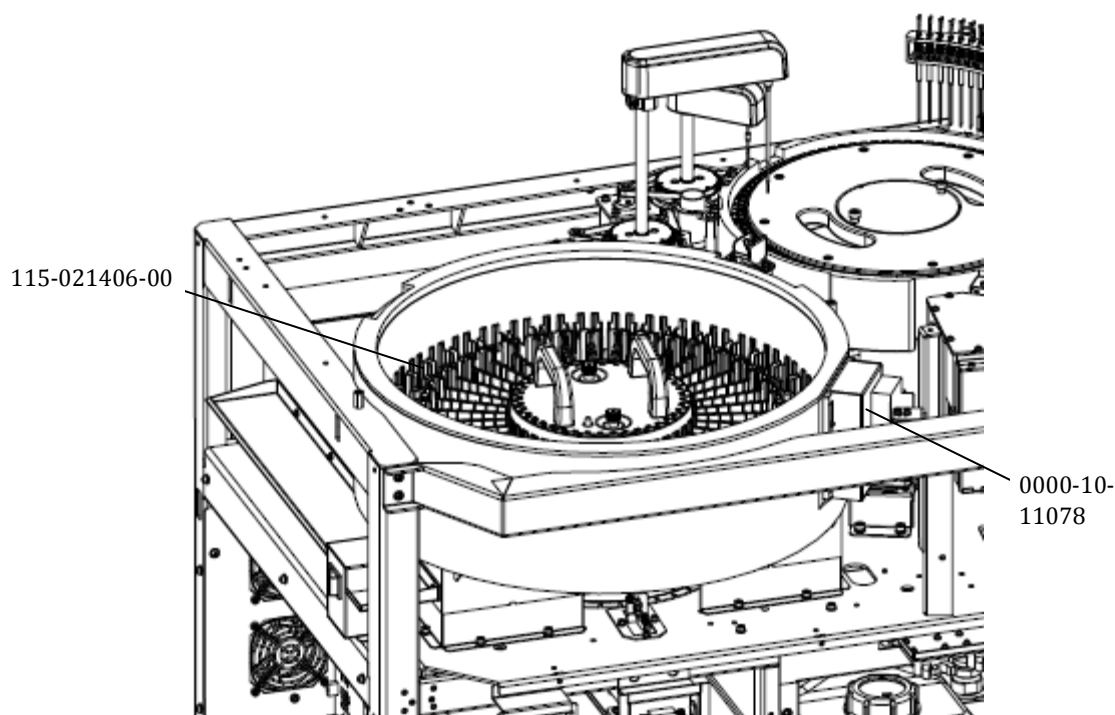
Reagent carousel assembly is situated at the left front of the analyzer, including the reagent carousel body assembly, reagent chamber, drive assembly and the reagent bar code reader. It is used to complete various actions, such as carrying and rotating the reagent bottle to the specified position, add samples together with sample addition assembly. It provides a refrigerating environment, and the reagents stored in such environment can be kept stable with little volatilization. For details, please refer to the following contents.

- 1 Holding reagent: Holding specified number of reagent bottles of certain specification, and cooperating with the dispenser to aspirate reagent from the bottles and dispense it into reaction cuvettes.
- 2 Refrigerating reagent: Provide 24-hour refrigeration and constant temperature environment (2-10° C) to keep the reagent stable with little volatilization..
- 3 Sequential feeding: With the drive assembly, the reagent carousel carries the specific reagent bottle to the aspirate position for aspiration in certain order. The outer and inner rings can rotate independently.
- 4 Input reagent data automatically: Bar code reader (optional).

5.5.2 Locations and FRU Details

The reagent carousel assembly is located in the front left of the instrument.

Figure 5.12 Reagent carousel assembly



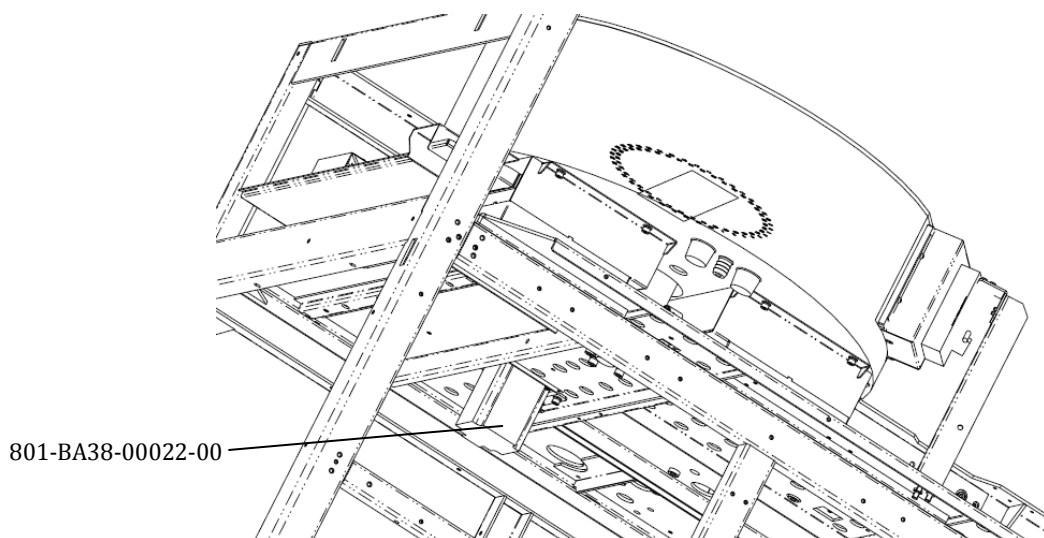


Table 5.4 List of materials

No.	FRU code and material code	Material Name	Remark
1	115-021406-00	Reagent carousel body assembly	FRU
2	009-002204-00	Sensor	FRU
3	043-000042-00 (This part can no longer be replaced after EBJ249-F, because the bottle base and carousel body will be molded as a whole.)	Base of reagent bottle	FRU
4	801-BA10-00037-00	Semi-conductive Peltier cable	FRU
5	801-BA38-00022-00	SST59D5300 motor (including belt pulley)	FRU
6	0000-10-11078	Built-in bar code reader, laser, with angle	FRU
7	BA40-30-61987	Anti-mist heating assembly of reagent carousel	FRU
8	801-BA30-00088-00	Cables of reagent refrigeration temperature sensor	FRU

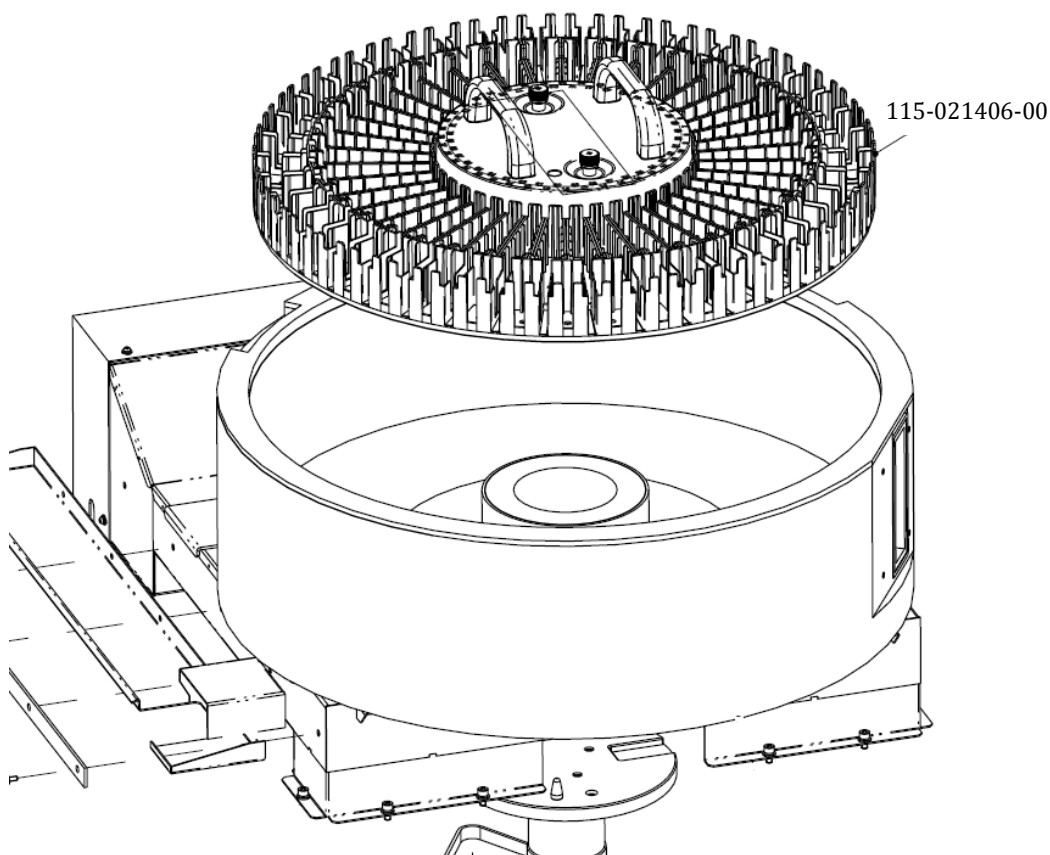
5.5.3 Replacement of Reagent Carousel Body Assembly

When to do

The reagent carousel body assembly should be replaced if it is damaged or cannot meet the operation requirements.

Tools

N/A

Exploded view for installation**Figure 5.13** Removing and Reinstalling Reagent Carousel Body Assembly**Removing/Reinstallation procedure**

- 1 Switch off the main power of the whole unit.
- 2 Remove reagent carousel cover.
- 3 Unscrew the two screws on the reagent carousel assembly by hands.
- 4 Lift the handlebar to remove the reagent carousel body assembly.
- 5 Align the positioning hole of the new reagent carousel body assembly with the stop bolt and install it onto the reagent carousel spindle. Tighten the two screws with hands.
- 6 Restore the reagent carousel cover.

Alignment and confirmation

N/A.

5.5.4 Replacement of Reagent Bottle Base**When to do**

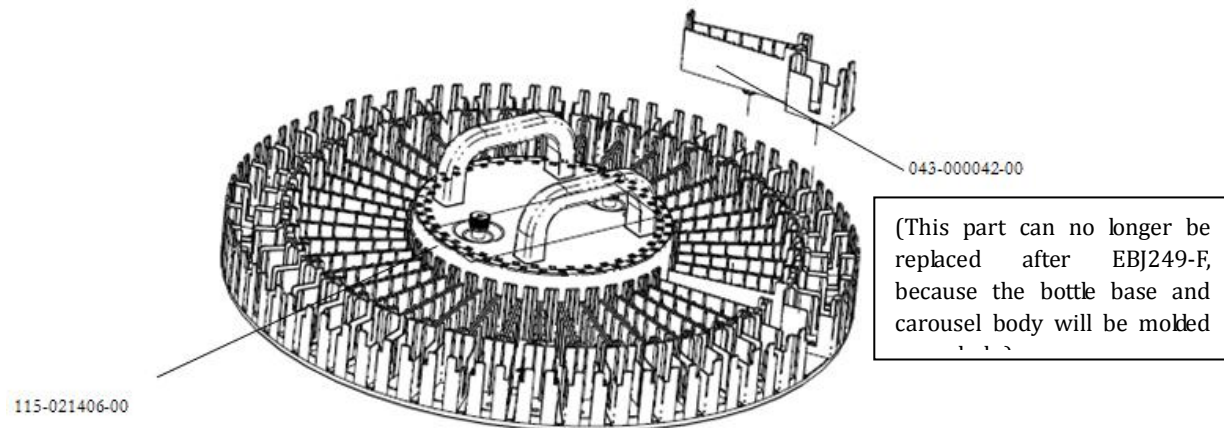
The base of reagent bottle should be replaced if it is damaged.

Tools

N/A.

Exploded view for installation

Figure 5.14 Replacement of Reagent Bottle Base



Removing/Reinstallation procedure

- 1 Refer to 5.5.3 Replacement of Reagent Carousel Body Assembly to remove the reagent carousel body assembly.
- 2 Remove the reagent bottle base.
- 3 Replace it with new one and tighten it.
- 4 Restore the reagent carousel body assembly.

Alignment and confirmation

N/A.

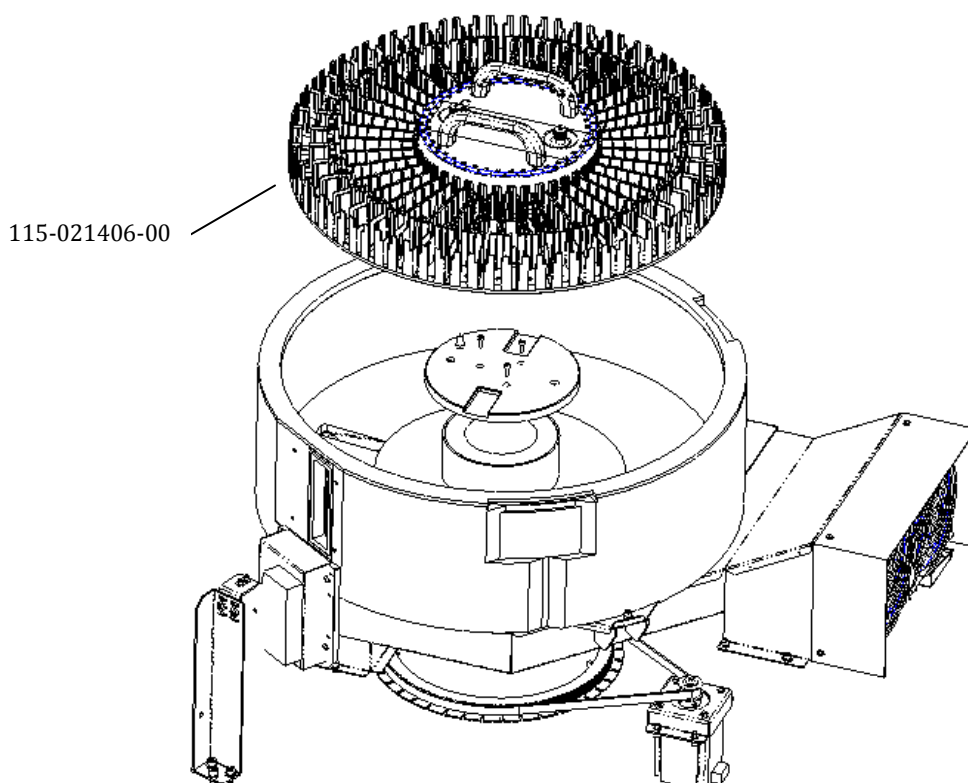
5.5.5 Replacement of Reagent Refrigeration Assembly

When to do

The reagent refrigeration assembly should be replaced if it is rusted seriously or has bad refrigeration effect.

Tools

Cross screwdriver, flathead screwdriver, hexagon wrench

Exploded view for installation**Figure 5.15** Removing and Reinstalling Reagent Refrigeration Assembly**Removing/Reinstallation procedure**

- 1 Switch off the power of the analyzer.
- 2 Remove the reagent carousel cover. Remove the panel around the **Reagent Refrigeration Assembly and the left side panel**.
- 3 Unscrew the two screws on the reagent carousel assembly by hands and remove it.
- 4 Unscrew the three 3*10 hexagon socket screws and remove the reagent carousel spindle to expose the transmission shaft.
- 5 Disconnect the drain tube from the reagent compartment.
- 6 Unplug the connectors of the refrigeration sensor, radiator and fan.
- 7 Remove the water proof shield on the Refrigerating air duct and the rubber cushion of the water proof shield.
- 8 Remove the ten M4*8 hexagon socket screws on the big bottom plate and Refrigerating air duct.
- 9 Remove the Reagent Refrigeration Assembly. Do not touch the reagent probe and wash well (If bar code reader is configured, loosen the screws on the bar code reader.)
- 10 Install the new Reagent Refrigeration Assembly and connect the radiator, reagent refrigeration sensor, fan cable and drain tubes. Adjust the reagent Refrigeration Assembly to make it concentric with the transmission shaft (If bar code reader is installed, check its position.)
- 11 Restore the reagent carousel spindle, reagent carousel body and the panels.

Alignment and confirmation

Refer to 8.13.7 to align the reagent Refrigeration Assembly. For bar code reader, refer to 8.11.4~8.11.6.

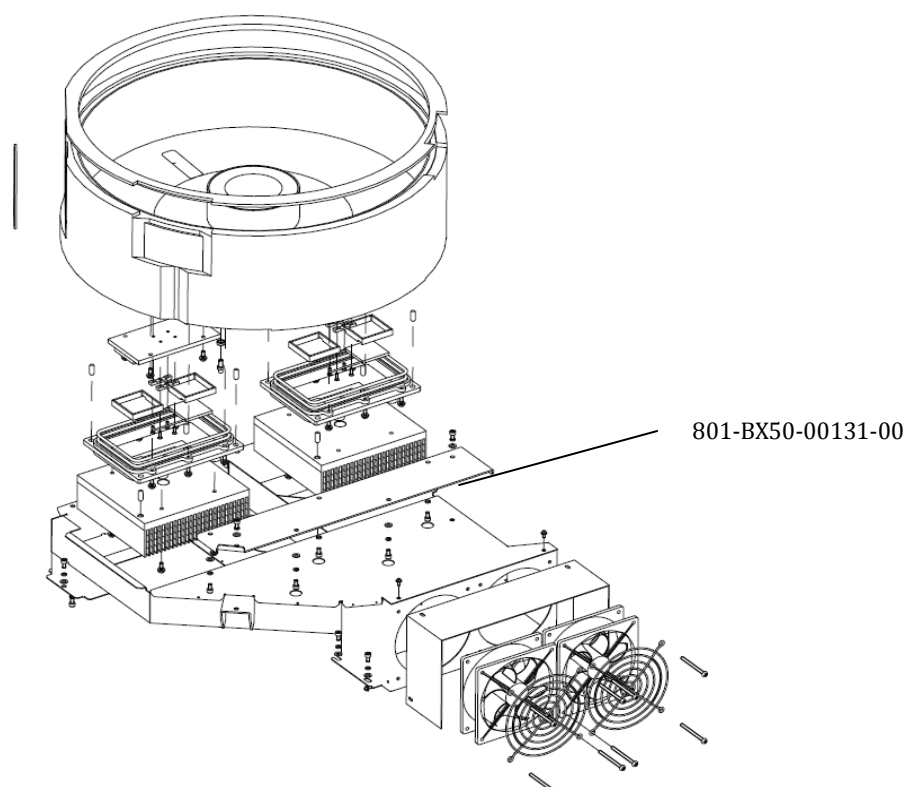
5.5.6 Replacement of Semi-conductive Radiator Cable

When to do

Replace the reagent refrigeration radiators when the alarm "Reagent refrigeration temperature is out of range." is given and the Peltier current is beyond 2~6.5A. If any cooler on the radiator is not qualified, change all the coolers on this radiator.

Tools

Cross screwdriver, flat head screwdriver, hexagon wrenches, scissors, Waterproof glue gun, Torque driver, AC impedance meter and knife.

Exploded view for installation**Figure 5.16** Removing and Reinstalling of the Semi-conductive cooler**Removing/Reinstallation procedure**

- 1 Enter the alignment screen and locate the failed cooler and mark it.
- 2 Switch off the power supply of the whole unit.
- 3 Remove the reagent carousel cover. Remove the panel around the **Reagent Refrigeration Assembly and the left side panel**.
- 4 Unscrew the two screws on the reagent carousel assembly by hands and remove it.
- 5 Unscrew the three 3*10 hexagon socket screws and remove the reagent carousel spindle to expose the transmission shaft.
- 6 Disconnect the drain tube from the reagent compartment.
- 7 Unplug the connectors of the refrigeration sensor, radiator and fan.
- 8 Remove the ten M4*8 hexagon socket screws on the reagent compartment and Refrigerating air duct. Remove reagent compartment. Do not touch the reagent probe and wash well (If bar code reader is configured, loosen the screws on the bar code reader.) Turn the compartment over for 180°, and put it on the desktop.
- 9 Use cross screwdriver to remove the 4 M4*12 cross screws on the radiator and remove the radiator. Use a single-side blade to clear the sealing glue around it.
- 10 Use cross screwdriver to remove the 6 M4*12 cross screws on the insulating plate and remove the insulating plate and square sealing ring.
- 11 Cut off the cables of the two coolers and remove the two coolers and use blade to clear the heat-conducting glue from its surface.
- 12 Use the AC impedance meter to measure the AC resistance of the cooler and mark the coolers by 1#~2#, 3#~4#;

- 13** Evenly daub KP-97 thermal paste on the contact surface of the coolers. Install the cooler frames, contact the no-text side of the coolers with the radiator.
- 14** Replace the square sealing ring and reinstall the insulating plate. Lead the cooler cables through the holes on the radiator, and apply some TSE382-W waterproof glue in the holes.
- 15** Restore the radiator and use the 4 M4×12 cross pan head combination screws removed in step 4 and tighten the screws alternately by using a torque driver with 4Kgf.cm, 10Kgf.cm and 12Kgf.cm torque.
Use an AC impedance meter to measure the AC impedance of the four coolers. If the impedance change is less than 5/1000 based on the value obtained in step 7, the coolers are deemed qualified.
- 16** Apply heat-conducting glue around the radiator that will contact the reagent compartment.
- 17** Restore the Reagent Refrigeration compartment and connect the radiator, reagent refrigeration sensor and drain tubes. Adjust the reagent Refrigeration Assembly to make it concentric with the transmission shaft (If bar code reader is installed, check its position.)
- 18** Restore the reagent carousel spindle, reagent carousel body and the panels.

**NOTE**

1. When replacing the Peltiers, store the removed screws properly.
2. When applying the TSE382-W waterproof glue, completely fill the space between the cable hole and Peltier cable.
3. Evenly daub KP-97 thermal paste on the contact surface of the Peltiers or on the Peltiers, and ensure that they are evenly coated with thermal paste.
4. When the cooler is changed, the square sealing ring between the insulating plate and radiator needs to be changed.

Alignment and confirmation

Refer to 8.13.7 to align the reagent Refrigeration Assembly. For bar code reader, refer to 8.11.4~8.11.6.

5.5.7 Replacement of Reagent Carousel Motor and Belt

When to do

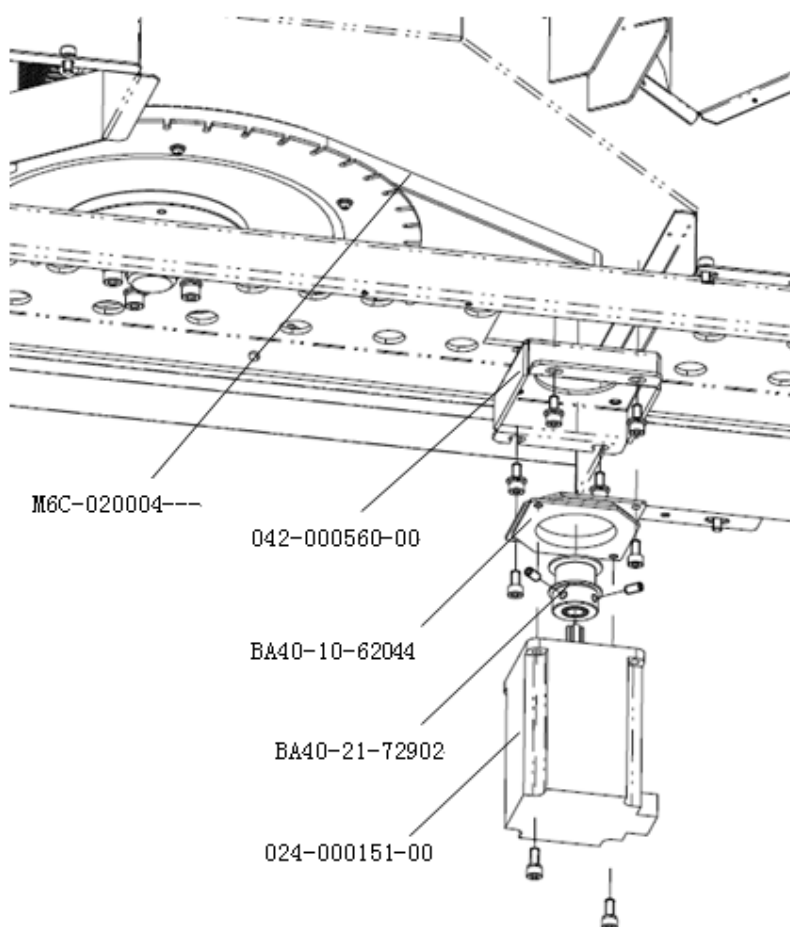
The reagent carousel motor and belt should be replaced when they are damaged.

Tools

Flathead screwdriver; hexagon wrench

Exploded view for installation

Figure 5.17 Replacement of Reagent Carousel Motor and Belt

**Removing/Reinstallation procedure**

Replace the motor:

- 1** Refer to 5.5.5 Replacement of Reagent Refrigeration Assembly to remove it.
- 2** Remove the rear panel 1 and unplug the motor cable.
- 3** Unscrew the 4 screws on the motor bracket and remove the belt from the wheel.
Remove the motor assembly.
Loosen the two retaining screws on the motor and remove the motor with the belt.
- 4** Fix a new motor with the pulley on the damping board using two M4X10 hexagon socket head screws with spring washer, and the motor should face the back of the analyzer.
- 5** Sleeve the belt on the wheel. Fix the motor assembly to the large bottom plate using four M5X12 hexagon socket cap head screws with spring washer. Tighten the belt and screws and connect the cable of the motor.
- 6** Restore the Reagent Refrigeration Assembly and the panel.

Change the synchronous belt:

- 1** Refer to 5.4.5 Replacement of Reagent Refrigeration Assembly to remove it.
- 2** Remove Rear panel 1
- 3** Unscrew the 4 screws on the motor bracket and remove the belt from the wheel.
- 4** Replace with new belt. Fix the motor assembly to the large bottom plate using four M5X12 hexagon socket cap head screws with spring washer. Tighten the belt and screws.
- 5** Restore the Reagent Refrigeration Assembly and the panel.

Alignment and confirmation

N/A

5.5.8 Replacement of Reagent Carousel Home Sensor and Coder Sensor Cables

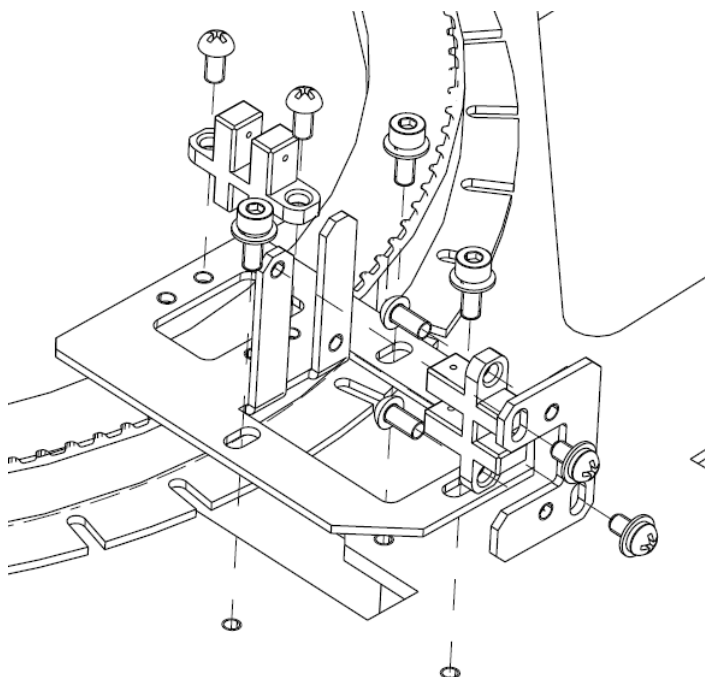
When to do

If the home position sensor and coder sensor or the reagent carousel become invalid, replace them immediately.

Tools

Hexagon wrench, cross head screwdriver

Exploded view for installation

Figure 5.18 Removing and Reinstalling Reagent Carousel Sensor Assembly**Removing/Reinstallation procedure**

- 1 Switch off the power of the analyzer.
- 2 Loosen the screws used to fix the front panel and remove the panel.
- 3 Disconnect the home position sensor and coder sensor.
- 4 Loosen the screws of the reagent carousel sensor and remove the sensor assembly.
- 5 Unscrew the screws on the sensor's mounting plate and remove the coder sensor cable with the sensor's mounting plate.
- 6 Unscrew the screws on the coder sensor, and remove the coder sensor cable from the mounting plate.
- 7 Fix the new coder sensor on the mounting plate using 2 M3×6 cross pan head screws.
- 8 Install the sensor's mounting plate together with cables of the encoder sensor onto the bracket of the sensor using two M3X6 cross pan head screws. Do not tighten them.
- 9 Unscrew the screws on the home position sensor of the reagent carousel, and remove the sensor from sensor bracket.
- 10 Fix the new cables of carousel home sensor onto the sensor bracket using two M3X6 cross pan head screws.
- 11 Fix the reagent carousel sensor assembly on the large bottom plate using three M3X8 hexagon socket cap head screws. Adjust the mounting plate and the sensor to ensure the coder is positioned in the middle of the coder sensor. Tighten the retaining screws of the mounting plate.
- 12 Connect the home position sensor and coder sensor.
- 13 After the position of the reagent carousel sensor assembly is aligned, fix the front panel with 3 M4*10 screws on top and 4 M4*8 cross pan head screws with washer in the front. Adjust the gap of the panels and then tighten the screws

Note:

To replace only the coder sensor cable, neglect step 9 and 10.

To replace only the home position sensor, neglect step 5 to 8.

Alignment and confirmation

Refer to 8.6 to check the reagent carousel position and refer to 8.8 to perform the Reagent Probe Rotary to Bottle alignment.

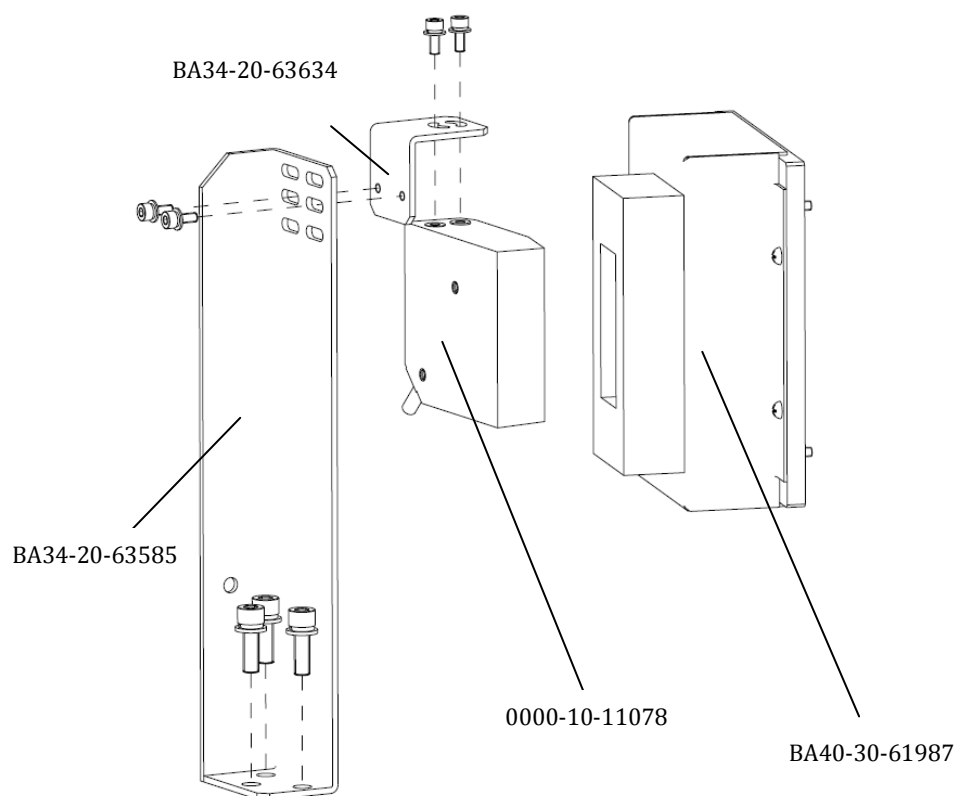
5.5.9 Replacement of Reagent Bar Code Reader

When to do

The reagent bar code reader should be replaced when it does not work normally.

Tools

Hexagon wrench

Exploded view for installation**Figure 5.19** Replacement of Reagent Bar Code Reader**Removing/Reinstallation procedure**

- 1** Remove the panel related to the reagent bar code reader.
- 2** Loosen the 2M3X10hexagon socket screws and remove the cable of the bar code reader to remove it.
- 3** Replace it with a new one and connect it in the reverse order of the above steps.

Alignment and confirmation

Refer to 8.11.4 to check the reagent bar code reader position.

5.6 Sample/reagent Probe Unit

5.6.1 Module Functions

Sample/reagent probe assembly includes sample probe movement assembly and reagent probe movement part.

Sample movement part can aspirate the sample from the sample tube and dispense it into the cuvette through the sample probe.

Reagent movement assembly can aspirate the reagent from the reagent bottle and dispense it into the cuvette through the reagent probe.

They both have the following functions: featuring level detection, horizontal/vertical bump detection and clog detection. The probe assembly has other functions, such as limiting of mechanical position, self-lock during power interruption, etc.

Sample/reagent probe position:

Sample probe assembly: wash well --> aspirate position on sample carousel --> dispense position on reaction carousel or ISE module.

Reagent probe assembly: wash well --> aspirate position on reagent carousel --> dispense position on reaction carousel.

5.6.2 Locations and FRU Details

Figure 5.20 Sample/reagent probe assembly

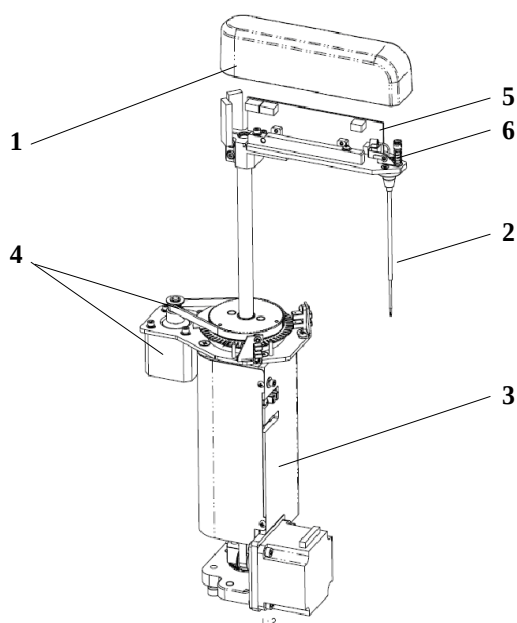


Table 5.5 List of FRU materials

No.	FRU code and material code	Material Name	Remark
1	043-002344-00	Probe arm cover (mould MR61400)	FRU
2	115-022013-00	BA49 Reagent probe assembly	FRU
	115-022012-00	BA49 Sample Probe Assembly	FRU

No.	FRU code and material code	Material Name	Remark
3	115-020536-00	Sample (ISE) probe drive assembly	FRU
	115-020537-00	Reagent probe drive assembly	FRU
4	115-015297-00	42 step motor 1 with belt pulley (horizontal)	FRU
5	801-BA80-00036-00	Sample Probe Level Sense Board	FRU
	051-000363-00	400 μ l reagent probe level sense board	FRU
6	115-016462-00	Guide Rod and Anti-Collision Spring	FRU

5.6.3 Replacement of Reagent Probe Assembly, Anti-collision Spring, and sample/reagent probe Arm Assembly

When to do

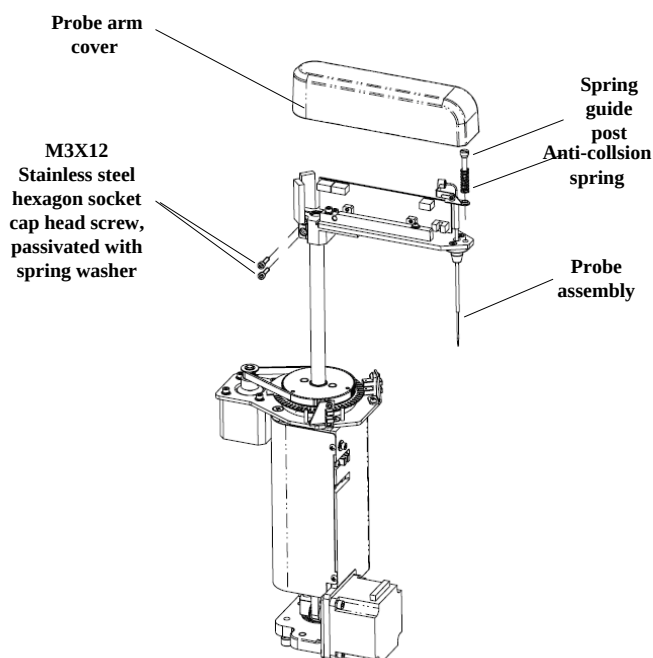
The probe should be replaced if it is bent or rusted; and the anti-collision spring should be replaced if it is impaired.

Tools

Flathead screwdriver, hexagon wrench and Cutting pliers

Exploded view for installation

Figure 5.21 Replacement of Reagent Probe Assembly, Anti-collision Spring, and sample/reagent probe Arm Assembly



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover.
- 3 Remove the powder screws on the probe cover and lift it up to remove it.

- 4 Loosen the tube connector and unplug the fluid level sensor cable. If you are going to replace the entire arm assembly, remove the connector from the sensor cable.
- 5 Remove the spring guide post and the anti-collision spring.
- 6 Remove Reagent Probe Assembly.
- 7 Loosen the two M3X12 hexagon socket cap head screws with washer on the sample/reagent probe arm assembly, and then remove the arm assembly.
- 8 Fix the new sample/reagent probe arm assembly on the spline shaft using two M3X12 hexagon socket cap head screws with spring washer.
- 9 Lead the new sample/reagent probe assembly through the mixer arm and support sleeve.
- 10 Fix the new anti-collision spring on the mixer arm using the spring guide post.
- 11 Refer to the above-mentioned steps in reversed order to install the tube connector, cable of level sensor board sensor and mixer arm cover.

Alignment and confirmation

Refer to 8.7 Sample Probe Unit and 8.8 Reagent Probe Unit to perform the alignment related to the sample/reagent probe.

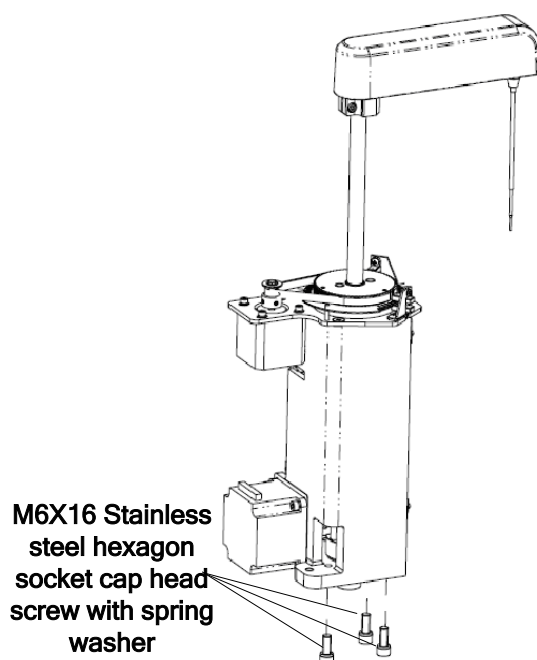
5.6.4 Removing and Reinstalling Sample/Reagent Drive Assembly

When to do

The sample/reagent probe drive assembly or its components should be replaced if they are impaired in functioning or damaged.

Tools

Cross screwdriver, hexagon wrench and Cutting pliers

Exploded view for installation**Figure 5.22** Removing and Reinstalling Sample/Reagent Drive Assembly**Removing/Reinstallation procedure**

- 1 Switch off the main power of the whole unit.
- 2 Open the shielding cover and left/right front doors, and then remove the front panels around the probe drive assembly.
- 3 Refer to 5.6.3 Replacement of Reagent Probe Assembly, Anti-collision Spring, and sample/reagent probe Arm Assembly to remove the sample/reagent probe assembly.
- 4 Unplug the motor cable, cable of level sense board sensor, and fluidic tube.
- 5 Loosen the three M6X16 hexagon socket head screws with spring washer on the probe drive assembly, and then remove the probe drive assembly.
- 6 Fix the new sample/reagent probe drive assembly on the large bottom plate using three M6X16 hexagon socket cap head screws with spring and flat washers. Make sure to lead the fluidic tube and sensor cable through the spline shaft prior to the installation.
- 7 Refer to the above-mentioned steps in reversed order to connect the motor cable, sensor cable and fluidic tube, and install the probe arm assembly, front panel and other parts.

Note:

Note: Connect connectors to the probe drive assembly sensor cable after leading the cable through the supporting flange.

Alignment and confirmation

Refer to 8.7 and 8.8 to perform the alignment related to the sample/reagent probe.

5.6.5 Removing and Reinstalling 42 Step Motor with Belt Pulley (horizontal) and Synchronous Cog Belt

When to do

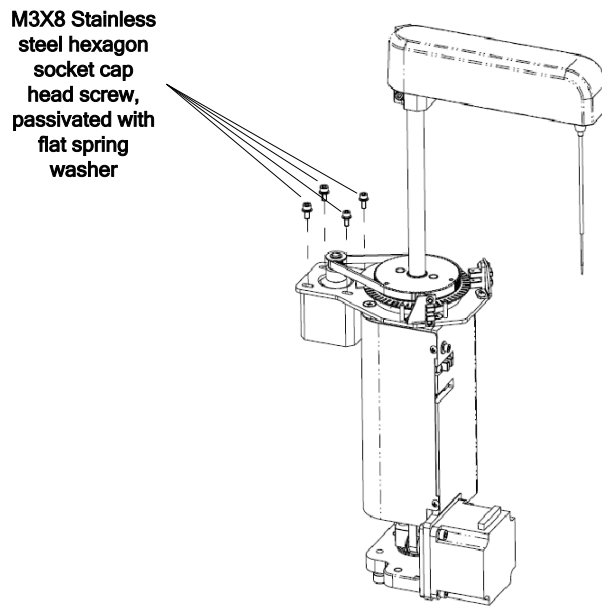
The probe horizontal motor assembly and synchronous cog belt should be replaced if they are impaired in functioning or damaged.

Tools

Cross screwdriver, hexagon wrenches and Synchronous belt tension alignment tool

Exploded view for installation

Figure 5.23 Removing and Reinstalling 42 Step Motor with Belt Pulley



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the shielding cover and remove the panels around the probe assembly.
- 3 Unplug the 42-step horizontal motor cable.
- 4 Loosen the four M3X12 socket head screws with spring and flat washers on the probe horizontal motor assembly, and then remove the motor assembly and synchronous cog belt.
- 5 Fix the new 42-step horizontal motor assembly on the support using four M3X12 hexagon socket cap head screws with spring and flat washers, and then sleeve the new cog belt on the synchronous pulley. Do not tighten the screws until you use the tension alignment tool to adjust the synchronous belt to the specified value (385~433Hz).
- 6 Refer to the above-mentioned steps in reversed order to install the 42-step horizontal motor cable, and panels.

Alignment and confirmation

N/A

5.7 Mixer Assembly

5.7.1 Module Functions

Mixer Assembly includes sample mixer and reagent mixer used to mix the reaction liquid after the sample or reagent is dispensed.

5.7.2 Locations and FRU Details

The reagent mixer assembly is located in the rear right of the instrument. The sample mixer assembly is located in the rear left of the instrument.

Figure 5.24 Mixer Assembly

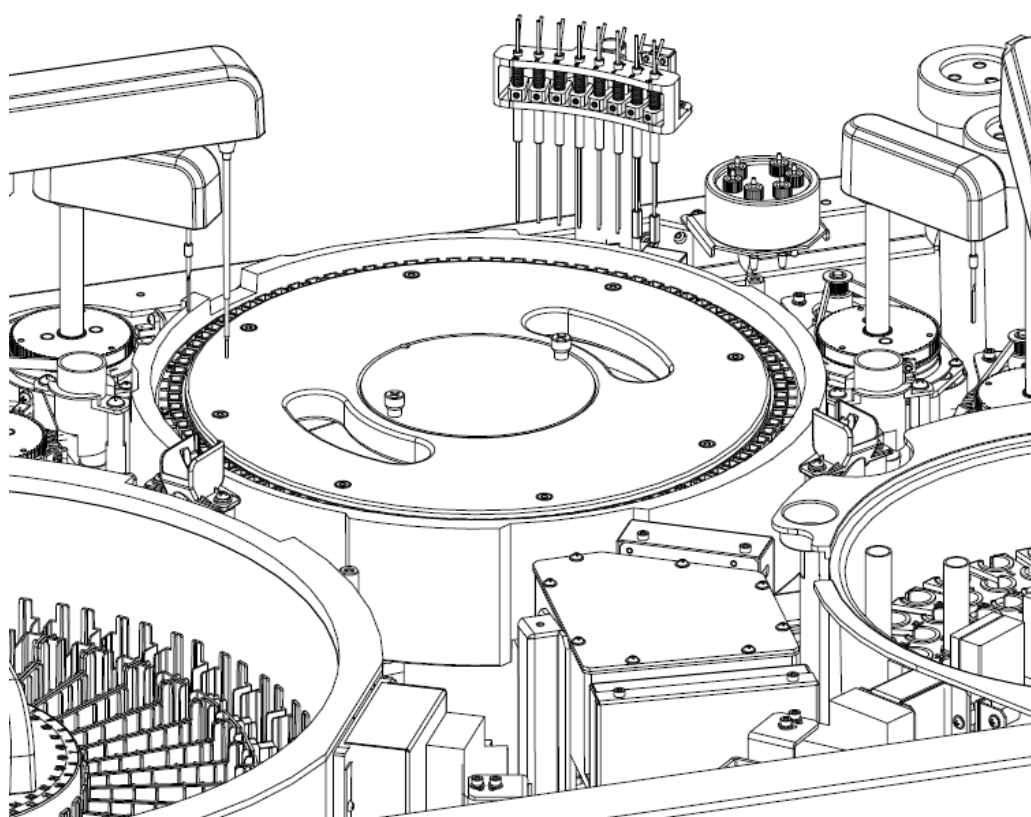


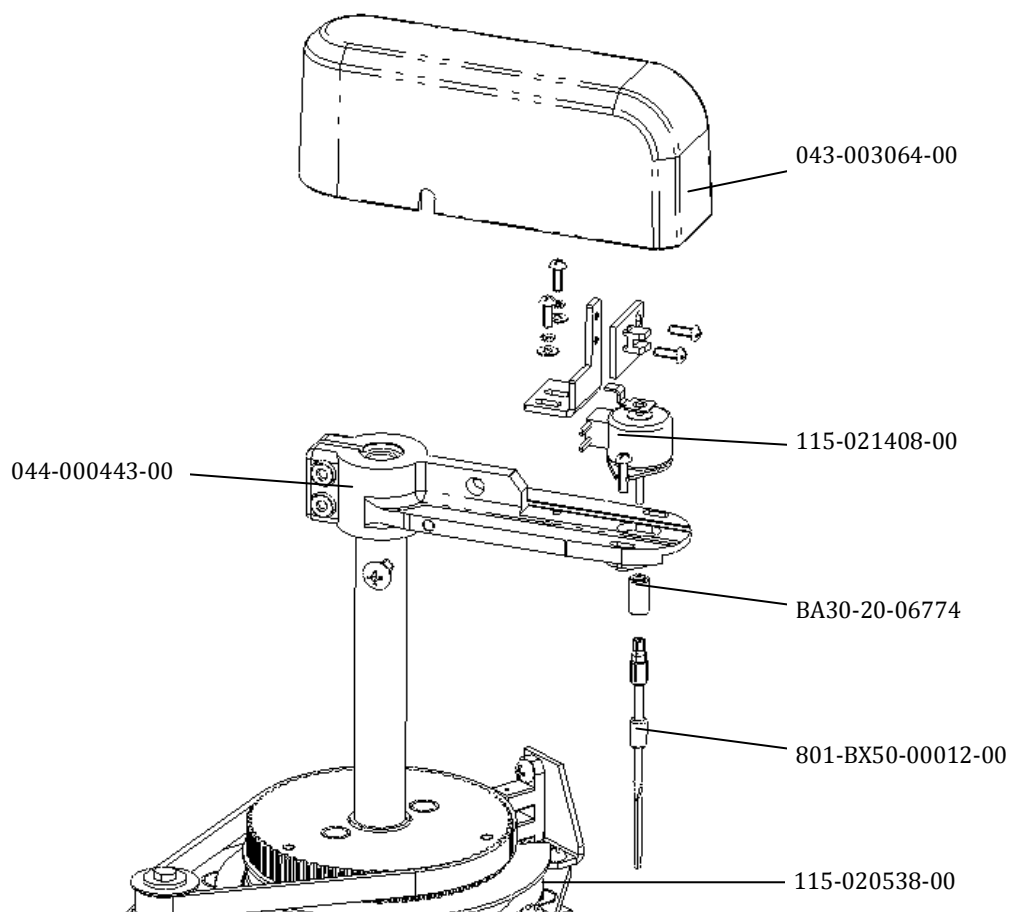
Table 5.6 List of materials

No.	FRU code and material code	Material Name	Remark
1	115-020538-00	Mixer drive assembly	FRU
2	801-BX50-00012-00	Mixer	FRU
3	115-021408-00	Mixer motor assembly	FRU
4	043-003064-00	Mixer arm cover (mould MR61400)	FRU
5	115-015297-00	42 step motor 1 with belt pulley (horizontal)	FRU
6	051-001620-00	Rotation speed optical coupler conversion board	FRU

5.7.3 Removing of mixer, mixer motor, optical coupler conversion board and arm assembly

Replacement of the sample mixer assembly is described in this section, and it is the same in steps and materials as reagent mixer assembly.

Figure 5.25 Removing of mixer, mixer motor, optical coupler conversion board and arm assembly



Replacement of Mixers

When to do

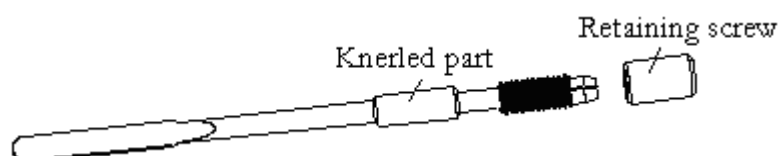
The mixer is bent or damaged.

Tools

N/A

Exploded view for installation

Figure 5.26 Removing and reinstalling mixer



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover, and pull the mixer arm to the highest point for operation.
- 3 Grab the knurled part of the mixer with one hand and unscrew the retaining screw counter-clockwise with the other hand to loosen the mixer, remove the mixer and retaining screw.
- 4 Reinstall the new mixer in the reverse order of above steps.

Note:

When installing the mixer, push it completely along the rotor. Force in other directions may damage the rotor or mixer.

Alignment and confirmation

N/A.

Replacement of Mixer Motor**When to do**

Mixer motor failed.

Tools

Cross screwdriver

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover, and pull the mixer arm to the highest point for operation.
- 3 Grab the knurled part of the mixer with one hand and unscrew the retaining screw counter-clockwise with the other hand to loosen the mixer, remove the mixer and retaining screw.
- 4 Remove the powder screws on the mixer cover and lift it up to remove it.
- 5 Disconnect the motor cable and loosen the two M2*8 screws to remove the mixer motor.
- 6 Refer to the above-mentioned steps in reversed order to install the cable of the mixer motor, and fix the motor.
- 7 Adjust the mixer motor position and install the mixer and mixer cover.

Note:

When installing the mixer, push it completely along the rotor. Force in other directions may damage the rotor or mixer.

Alignment and confirmation

Refer to 8.9 Mixer Unit to perform the alignment of optical coupler conversion board and Sample Mixer Rotary to Cuvette.

Replacement of the optical coupler conversion board**When to do**

Optical coupler conversion board failed.

Tools

Cross screwdriver

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover, and pull the mixer arm to the highest point for operation.
- 3 Remove the powder screws on the mixer cover and lift it up to remove it.
- 4 Loosen the two M2*6 screws on the optical coupler conversion board and remove the board.
- 5 Connect the optical coupler conversion board in the reverse order of the above steps and after adjusting the position of the optical coupler conversion board, install back the mixer cover.

Alignment and confirmation

Refer to 8.9 Mixer Unit to adjust the position of the optical coupler conversion board.

Replacement of Arm Assembly

When to do

The motor should be replaced when going wrong, the mixer replaced when bent or damaged, and the arm assembly replaced when damaged.

Tools

Flathead screwdriver, hexagon wrench

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the front shielding cover, and pull the mixer arm to the highest point for operation.
- 3 Remove the powder screws on the mixer cover and lift it up to remove it.
- 4 Disconnect the mixer motor cable. Loosen the two hexagon socket cap head screws with spring washer on the mixer arm assembly, and then remove the arm assembly.
- 5 Fix the new arm assembly on the spline shaft using two hexagon socket cap head screws with spring washer. Note to lead the motor cable through the spline shaft first.
- 6 Refer to the above-mentioned steps in reversed order to install the mixer motor cable and mixer arm cover.

Alignment and confirmation

Refer to 8.9 Mixer Unit to adjust the mixer position.

5.7.4 Replacement of Sample/Reagent Mixer Drive Assembly

Replacement of the sample mixer assembly is described in this section, and it is the same in steps and materials as reagent mixer assembly.

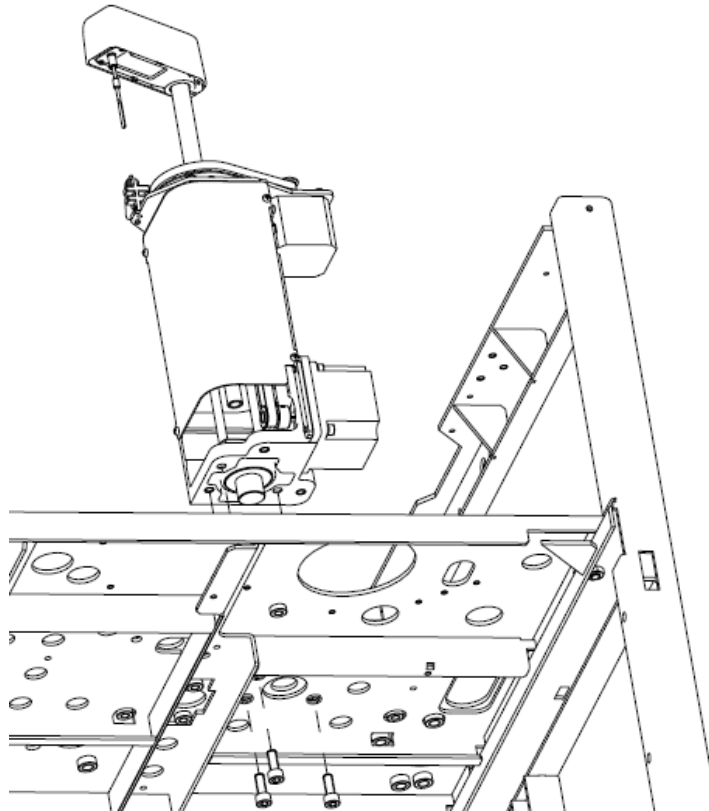
When to do

The mixer drive assembly should be replaced when it is damaged or impaired in functioning.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Open the shielding cover, and remove the panels around the probe drive assembly and rear panel.
- 3 Refer to 5.6.3 to remove the mixer arm assembly.
- 4 Loosen the three M6X16 hexagon socket head screws with spring washer on the mixer drive assembly, and then remove the mixer drive assembly.
- 5 Fix the new mixer drive assembly on the large bottom plate using three M6X16 hexagon socket cap head screws with spring and flat washers. Make sure to lead the motor cable through the spline shaft prior to the installation.
- 6 Refer to the above-mentioned steps in reversed order to connect the arm assembly, mixer motor and optical coupler cable, and install the mixer cover assembly, panels and other parts.

Alignment and confirmation

Refer to 8.9 Mixer Unit to perform the alignment of optical coupler conversion board and Sample Mixer Rotary to Cuvette.

5.7.5 Removing and Reinstalling 42 Step Motor with Belt Pulley (horizontal) and Synchronous Cog Belt

When to do

The mixer horizontal motor assembly and synchronous cog belt should be replaced if they are impaired in functioning or damaged.

Tools

Flathead screwdriver, hexagon wrench

Removing/Reinstallation procedure

5 Modules and Units

The steps are same as 5.6.5 except those of arm

Alignment and confirmation

N/A

5.8 Wash syringe assembly

5.8.1 Module Functions

The wash syringe assembly consists of the probe interior wash syringe assembly and the cuvette wash syringe assembly, which are identical in mechanical structure and share the same syringe assembly, but are different in timing and fluidics.

5.8.2 Locations and FRU Details

The two parts of the wash syringe assembly are located in the lower-right side of the instrument. The wash syringe assembly is composed of the drive assembly, wash syringe, etc.

Figure 5.27 Removing/Installing Wash Syringe

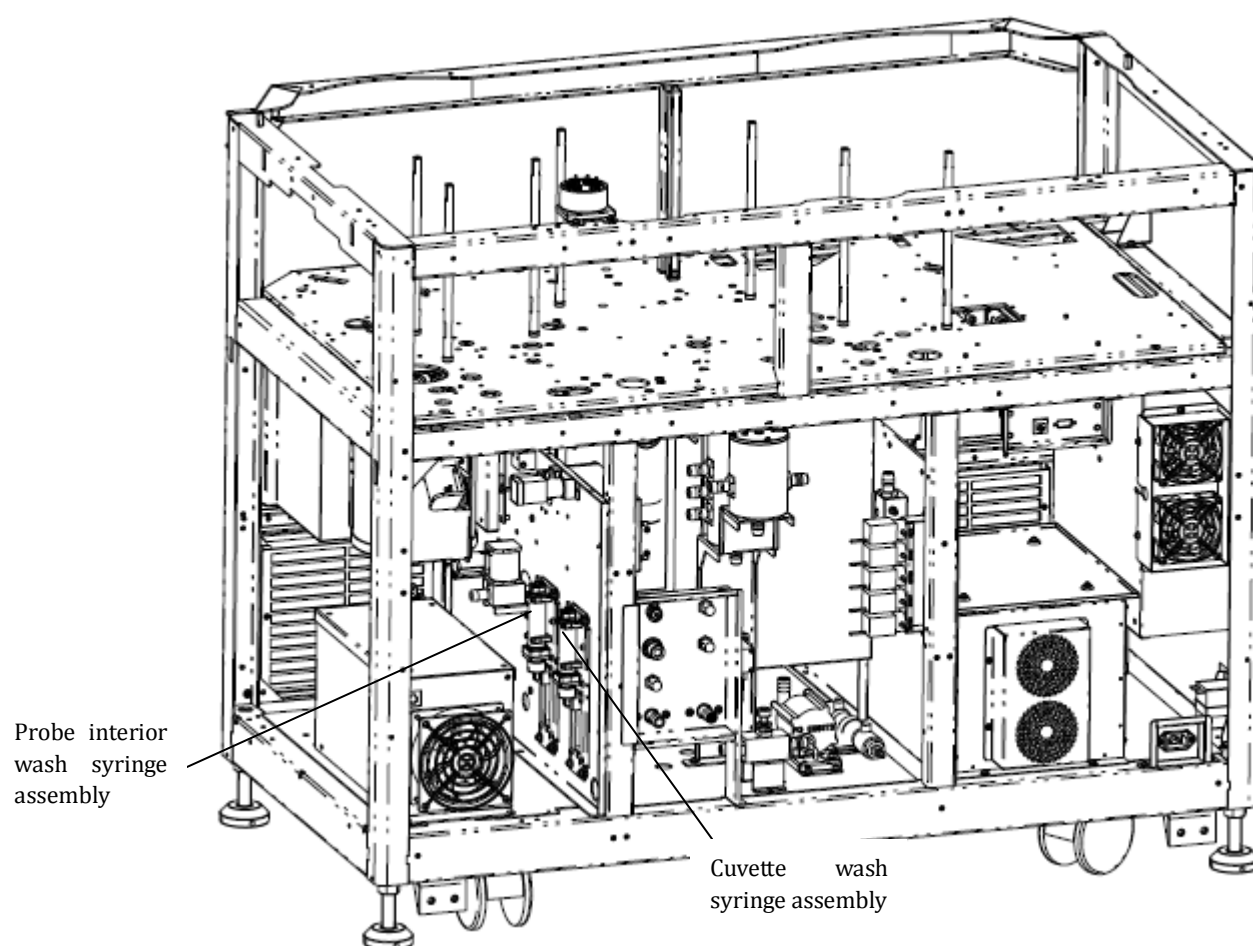


Table 5.7 List of materials

No.	FRU code and material code	Material Name	Remark
1	115-013114-00	Wash syringe assembly	FRU
2	009-002204-00	Sensor	FRU
3	024-000366-00	Wash syringe motor	FRU

5.8.3 Removing and Reinstalling Wash Syringe Assembly

Removing and reinstalling of cuvette wash syringe assembly is described in this section, which is the same as the probe interior wash syringe assembly.

When to do

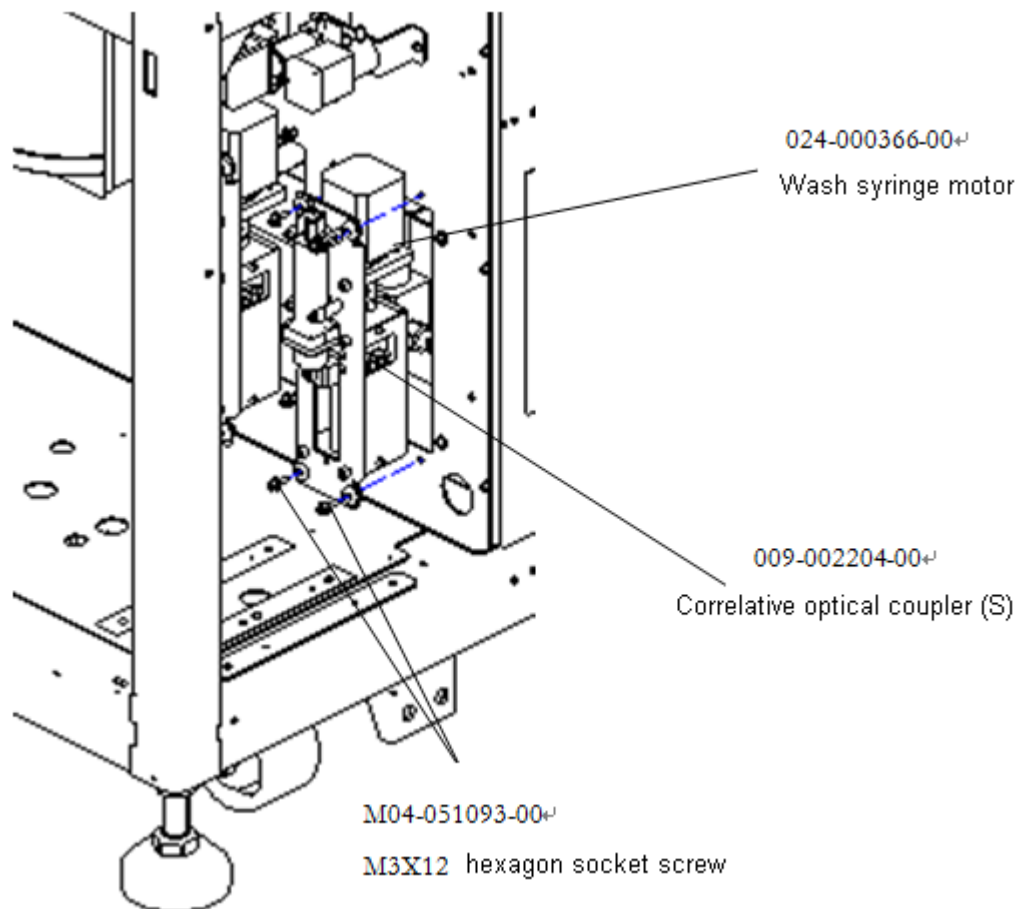
The wash syringe assembly should be removed when replacing the entire assembly, or the motor, sensor and other parts.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation

Figure 5.28 Removing/Installing Wash Syringe



Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Remove the right plate and rear plate. Remove the vacuum pump assembly before removing or installing the wash syringe.
- 3 Unplug the connectors of sensor 3 and wash syringe motor 4, and the wash syringe tube.
- 4 Loosen the four hexagon socket cap head screws to remove the wash syringe assembly.
- 5 Install the new drive assembly, and then the removed parts such as the four screws, vacuum pump assembly, syringe tube, motor cable, right plate, and rear panel.

Alignment and confirmation

N/A.

5.8.4 Removing and Reinstalling Wash Syringe Sensor

When to do

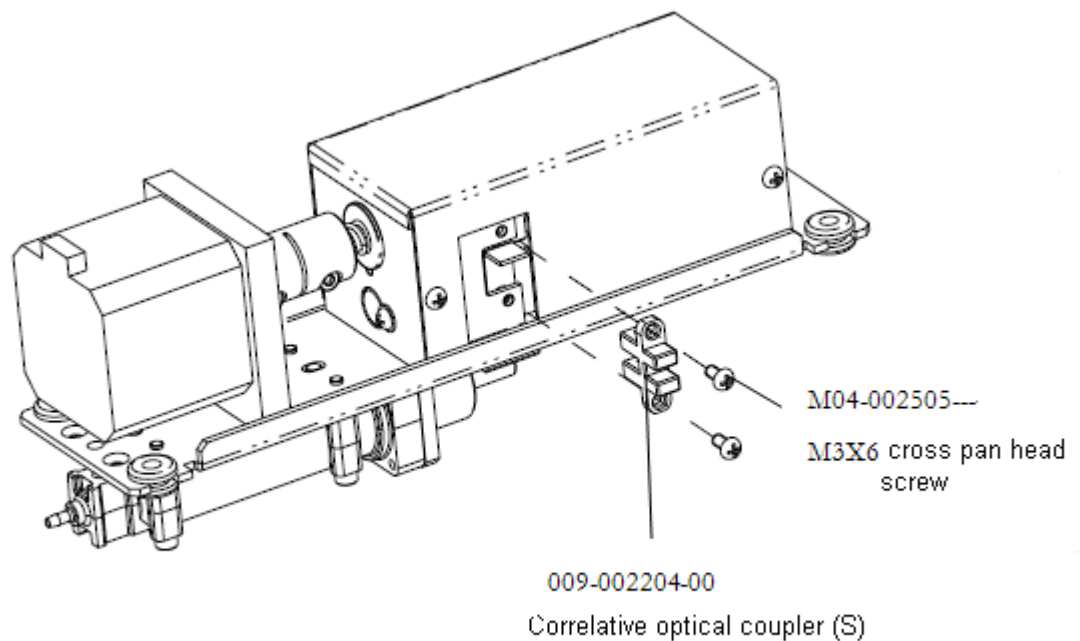
The wash syringe sensor should be removed when it is abnormal or damaged by external force.

Tools

Cross screwdriver

Exploded view for installation

Figure 5.29 Removing/Installing Wash Syringe Sensor



Removing/Reinstallation procedure

- 1 Refer to 5.8.3 Removing and Reinstalling Wash Syringe Assembly to remove the wash syringe sensor assembly.
- 2 Remove the two cross head screws 6 on the wash syringe sensor, unplug the sensor connector and remove the sensor cable.
- 3 Install the new sensor and cable connector, and then install the wash syringe assembly and vacuum pump assembly by referring to 5.8.3 Removing and Reinstalling Wash Syringe Assembly.

Alignment and confirmation

N/A

5.8.5 Removing and Reinstalling Wash Syringe Motor

When to do

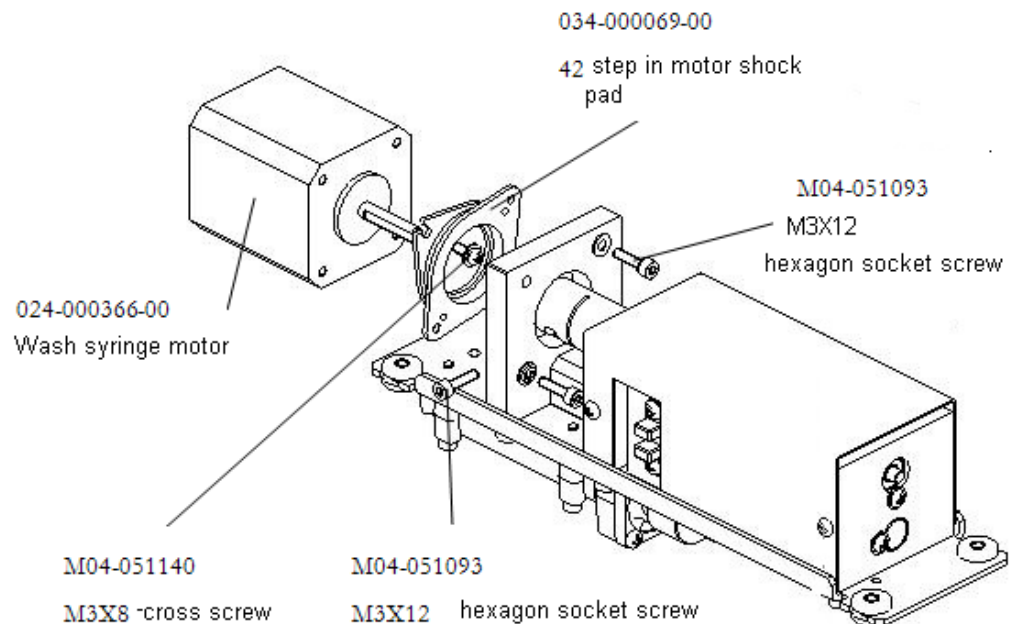
The wash syringe motor should be removed when it does not work normally or damaged.

Tools

Flathead screwdriver, hexagon wrench

Exploded view for installation

Figure 5.30 Removing/Installing Wash Syringe Motor



Removing/Reinstallation procedure

- 1 Refer to 5.8.3 Removing and Reinstalling Wash Syringe Assembly to remove the wash syringe sensor assembly.
- 2 Loosen the screw (with glue) on the shaft by the motor side, and the four screws on the motor.
- 3 Install the four screws on the motor, tighten the screw on the shaft and then the four screws, apply glue on the shaft screw and lock it tight.
- 4 Install the wash syringe assembly and vacuum pump assembly by referring to 5.8.3 Removing and Reinstalling Wash Syringe Assembly.

Alignment and confirmation

N/A

5.8.6 Removing and Reinstalling Wash Syringe

For replacement of the syringe, refer to 7.5.2 Probe wash module.

5.9 Photometer Unit

5.9.1 Functions and Parameters

The BS-480/BS-490 has inherited the mature concave gratings light-splitting technology of the BS series, which can not only simplify the optical design to compact the optical structure, but also eliminate the stray light. A combined light passing through the entrance slit projects on the PDA (Photodiode Array) via the concave flat-field gratings. One or multiple light-activated elements on different positions receive the monochromatic light at certain wavelength. During operation, the photometric system controlled by the computer receives the electric signals produced by the light-activated elements of corresponding wavelength and then converts them into absorbance. In case the spectrum is defined, the absorbance at multiple wavelengths can be calculated quickly.

Technical Parameters

- Optical system: reversed optics of holographic concave flat-field gratings, with each wavelength detected by a Photodiode array.
- Wavelength: 12 wavelengths, 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm, and 800nm
- Wavelength accuracy: $\pm 2\text{nm}$
- Light source: 12V/20W, tungsten-halogen lamp, transmitting light through fiber bundle
- Minimum reaction volume: 120 μL
- Absorbance range: 0-3.3A, 10mm light pathlength
- Light pathlength of cuvette: 5mm
Number of cuvettes: 90 cuvettes, 5mm $\times$ 5mm $\times$ 29mm
- Cuvette material: plastic cuvette

Terms of Optics

- AD value: the value converted from photoelectric signal (voltage) through AD converter.
- Water blank: the AD value of a cuvette measured when the wash station dispenses water in phase 6. Water blank is the base point for calculation absorbance, that is, the 0 point of absorbance.
- Cuvette blank out of range: means the water blank of phase 6 is less than the light intensity low limit. This alarm is used to monitor the energy level of the optical system to ensure normal signal-to-noise ratio.
- Cuvette blank out of range (10X): means the relative change of continuous phase-6 water blanks for 10 cuvettes is greater than 3% comparing with the history data. This alarm is used to prevent light fluctuation or erroneous result due to cuvette overflowing and to remind the operator of these phenomena.

5.9.2 Composition and Structure of Optical Assembly

The BS-480/BS-490 applies the holographic concave flat-field gratings and PDA for photometric measurement. See the figure below. The front lens converge the light beam sent from the tungsten-halogen lamp to the reaction cuvette via the fiber bundle. The light beam passes through the reaction cuvette and then converges at the entrance slit via the lens group 2. The gratings divide the light beam from the entrance slit and then converges them to the slit array. Finally the PDA behind the slit array converts the light signals into electric signals and then outputs them.

Figure 5.31 Photometer light path

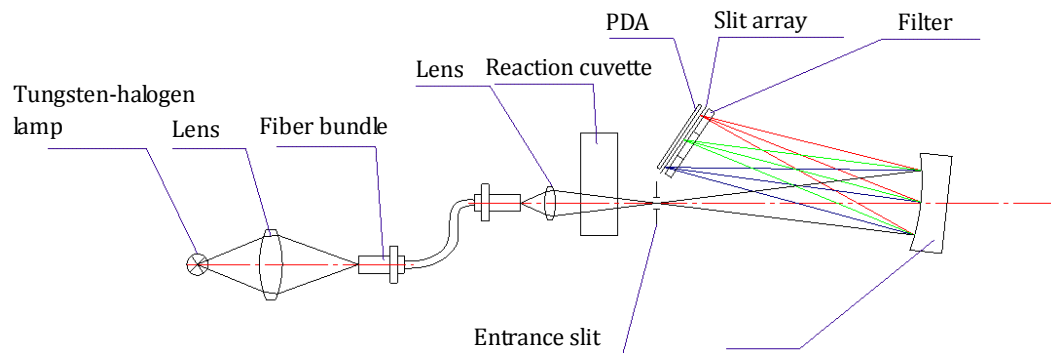
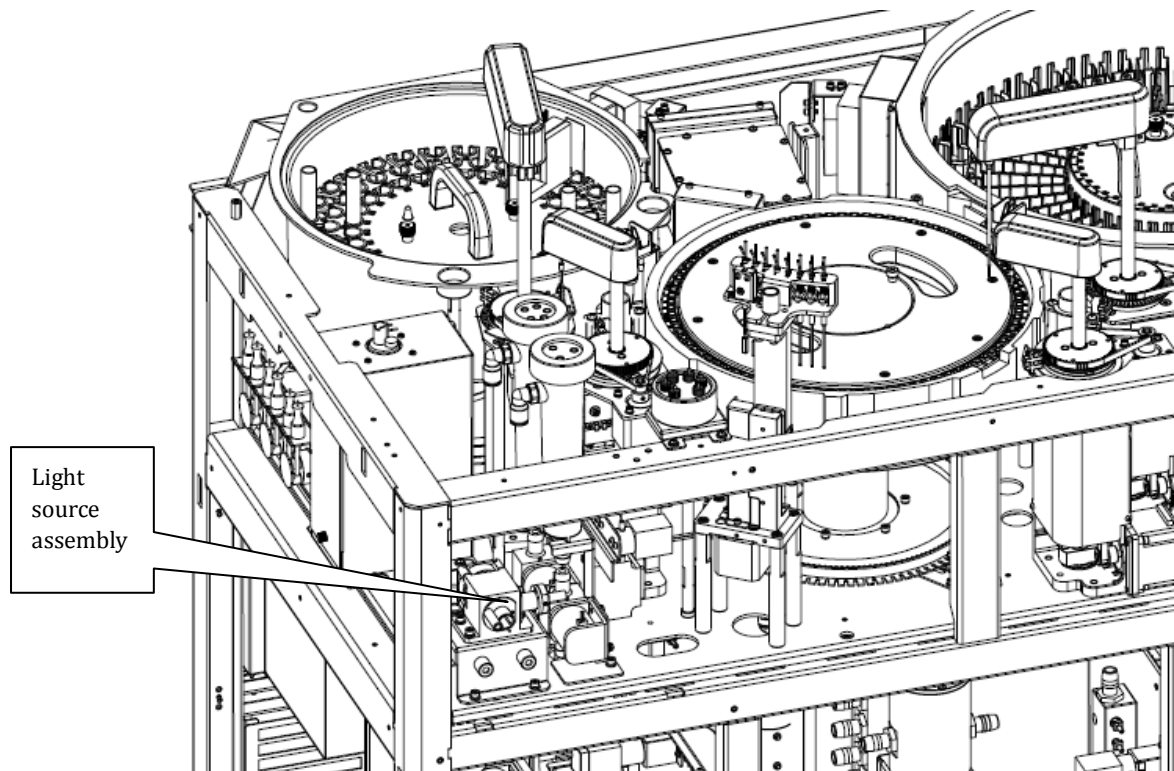
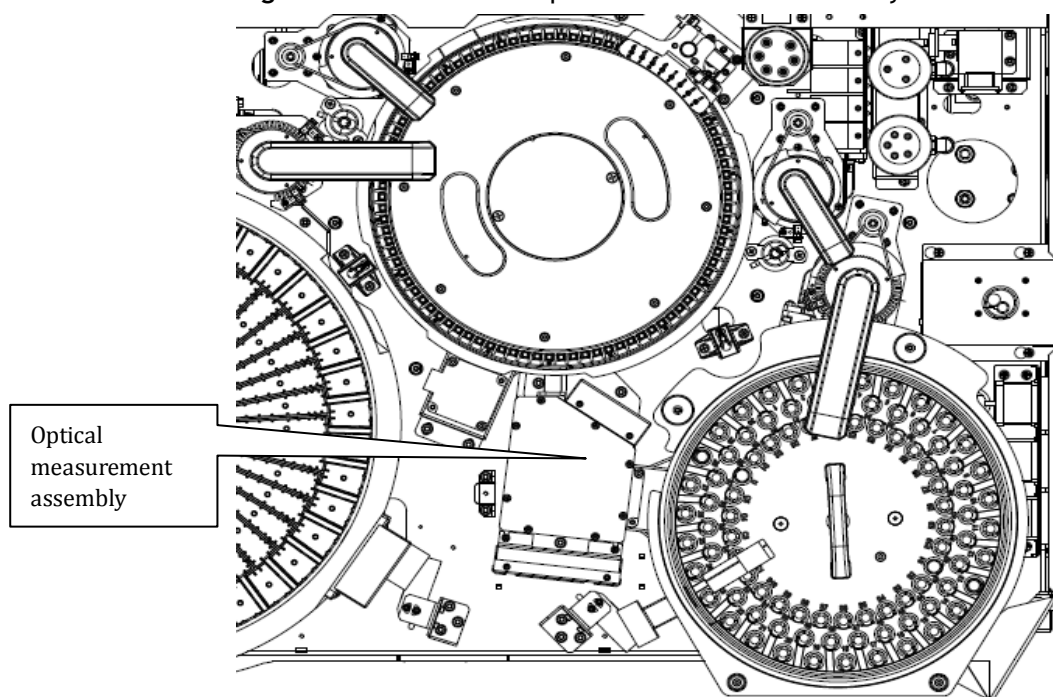


Figure 5.32 Location of Light Source Assembly



The optical measurement assembly consists of the light-splitting assembly and AD housing assembly.

Figure 5.33 Location of Optical Measurement Assembly**Table 5.8** List of materials

No.	FRU code and material code	Material Name	Remark
1	801-BA80-00222-00	20W lamp (FRU)	FRU
2	801-BA40-00167-00	Fiber bundle	FRU
3	115-004014-00	Plastic cuvette	FRU
4	115-022370-00	Optical measurement assembly (BA49)	FRU
5	115-022371-00	PDA assembly (BA49)	FRU

5.9.3 Replacement of Lamp

When to do

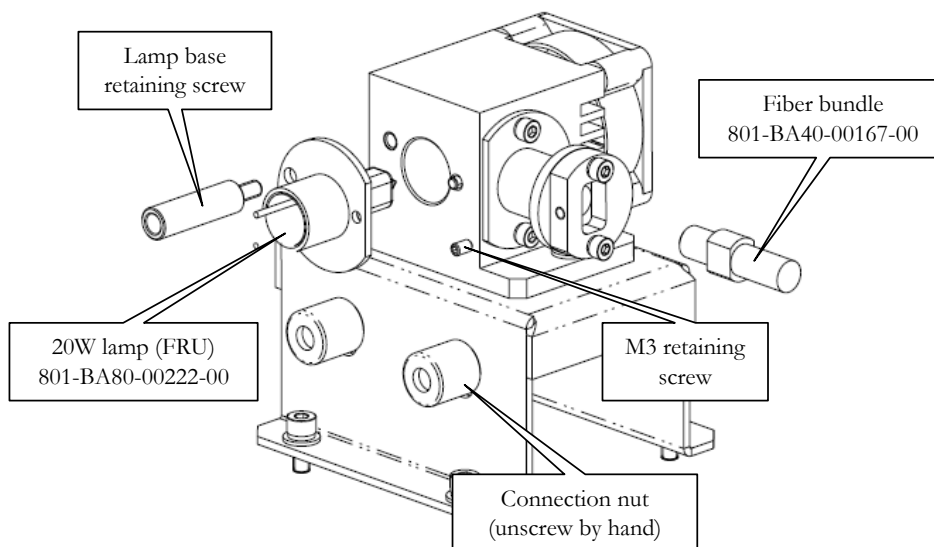
The lamp should be replaced when an alarm is triggered indicating that the lamp has insufficient intensity, or not turned on, or has been used for over 2000 hours, or has cuvette blank out of range (10X).

Tools

N/A.

Exploded view for installation

Figure 5.34 Exploded view of lamp and fiber bundle for installation



NOTE

Replace the lamp according to the instructions on the screen.

Please wait for at least 5 minutes (if the lamp fan works properly) after turning off the lamp to avoid injury.

Do not touch the bulb when installing the new lamp. In case of touching, please clean the bulb with fibre cloth or ethanol-moistened gauze.

Removing/Reinstallation procedure

- 1 Select Utility -> Maintenance -> Maintenance -> Biochemistry Maintenance, and select Replace Lamp.
- 2 Follow the instructions prompted by the software and wait for five minutes until the lamp cools down.
- 3 Loosen the screws on the back cover of the lamp and remove the lamp rear cover.
- 4 Unscrew the cable connectors and the lamp base retaining screws with hands, to remove the lamp.
- 5 Install the lamp and rear panel in a reversed order. Please ensure that the lamp has been installed properly in the right place and there is no space between the lamp and the lamp base before mounting the panel.
- 6 Select **Continue** and wait for ten minutes to start **Lamp Check** to check if the lamp can work properly.

Alignment and confirmation

Refer to Section 8.3 Photometer Unit.

5.9.4 Replacement of Fiber Bundle**When to do**

The fiber bundle should be replaced when it is broken or the light intensity is insufficient.

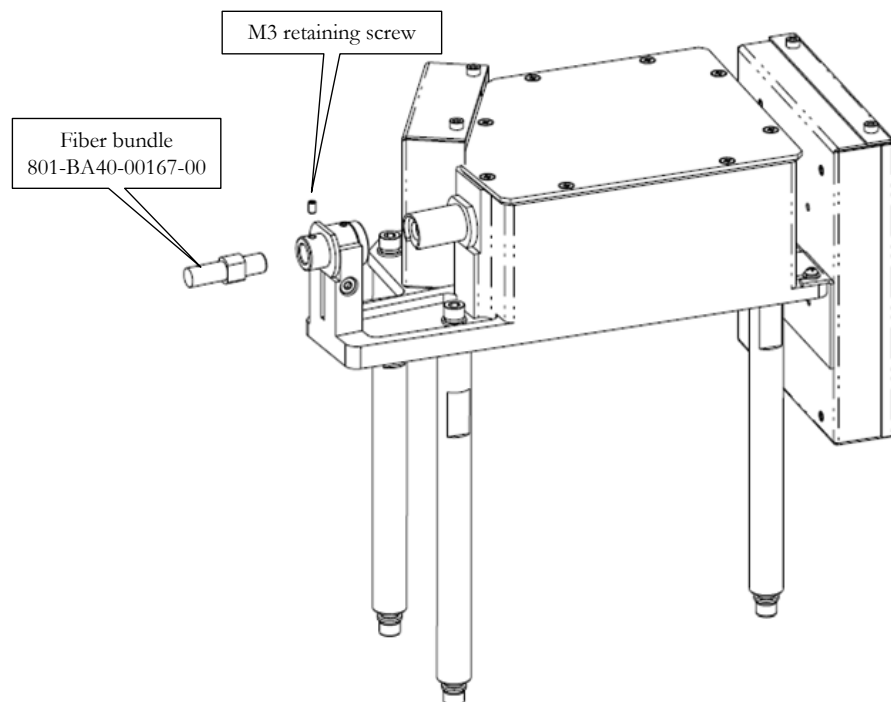
Tools

Hexagon wrench

Removing/Reinstallation procedure

- 1 Loosen the screws on the back cover of the lamp and remove the lamp rear cover.
- 2 According to the figure above, use a hexagon wrench to remove the M3 socket set screws with cup point on the light source assembly.
- 3 Refer to 5.3.3 Replacement of Reaction Carousel Body Cables to remove the reaction carousel body assembly.
- 4 According to the figure below, use a hexagon wrench to remove the M3 socket set screws with cup point on the optical measurement assembly.
- 5 Install the new fiber bundle, and tighten the screws on the light source assembly and the optical measurement assembly. Note to avoid bending the fiber bundle with radius less than 70mm.

Figure 5.35 Exploded view of fiber bundle on optical measurement assembly for installation

**Alignment and confirmation**

Refer to Section 8.3 Photometer Unit.

5.9.5 Replacement of Optical Measurement Assembly**When to do**

The optical assembly should be replaced if some of functions are disabled or its performance cannot meet the requirements.

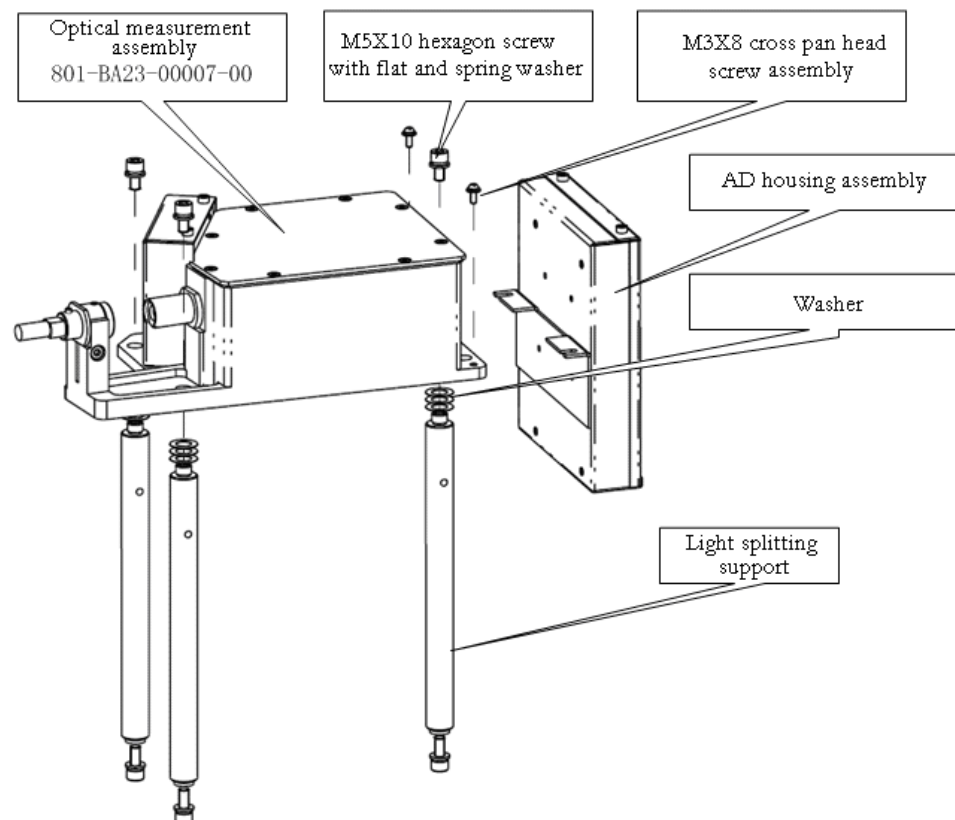
Tools

Flathead screwdriver, hexagon wrench

Removing/Reinstallation procedure

- 1 Switch off the main power of the whole unit.
- 2 Loosen the screws on the back cover of the lamp and remove the lamp rear cover.
- 3 According to the figure above, use a hexagon wrench to remove the M3 socket set screws with cup point on the light source assembly.
- 4 Remove all panels around the reaction carousel and then remove the reaction carousel body assembly.
- 5 Loosen the drain line of the reaction carousel heat chamber and remove the heat chamber.
- 6 Loosen the cross pan head screws on the AD housing, and then remove the AD housing assembly.
- 7 Loosen the three socket cap head screws on the optical measurement assembly to remove it.
- 8 Install the new optical measurement assembly following the above steps reversely.

Figure 5.36 Exploded view of optical measurement assembly for installation



NOTE

When removing the optical measurement assembly, keep the washers with the supports and ensure the washers are of equal amount on each support.

Alignment and confirmation

Refer to Section 8.3 Photometer Unit.

5.9.6 Replacement of Reaction Cuvette

When to do

Replace a cuvette if,

- it is detected abnormal through the Cuvette Check procedure; or
- it still cannot be used after being cleaned through the Clean Cuvettes procedure; or
- scratches or cracks are found on the optical surface of the cuvette.



WARNING

While installing the reaction cuvettes, exercise caution to avoid scratching them. Do not touch the optical surface of the reaction cuvettes. If the optical surface is polluted, the obtained absorbance may be inaccurate.

While installing the reaction cuvettes, make sure that the optical surface is confronted with the outside of the reaction carousel.

Wear gloves free of fibre and powder to avoid polluting the optical surface of the reaction cuvettes.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



NOTE

If a cuvette cannot be removed from the reaction carousel, remove 1 or 2 cuvettes to the right of the cuvette, use a knife to remove the metal plate next to it, and then use your hands or tweezers to take out the cuvette.

Tools

Fiber-free gloves, dry cloth or gauze, reaction cuvettes, and concentrated wash solution manufactured by our company

Removing/Reinstallation procedure

- 1 Select Utility-Maintenance-Maintenance- Biochemistry Maintenance.
- 2 Choose Replace Cuvette.
- 3 Click Continue and the maintenance procedure starts.
- 4 Remove the reaction carousel cover.
- 5 Type in the position number of the cuvette you want to replace.
The input range is 1-90. Only one position number can be entered each time.
- 6 Select Replace.
- 7 Use your thumb and forefinger to take out cuvettes, and correspond them to the numbers on the reaction carousel.
- 8 Place the clean cuvette or the new one into the reaction carousel, and make sure that the cuvette bottom attaches completely to the reaction cuvette.
- 9 Cover the reaction carousel.
- 10 Select Done. The system resets mechanically.

- 11 Perform the Cuvette Check procedure to check if the new cuvettes meet the requirements.
- 12 Select Utility-Maintenance-Maintenance, and then select Scheduled Maintenance-Other.
- 13 Mark the Select checkbox in the same row as Replace Cuvettes.
- 14 Select Approve to refresh the current date as the performance date.
- 15 Select Log and then record comments and other important information for the procedure.
- 16 Select OK to save your input information.

5.9.7 Photometer Lens Maintenance

When to do

The lens of the photometer is contaminated or the maintenance time is above 1 year.

Materials required

Lens cleaning paper, Cotton swabs and Ethanol



NOTE

When moving the sample/reagent probe and mixer, do not bend or collide with them.
Do not leave the cotton tissue on the lens.

Removing/Reinstallation procedure

- 1 Place the analyzing unit power to the OFF position.
- 2 Loosen the nuts on the wash station, remove it and put in a clear box.
- 3 Remove the reaction carousel cover gently.
- 4 Loosen the screws of the reaction carousel and the cable of the slip ring. Remove the reaction carousel.
- 5 Use cotton swabs dipped with ethanol to wipe the exposed lens of the front and rear lens assembly.



- 6 Install the reaction carousel and insert the cable of the slip ring. Tighten the screws.
- 7 Cover the reaction carousel.
- 8 Install back the wash station and tighten the screws.
- 9 Turn on the analyzing unit.

Alignment and confirmation

Select Utility-Maintenance-Maintenance- Biochemistry Maintenance-Cuvette Check. If the dirty cuvettes are found, replace them.

5.9.8 Replacing PDA Assembly

When the alarm "Light intensity is too weak" is given due to the attenuated photometric system, perform the following maintenance procedures:

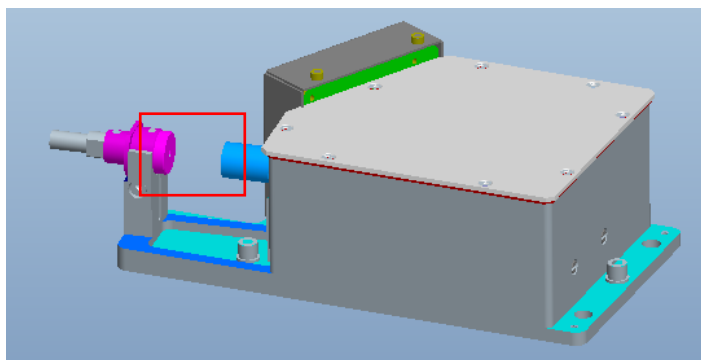
- Wipe the lens clean
- Replace the cuvettes (If necessary)
- Adjust photoelectric gain

Check the water blank and judge to adjust the photoelectric gain and replace the PDA assembly. After replacing the PDA assembly, recalibrate all chemistries.

Wipe the lens clean

How to do

Use cotton swab to dip medical ethyl alcohol and wipe the lens of the optical module as indicated by the red box in the following figure. Use clean dust-free fine-fiber cloth(A90-000043---) to clean the lens.



Precautions:

- ① After dipping the ethyl alcohol, please swing the cotton to remove extra ethyl alcohol, in case too much ethyl alcohol is penetrating into the inside of the lens assembly.
- ② The cotton must be dampened with ethyl alcohol and then used to wipe the lens in case cotton tissue is left on the surface of the lens.
- ③ You must dry the lens with the cloth in case water residue or cotton tissues are left on the surface of the lens.

Notes:

Clean the lens every year. If the analyzer is installed in dusty environment, wipe it every half year.

Replace the cuvettes

If the exterior of the cuvette is obviously dirty or its interior is dirty, which cannot be cleaned through special wash, then replace the cuvettes.



WARNING

- While removing or installing the reaction cuvettes, exercise caution to avoid scratching them. Do not touch the optical surface of the reaction cuvettes. If the optical surface is polluted, the obtained absorbance may be inaccurate.

- Wear gloves free of fiber and powder to avoid polluting the optical surface of the reaction cuvettes.
- While installing the reaction cuvettes, make sure that the optical surface is confronted with the outside of the reaction carousel.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



NOTE

If a cuvette cannot be removed from the reaction carousel, first remove 1 or 2 cuvettes to its right, use a knife to remove the metal plate next to it, and then use your hands or tweezers to take out the cuvette.

Adjust the gain parameters

How to do

Select Maintenance > Parameters, select Middle-layer Unit, and adjust the gain parameter of 340nm channel to 85.

Precautions:

The gain parameter must not be less than 85. When it is less than 85, the alarm limit will change with the gain parameter accordingly.

Note: After the previous 3 steps are performed, there is no need to perform calibration, because absorbance test for the biochemistry analyzer is a relative test, and the energy change of the optical system does not affect the absorbance or response result.

Clear the water blank

After perform the above procedures, remember to clear water blank. Otherwise the alarm "Water blank out of range (10X)" will be given.

How to do

- 1 Select Maintenance > Biochemistry Maintenance.



- 2 Perform the Cuvette Check procedure.
- 3 If the lamp is just replaced, you can perform the Replace Lamp procedure again (without changing the lamp practically) to clear the water blank.

Acquire the water blank data

How to do

- 1 Replace the lamp
- 2 Select Maintenance > Biochemistry Maintenance.
- 3 Perform the Photometer Check procedure.



- 4 After photometer check procedure is complete, the results shown in the figure below are obtained.

Photometer Check

Start Stop

Current Result Last Result

Test Date 08/25/2015 Test Date 01/08/2014

Wavelength	Absorbance
340	15084
380	10685
412	7508
450	6930
505	6427
546	6240
570	6284
605	6154
660	6103
700	6084
740	5906
800	5749

Wavelength	Absorbance
340	6689
380	6500
412	6139
450	6157
505	6069
546	5952
570	5937
605	5854
660	5979
700	5953
740	5933
800	5736

Current Status OK

If the light intensity is too weak, replace the lamp.

Print Exit

Judge if gain parameter is further adjusted

Make judgment according to the water blank data in step 1 and the following rules:

A. For 340nm channel, when the gain parameter is 85, the absorbance is no less than 4838. If it is less than 4838, you should adjust the gain parameter to a bigger value to make the absorbance greater than 4838.

B. For other channels other than 340nm, if the absorbance is within 4376~12041, then there is no need to adjust the gain parameter; otherwise adjust it to make the absorbance within 4376~12041.

Judge if PDA assembly should be replaced

Make judgment according to the absorbance in step 1 and the following rules:

After performing the maintenance procedures of the optical assembly, if the absorbance of 340nm channel of the new lamp is larger than 12041, then it is recommended to replace the PDA assembly.

Notes:

- The corresponding water blank AD value of absorbance 12041 is 24000.
- When the gain parameter of 340nm channel is 85, the AD value for the alarm limit of "Light intensity is weak" is 12000.
- The energy of the lamp in its last service life is about 60%~70% of its initial value.
- Because the lens and cuvettes get dirty unavoidably and the energy of the lamp may differ, if maintenance procedure of 5.2.1 is performed. the water blank AD value of the 340nm channel is smaller than 24000 or absorbance is larger than 12041, then frequent alarms of Lamp is weak may occur and the service life of the lamp may be affected.

Replace PDA assembly

PN:115-022371-00

Name: PDA assembly(BA49)

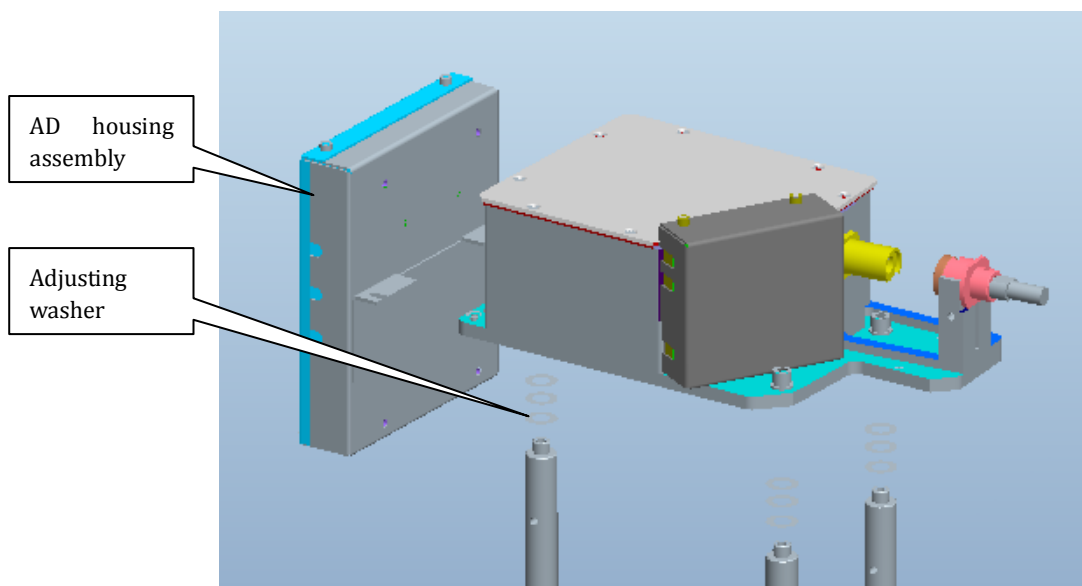


NOTE

- Use a new lamp to align gains of each channel after replacing the PDA assembly, in order to avoid strong light of the lamp.
- When removing the optical assembly, exercise caution not to lose the washers on the supporting rods. The washers on each supporting rod are of equal number.
- The signal cable must be connected correctly.
- Tighten the PDA assembly in place until the inner hexagon wrench cannot screw. Ensure that the two surfaces contact closely.

Replacement steps:

- 1 Switch off the main power of the whole unit.
- 2 Remove the wash station.
- 3 Remove the reaction carousel cover and all panels around it. Rotate the reagent probes and sample probe to above the reagent carousel and sample carousel, and the mixers to the wash wells, for the purpose of conveniently removing the reaction carousel and heat chamber.
- 4 After removing the reaction carousel, loosen the drain tube to remove the heat chamber.
- 5 Remove the AD housing and open its upper cover. Unplug the pre-amplification-AD-collection cable and power cable from the AD collection board.



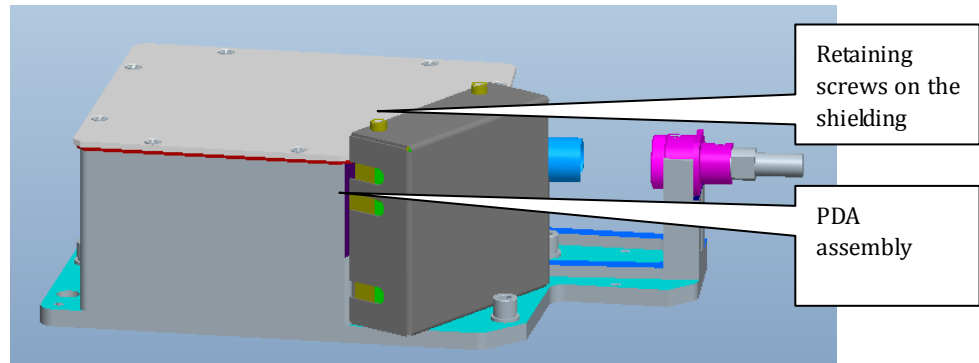
- 6 Loosen the screws on the light source assembly to remove the fiber bundle.
- 7 Loosen the three M5 hexagon screws to remove the light-splitting assembly together with the fiber bundle.

Precautions:

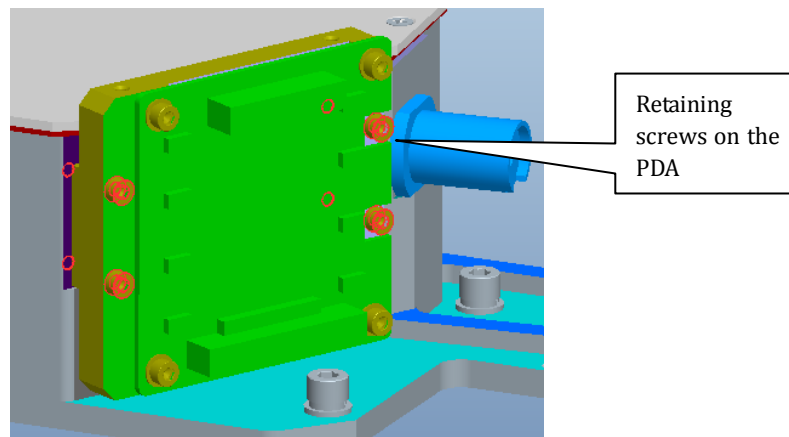
5 Modules and Units

- Do not bend the fiber bundle too much (less than R60MM for the best) during removing. When leaving the factory, the fiber bundle is fixed in the middle by a cable tie on the heat chamber posts. You can remove the cable tie and fiber bundle by sliding them upwards. (You can also directly cutting off the cable tie and use a new one in the spare parts kit instead.)
- Protect the exposed optical part on the head of the fiber bundle from scratching and contamination.
- There are washers on each post used to adjust the height of the optical axis. Keep them properly and do not drop them while removing the optical assembly.

- 8** Loosen the two M3 screws to remove the shielding cover of the PDA assembly.

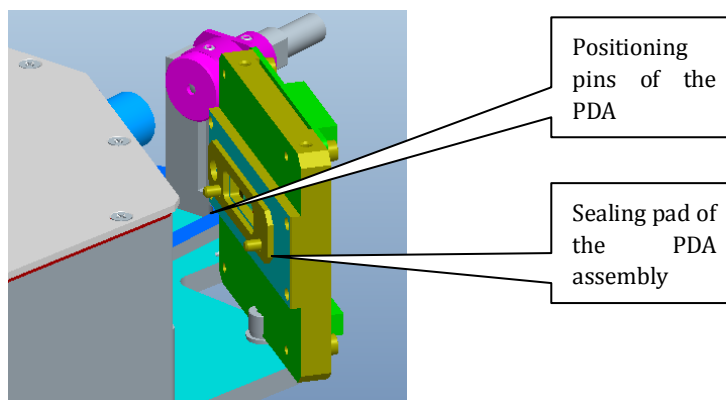


- 9** Disconnect the signal cable on the pre-amplification board and power cable. Remove the 4 M3 hexagon screws highlighted in red in the following figure.



Note: Keep the four screws with the sealing adhesive. Use the original screws when installing the new PDA assembly.

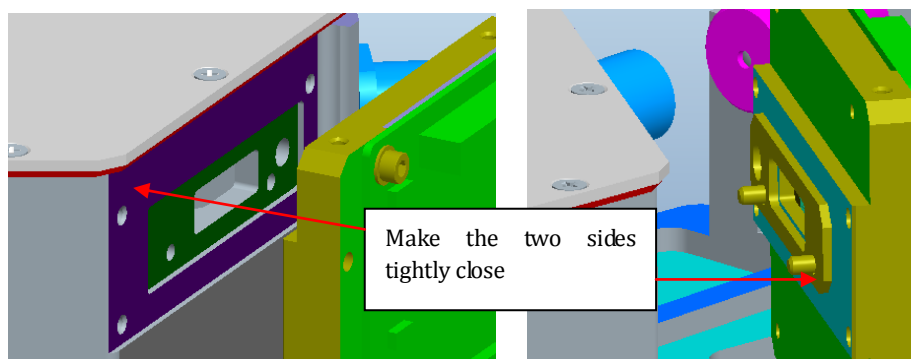
- 10** Remove the PDA assembly. If the PDA assembly is tight on the positioning pins, you can remove it by swinging. Make sure to also remove the sealing pad.



11 Install a new PDA assembly in the reversed order of step 8-10 and pay attention to the following three points:

- Insert the signal cables correctly as J2 to J2 and J3 to J3.
- Check if the sealing pad is in place when the new PDA assembly is installed.
- Tighten the PDA assembly in place until the inner hexagon wrench cannot screw. Ensure the surface showed in purple and the one in green contact closely. Since the sealing pad is elastic, tighten the four screws alternately for two or three times.

Note: Use the original screws when installing the new PDA assembly.



12 Install the light-splitting assembly in the reversed order of step 2-7 and pay attention to the following points:

- 1) Ensure that the same number (or thickness) of washers are installed on the supporting rods as before the removing.
- 2) Insert the signal cables correctly on the AD collection board, as J2 to J2 and J3 to J3.
- 3) Do not bend the fiber bundle too much (less than R60MM for the best) during installation. Fix the fiber bundle with a cable tie on the heat chamber posts in the same way when it leaves the factory.

13 After replacement, switch on the power of the analyzer and the analyzing unit, and run the Test and Maintenance Software. When the lamp becomes steady, readjust the gain parameters of the channels according to "Adjust the gain parameters".

14 Check the stability of the instrument by running tests with water as sample and reagent. Check that the reaction curve is flat and straight.

Recalibrate

After replacing the PDA assembly, run calibration tests of all clinical chemistries.

5.10 ISE Unit (optional)

5.10.1 Module Functions

The ISE module is an optional module for the fully-automated chemistry analyzer and designed to measure the concentration of K⁺, Na⁺ and Cl⁻ in serum, plasma and diluted urine. The sample volume for measurement is: 70μl for serum and plasma, 140μl for urine diluted at the ratio of 1:9 (1 for urine and 9 for diluent).

5.10.2 Structure and FRU list

The ISE unit consists of the ISE module, pump module and reagent module, functions of which are introduced in Table below.

Name	Functions
ISE module	The ISE module has the following electrodes: Na ⁺ , K ⁺ , Cl ⁻ , and reference. It analyzes and measures sample dispensed through the injection port.
Pump module	Three peristaltic pumps are contained to transfer reagent and liquid waste.
Reagent module	The reagent module integrates calibrator A, calibrator B, waste container, and a chip used for monitoring reagent inventory. It provides reagent for sample analysis and stores the liquid waste herein.

The figure below shows the composition of the module.

Figure 5.37 Exploded view for installation of Measuring module

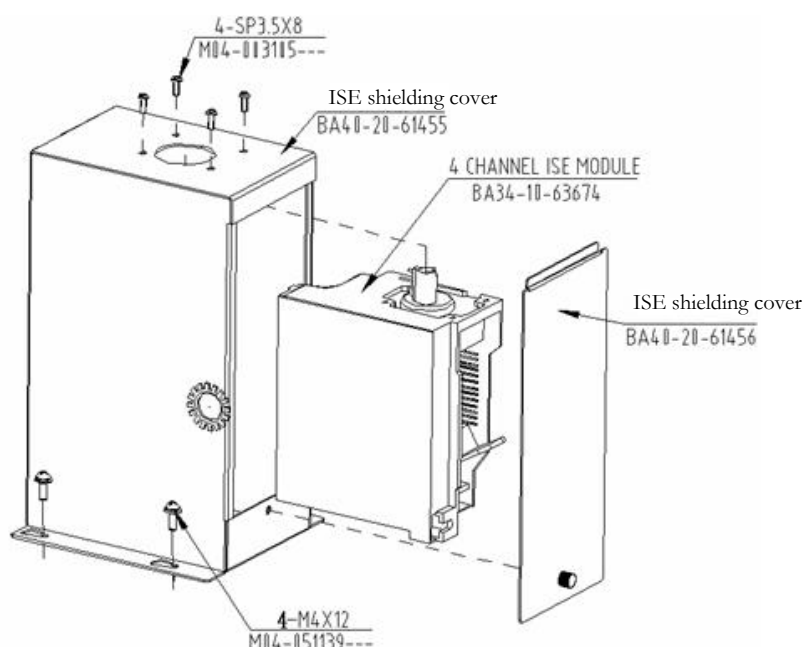


Figure 5.38 Exploded view for installation of pump module

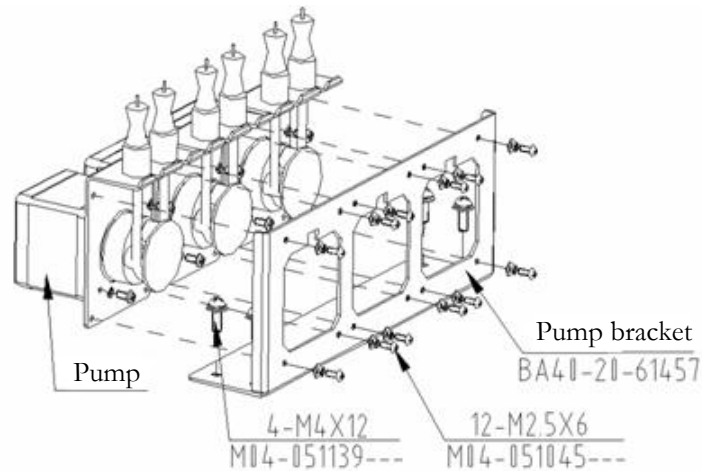


Figure 5.39 Exploded view for installation of reagent module

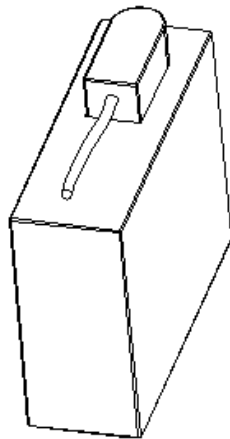


Figure 5.40 ISE measuring system tube connection

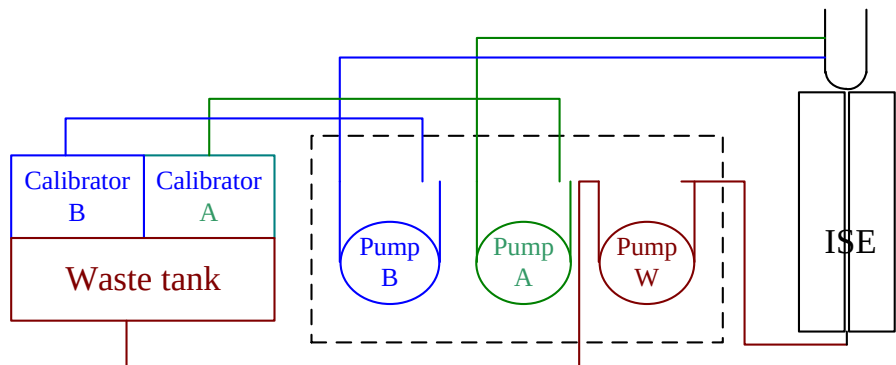


Table 5.9 List of materials

No.	FRU code and material code	Material Name	Remark
1	115-004626-00	K electrode	FRU
2	115-004627-00	Na electrode	FRU
3	115-004630-00	Cl electrode	FRU
4	115-004625-00	Reference Electrode	FRU

No.	FRU code and material code	Material Name	Remark
5	115-004628-00	spacer	FRU
6	BA34-10-63657	Main control board of the ISE module	FRU
7	BA34-10-63659	Pre-amplification board of the ISE module	FRU
8	BA34-10-63658	Compression plate	FRU
9	801-BA34-00104-00	ISE module bubble detector	FRU
10	801-BA34-00103-00	Sample tube of ISE module	FRU
11	BA34-10-63663	O ring of ISE module(3)	FRU
12	BA34-10-63668	O ring of ISE electrode	FRU
13	082-000684-00	Peristaltic Pump	FRU
14	801-BA34-00105-00	The peristaltic pump tube(3 pcs)	FRU
15	BA34-10-63666	ISE waste tube connector	FRU
16	BA34-10-63667	Tube connector of ISE (4)	FRU
17	801-BA34-00106-00	ISE module plastic tube for A ,B and W liquid	FRU
18	BA34-10-63812	ISE module wand	FRU

5.10.3 Unclogging Waste Tubes

When to do

If the sample contains insoluble substances like fibrin or other substances, they can be accumulated in the waste tube after long time use, causing clogging of the waste tube.

Maintenance tools

Name	Code	Quantity
Unclogging tool for ISE	115-023372-00	1



WARNING

Do not spill liquid on the analyzer. Liquid ingress may cause equipment damage.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



NOTE

Excessive bleach and DI water flushed into the ISE reagent pack waste bag may cause waste bag over expansion and clog the Cal A & Cal B reagent flow.

To prevent this problem, connect to an old used-up reagent pack or use the connector of the used-up reagent.

How to do

- 1 Ensure the analyzer is on idle (standby) condition. Open the ISE cover on the right side analyzer panel.

- 2 Remove the electrode housing cover. Remove the waste tube fitting from the bottom of the right angle adaptor. Remove waste peri-pump tube from the pump bracket. Refer to pictures below.
- 3 Connect the waste tube fitting to a syringe and unclogging tool with 5 mL of undiluted household bleach. Refer to pictures below.
- 4 Press the wand release button to remove the wand from the current in use ISE reagent pack and keep it in a save place. Engage the wand to an old used-up reagent pack.
- 5 Inject bleach into the ISE waste tube and soak the tube for 5 minutes. Discharge the waste into the reagent pack.

Note: When the bleach cannot be injected into the ISE pack, remove the wand and push down to open the waste valve manually with a sharp object, and then inject again. If bleach can go through this time, the waste bag was clogged and cannot be used. If bleach still cannot be injected, replacing the ISE wand is recommended.

- 6 Repeat this step with 5 mL of DI water without the 5 minutes of soaking time.
- 7 Remove the wand from the old use-up pack and re-install it back to the current in use ISE pack. Re-install the waste tube fitting back to the ISE electrode housing right angle adaptor and waste peri-pump tube back to the pump bracket. Re-install the housing cover.

Alignment and confirmation

Perform ISE pump calibration and if it succeeds, the tube has been successfully unclogged.

5.10.4 Replace Pump Tube

When to do

Pump tube is aged or leakage occurs or maintenance time exceeds half year.

Maintenance tools

Cross screwdriver, pump tube.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Ensure the analyzer is on idle (standby) condition. Open the ISE cover on the right side analyzer panel.
- 2 Select **Utility-Maintenance-Maintenance-ISE-Remove reagent pack.**
- 3 Select **Utility-Maintenance-Maintenance-ISE-Replace Pump Tube and Calibrator Tube.**
- 4 Select Pump Tube.
- 5 Replace it with new one. Make sure the connectors and tubes are correctly connected; otherwise ISE module failure may occur.
- 6 Load the reagent pack and the procedure is completed.
- 7 Install back the right side panel.
- 8 Record the log.

Alignment and confirmation

Perform ISE pump calibration and if it succeeds, the tube has been successfully replaced.

5.10.5 Replace Calibrator Tube

When to do

Calibrator tube is aged or leakage occurs or maintenance time exceeds one year.

Maintenance tools

Cross screwdriver, calibrator tube

Precautions:

BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Ensure the analyzer is on idle (standby) condition. Open the ISE cover on the right side analyzer panel.
- 2 Select **Utility-Maintenance-Maintenance-ISE-Remove reagent pack.**
- 3 Select **Utility-Maintenance-Maintenance-ISE-Replace Pump Tube and Calibrator Tube.**
- 4 Select Calibrator Tube.
- 5 Replace it with new one. Make sure the connectors and tubes are correctly connected; otherwise ISE module failure may occur.
- 6 Load the reagent pack and the procedure is completed.
- 7 Install back the right side panel.
- 8 Record the log.

Alignment and confirmation

Perform ISE pump calibration and if it succeeds, the tube has been successfully replaced.

5.10.6 Notes for Calculation of Reagent Pack Inventory

Reagent pack as one part of the ISE module should be installed as instructed by this manual. You should install reagent pack while the analyzer is powered off, and then check the reagent inventory and prime the ISE module with Calibrator A and B. If the reagent pack is not properly installed, error may be produced in inventory calculation, and the actual volume may differ from that displayed on the screen.

The following operations may result in the actual inventory of the reagent pack less than the displayed one.

- When the analyzer is turned on and the ISE module is in failure status, the reagent pack is changed, without recovering the module failure or checking the inventory immediately.

Operation explanation: When the ISE module is in Stopped status, the software will stop writing volume information into the reagent pack chip. However, the analyzer will automatically execute the sip action of the ISE module during startup to consume certain amount of reagent. Therefore, when the reagent pack is replaced, but the ISE module is not restored or the screen display refreshed, the software will not calculate the reagent consumption volume until reading the new reagent pack. At this moment, the actual reagent volume = displayed volume - consumption of Sip after replacing the reagent pack, which means that the actual reagent volume is less than the displayed volume on the screen.

- When the analyzing unit is powered on but the software is exited or the PC is shut down, the ISE reagent pack is changed.

Operation explanation: During shutdown, the analyzer will automatically execute the sip action of the ISE module to consume certain amount of reagent. When the analyzer is started up, the software will detect a new reagent pack and zero the consumption that is not written into the reagent pack chip, and then display the volume read from the reagent pack chip. At this moment, the actual reagent volume = displayed volume - consumption of Sip after replacing the reagent pack, which means that the actual reagent volume is less than the displayed volume on the screen.

- When the analyzing unit is powered on but the software is exited or the PC is shut down, the ISE reagent pack is removed to be used on another analyzer and install back on the initial analyzer after its usage.

Operation explanation: During shutdown process, the analyzer cannot detect if the reagent pack is normal. If the reagent pack is installed on the current instrument after being used on another one, the analyzer, when started up, will detect that the recorded reagent volume on the chip is less than the original one, and the software will zero the reagent consumption of Sip during shutdown and display the read volume on the screen. At this moment, the actual reagent volume = displayed volume - consumption of Sip during shutdown (with reagent pack installed), which means that the actual reagent volume is less than the displayed volume on the screen.

- The ISE module is de-configured, but the reagent pack is still on the instrument.

Operation explanation: Though the ISE module is de-configured, the original reagent pack information is not cleared from the database. When the ISE

module is configured again with the same reagent pack, the Sip period without ISE module will be included in calculation of the reagent consumption.

Note

- In condition 1, restore the ISE module and check the reagent inventory immediately to minimize or eliminate the calculation error.
- In condition 2 and 3, before running the operating software to start up the analyzer, remove the reagent pack and then run the operating software. After the startup procedure is complete, re-install the reagent pack and inquire the reagent volume immediately to minimize or eliminate the calculation error.
- In condition 4, remove the reagent pack and then de-configure the ISE module.

The following operations may result in the actual inventory of the reagent pack more than the displayed one.

- When the analyzer is turned on and the ISE reagent pack is changed, ISE inventory is not inquired immediately.

Operation explanation: The analyzer will automatically refresh the ISE reagent consumption, by subtracting the consumption of calibrator A or B from the known volume (of the original reagent pack) and writing it into the reagent pack chip, when calibrator A or B is consumed for 1%. The written volume is based on the original reagent pack before replacement, and the volume of a used reagent pack is often less than that of a new one. So the volume recorded on the chip of the new reagent pack is less than the actual volume, and the actual volume is greater than the displayed volume on the screen.

- When the analyzing unit is powered on but the software is exited or the PC is shut down, the ISE reagent pack is removed but not used on another analyzer and then it is installed back before the analyzer is turned on.

Operation explanation: During shutdown process, the analyzer cannot detect if the reagent pack is normal. If the reagent pack is loaded before the analyzer is started up, the analyzer, when started up, will detect that the reagent volume recorded on the chip is same as that recorded by the software, but the software will confirm by mistake the reagent consumption of sip during shutdown. (Actually, no reagent is consumed.) Therefore, after calculating the reagent consumption, the software will write the new volume information onto the chip and display it on the screen. At this moment, the actual reagent volume = displayed volume + consumption of Sip during shutdown (with reagent pack installed), which means that the actual reagent volume is greater than the displayed volume on the screen.

- After the operating software is exited, the analyzer is powered off and then it is powered on before the operating software is run.

Operation explanation: When closed normally, the operating software cannot detect if the ISE module is working. When powered on before the software is run, the analyzer, when started up, will detect that the reagent volume recorded on the chip is same as that recorded by the software, but the software will confirm by mistake the reagent consumption of sip during shutdown. (Actually, no reagent is consumed because the ISE module is not working while the analyzer is powered off.) Therefore, after calculating the reagent consumption, the software will write the new volume information onto the chip and display it on the screen. At this moment, the actual reagent volume = displayed volume +

consumption of Sip during shutdown (with reagent pack installed), which means that the actual reagent volume is greater than the displayed volume on the screen.

Note

- In condition 1, inquire the ISE reagent volume immediately to minimize or eliminate the calculation error.
- In condition 2 and 3, before running the operating software to start up the analyzer, remove the reagent pack and then run the operating software. After the startup procedure is complete, re-install the reagent pack and inquire the reagent volume immediately to minimize or eliminate the calculation error.

5.10.7 Calculation of Reagent Pack Inventory

The reagent volume decreases as ISE instructions are executed, and it is updated timely on the software. The table below shows reagent volumes consumed for different instructions:

Table 5.1 Reagent volume consumptions

Module instructions	Cal A volume (μl)	Cal B volume (μl)
Calibration	360	360
Serum test	180	/
Urine test	100	230
Wash	180	/
Purge A	100	/
Purge B	/	100
Pump check	475	175
Air bubble detector calibration	75	/
Sip period	130	130

The software calculates the consumption volume (μl) of calibrator A and B by multiplying the total execution times of each ISE instruction and the reagent volume per time.

When started up, the software writes the volume information into the reagent pack as the calibrator A and B are consumed for 1%. When they are consumed for 100%, the software marks reagent exhausted on the reagent pack chip.

The Sip period is special. The instrument equipped with the ISE module should not be powered off, because the Sip period instruction is executed automatically for every 30 minutes even the ISE module is in standby status (without running any tests). During biochemistry tests, the Sip period instruction will be also executed as long as the ISE module is continuously standby for over 30 minutes. Thus the software will record the standby time of the instrument and the shutdown/startup time of itself. For the shutdown period, the times of Sip period are calculated by dividing the total time with 30 minutes. For the running period, the times of Sip period are recorded based on the ISE module's continuous standby periods that are over 30 minutes. Sum of the two numbers is the final Sip period number, and it can be used to determine the reagent consumption of the Sip period instruction.

6

Hardware Circuits

6.1 Overview

This chapter describes the functions of the printed circuit board assemblies (PCBAs) used on the BS-480/BS-490 Chemistry Analyzer.

6.2 Summary of Hazards



NOTE

While the instrument is working, do not touch the hardware circuit boards with your hands or other objects.

To remove a circuit board, you should first disconnect the instrument from the (AC) power supply. Please wear a pair of anti-static gloves or take other measures to prevent static electricity prior to removing a circuit board.

6.3 Summary of PCBAs

Sequence number and function description of PCBAs

The table below provides a summary of the PCBAs used on the BS-480/BS-490 Chemistry Analyzer and briefly describes their functions.

PCBA (PCB)	Functions	No.
Main control board 051-001799-00	The main control board is the control center of the instrument. It is mainly used to fulfill the following tasks: 1). communicating with a computer through the RS232 serial port to transmit data and instructions; 2) communicating with the smart modules, including the ISE module, through the extended serial ports to transmit data and instructions; and 3) controlling digital potentiometer adjustment and photoelectric data collection of the AD collection board and receiving the photoelectric data. It provides the BDM/JTAG interface for debugging software and downloading application programs. The application software of the main control board can be updated through the RS232 serial port.	#1
Communication Conversion Board 801-BA40-00021-00	The communication conversion board is the bridge used for communication between the main control board and a computer. It provides the serial port for connecting the main control board, and the standard DB9 serial port for connecting a computer. Note: Only the serial port is used now, and other ports are reserved.	#2
Reaction carousel temperature collection board 801-051-000817-00	The reaction carousel temperature collection board adjusts the signal of the reaction carousel temperature sensor and controls it for AD conversion. It provides a serial peripheral interface (SPI) and a power supply jack for the reaction carousel heater and connects with the wash solution temperature control board via the slip ring.	#3
Cleaning fluid temperature collection board 801-BA38-00033-00	The cleaning fluid temperature collection board adjusts the signals of two cleaning fluid temperature sensors and one environment temperature sensor and controls them for AD conversion. It provides a SPI interface for connecting with the wash temperature control board.	#4
Three-carousel drive board 051-001801-00	The three-carousel, temperature control and cuvette wash drive board is used to drive and control the sample carousel, reagent carousel, reaction carousel, cuvette wash station, wash syringes, and related components.	#5

PCBA (PCB)	Functions	No.
Three-probe drive board 051-001800-00	The three-probe drive board is used to control and drive the sample probe, reagent probe, sample mixer, reagent mixer, and related components.	#6
Pump/Valve drive board 051-000957-00	The pump/valve drive board drives the auto wash pumps and valves, and transmits floater signal.	#7
Reagent Refrigeration Board 051-001802-00	The reagent refrigeration board is used to drive the reagent refrigeration circuit (including radiators), fans, and demisting circuit of the reagent bar code reader.	#8
Sample Probe Level Sense Board 801-BA80-00036-00 Reagent probe level sense board 051-000363-00	The two boards respectively sense the level of samples and reagents, and detect/convert the signals of the two vertical obstruct detection sensors.	#9
Clog detection board 801-BA80-00048-00	The clog detection board detects clogs inside the sample probe by checking the pressure of the fluidic tube when the sample probe aspirates samples.	#10
AD collection board BA40-30-61365 (BA40-20-61364)	The AD collection board adjusts the photoelectric signals output by the preamplifier board and controls it for AD conversion. It provides an SPI interface for connecting with the main control board.	#11
Preamplifier Board BA40-30-61363 (BA40-20-61362)	The preamplifier board converts the signals of the discrete photodiode array from analog to digital.	#12
Pressure detection board 051-001150-00	The pressure detection board processes and converts the pressure signals detected on the pneumatic path of the vacuum compression pump. It provides a serial port for communication and transmitting pressure data.	#13
Wash probe obstruction optical coupler conversion board 051-001147-00	The wash bump optical coupler conversion board is used to shift the cable of the wash bump optical coupler.	#14
Mixer drive board 051-001151-00	feedback the reagent/sample mixing speed	#15
Rotation speed optical coupler conversion board 051-001620-00	Connect the optical coupler of the mixer rotation speed	#16
Analog power supply conversion board 801-BA38-00005-00	The analog power supply conversion board converts the A5V DC/DC digital of the BA40 power supply assembly into +/-12V analog and then provides it for the AC collection board and preamplifier board.	#17

6 Hardware

PCBA (PCB)	Functions	No.
Power supply assembly for water supply module(Optional) 801-BA40-00053-00	provide the power supply for the water supply module	#18
Control board of water drainage module(Optional) BA40-30-61869	Control the start and stop of the diaphragm pump of the drainage module	#19
12V power board 801-BA40-00029-00	These boards are used to power the whole unit, outputting the A12V, A5V digital, D12V/E12V analog, C12V power, B24V power, and the B12V power for partial radiators.	#20
24V power supply board 801-BA40-00030-00		
Power supply conversion board 801-BA40-00031-00		
110V conversion board 051-000958-00	The 110V conversion board connects the 110V power supply to the 110V transformer, and converts 110V to 220V to power the vacuum pump.	#20

6.4 Functions of PCBA

6.4.1 Control Structure

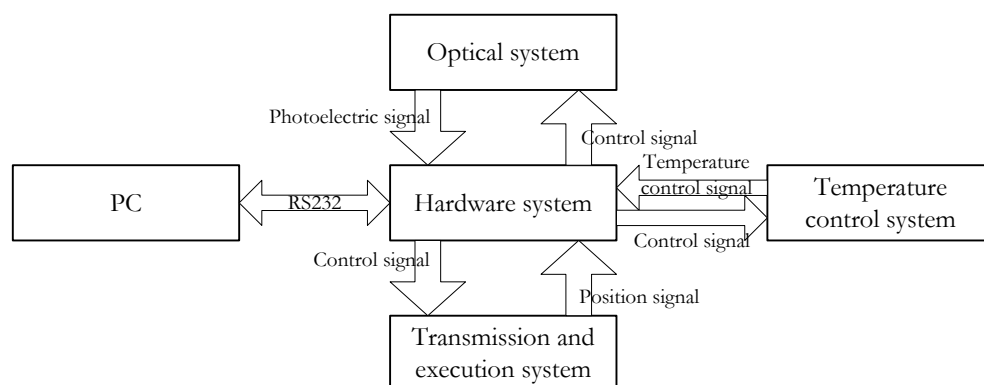
The BS-480 Chemistry Analyzer consists of the analyzing unit (analyzer), operation unit (computer), and output unit (printer).

The analyzing unit (analyzer) is composed of the temperature control system, reaction system (including ISE module), photometric system, sample/reagent handling system, mixer system, and cuvette wash station.

The general control structure of the BS-480/BS-490 is shown in the figure below.

The hardware system has the following functions:

- Communicating with a computer through the serial port, and receiving/sending commands, responses, and data.
- Controlling data collection of the photometric system.
- Controlling movement and status signal collection of execution units.
- Controlling working and temperature control signal collection of the temperature control system.



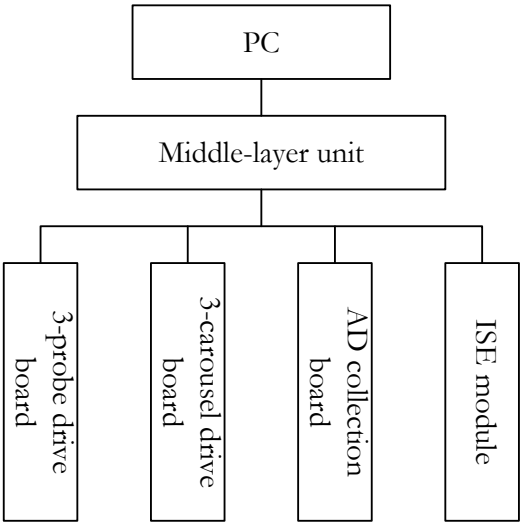
6.4.2 Main Control Board

Functions and principles

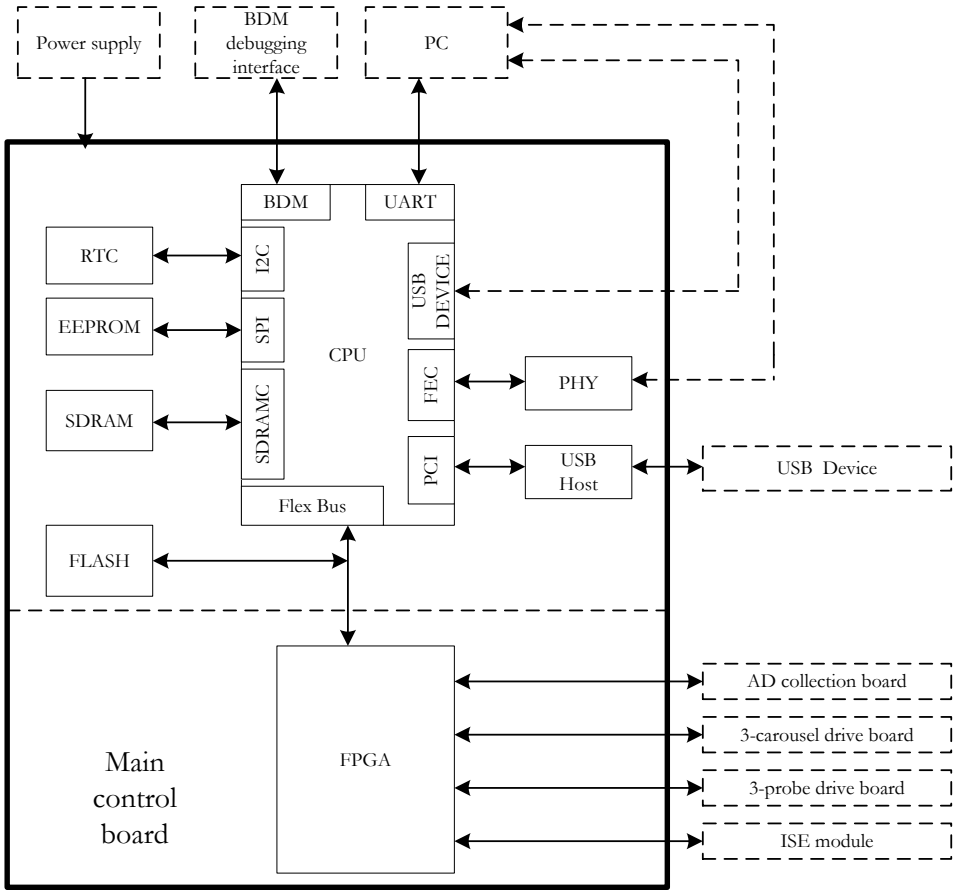
The main control board (051-000742-00) is used to fulfill the following functions:

- Communicating with a computer through the serial port to transmit data and instructions and to update the application programs of the PCBA.
- Communicating with the smart modules, including the ISE module, through the extended serial ports to transmit data and instructions.
- Controlling digital potentiometer adjustment and photoelectric data collection of the AD collection board and receiving the photoelectric data.
- Providing the BDM/JTAG interface for debugging software and downloading application programs.

The figure below shows the relation between the main control board and other PCBAs.



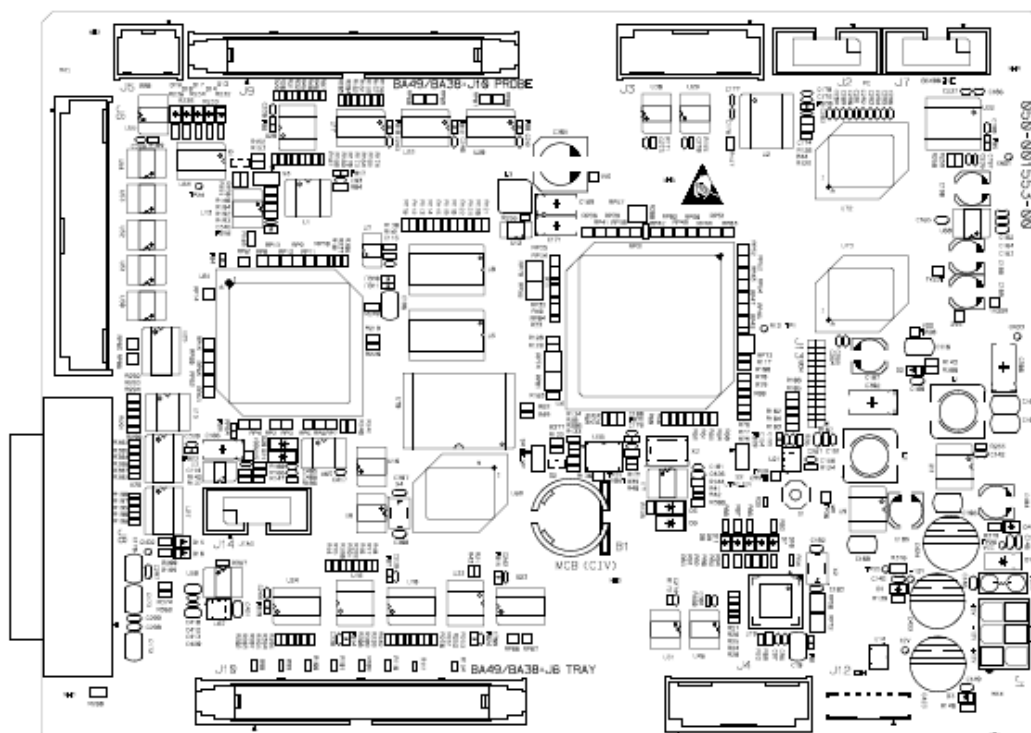
The functional diagram of the main control board is as shown below.



Description

PCB layout

The PCB layout of the main control board is as shown below.



Connectors

The main control board includes the following connectors.

Power supply:

Power supply input (J1): 6-pin, providing +/-12V analog and +5V digital for PCBAs.

Pin No.	Signal	Reference Value
1	+12V	11.4 ~12.6V
2	-12V	-11.4 ~-12.6V
3	+5V	4.75 ~5.25V
4	12VGND	/
5	12VGND	/
6	GND	/

Connectors for sending/receiving communication signals:

- Connector (J2) for PC: 10-pin, RS232, used for communication with a computer.
- Connector (J9) for smart module: 34-pin, TTL, used for communication with the three-probe drive board.
- Connector (J10) for smart module: 34-pin, TTL, used for communication with the three-carousel drive board.
- Connector (J8) for AD collection board: DB25, used for communication with the AD collection board and providing power supply for it.
- Connector (J7) for ISE module: 10-pin, RS232, used for communication with the ISE module.
- Connector (J6) for mixer motor: 40-pin, TTL, used for communication with the mixer motor drive board.
- Connector (J5) for reagent refrigeration board: 8-pin, RS422, used for communication with the reagent refrigeration board.

Connectors for debugging:

- BDM connector (J13): 26-pin, used for debugging the CPU software.
- JTAG connector (J1): 10-pin, used for debugging the FPGA.

Switches and jumpers

The main control board contains the following switches and jumpers.

RST key (S1): used to reset the CPU of the PCBA.

Indicators

The main control board contains the following indicators.

- +12V power supply indicator (D3): green. It is lit when the analyzer power switch is turned on, indicating that the +12V power supply has been connected.
- -12V power supply indicator (D1): green. It is lit when the analyzer power switch is turned on, indicating that the -12V power supply has been connected.
- +5V power supply indicator (D4): green. It is lit when the analyzer power switch is turned on, indicating that the +5V power supply has been connected.
- +3.3V power supply indicator (D2): green. It is lit when the analyzer power switch is turned on, indicating that the +3.3V power supply has been connected.
- D13: orange FPGA configuration indicator. On indicates FPGA is successfully configured.
- D14: Green FPGA working status indicator. LED is flashing every second, indicating FPGA works normally.
- D17: Green Reserved. Constantly lit.
- D18: Green. When the photoelectric signal collecting is interrupted, the indicator is lit.
- D19: Green. When communication with the lower layer unit is interrupted, the indicator is lit.

Test points

In the following positions of the main control board can signal tests be performed.

- 12V: +12V power supply input. Normal range: 11.4 - 12.6V.
- -12V: -12V power supply input. Normal range: -11.4 - -12.6V.
- VCC: +5V power supply input. Normal range: 4.75 - 5.25V.
- VDD: +3.3V power supply. It is secondary power supply used to power the major digital circuits of the PCBA. Normal range: 2.97 - 3.63V.
- 2V5: +2.5V power supply. It is secondary power supply used to power the DDR memory. Normal range: 2.25 - 2.75V.
- 1V5: +1.5V power supply. It is secondary power supply used to power the CPU core. Normal range: 1.35 - 1.65V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the PCBA.

Check the connectors with clamps and ensure that the clamps have been locked properly.

Check other connectors and ensure that they are inserted into the end of the slots.

It requires great force to plug/unplug the J3~J4, J10~J11 connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

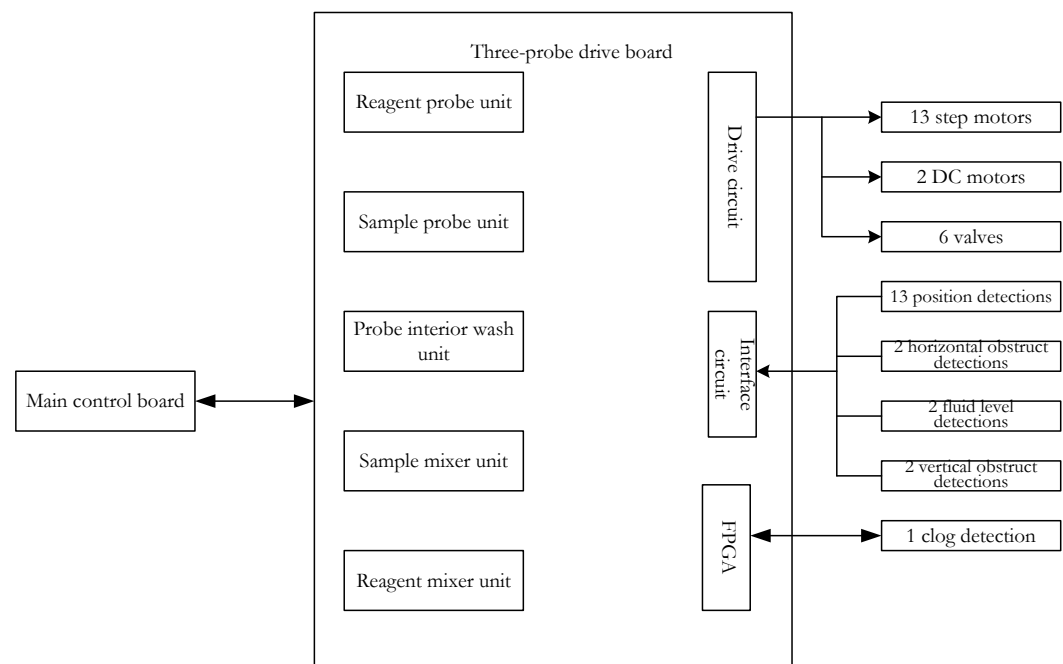
After connecting J7 connector (DB25), tighten the retaining screws on two sides of it.

6.4.3 Three-probe drive board

Functions

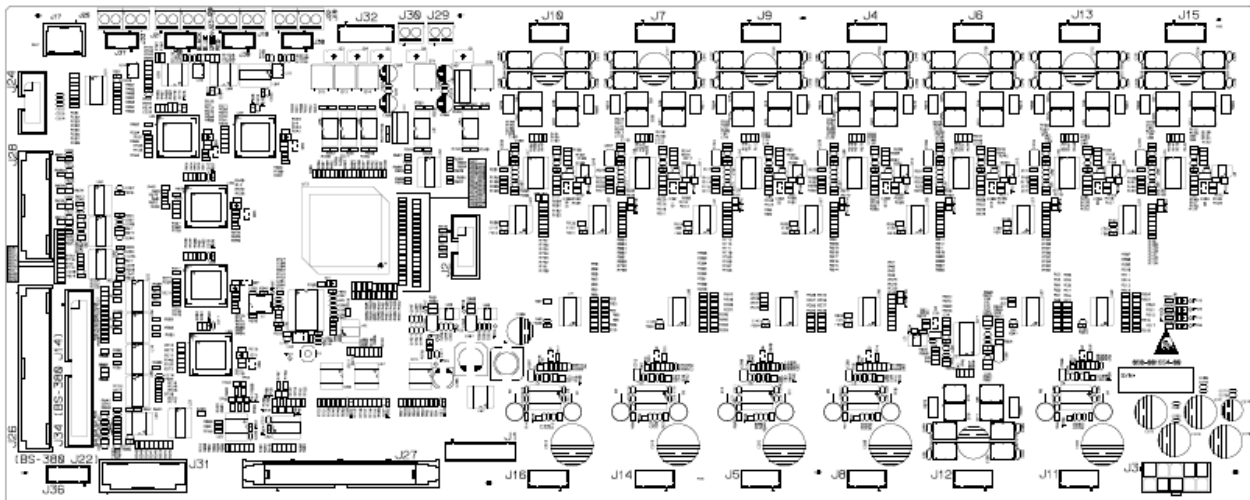
- The three-probe drive board controls and drives the step motors, DC motors and valves, and detects the signal of relevant sensor.
- Receiving instructions from or sending data to the main control board through the relevant interface.
- Controlling and driving the probe interior wash syringe, reagent probe, sample probe, sample mixer, reagent mixer, valves, syringes, etc.
- Detecting position signal and horizontal obstruct signal.
- Detecting level sense signal and vertical obstruct signal through the interface for the level sense board.
- Setting up level detection sensibility through the serial port for the level sense board.
- Detecting sample probe clog signal through the interface for the sample probe clog detection board, and reading clog pressure through the relevant serial port.

The functional diagram of the three-probe drive board is as shown below.



Description

The PCB layout of the three-probe drive board is as shown below.



Connectors

The three-probe drive board contains the following connectors.

Power supply:

Power supply input (J3): 10-pin, providing A12V analog, +/-12V, +5V digital, and 24V for the PCBAs.

Pin No.	Signal	Reference Value
1	VCC	4.85 ~5.25V
2	+24V	23.5 ~26V
3	+24V	23.5 ~26V
4	+12V	11.4 ~12.6V
5	-12V	-11.4 ~-12.6V
6	GND	/
7	GND	/
8	GND	/
9	+12VA	11.4 - 12.6
10	AGND	/

Connectors for sending/receiving communication signals:

- Main control board communication serial port (J27): 34-pin, TTL, used for communicating with the main control board.
- Reagent level sense communication serial port (J37): 4-pin, TTL, used for communicating with the reagent and sample level sense boards.
- Clog detection communication serial port (J17): 8-pin, TTL, used for communicating with the clog detection board.

Connectors for debugging:

JTAG connector (J2): 10-pin, used for debugging the FPGA.

Detection and control connectors:

- Sample level sense signal connector (J40): 4-pin, used for detecting sample probe level sense signal.
- Sample probe vertical obstruct sensor connector (J40): 4-pin, used for detecting sample probe obstruct in vertical direction.

- Reagent level sense signal connector (J39): 2-pin, used for detecting reagent probe level sense signal.
- Reagent probe vertical obstruct sensor connector (J39): 2-pin, used for detecting reagent probe obstruct in vertical direction.
- Clog signal connector (J24): 10-pin, used for detecting clog signal and empty aspirate signal.
- Optical coupler detection connector 1 (J26): 40-pin, used for detecting home position of probe interior wash syringe, sample syringe, reagent syringe, sample mixer assembly, and reagent mixer assembly.
- Optical coupler detection connector 2 (J28): 26-pin, used for detecting home position and horizontal obstruct of sample probe assembly and reagent probe assembly.
- Probe exterior wash valve connector (J31): 20-pin, used for controlling exterior wash valve of sample probe and reagent probe.
- Probe interior wash drive connector (J32): 6-pin, used for driving interior wash valve of sample probe and reagent probe.
- Sample horizontal motor drive connector (J11): 4-pin, used for driving the sample horizontal motor.
- Sample vertical motor drive connector (J10): 4-pin, used for driving the sample vertical motor.
- Sample syringe motor drive connector (J12): 4-pin, used for driving the sample syringe motor.
- Reagent horizontal motor drive connector (J8): 4-pin, used for driving the reagent horizontal motor.
- Reagent vertical motor drive connector (J7): 4-pin, used for driving the reagent vertical motor.
- Reagent syringe motor drive connector (J9): 4-pin, used for driving the reagent syringe motor.
- Probe interior wash syringe motor drive connector (J6): 4-pin, used for driving the probe interior wash syringe motor.
- Sample mixer horizontal motor drive connector (J16): 4-pin, used for driving the sample mixer horizontal motor.
- Sample mixer vertical motor drive connector (J15): 4-pin, used for driving the sample mixer vertical motor.
- Reagent mixer horizontal motor drive connector (J14): 4-pin, used for driving the reagent mixer horizontal motor.
- Reagent mixer vertical motor drive connector (J13): 4-pin, used for driving the reagent mixer vertical motor.

Switches and jumpers

The main control board contains the following switches and jumpers.

RST key (S1): used to reset the CPU of the PCBA.

Indicators

The three-probe drive board contains the following indicators.

Power supply:

- +12V Power supply (D8): green It is lit when the analyzer power switch is turned on, indicating that the +12V power supply has been connected.
- -12V power supply indicator (D7): green. It is lit when the analyzer power switch is turned on, indicating that the -12V power supply has been connected.

- +5V Power (D9): Green It is lit when the analyzer power switch is turned on, indicating that the +5V power supply has been connected.
- +24V power supply indicator (D10): green. It is lit when the analyzer power switch is turned on, indicating that the +24V power supply has been connected.
- FPGA working indicator (D11): green. It is flashing when FPGA works normally.

Test points

In the following positions of the main control board can signal tests be performed.

- J1.1: +12V power supply input. Normal range: 11.4 - 12.6V.
- J1.2: -12V power supply input. Normal range: -11.4 - -12.6V.
- J1.3: +5V power supply input. Normal range: 4.75 - 5.25V.
- TP20: +3.3V power supply. It is secondary power supply used to power the major digital circuits of the PCBA. Normal range: 2.97 - 3.63V.
- TP17: +2.5V power supply. It is secondary power supply used to download the FPGA. Normal range: 2.25 - 2.75V.
- TP18: +1.5V power supply. It is secondary power supply used to power the FPGA core. Normal range: 1.35 - 1.65V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the PCBA.

Check the connectors with clamps and ensure that the clamps have been locked properly.

Check other connectors and ensure that they are inserted into the end of the slots.

It requires relatively great force to plug/unplug connectors J3, J26, J28, J31, and J27. Hold the PCBA by its edge while plugging/unplugging these connectors to prevent it from being deformed or damaged.

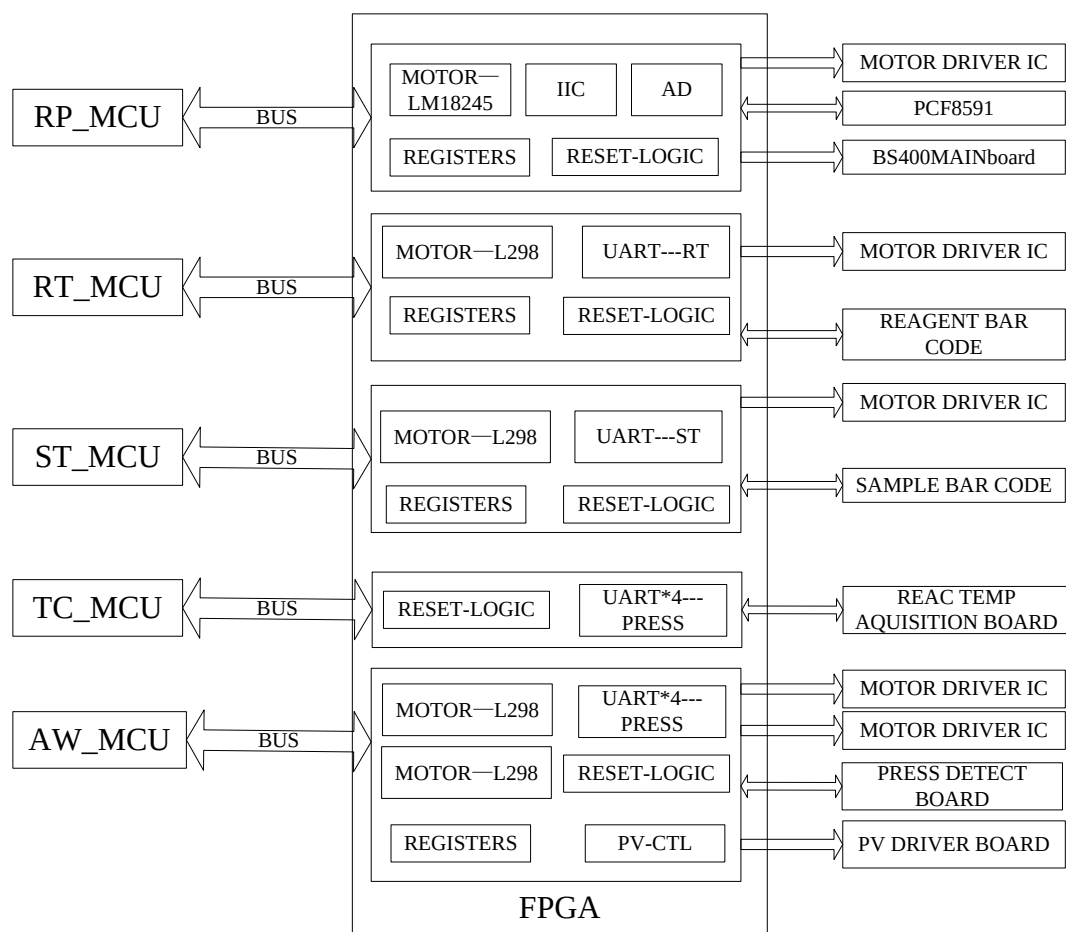
6.4.4 Three-carousel drive board

Functions

The three-carousel drive board receives commands from the main control board and then analyzes them; detects position of the three carousels and cuvette wash station, and drives these units; detects the temperature of the temperature sensors and controls the heaters. The major functions of the PCBA include:

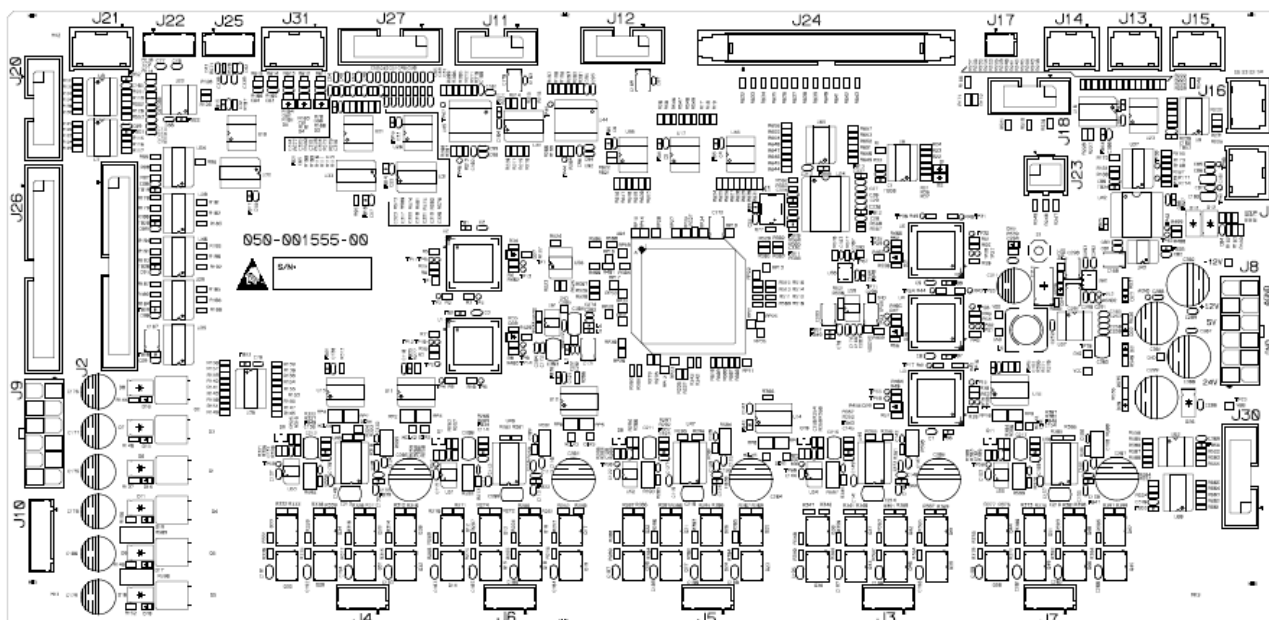
- Supporting communication through the serial port. Making the CPUs communicate with the main control board, receiving and analyzing the instructions from it.
- Making the CPUs output control signals to each execution unit.
- Receiving position sensor signals of relevant execution units, auto wash bump prevention signals, and other status signals.
- Detecting the temperature of reaction carousel solid heat, cleaning fluid preheat and the environment, and controlling the heaters.
- Receiving fluid level sensor signals from the hydropneumatic system, and controlling the fluidic valves.

The functional diagram of the three-carousel drive board is as shown below.



Description

The PCB layout of the three-carousel drive board is as shown below.



Connectors

The three-carousel drive board contains the following connectors.

Power supply:

Power supply input (J8): 12-pin, providing +/-12V analog, +5V digital, and 24V for the PCBAs.

Pin No.	Signal	Reference Value
1	+24V	23.5 ~26V
2	+24V	23.5 ~26V
3	+24V	23.5 ~26V
4	+VCC	4.85 ~5.25V
5	+12V	11.4 ~12.6V
6	-12V	-11.4 ~-12.6V
7	GND	/
8	GND	/
9	GND	/
10	GND	/
11	AGND	/
12	AGND	/

Connectors for sending/receiving communication signals:

- Main control board communication serial port (J24): 34-pin, TTL, used for communicating with the main control board
- Pressure detection communication serial port (J16): 8-pin, TTL, used for communicating with the pressure detection board
- Reaction carousel temperature collection communication serial port (J13): 8-pin, TTL, used for communicating with the reaction carousel temperature collection board
- Reagent bar code communication serial port (J12): 10-pin, RS-232, used for communicating with the reagent bar code reader.
- Sample bar code communication serial port (J11): 10-pin, RS-232, used for communicating with the sample bar code reader.
- Cleaning fluid temperature collection communication serial port (J20): 14-pin, SPI, used for communicating with the cleaning fluid temperature collection board
- Connectors for debugging:
- JTAG connector (J18): 10-pin, used for debugging the FPGA.

Detection and control connectors:

- Lamp and vacuum pump control signal connector (J1): 8-pin, used for controlling the lamp brightness and vacuum pump.
- Cuvette wash syringe optical coupler connector (J15): 9-pin, used for detecting the home position of cuvette wash syringe.
- Sample carousel indicator connector (J17): 2-pin, used for controlling the sample carousel indicator.
- Floater connector (J27): 14-pin, used for detecting the status of fluidic floaters.
- Vacuum pump and refrigeration fan jam detection interface (J22): 4-pin, used for detecting jams of the vacuum pump and refrigeration fans.
- Optical coupler connector (J26): 34-pin, used for detecting the optical couplers of the sample carousel, reagent carousel, reaction carousel, cuvette wash station, and wash syringes.

- Cuvette wash pump/valve control connector (J2): 34-pin, used for controlling the pumps and valves of the cuvette wash unit.
- Cleaning fluid heater drive connector (J9): 12-pin, used for driving the cleaning fluid heater and wash solution heater.
- Reaction carousel heater drive connector (J10): 6-pin, used for driving the reaction carousel heater.
- Reagent carousel motor drive connector (J4): 4-pin, used for driving the reagent carousel motor.
- Sample carousel motor drive connector (J5): 4-pin, used for driving the sample carousel motor.
- Cuvette wash motor drive connector (J6): 4-pin, used for driving the cuvette wash motor.
- Reaction carousel motor drive connector (J3): 4-pin, used for driving the reaction carousel motor.
- Cuvette wash syringe motor drive connector (J7): 4-pin, used for driving the cuvette wash syringe motor.

Switches and jumpers

The main control board contains the following switches and jumpers.

RST key (S5): used to reset the CPU of the PCBA.

Indicators

The three-carousel board contains the following indicators.

- D19: Green. It is lit when the analyzer power switch is turned on, indicating that the 3.3V power supply has been connected.
- D20: Green. It is lit when the analyzer power switch is turned on, indicating that the 5V power supply has been connected.
- D21: Green. It is lit when the analyzer power switch is turned on, indicating that the +12V power supply has been connected.
- D22: Green. It is lit when the analyzer power switch is turned on, indicating that the 24V power supply has been connected.
- D23: Green. It is lit when the analyzer power switch is turned on, indicating that the -12V power supply has been connected.
- D32: Green. It indicates the working status of cleaning fluid heater 2. It is lit when the heater is turned on.
- D33: Green. It indicates the working status of cleaning fluid heater 3. It is lit when the heater is turned on.
- D36: Green. It indicates the working status of reaction carousel heater 1. It is lit when the heater is turned on.
- D38: Green. It indicates the status of the high-concentration waste tank. It is lit when the waste tank is full.
- D39: Green. It indicates the status of the waste buffer tank. It is lit when the waste buffer tank is full.
- D2: Green. It is flashing when FPGA works normally.

Test points

In the following positions of the three-carousel board can signal tests be performed.

- VBB: +24V power supply input. Normal range: 11.4 - 12.6V.
- VCC: +5V power supply input. Normal range: 4.75 - 5.25V.

- -12V: -12V power supply input. Normal range: -11.4 - -12.6V.
- VDD: +3.3V power supply. It is secondary power supply used to power the major digital circuits of the PCBA. Normal range: 2.97 - 3.63V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the PCBA.

Check the connectors with clamps and ensure that the clamps have been locked properly.

Check other connectors and ensure that they are inserted into the end of the slots.

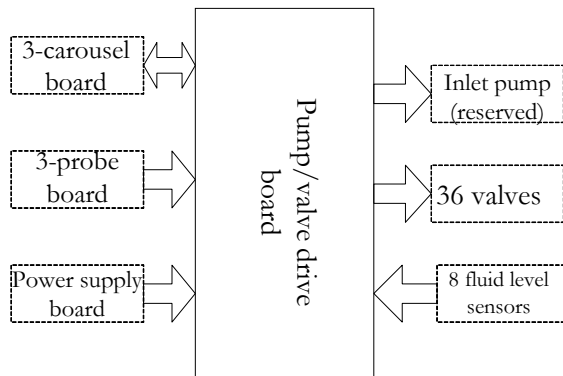
It requires relatively great force to plug/unplug connectors J2, J8, J9, J28, J26, and J27. Hold the PCBA by its edge while plugging/unplugging these connectors to prevent it from being deformed or damaged.

6.4.5 Pump/Valve drive board

Functions

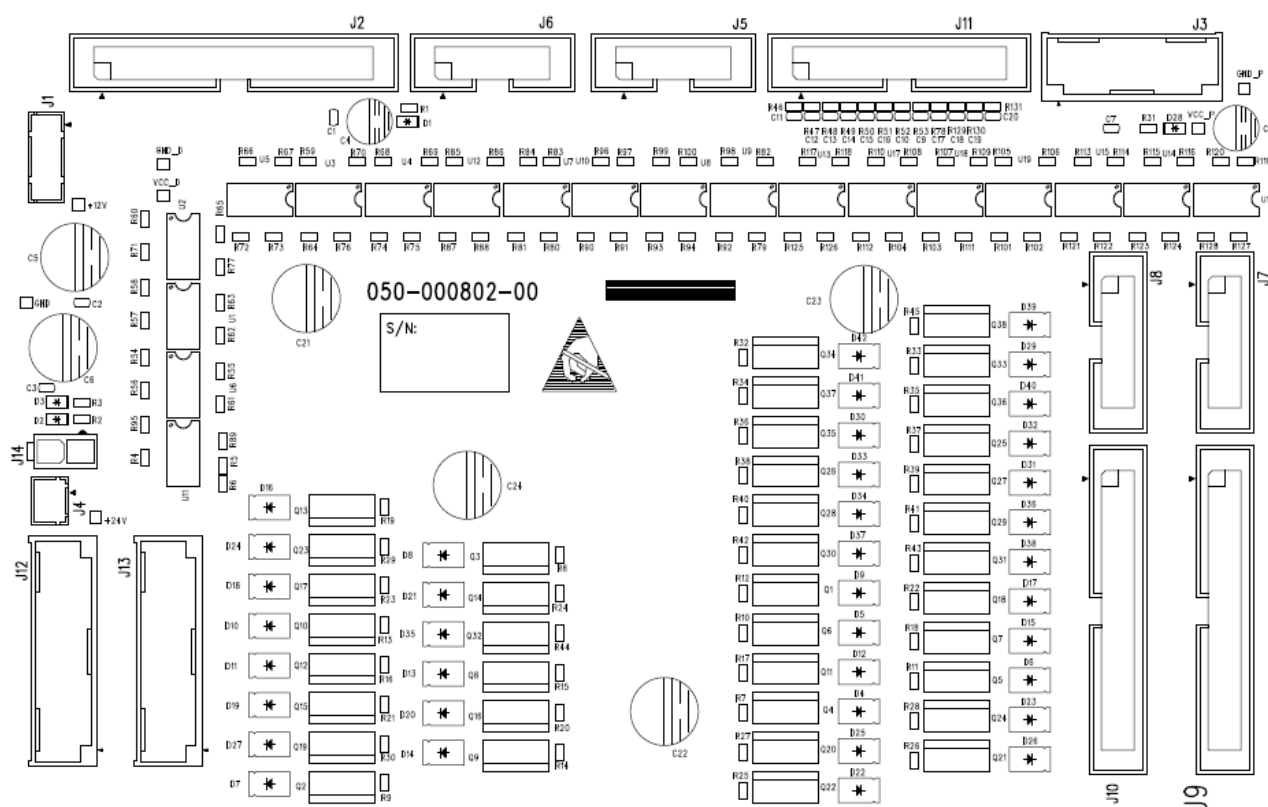
The pump/valve drive board receives pump/valve control signals from the three-carousel control drive board and three-probe control drive board, drives the water inlet pump (reserved) and 36 valves (16 of which are reserved), and transmits the 5 fluidic sensors' signals to the three-carousel control drive board.

The functional diagram of the pump/valve drive board is as shown below.



Description

The PCB layout of the pump/valve drive board is as shown below.



Connectors

The pump/valve drive board contains the following connectors.

Power supply:

Power supply input (J1): 12-pin, providing +5V digital, 12V and 24V for the PCBAs.

Pin No.	Signal	Reference Value
1	+12V	11.4 ~12.6V
2	GND	/
3	+12V	11.4 ~12.6V
4	GND	/
5	+12V	11.4 ~12.6V
6	GND	/
7	+12V	11.4 ~12.6V
8	GND	/
9	+24V	23.5 ~26V
10	GND	/
11	+24	23.5 ~26V
12	+24	23.5 ~26V

Detection and control connectors:

- Cuvette wash pump/valve control connector (J2): 34-pin, used for controlling the pumps and valves of the cuvette wash unit.
- Floater conversion connector (J5): 14-pin, used for feeding back status of fluidic floaters to the three-carousel drive board.
- Floater connector (J11): 26-pin, used for detecting the status of fluidic floaters.

- Pump/valve control connector 1 (J3): 20-pin, used for receiving pump/valve control signal transmitted from the 3-probe drive board and 3-carousel drive board and the 24V pump control signal.
- 24V pump drive connector (J7): 16-pin, used for driving the deionized water circulating pump and concentrated wash solution pump.
- Valve drive connector 1 (J9): 34-pin, used for driving the exterior wash valves of reagent probe, sample probe, sample mixer, and reagent mixer, water inlet valve, cuvette phase 1-2 injection valve, cuvette phase-6 injection valve, high-concentration waste discharge valve, dilution valve, and cuvette phase-5 injection valve.
- Valve drive connector 2 (J12): 26-pin, used for driving the cuvette phase-3 injection valve, vacuum release valve, waste suction valve, low-concentration waste discharge valve, vacuum container waste discharge valve, waste pump, external vacuum valve, and cuvette phase-4 injection valve.

Indicators

The pump/valve drive board contains the following indicators.

- D1: Green. It is lit when the analyzing unit power switch is turned on, indicating that the 5V power supply for three-carousel drive board has been connected.
- D2: Green. It is lit when the analyzer power switch is turned on, indicating that the 12V power supply has been connected.
- D3: Green. It is lit when the analyzer power switch is turned on, indicating that the 24V power supply has been connected.
- D28: Green. It is lit when the analyzing unit power switch is turned on, indicating that the 5V power supply for three-probe drive board has been connected.

Test points

In the following positions of the pump/valve drive board can signal tests be performed.

- +12V: +12V power supply input. Normal range: 11.4 - 12.6V.
- +24V: +24V power supply input. Normal range: 23.5 - 26V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the PCBA.

Check the connectors with clamps and ensure that the clamps have been locked properly.

Check other connectors and ensure that they are inserted into the end of the slots.

It requires relatively great force to plug/unplug connectors J2, J3, J5, J9, and J12. Hold the PCBA by its edge while plugging/unplugging these connectors to prevent it from being deformed or damaged.

6.4.6 Rotation speed optical coupler conversion board

Functions

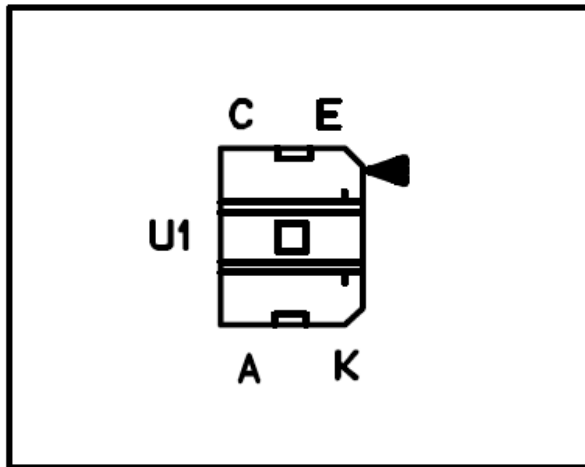
The communication conversion board is intended for the following functions:

- Use to connect the optical coupler of the mixer rotation speed.
- Providing the signal of the mixer rotation speed and the mixer drive board.

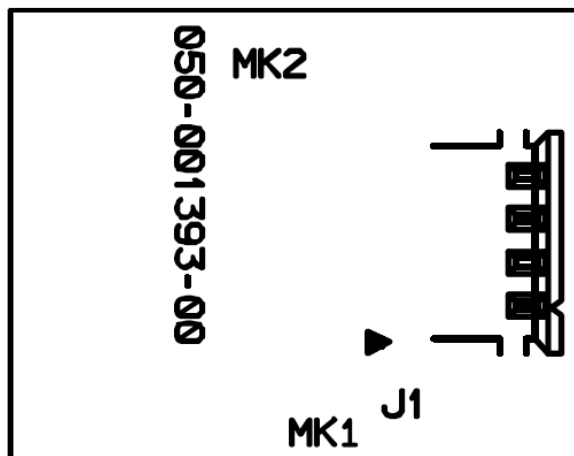
Description

The PCB layout of the communication conversion board is as shown below.

Top view:



Bottom view:



Connectors

- J1: connector for the optical coupler of the mixer rotation speed and mixer drive board.

Indicators

N/A

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

6.4.7 Communication Conversion Board

Functions

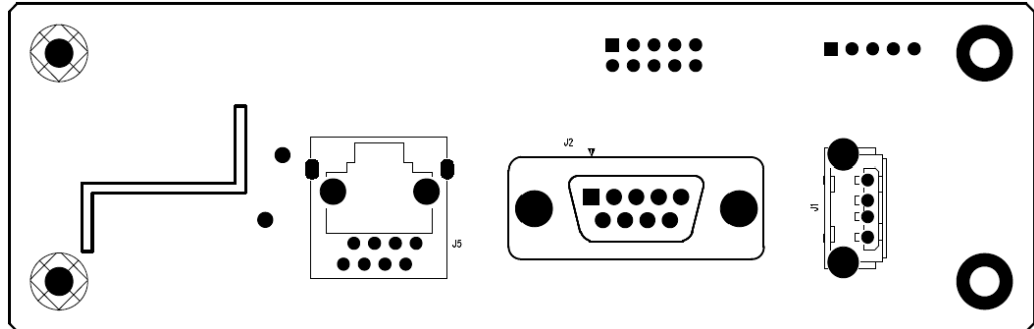
The communication conversion board is intended for the following functions:

- Providing connectors for the main control board: serial port, network interface, and USB interface.
- Providing connectors for the control console (computer): standard DB9 serial port, RJ45 network interface, and standard USB interface.
- Shifting, isolating and protecting the signals received at the connectors connecting the main control board and the control console.

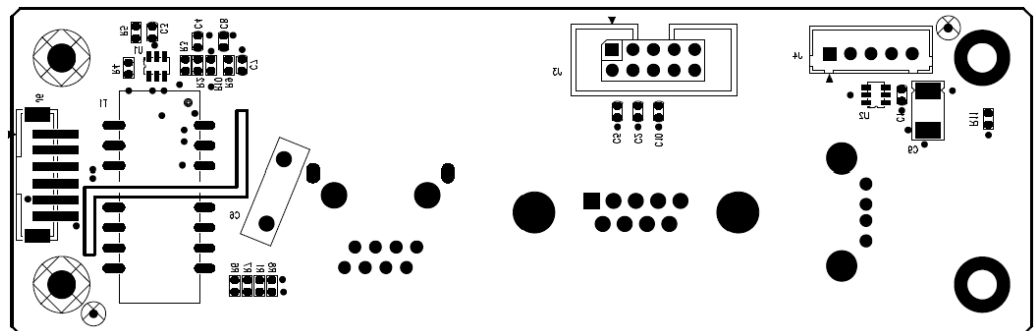
Description

The PCB layout of the communication conversion board is as shown below.

Top view:



Bottom view:



Connectors

- J1: USB interface to connect a computer. (Reserved)
- J2: DB9 serial port to connect a computer.
- J3: serial port communication interface to connect the main control board
- J4: USB interface to connect the main control board
- J5: standard RJ45 network interface to connect a computer.
- J6: Ethernet interface to connect the main control board

Indicators

N/A

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA. Check if the PCBA has been locked tightly.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

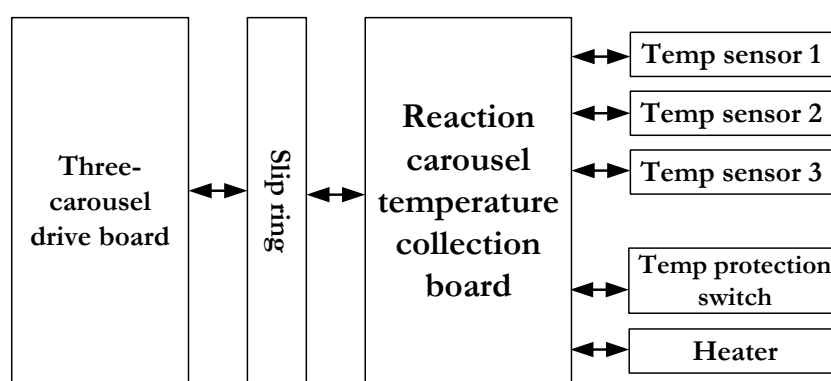
6.4.8 Reaction Carousel Temperature Collection Board

Functions and principles

The instrument has one reaction carousel temperature collection board, which is used to:

- Collect/Convert the reaction carouse temperature sensor signals and output them to the wash temperature control board
- Provide the SPI interface and reaction carousel heater power supply connector, and connect the wash temperature control board via the slip ring to control the heater.

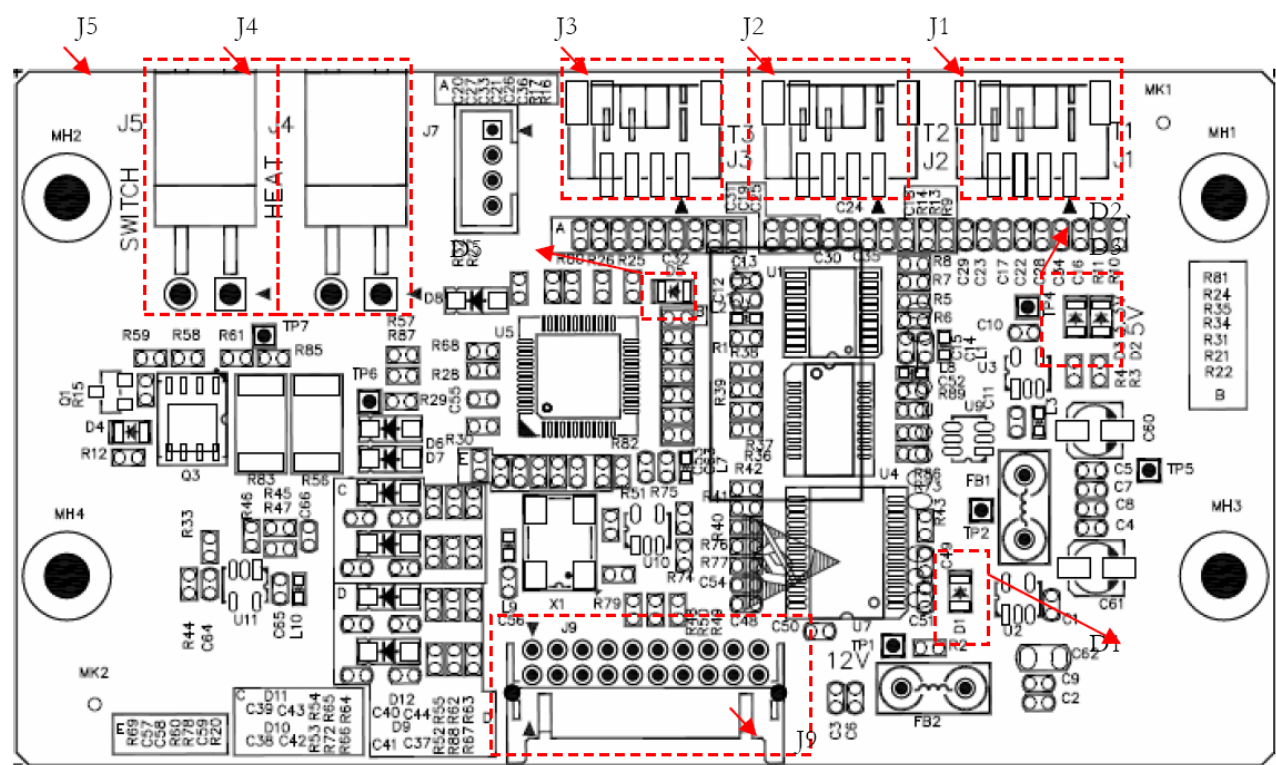
The functional diagram of the reaction carousel temperature collection board is as shown below.



Description

PCB layout

The PCB layout of the reaction carousel temperature collection board is as shown below.



Connectors

The reaction carousel temperature collection board contains the following connectors.

Connectors 1-3 (J1-J3) for reaction carousel temperature sensor: 3-pin. Used to connect the reaction carousel temperature sensor with the temperature detection circuit.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
1	SHIELD	/	Shielded wire of sensor	/
2	REF	I	Signal cable of sensor	/
3	PT1000	I	Signal cable of sensor	/

Heater connector (J4): 2-pin. Used to connect the heater with the temperature detection circuit.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
1	+24V	/	Power supply of heater	/
2	SW	/	Connecting with the temperature protection switch	/

Temperature protection switch connector (J5): 2-pin. Used to connect the temperature protection switch with the temperature detection circuit.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
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1	SW	/	Connecting with the temperature protection switch	/
2	HEATER	/	Heater control signal	/

Connector for connecting the reaction carousel temperature collection board and the slip ring (J9): 20-pin. Used to connect the reaction carousel temperature collection board and the slip ring.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
1	+24V	/	Positive pole of heater	/
2	+24V	/	Positive pole of heater	/
3	+24V	/	Positive pole of heater	/
4	+24V	/	Positive pole of heater	/
5	HEATER	/	Negative pole of heater	/
6	HEATER	/	Negative pole of heater	/
7	HEATER	/	Negative pole of heater	/
8	HEATER	/	Negative pole of heater	/
9	SYNC_ADT_I	I	Synchronous input signal of AD	5V TTL
10	DIN_ADT_I	I	Data input signal of AD	5V TTL
11	CLK_ADT_I	I	Clock input signal of AD	5V TTL
12	DOUT_ADT_O	O	Data output signal of AD	5V TTL
13	BUSY_ADT_O	O	Busy output signal of AD	5V TTL
14	NC	/	/	/
15	NC	/	/	/
16	S2_B_I	I	Control signals of multiplexer switch	5V TTL
17	S2_A_I	I	Control signals of multiplexer switch	5V TTL
18	VPP	/	12V power supply input	/
19	AGND	/	Analog ground	/
20	SHIELD	/	Shielding ground	5V TTL

Indicators

The reaction carousel temperature collection board contains the following indicators.

- +12V power supply indicator (D1): green. When it is lit, it indicates that the +12V power supply has been connected.
- +5V power supply indicator (D2): green. When it is lit, it indicates that the +5V power supply has been connected.
- +3.3V power supply indicator (D3): green. When it is lit, it indicates that the +3.3V power supply has been connected.
- Heater switch (D4): red. When it is lit, it indicates that heater switch on the temperature collection board is ready.
- PCB status (D5): orange. When the PCB is normal, the indicator will flash shortly for 4 times and then go out for a relatively long time, with the flashing and going out repeated.

Test points

In the following positions of the reaction carousel temperature collection board can signal tests be performed

- TP1: +12V power supply input. Normal range: $12V \pm 5\%$, that is, 11.4 - 12.6V.
- TP2: +5V power supply, secondary power supply. Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.
- TP4: +3.3V power supply, secondary power supply. Normal range: $3.3V \pm 5\%$, that is, 3.14 - 3.47V.
- TP5: reaction carousel temperature collection board ground.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA. Check if the PCBA has been locked tightly.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

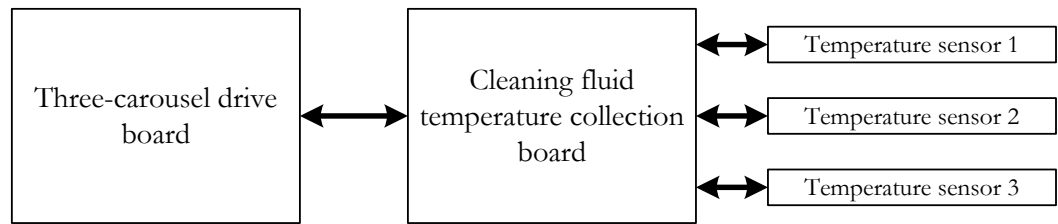
6.4.9 Cleaning fluid temperature collection board

Functions

The instrument has one cleaning fluid temperature collection board, which is mainly used to:

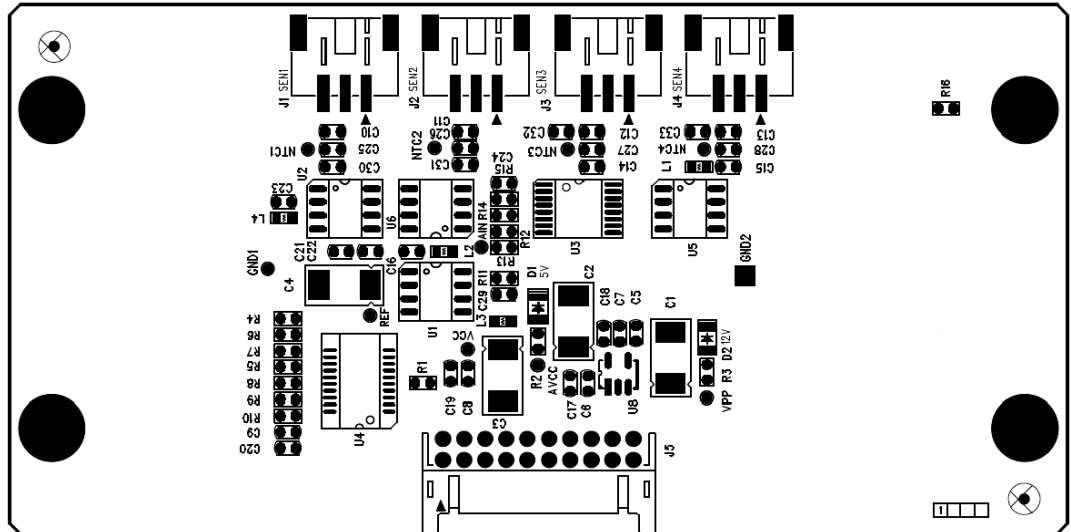
Process, collect and AD-convert signals of two cleaning fluid temperature sensors and of one environment temperature sensor. The cleaning fluid temperature collection board is connected with the wash temperature control board.

The functional diagram of the cleaning fluid temperature collection board is as shown below.



Description

The PCB layout of the cleaning fluid temperature collection board is as shown below.



Connectors

The cleaning fluid temperature collection board contains the following connectors:

Temperature sensor connectors 1-4 (J1-J4): 3-pin, used to connect the temperature sensor with the temperature detection circuit.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
1	SHIELD	/	Shielded wire of sensor	/
2	AGND	/	Signal cable of sensor (analog ground)	/
3	NTC	I	Signal cable of sensor	Analog

Connector J5 for wash temperature control board: 20-pin, used to power the PCBA and transmit/control the signals.

Pin No.	Signal Name	I/O	Description	Valid Level and Type
1	SGND	/	Enclosure ground	/
2	SGND	/	Enclosure ground	/
3	CS	I	AD chip selection input	TTL, valid in the case of low level.
4	NC	/	NC	/
5	SDO	O	AD chip data output	TTL
6	SCLK	I	AD chip clock input	TTL
7	NC	/	NC	/

Pin No.	Signal Name	I/O	Description	Valid Level and Type
8	NC	/	NC	/
9	NC	/	NC	/
10	MUX_A	I	Channel A	TTL
11	VPP	/	12V power supply input	/
12	MUX_B	I	Channel B	TTL
13	GND	/	Ground	/
14	GND	/	Ground	/
15	GND	/	Ground	/
16	NC	/	NC	/
17	NC	/	NC	/
18	NC	/	NC	/
19	NC	/	NC	/
20	NC	/	NC	/

Indicators

The cleaning fluid temperature collection board contains the following indicators.

- +12V power supply indicator (D2): green. When it is lit, it indicates that the +12V power supply has been connected.
- +5V power supply indicator (D1): green. When it is lit, it indicates that the +5V power supply has been connected.

Test points

- In the following positions of the cleaning fluid temperature collection board can signal tests be performed.
- VPP: +12V power supply input. Normal range: $12V \pm 5\%$, that is, 11.4 - 12.6V.
- AVCC: +5V power supply. It is secondary power supply used to power the analog parts of the PCBA. Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.
- VCC: +5V power supply. It is secondary power supply used to power the digital parts of the PCBA. Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.
- GND1 and GND2: grounding terminals of the cleaning fluid temperature collection board.
- AIN: AD collection signal test point.
- REF: reference voltage.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

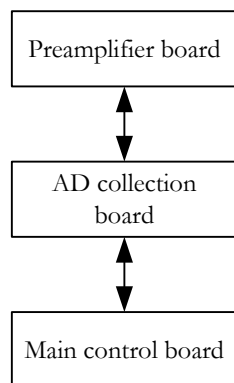
6.4.10 AD collection board

Functions and principles

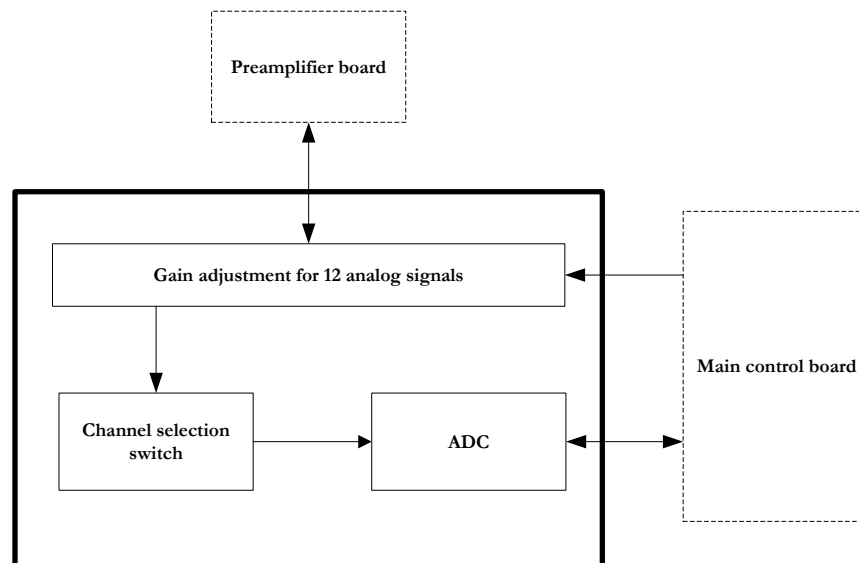
The AD collection board is intended to implement the following functions:

- Adjusting gains of 12 photoelectric signals via a digital potentiometer.
- Controlling the channel selection switch to switch among 12 channel signals at different time.
- Converting photoelectric analog signals into digital signals via an AD converter and then outputting the signals.

The figure below shows the relation between the AD collection board and other PCBAs.



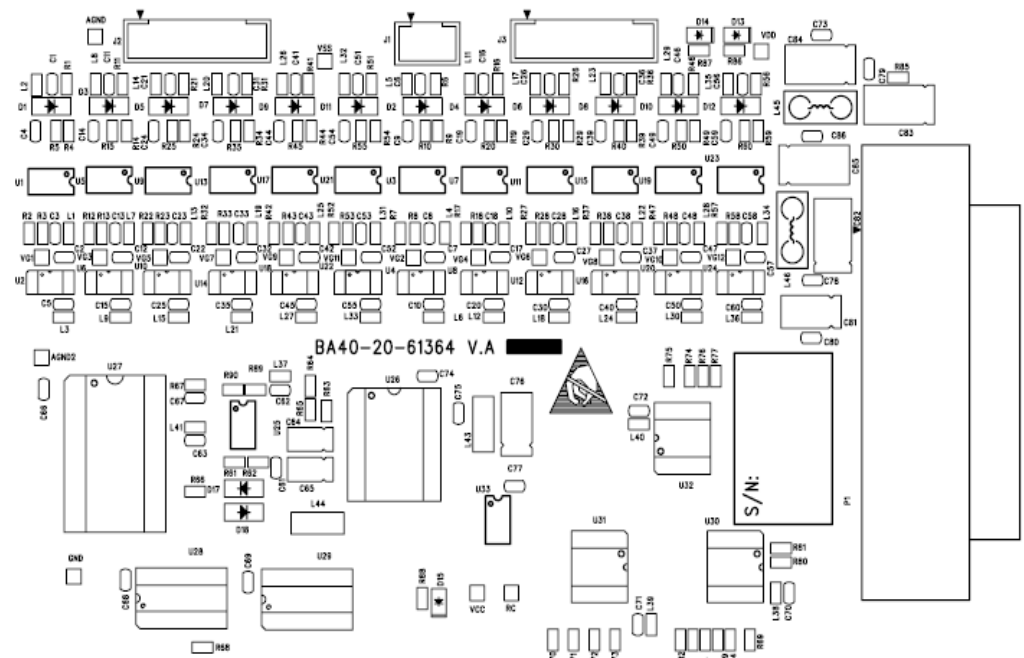
The functional diagram of the AD collection board is as shown below.



Description

PCB layout

The PCB layout of the AD collection board is as shown below.



Connectors

The AD collection board contains the following connectors.

Power supply:

Power supply output connector (J1): 3-pin, used to provide +12V and -12V power supplies for the preamplifier board.

Pin No.	Signal	Reference Value
1	+12V	11.4 ~12.6V
2	AGND	/
3	-12V	-11.4 ~-12.6V

Connectors for sending/receiving communication signals:

Connector (J2) for preamplifier board signal input: 8-pin, used to receive photoelectric analog signals of six channels.

Pin No.	Signal	Reference Value
1	AGND	/
2	Signal1	0.25V-2.5V (water blank condition)
3	Signal3	0.25V-2.5V (water blank condition)
4	Signal5	0.25V-2.5V (water blank condition)
5	Signal7	0.25V-2.5V (water blank condition)
6	Signal9	0.25V-2.5V (water blank condition)
7	Signal11	0.25V-2.5V (water blank condition)
8	AGND_SHELD	/

Connector (J3) for preamplifier board signal input: 8-pin, used to receive photoelectric analog signals of six channels.

Pin No.	Signal	Reference Value
1	AGND	/

2	Signal2	0.25V-2.5V (water blank condition)
3	Signal4	0.25V-2.5V (water blank condition)
4	Signal6	0.25V-2.5V (water blank condition)
5	Signal8	0.25V-2.5V (water blank condition)
6	Signal10	0.25V-2.5V (water blank condition)
7	Signal12	0.25V-2.5V (water blank condition)
8	AGND_SHELD	/

Main control board connector (P1): 25-pin, used for communication with the main control board.

Switches and jumpers

The AD collection board has no switches or jumpers.

Indicators

The AD collection board contains the following indicators.

Power supply:

- +12V power supply indicator (D13): green. When it is lit, it indicates that the +12V power supply has been connected.
- -12V power supply indicator (D14): green. When it is lit, it indicates that the -12V power supply has been connected.
- +5V power supply indicator (D15): green. When it is lit, it indicates that the +5V power supply has been connected.

Test points

In the following positions of the AD collection board can signal tests be performed

- VG1: signal output of channel 340nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG2: signal output of channel 380nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG3: signal output of channel 412nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG4: signal output of channel 450nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG5: signal output of channel 505nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG6: signal output of channel 546nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG7: signal output of channel 570nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG8: signal output of channel 605nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG9: signal output of channel 660nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG10: signal output of channel 700nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG11: signal output of channel 740nm. Normal range: varies with the signal strength and lies between 0-5V.
- VG12: signal output of channel 800nm. Normal range: varies with the signal strength and lies between 0-5V.
- VDD: +12V power supply input. Normal range: $12V \pm 5\%$, that is, 11.4 - 12.6V.
- VSS: -12V power supply input. Normal range: $-12V \pm 5\%$, that is, -11.4 - -12.6V.
- VCC: +5V power supply input. Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the PCBA.

Check the connectors with clamps and ensure that the clamps have been locked properly.

Check other connectors and ensure that they are inserted into the end of the slots.

It requires relatively great force to plug/unplug connectors J2 and J3. Hold the PCBA by its edge while plugging/unplugging these connectors to prevent it from being deformed or damaged.

After connecting P1 connector (DB25), tighten the retaining screws on two sides of it.

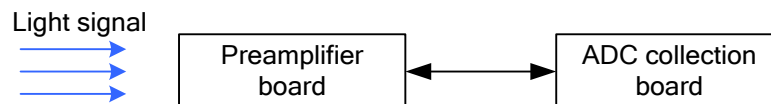
6.4.11 Preamplifier Board

Functions and principles

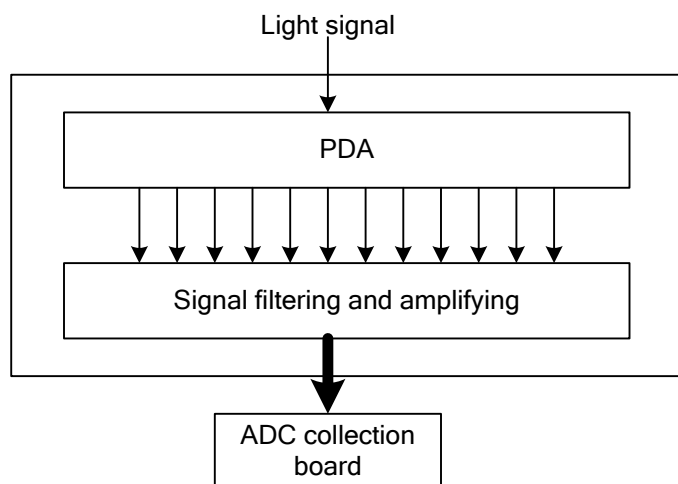
The preamplifier board is intended to implement the following functions:

- Converting optical signals into electric signals via a photodiode array.
- With the help of the back circuit, filtering signals and transmitting them to the AD collection board after amplification.

The figure below shows the relation between the preamplifier board and other PCBAs.



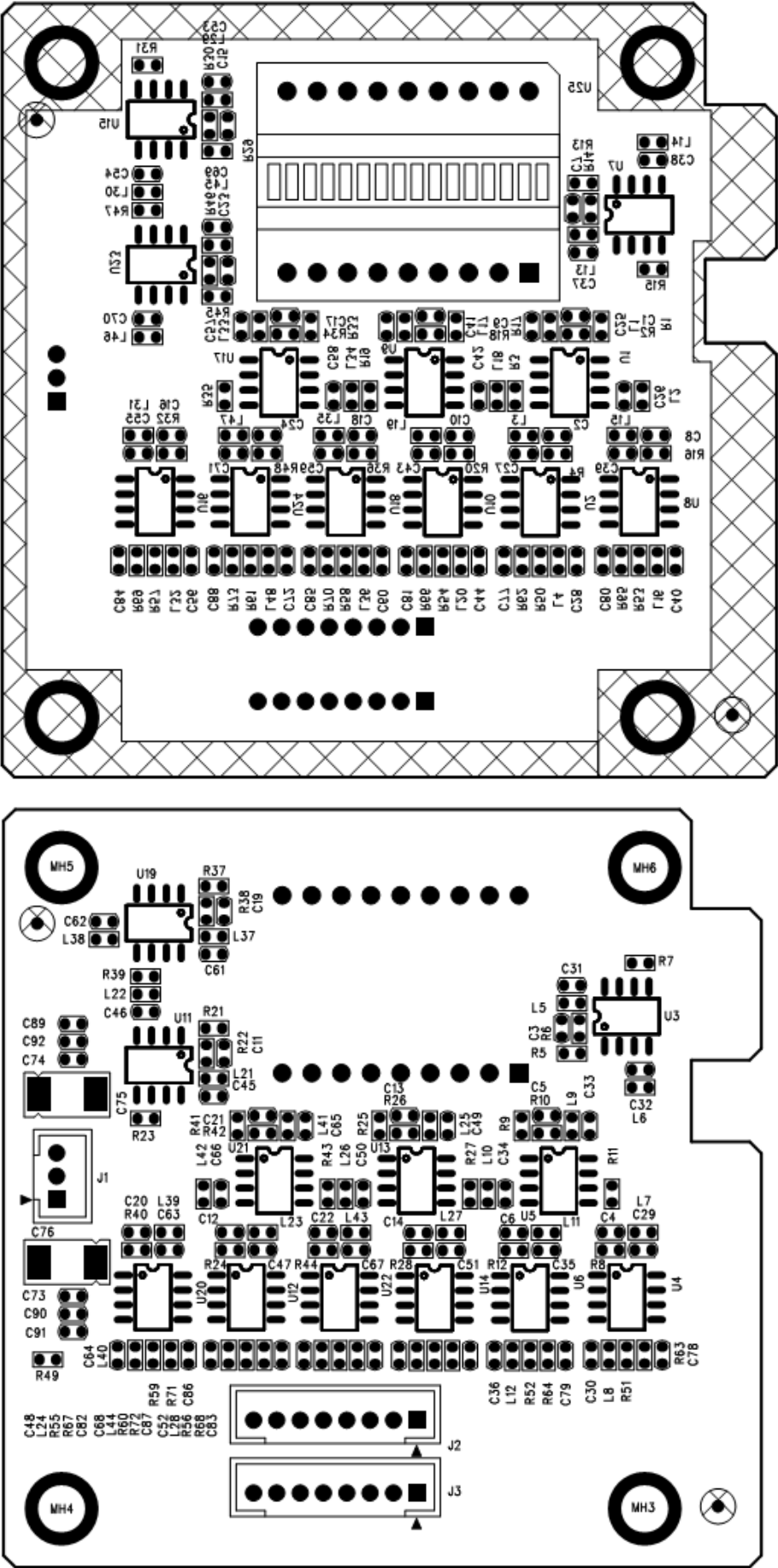
The functional diagram of the preamplifier board is as shown below.



Description

PCB layout

The PCB layout of the preamplifier board is as shown below.



Connectors

The preamplifier board includes the following connectors.

Power supply:

Power supply input connector (J1): 3-pin, used to provide +12V and -12V power supplies for the preamplifier board.

Pin No.	Signal	Reference Value
1	+12V	11.4 ~12.6V
2	AGND	/
3	-12V	-11.4 ~-12.6V

Connectors for sending/receiving communication signals:

Connector (J2) for preamplifier board signal output: 8-pin, used to receive photoelectric analog signals of 6 channels.

Pin No.	Signal	Reference Value
1	AGND	/
2	Signal1	Depending on the actual measuring light intensity.
3	Signal3	Depending on the actual measuring light intensity.
4	Signal5	Depending on the actual measuring light intensity.
5	Signal7	Depending on the actual measuring light intensity.
6	Signal9	Depending on the actual measuring light intensity.
7	Signal11	Depending on the actual measuring light intensity.
8	AGND_SHELD	/

Connector (J3) for preamplifier board signal output: 8-pin, used to receive photoelectric analog signals of six channels.

Pin No.	Signal	Reference Value
1	AGND	/
2	Signal2	Depending on the actual measuring light intensity.
3	Signal4	Depending on the actual measuring light intensity.
4	Signal6	Depending on the actual measuring light intensity.
5	Signal8	Depending on the actual measuring light intensity.
6	Signal10	Depending on the actual measuring light intensity.
7	Signal12	Depending on the actual measuring light intensity.
8	AGND_SHELD	/

Switches and jumpers

The preamplifier board has no switches or jumpers.

Indicators

The preamplifier board is encapsulated inside the Optical measurement assembly and has no indicators.

Test points

The preamplifier board is encapsulated inside the Optical measurement assembly and does not need maintenance, and it therefore, has no test points.

Installation methods and precautions

Since encapsulated inside the Optical measurement assembly, the preamplifier board should be maintained together with the Optical measurement assembly rather than maintained independently.

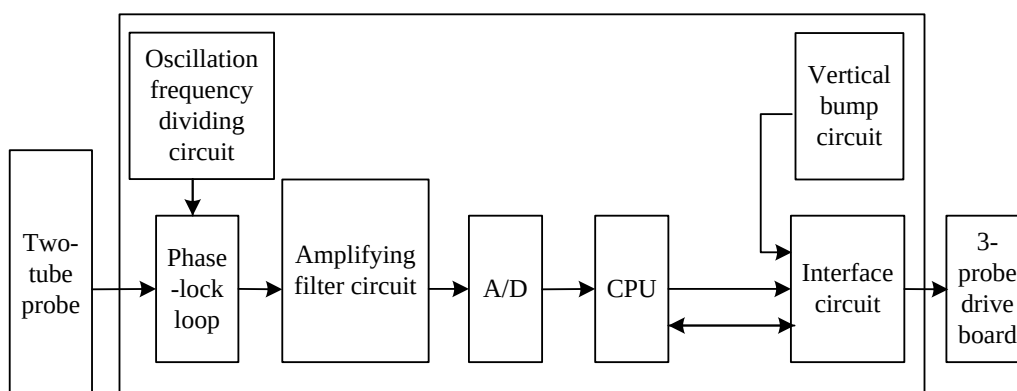
6.4.12 Level sense board

Functions and principles

The instrument has two level sense boards, one for detecting reagent level and the other for detecting sample level.

- The level sense boards are intended for implement the following functions: Sensing the reagent and sample levels. The three boards are identical in circuit structure and interfaces and capable of detecting the fluid level steadily and reliably, especially allowing the sample probe to correctly detecting the fluid level inside reaction cuvettes. •
- When the probe touches the fluid level, the fluid level signal is output through the five-probe conversion board to the control drive board.
- Providing vertical obstruction detection and outputting the detection signals to the control drive board through the five-probe/mixer conversion board.

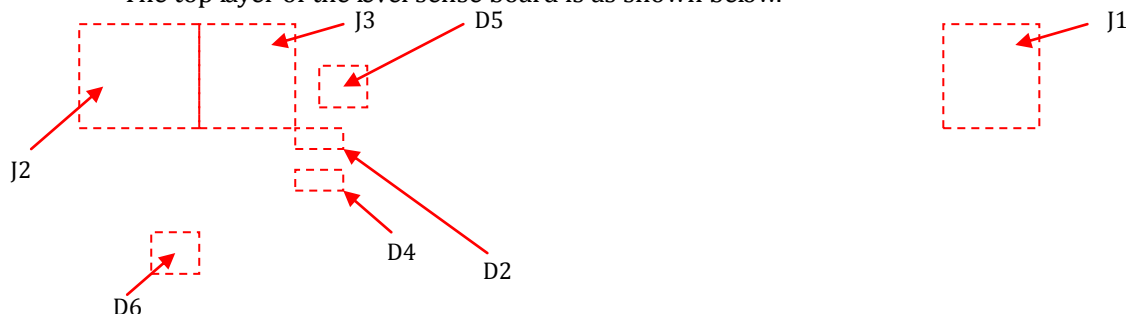
The functional diagram of the level sense board is as shown below.

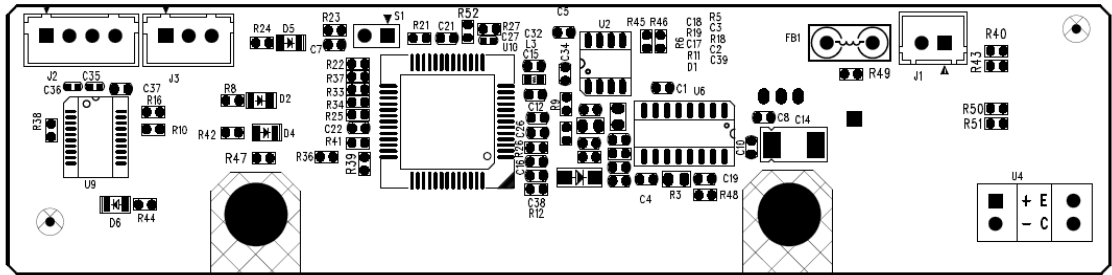


Description

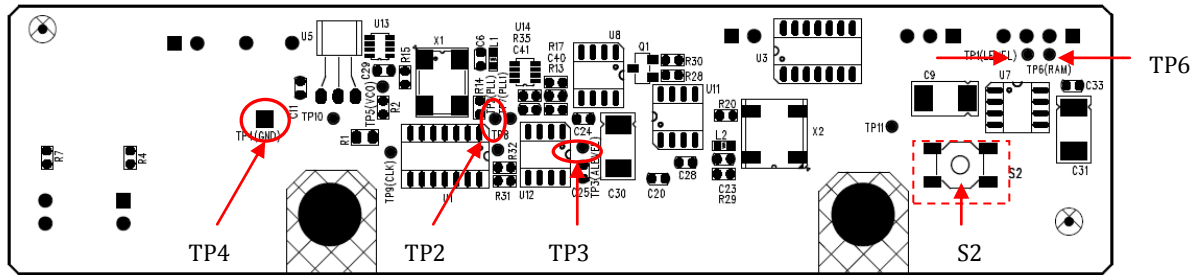
PCB layout

The top layer of the level sense board is as shown below.





The bottom view of the level sense board is as shown below.



Connectors

The level sense board contains the following connectors.

Probe connector (J1): 2-pin, used for connecting the sample probe or reagent probes with related circuit.

Pin No.	Signal
1	GND
2	Probe capacitor signal input

Power supply and signal output connector (J2): 4-pin, used to provide power supply for the level sense board and output the level sense signal and vertical obstruction signal.

Pin No.	Signal	Reference Value
1	GND	/
2	Vertical obstruction signal output	High level (about 4V) is output when no vertical obstruction occurs, and low level (about 0V) is output when vertical obstruction happens.
3	Level sense signal output	Low level (about 0V) is output when the probe fails to detect the fluid level, and high level (about 4V) is output when the probe detects the fluid level.
4	+12V	11.4 ~12.6V

Serial port communication cable connector (J3): 3-pin, used for communication between the level sense board and the control drive board.

Pin No.	Signal
1	RXD
2	RST
3	TXD

Switches and jumpers

The level sense board contains the following switches and jumpers.

- PSEN enable jumper (S1): used for downloading application programs.
- Calibrate key (S2): used for manual self-calibration of fluid level detection.

Jumper S1 must be disconnected in normal conditions.

Indicators

The level sense board contains the following indicators.

- Level sense system calibration indicator (D2): yellow. It is extinguished when the self-calibration of the level sense system fails, and vice versa. If it is lit, it indicates that the calibration is complete and the level sense system is ready for measurement. When the calibration is complete, the indicator will be lit all the time.
- Sensitivity switch indicator (D4): red. It is extinguished when the default sensitivity is being used, and vice versa, which means level sensing is performed in special positions.
- Level sense indicator (D5): green. It is extinguished when the probe fails to detect the fluid level, and vice versa.
- Vertical obstruction indicator (D6): green. It is extinguished when no vertical obstruction occurs, and vice versa.

Test points

In the following positions of the level sense board can signal tests be performed

- TP1 (LEVEL): level sense signal output. Normal condition: Low level (about 0V) is output when the probe fails to detect the fluid level, and high level (about 4V) is output when the probe detects the fluid level.
- TP2 (PLL): working point voltage of the level sense system. Normal range: $3.8 \pm 0.5V$.
- TP3 (ALEVEL): analog fluid level. Analog level sense signal, based on which whether the fluid level is detected or not is determined.
- TP4 (GND): grounding terminal of the level sense board.
- TP5 (VCO): reserved.
- TP6 (RAM): vertical obstruction signal output. Normal condition: High level (about 4V) is output when no vertical obstruction occurs, and low level (about 0V) is output when vertical obstruction happens.
- TP7 (PLL1): reserved.
- TP8: voltage monitoring signal for self-calibration of the level sense system.
- TP9 (CLK): reserved.
- TP10: +9V power supply. It is secondary power supply used to power the internal phase-locked loop and some operational amplifiers of the PCBA. Normal range: $9V \pm 5\%$, that is, 8.55 - 9.45V.
- TP11: +5V power supply. It is secondary power supply used to power most components inside the PCBA, such as micro control unit (MCU). Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

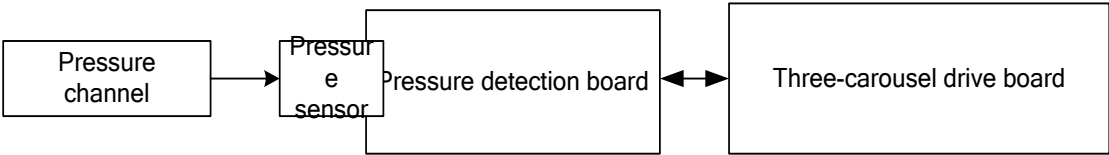
6.4.13 Pressure detection board

Functions and principles

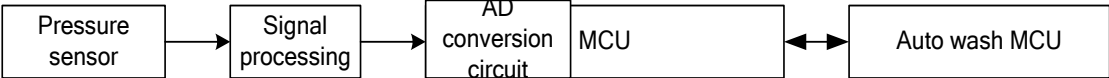
The pressure detection board is intended to implement the following functions:

Detecting vacuum pressure, converting the detection results into digital signals, and then sending them to the cuvette wash unit via the serial port.

The figure below shows the location of the pressure detection board in the hardware system.

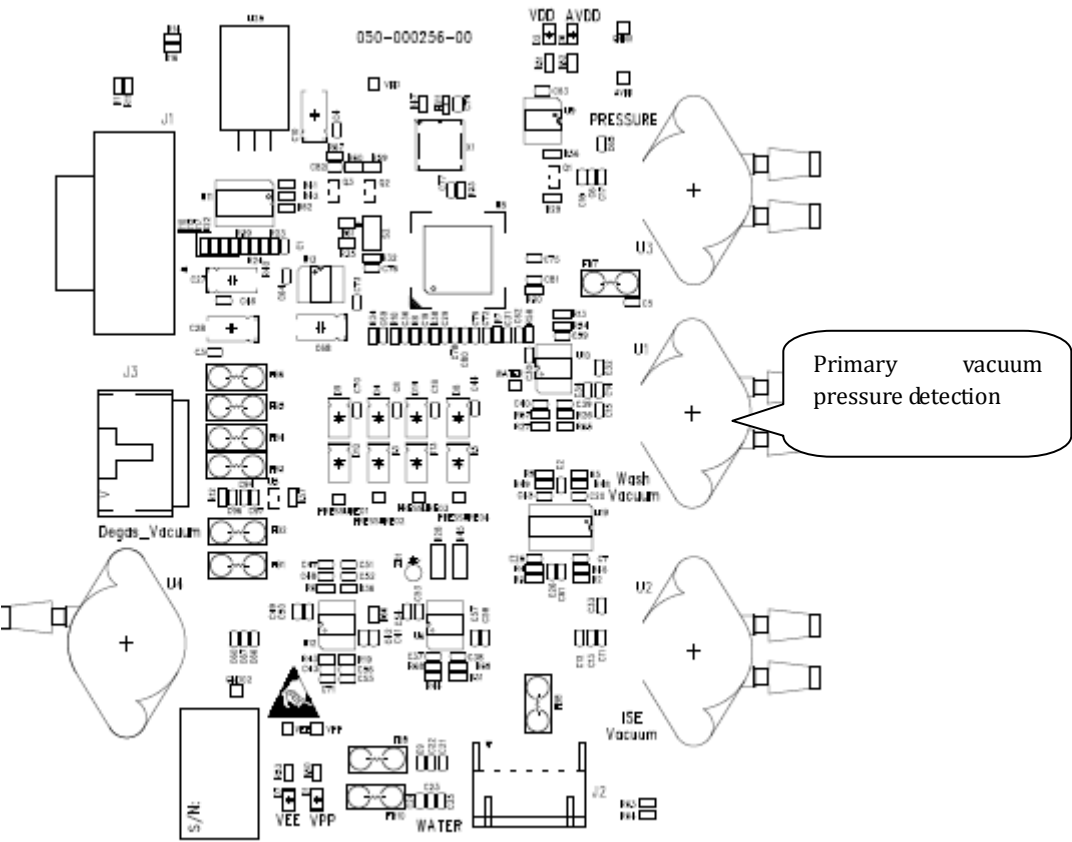


The functional diagram of the pressure detection board is as shown below.



Description

PCB layout



Connectors

Power supply and communication connector (J1): DB9 connector, used for connecting the wash temperature control board, providing +12V and -12V power supplies for the PCBA, and serial port communication with the upper-level PCBA.

Pin No.	Description	Applicable Test Methods
1	+12V power supply	The voltage at this point lies between 11.4 - 12.6V when the PCBA is working normally.
2	Serial port input signal	The voltage at this point is greater than 2.4V when the system status is Standby.
3	Serial port output signal	The voltage at this point is greater than 2.4V when the system status is Standby.
4	Online-downloading enable signal	The voltage at this point is greater than 2.4V when the system status is Standby.
5	Ground	/
6	-12V power supply	The voltage at this point lies between -12.6 - -11.4V when the PCBA is working normally.
7	Reset signal	The voltage at this point is greater than 2.4V when the system status is Standby.
8	Ground	/

Pin No.	Description	Applicable Test Methods
9	Ground	/

Pin No.	Description	Test Methods
1	+12V power supply	The voltage at this point lies between 11.4 - 12.6V when the PCBA is working normally.
2	-12V power supply	The voltage at this point lies between -11.6 - -11.4V when the PCBA is working normally.
4	Power supply ground	/
5	Power supply ground	/
6	Shielded layer of cable, connected with the grounding terminal of the PCBA	/

Reserved pressure sensor connector (J3): reserved connector.

Switches and jumpers

MCU downloading enable jumper (S2): When it is short-circuited, it indicates that the MCU is downloading something. The jumper must be disconnected when the PCBA is working normally.

Indicators and test points

Indicators:

- +12V power supply indicator (VPP): green. It is lit when the +12V power supply is working normally.
- -12V power supply indicator (VEE): green. It is lit when the -12V power supply is working normally.
- +5V digital power supply indicator (VDD): green. It is lit when the +5V digital power supply is working normally.
- +5V analog power supply indicator (AVDD): green. It is lit when the +5V analog power supply is working normally.

Test points:

- +12V power supply (VPP): with normal range of 11.4 - 12.6V.
- -12V power supply (VEE): with normal range of -12.6 - -11.4V.
- +5V digital power supply (VDD): with normal range of 4.8 - 5.2V.
- +5V analog power supply (AVDD): with normal range of 4.8 - 5.2V.
- Auto cleaning negative pressure test point (PRESSURE03): The voltage at this point lies between 0.1-0.3V when the pressure sensor is connected to the air.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Tighten the two retaining screws on two sides of connector J1, and insert connector J2 properly into the PCBA.

When connecting the pressure sensors (U1) to the gas circuit, press them slightly and connect them with proper force to avoid breaking the connectors.

Install the shielded cover after installing or repairing the PCBA.

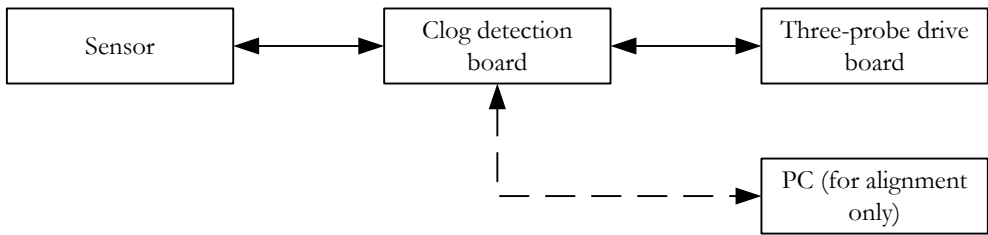
6.4.14 Clog detection board

Functions and principles

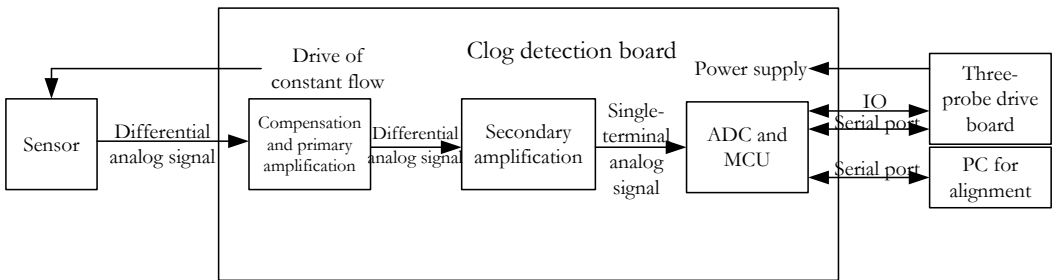
The clog detection board possesses the following functions:

- Collecting pressure by connecting with the pressure sensor.
- Checking for clogs and empty aspirate under the control of the sample control drive board.
- Sending debugging information to the debugging computer.

The figure below shows the location of the clog detection board in the hardware system.



The circuit diagram of the clog detection board is as shown below.



Pin No.	Description	Test Methods
6	Interruption control signal	The voltage at this point is less than 0.4V when the system status is Standby.
7	Interruption enable signal	The voltage at this point is less than 0.4V when the system status is Standby.
8	Ground	/
9	Reserved input	/
10	Reserved output	/

Clog detection board communication connector (J3): connected with the sample control drive board and used for serial port communication between the clog detection board and the sample probe unit.

Pin No.	Description	Test Methods
1	+12V power supply	The voltage lies between 11.4-12.6V when the PCBA is working normally.
2	-12V power supply (reserved)	/
3	Serial port input signal	The voltage at this point is greater than 2.4V when the system status is Standby.
4	Serial port output signal	The voltage at this point is greater than 2.4V when the system status is Standby.
5	Online-downloading enable signal	The voltage at this point is less than 0.4V when the system status is Standby.
6	Reset signal	The voltage at this point is greater than 2.4V when the system status is Standby.
7	Ground	/
8	Ground	/

Clog detection board debugging connector (J4): connected with the serial port of a computer and used for debugging the clog detection system. This connector is temporarily not used.

Pin No.	Description	Test Methods
1	+12V power supply	/
2	-12V power supply (reserved)	/
3	Serial port input signal	/
4	Serial port output signal	/
5	Ground	/

Switches and jumpers

MCU downloading enable jumper (S2): When it is short-circuited, it indicates that the MCU is downloading something. The jumper must be disconnected when the PCBA is working normally.

Indicators and test points

Indicators

- Clog signal indicator (D3): red. It is lit when the PCBA outputs a clog signal.
- Empty aspirate signal indicator (D4): It is lit when the PCBA outputs an empty aspirate signal. •

- Possible clog signal indicator (D5): It is lit when the PCBA outputs a possible clog signal.
- +5V power supply indicator (D6): green. It is lit when the +5V power supply is working normally.
- +12V power supply indicator (D7): green. It is lit when the +12V power supply is working normally.

Test points

- +12V power supply (VPP): The voltage at this point lies between 11.4-12.6V when the PCBA is working normally.
- -12V power supply (VEE): reserved test point.
- Ground (GND): This test point is connected with the grounding terminal of the PCBA.
- Pressure sensor output signal (ANALOG): It is an analog signal amplified by the pressure sensor. When the pressure sensor is connected to the air, the voltage at this point lies between 0.8-1.3V.
- Clog signal (CLOG_Y): It is a test point of clog signal output. When the PCBA outputs a clog signal, indicator D3 will be lit and the voltage at this point greater than 2.4V; when the PCBA outputs no clog signal, indicator D3 will be extinguished and the voltage at this point less than 0.4V.
- Empty aspirate signal (NO_SAM): When the PCBA outputs an empty aspirate signal, indicator D4 will be lit and the voltage at this point greater than 2.4V; when the PCBA outputs no empty aspirate signal, indicator D4 will be extinguished and the voltage at this point less than 0.4V.
- Debugging test point (TEST): used for test during debugging.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Insert connectors J2 and J3 tightly into the PCBA. When inserting connector J1, you will hear a click, which means that the connector is inserted in place into its slot.

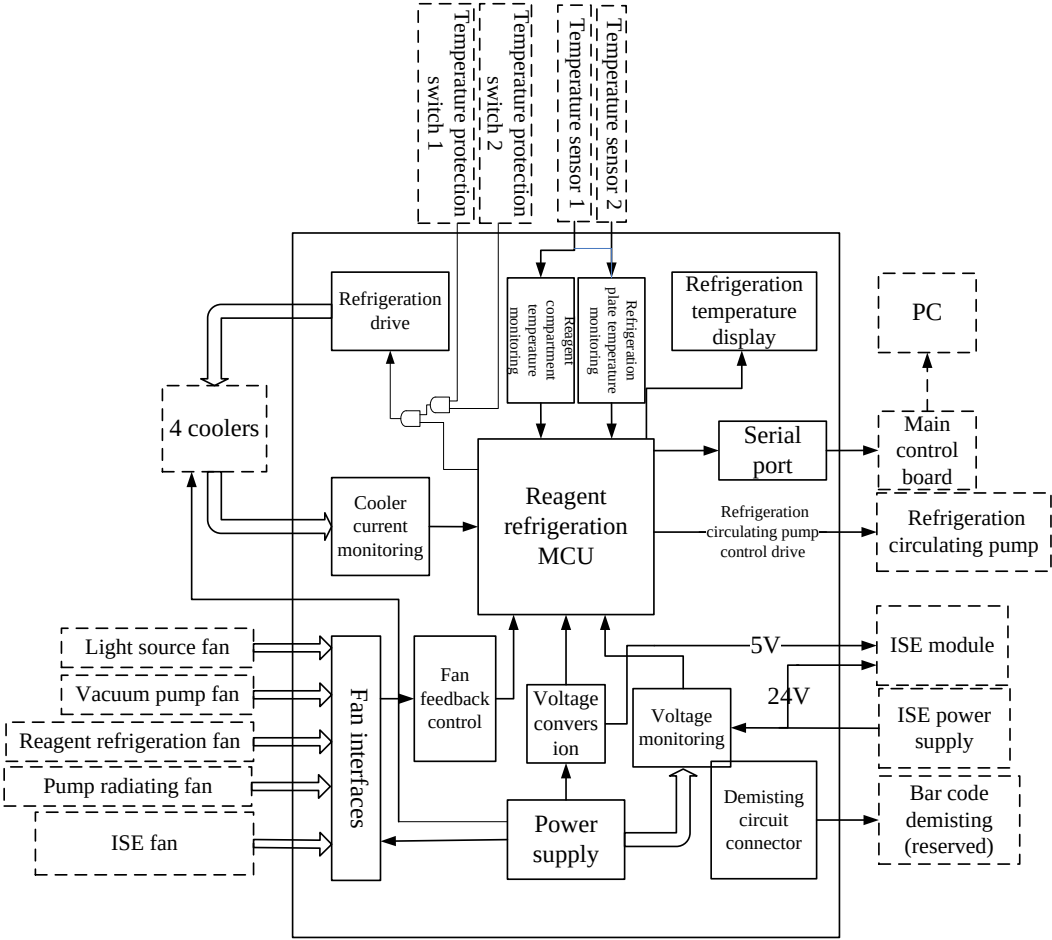
6.4.15 Reagent Refrigeration Board

Functions and principles

The reagent refrigeration board is relatively independent in the entire hardware system and has the following major functions:

- Refrigeration control and drive: controlling the switch of the radiators and refrigerating the reagent compartment to keep the temperature within certain range.
- Control of the refrigeration temperature indicator.
- Control of cooling fans and demisting.
- Monitoring working voltage of the PCBA.

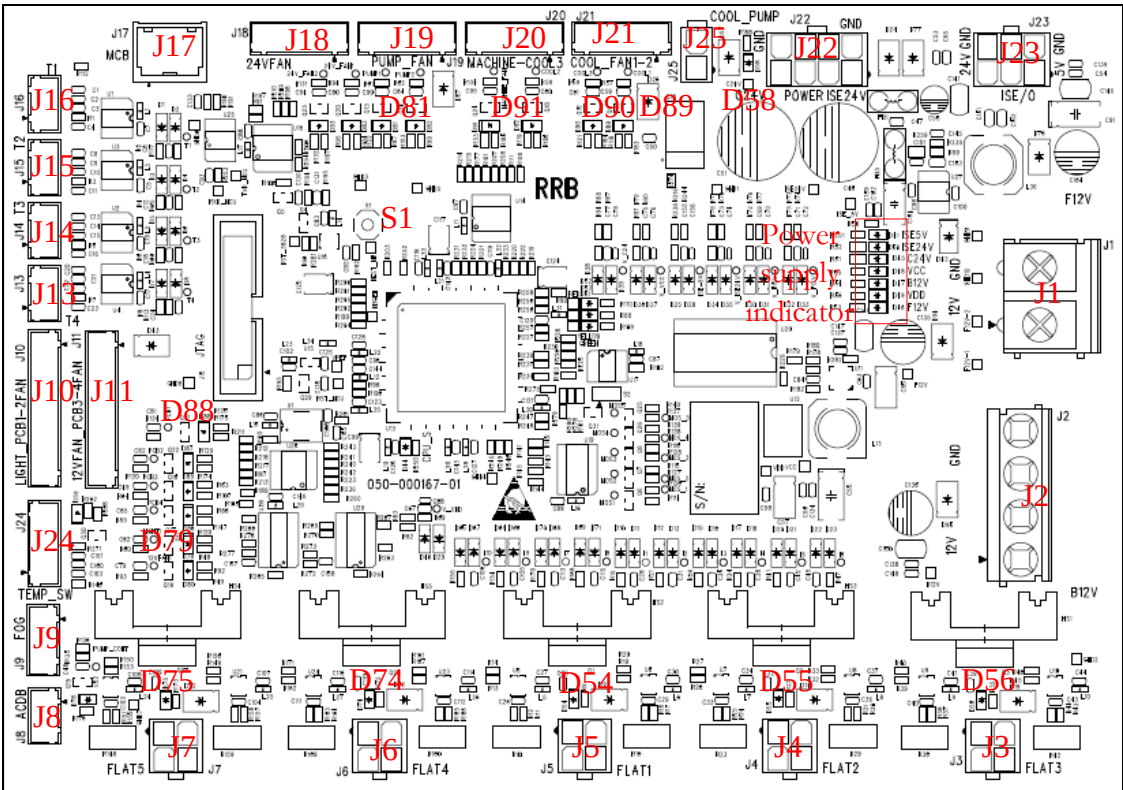
The functional diagram of the reagent refrigeration board is as shown below.



Description

PCB layout

The PCB layout of the reagent refrigeration board is as shown below.



Connectors

The reagent refrigeration board includes the following connectors.

Power supply:

Power supply input connectors (J1, J2, J3): providing B12V refrigeration, and C24V fan for the PCBAs.

Connector	Pin No.	Signal	Reference Value	Description
J1	1,2	B12V	11.4 ~12.6V	Refrigeration power supply input: Used for loading the corresponding radiators J5/J6/J7.
	3, 4	GND	0	
J2	1,2	B12V	11.4 ~12.6V	Refrigeration power supply input: used for loading the corresponding radiators J3/J4 and the PCBA.
	3, 4	GND	0	
J22	4	C24V	23.8 ~25.2V	24V power supply input for cooling fans.
	8	GND	0	

Cooler connectors (J4-J7): providing power supply for the cooler.

Connector	Pin No.	Signal	Reference Value
J4~J7	1	1 cooler	0V for ON and 12V for OFF.
	3		12V

Demisting circuit connector (J9): providing power supply for the demisting heater.

Connector	Pin No.	Signal	Reference Value
J9	1	1 demisting heaters	12V
	2		/
	3		/
	4		0

Fan connector (J10): providing 12V power supply for the cooling fans and checking the fan status.

Connector	Pin No.	Signal	Reference Value
J10	1	F12V	12V
	2	/	/
	3	Gnd	0V
	4	F12V	12V
	5	/	/
	6	Gnd	0V
	7	F12V	12V
	8	Light source fan's feedback signal	below 0.5V
	9	Gnd	0V

Temperature sensor connector (J16): checking temperature of the refrigeration compartment and refrigeration board.

Connector	Pin No.	Signal	Reference Value
J16	1	Shielding ground	0V
	2	GND	0V
	3	Temperature sensor signal of the refrigeration compartment	/

Fan connector (J19): providing 24V power supply for the 1air pump fan and checking the fan status.

Connector	Pin No.	Signal	Reference Value
J19	1	/	/
	2	/	/
	3	/	/

Connector	Pin No.	Signal	Reference Value
	4	C24V	24V
	5	Air pump radiating feedback signal	below 0.5V
	6	Gnd	0V

Fan connector (J21): providing 24V power supply for the cooling fans and checking the fan status.

Connector	Pin No.	Signal	Reference Value
J21	1	C24V	24V
	2	Feedback signal of cooling fan 1	below 0.5V
	3	Gnd	0V
	4	C24V	24V
	5	Feedback signal of cooling fan 2	below 0.5V
	6	Gnd	0V

Connectors for sending/receiving communication signals:

Reagent refrigeration board serial port (J17): 8-pin, used for communicating with the main control board.

Switches

The reagent refrigeration board contains the following switches.

RST key (S1): used to reset the CPU of the PCBA.

Indicators

The reagent refrigeration board contains the following indicators.

Power supply:

No.	Description	Color	Description
D45	C24V	Green	When it is lit, it indicates that the power supply works normally.
D48	VCC	Green	When it is lit, it indicates that the power supply works normally.
D47	B12V	Green	When it is lit, it indicates that the power supply works normally.
D49	VDD	Green	When it is lit, it indicates that the power supply works normally.
D46	F12V	Green	When it is lit, it indicates that the power supply works normally.

Indicators to indicate PCBA functions:

No.	Description	Color	Description
D52	Indicating the reagent compartment temperature	Red	When the reagent compartment temperature exceeds the high limit
D51	Indicating the reagent compartment temperature	Green	When the reagent compartment temperature is within the target range
D53	Indicating the reagent compartment temperature	Yellow	When the reagent compartment temperature is lower than the low limit
D55	Indicating status of the cooler	Green	When it is lit, it indicates that the cooler is working.
D54	Indicating status of the cooler	Green	When it is lit, it indicates that the cooler is working.
D74	Indicating status of the cooler	Green	When it is lit, it indicates that the cooler is working.
D75	Indicating status of the cooler	Green	When it is lit, it indicates that the cooler is working.
D79	Indicating status of the lamp fan	Red	When it is lit, it indicates that the fan goes wrong.
D81	Indicating status of the vacuum pump fan	Red	When it is lit, it indicates that the fan goes wrong.
D89	Indicating status of cooling fan 1	Red	When it is lit, it indicates that the fan goes wrong.
D90	Indicating status of cooling fan 2	Red	When it is lit, it indicates that the fan goes wrong.

Installation methods and precautions

- Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.
- Make sure that the connectors are inserted properly into the PCBA.
- Check the connectors with clamps and ensure that the clamps have been locked properly.
- Check other connectors and ensure that they are inserted into the end of the slots.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

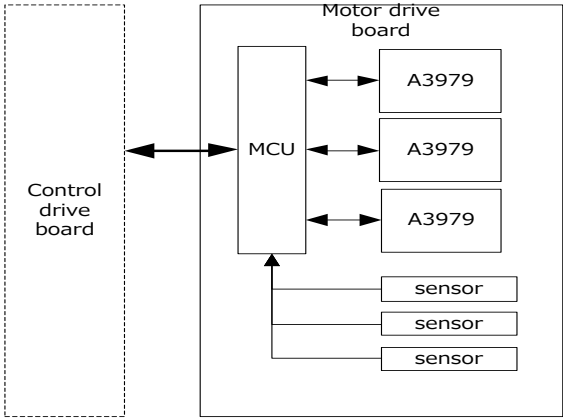
6.4.16 Mixer drive board

Functions and principles

The mixer motor drive board is used to:

- Mix the sample and reagent and send the status of the mixer motor to control drive board.

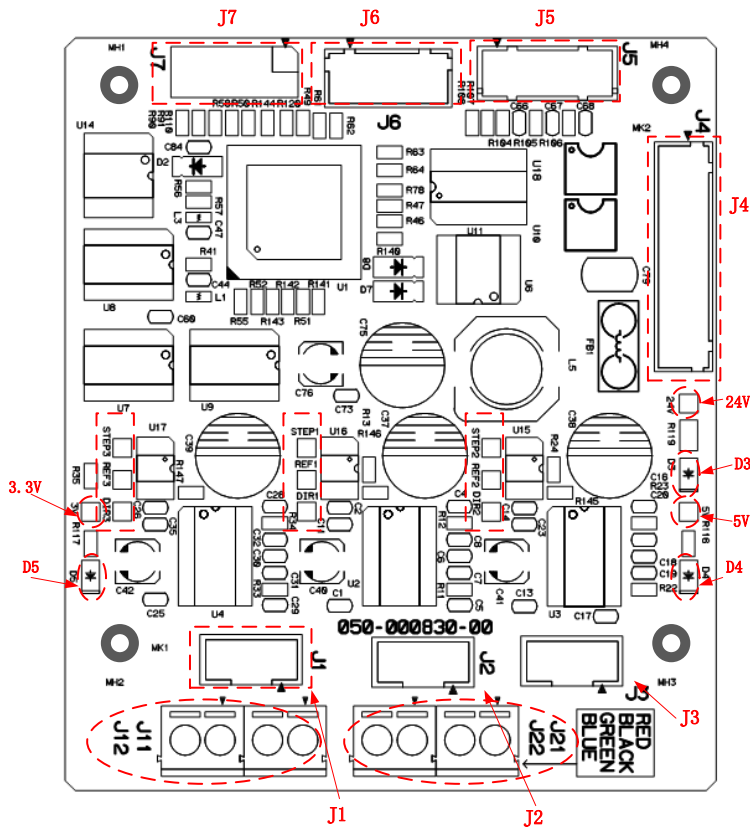
The functional diagram of the mixer motor drive board



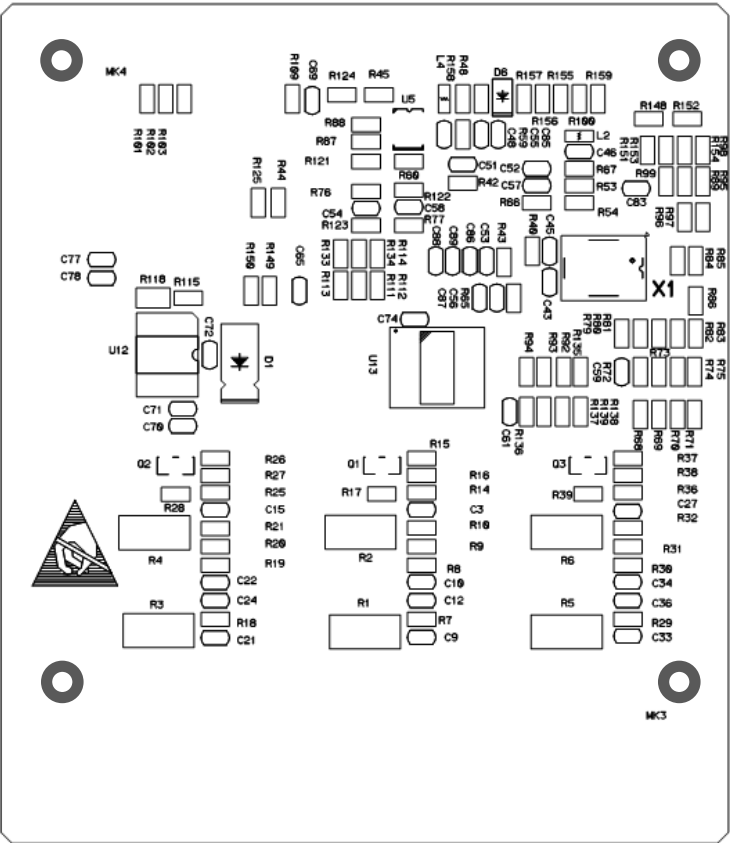
Description

PCB layout

The PCB layout (TOP layer) of the mixer motor drive board is as shown below.



The PCB layout (BOTTOM layer) of the mixer motor drive board is as shown below.



Connectors

The mixer motor drive board contains the following connectors.

Motor connectors (J1-J3): 4-pin, used for driving the mixer motors.

Pin No.	Signal
1	Motor drive signal A+
2	Motor drive signal A-
3	Motor drive signal B+
4	Motor drive signal B-

Power supply and signal output connector (J4): 8-pin, used to provide power supply for the mixer motor drive board and communicate through serial port with the main control board.

Pin No.	Signal	Reference Value
1	RXD	Signal received through serial port
2	RST	Reset signal
3	TXD	Signal sent through serial port
4	ISP	Relevant signal downloaded online
5	GND	/
6	GND	/
7	24V	22.8V-25.2V
8	24V	22.8V-25.2V

Optical coupler connector (J5): 12-pin, used to control and receive optical coupler signal of the mixer motor drive board.

Pin No.	Signal	Reference Value
1	VCC	4.75V-5.25V
2	PHO1	Blocked: 4V-5V; unblocked: 0V-1V
3	GND	/
4	GND	/
5	VCC	4.75V-5.25V
6	PHO2	Blocked: 4V-5V; unblocked: 0V-1V
7	GND	/
8	GND	/
9	VCC	4.75V-5.25V
10	PHO3	Blocked: 4V-5V; unblocked: 0V-1V
11	GND	/
12	GND	/

Optical coupler signal cable (J6): 4 pin, reserved

Pin No.	Signal	Reference Value
1	PHO1_A	/
2	PHO1_B	/
3	PHO2_A	/
4	PHO2_B	/

JTAG debugging connector (J7) 10 pin used to debug or download the PCB program.

Pin No.	Signal	Reference Value
1	VDD	3.13V-3.47V
2	TMS	/
3	GND	/
4	TCK	/
5	GND	/
6	TDO	/
7	RTCK	/
8	TDI	/
9	GND	/
10	TRST	/
11	VDD	3.13V-3.47V
12	TMS	/

Reserved connector(J11-12, J21-22): 2 pin.

Switches and jumpers

The mixer motor drive board has no switches or jumpers.

Indicators

The mixer motor drive board contains the following indicators.

- 24V power supply indicator (D4): green. When the board is connected to power supply, this indicator is always lit; going out means the 24V power supply is abnormal.
- 5V power supply indicator (D5): green. When the board is connected to power supply, this indicator is always lit; going out means the 5V power supply is abnormal.
- 3.3V power supply indicator (D6): green. When the board is connected to power supply, this indicator is always lit; going out means the 3.3V power supply is abnormal.

Test points

In the following positions of the mixer motor drive board can signal tests be performed.

- 24V: 24V power supply signal. Normal condition: 22.8V-25.2V.
- 5V: 5V power supply signal. Normal condition: 4.75V-5.25V.
- 3.3V: 3.3V power supply signal. Normal condition: 3.13V-3.47V.
- REF: The reference voltage signal of the motor drive. The voltage fluctuates within 0-5V according to the working status of the motor .
- STEP: Motor pulse drive signal. One pulse indicates the motor moves one micro-step.

- DIR: Motor moving direction signal. The low level and high level signal indicate different moving direction of the motor.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

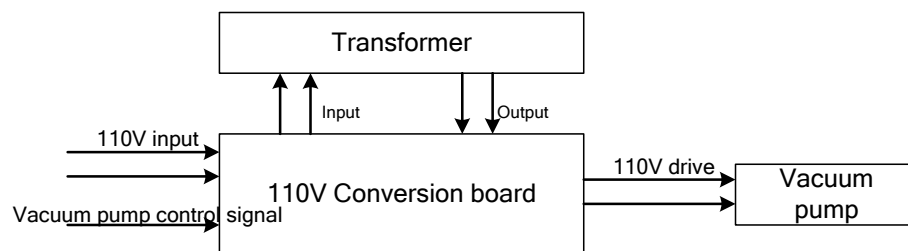
Make sure that the connectors are inserted properly into the end of the slots on the PCBA.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

6.4.17 110V Conversion Board

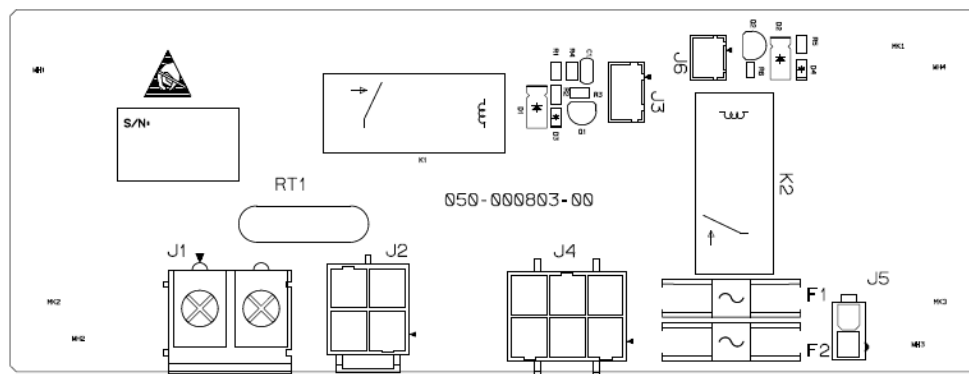
Functions

- When the power supply is 110V, 110V conversion board is used to connect the transformer to convert the drive power of vacuum pump to 220V.
- The following figure shows the circuit of the 110V conversion board:



Description

PCB layout



Connectors

- 110V power supply input connector (J1):2 PIN, used to connect 110V power supply.
- Transformer primary input connector(J2):4 PIN, used to connect the primary electric coil to load 110V power on it.
- Transformer secondary output connector(J4):6PIN, used to connect the secondary electric coil and input the converted 220V power to the vacuum pump.
- Vacuum pump control connector(J6):2PIN, used to control the on-off of the vacuum pump.

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- Vacuum pump drive connector(J5):2PIN, used to drive the vacuum pump.
- RT1 protection connector(J3):3PIN, used to protect the thermistor circuit.

Indicators

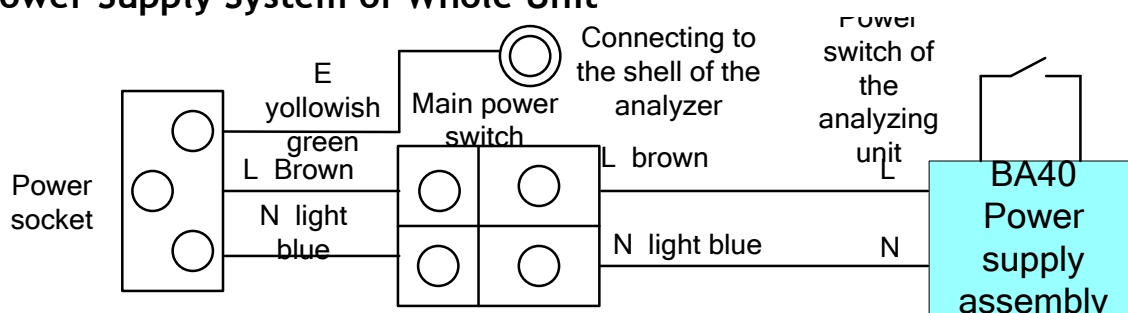
- D3:green, when the thermistor circuit works normally, it is lit.
- D4:green: when the vacuum pump works normally, it is lit.

Installation methods and precautions

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

6.5 Power Supply System

6.5.1 Power Supply System of Whole Unit

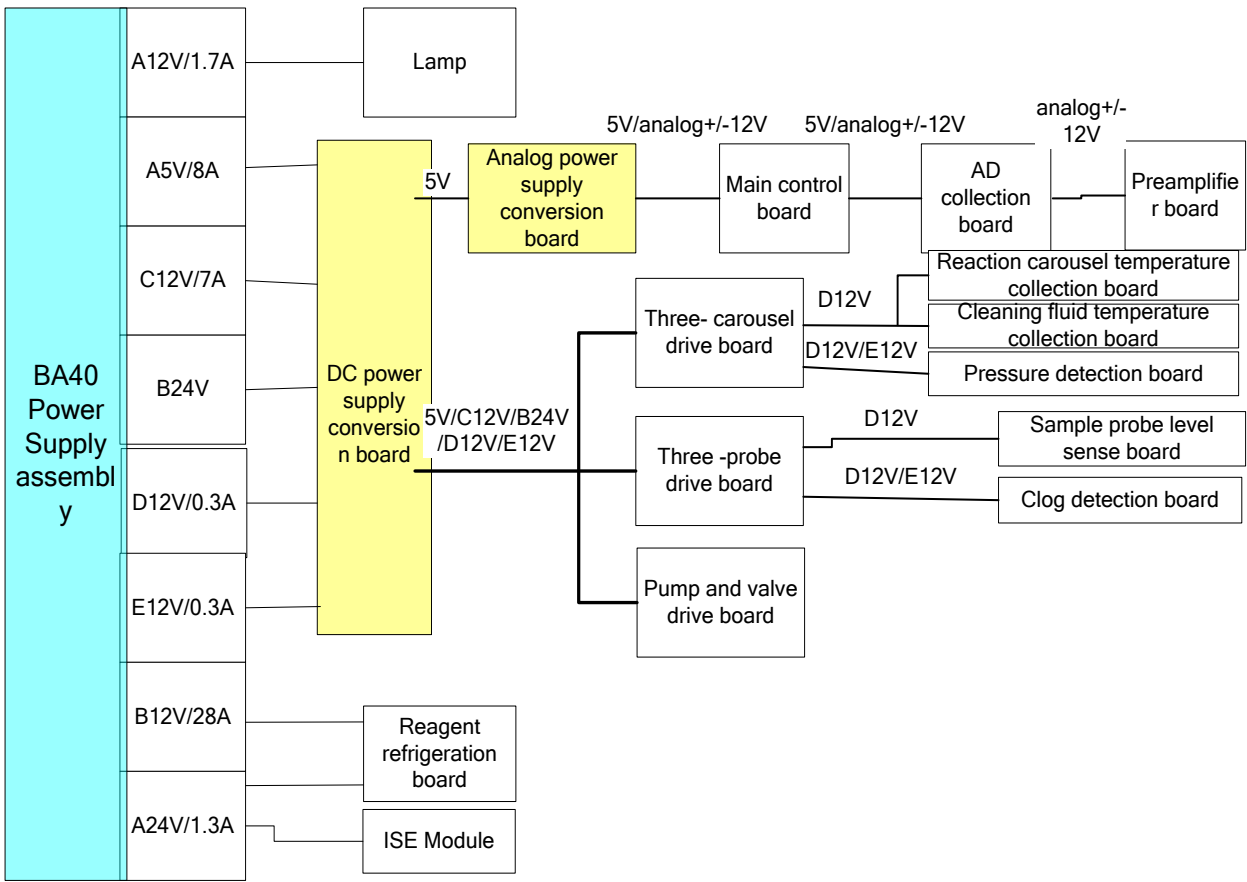


BA40 power supply assembly

The BA40 power supply assembly, as the major part of the entire power supply system, consists of three boards, which are 24V power supply board, 12V power supply board and power supply conversion board. It provides power supply for the lamp, DC power supply for major control boards, and power supply for refrigeration modules. After passing through an analog power supply conversion board, the 5V power is converted into $\pm 12V$ and then provided for the AD collection board and preamplifier board. The A5V, C12V, B24V, D12V and E12V power goes through a DC power supply conversion board and then is provided for the wash temperature control board, sample control drive board and reagent control drive board. The B12V and A24V power are input to the reagent refrigeration board for powering the radiators and ISE module. Other boards, such as reaction carousel temperature collection board, cleaning fluid temperature collection board, pressure detection board, clog detection board and level sense board, are powered by those boards connected to them. See the figure below.

- The 24V power supply board converts the AC input into A24V and B24V and into 24VFAN for the cooling fans of radiators.
- The 12V power supply board converts the AC input into B12V, A5V, $\pm 14V$, and 24VLAMP. The 24VLAMP power supply is transformed to A12V through the power supply conversion board and used for powering the lamp.
- The power supply conversion board is used to shift the AC voltage, control the vacuum pump AC voltage, convert the $\pm 14V$ power into $\pm 12V$, control the C12V power, and shift various output voltages.

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6.5.2 Performance Indices

Power supply input

- AC voltage input: 90-264VAC.
- AC voltage frequency: 50/60±3Hz
- AC input power: 2KVA

Power supply output

No.	Name	Rated Output	Lower-level PCBA	Usage	Control Description
1	A5V	5V/8A		5V digital	Controlled by analyzer power switch
2	A12V	12V/4.2A		Light source	Controlled by operating software
3	B12V	12V/28A		Reagent refrigeration	Controlled by main power switch
4	C12V	12V/7A		12V drive	Controlled by analyzer power switch
5	D12V	12V/0.3A		+12V analog	
6	E12V	-12V/0.3A		-12V analog	
7	F12V	12V/30A		Reagent refrigeration	Controlled by main power switch

No.	Name	Rated Output	Lower-level PCBA	Usage	Control Description
8	A24V	24V/1.3A		ISE module	Controlled by main power switch
9	B24V	24V/21A		24V drive and heater	Controlled by analyzer power switch
10	24VFAN	24V/1A	/	Reserved	Controlled by main power switch
11	C24V	24V/6A		Reagent refrigeration fan 1	Controlled by main power switch

Notes:

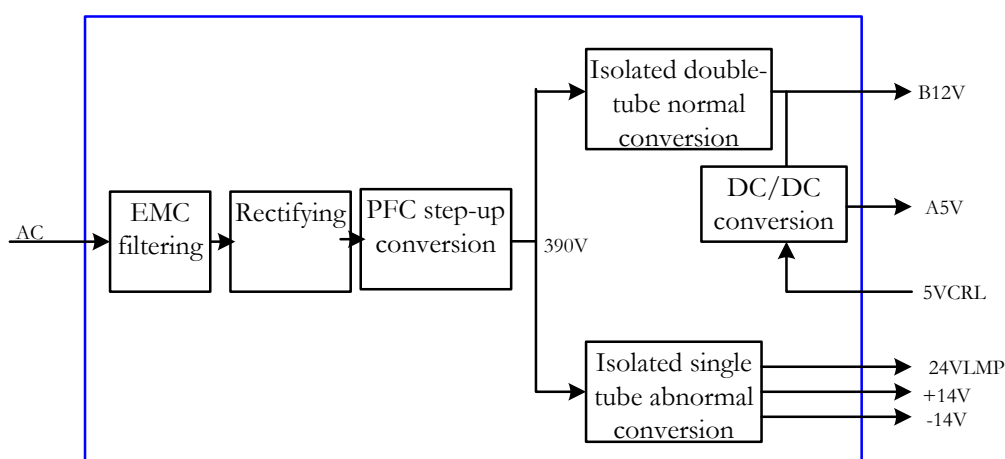
- Control by operating software means that the lamp voltage is controlled by the operating software while the main power switch and analyzer power switch are turned on.
- Control by analyzer power switch means that the power supply is controlled by the analyzer power switch with the prerequisite that the main power switch is turned on.
- Control by main power switch means that the power supply works normally when the analyzer power switch is turned on, and vice versa.

6.5.3 BA40 power supply assembly

The BA40 power supply assembly is composed of the 12V board, 24V board and conversion board. Refer to the functional diagram of the power supply system for the internal structure of the assembly. The functions and principles of the three boards are described on the following pages.

12V power supply board

The functional diagram of the board is as shown below.

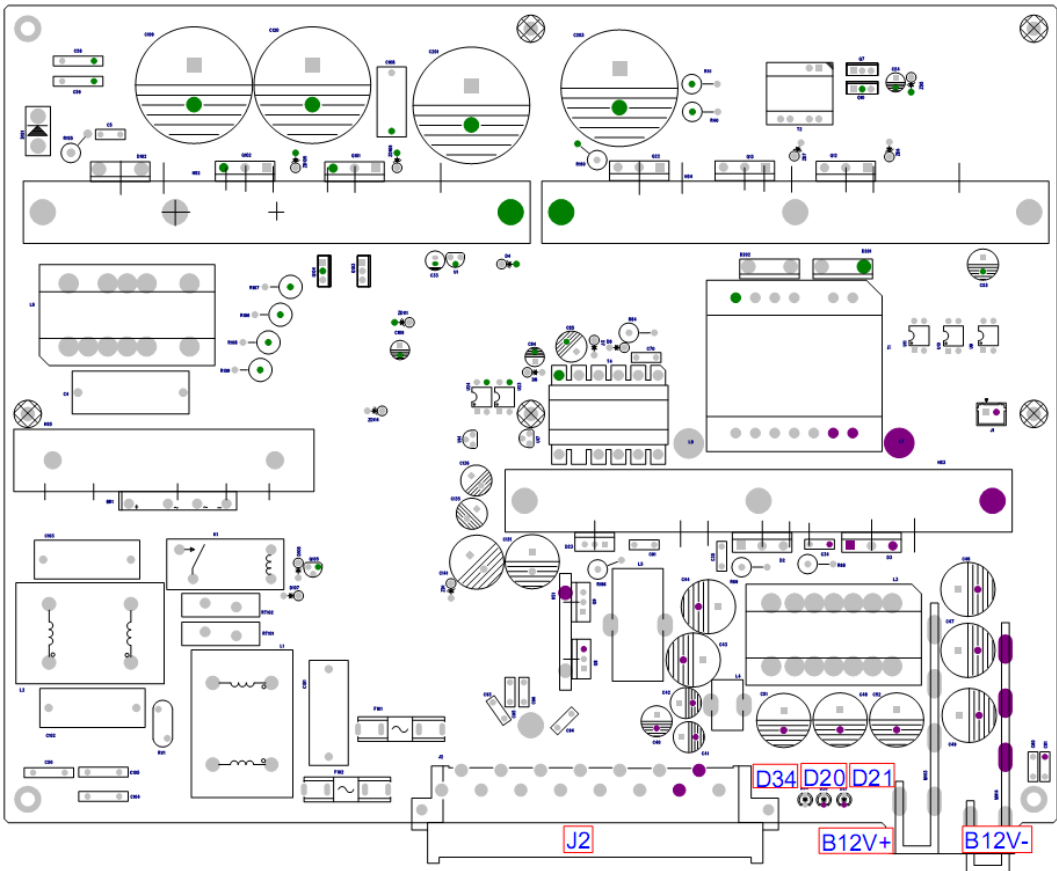


The 12V power supply board converts the AC input into the following outputs:

- B12V: used for powering the radiators.
- A5V: used for powering the PCBA and controlled by the analyzer power switch.
- 24VLMP: outputs constant 24V, which is then converted into lamp voltage A12V in the way of DC/DC through the power supply conversion board.

- $\pm 14V$: The two outputs are converted into $\pm 12V$ (D12V and -12V) through the voltage stabilizer on the power supply conversion board and then used for powering analog circuits. The PCB layout of the 12V board is as shown below.

The PCB layout of the 12V power supply board is as shown below.



The 12V power supply board is located inside the power supply box. Indicators D20, D21 and D34 indicate voltage output during debugging and repairing of the PCBA and cannot be seen from the outside of the instrument.

Indication of voltage outputs:

LED	Output Voltage	Indication Status	Controlled By
D20	A5V	When it is lit, it indicates that the PCBA outputs an A5V power.	Controlled by analyzer power switch
D21	B12V	When it is lit, it indicates that the PCBA outputs a B12V power.	Controlled by main power switch
D34	A24V	When it is lit, it indicates that the PCBA outputs an A24V power.	Controlled by main power switch

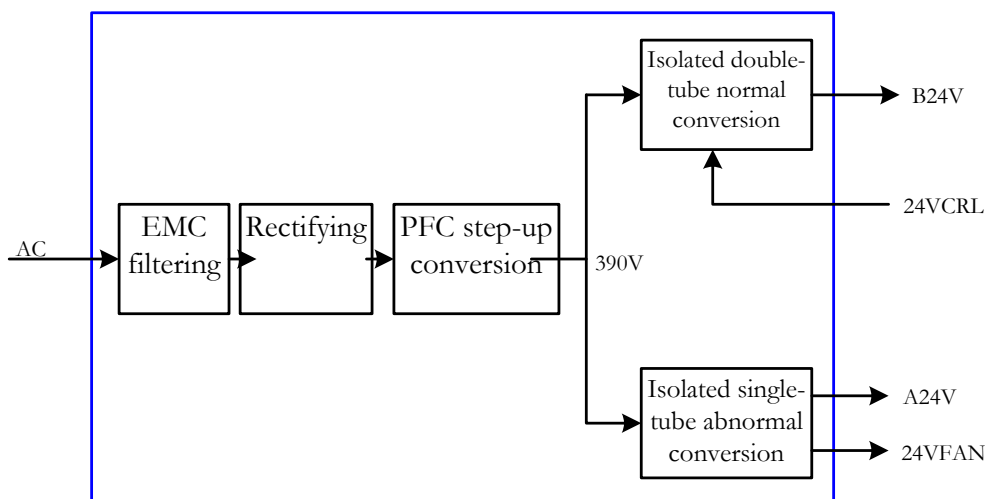
Notes:

- Control by operating software means that the lamp voltage is controlled by the operating software while the main power switch and analyzer power switch are turned on.

- Control by analyzer power switch means that the power supply is controlled by the analyzer power switch with the prerequisite that the main power switch is turned on.
- Control by main power switch means that the power supply works normally when the analyzer power switch is turned on, and vice versa.

24V power supply board

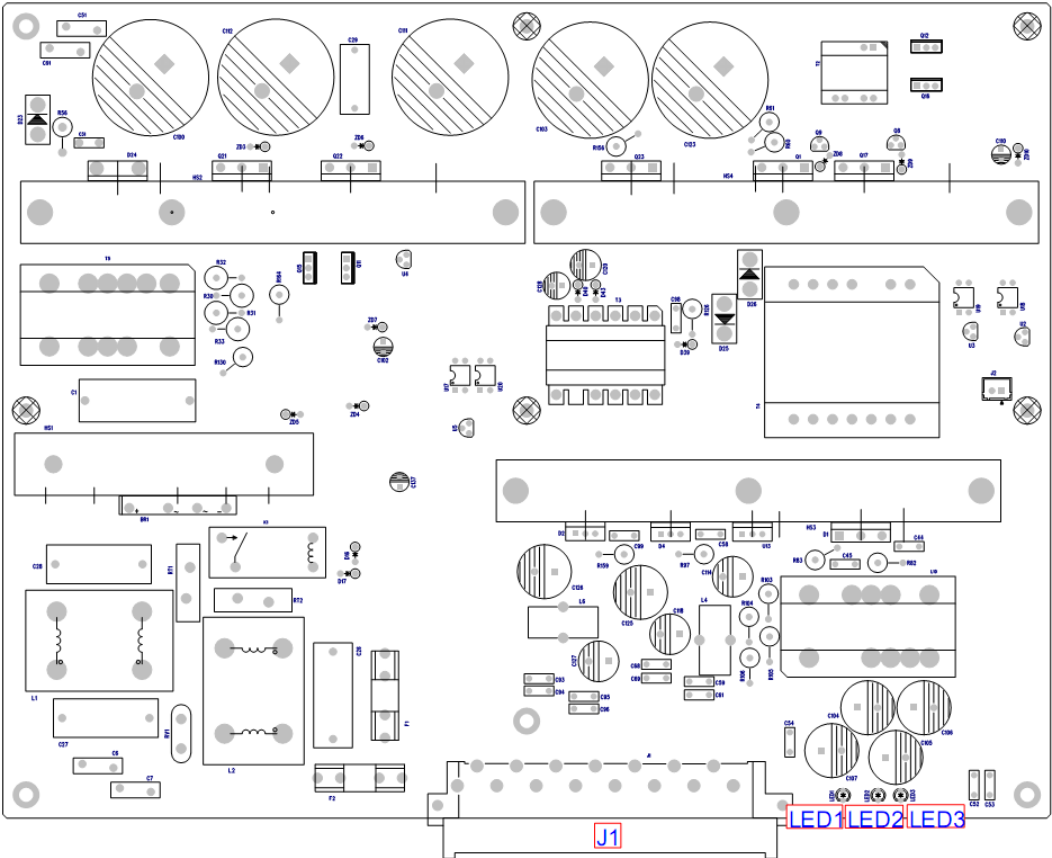
The functional diagram of the board is as shown below.



The 24V board converts the AC input into the following outputs:

- A24V: used for powering the ISE module.
- 24VFAN: used to power the reagent refrigeration fan.
- B24V: used for powering the moving parts, motors and heaters, and controlled by the analyzer power switch.

The PCB layout of the 24V power supply board is as shown below.



LED1, LED2 and LED3 in the figure above are indicators of the A24V, 24VFAN and B24V outputs. Since located inside the power supply box, the three indicators cannot be seen from the outside of the instrument. They are used to indicate the voltage output during debugging and repairing of the PCBA. There are three indicators on the power supply conversion board, and they are used for indicating the voltage outputs of the 24V board.

Indication of voltage outputs on the 24V board:

LED	Output Voltage	Indication Status	Controlled By
LED1	A24V	When it is lit, it indicates that the PCBA outputs an A24V power.	Controlled by main power switch
LED2	24VFAN	When it is lit, it indicates that the PCBA outputs a 24VFAN power.	Controlled by main power switch
LED3	B24V	When it is lit, it indicates that the PCBA outputs a B24V power.	Controlled by analyzer power switch

Notes:

- Control by operating software means that the lamp voltage is controlled by the operating software while the main power switch and analyzer power switch are turned on.
- Control by analyzer power switch means that the power supply is controlled by the analyzer power switch with the prerequisite that the main power switch is turned on.
- Control by main power switch means that the power supply works normally when the analyzer power switch is turned on, and vice versa.

There are 9 voltage indicators on the power supply conversion board indicated by a dotted frame. They are from left to right: D20, D11, D14, D10, D15, D12, D13, D9, and D21, which are used to indicate the following voltage outputs.

LED	Output Voltage	Indication Status	Controlled By
D20	D12V	When it is lit, it indicates that the PCBA outputs a D12V power.	Controlled by analyzer power switch
D11	B12V	When it is lit, it indicates that the PCBA outputs a B12V power.	Controlled by main power switch
D14	-12V	When it is lit, it indicates that the PCBA outputs a -12V power.	Controlled by analyzer power switch
D10	A12V	When it is lit, it indicates that the PCBA outputs an A12V power.	Controlled by analyzer power switch
D15	A24V	When it is lit, it indicates that the PCBA outputs an A24V power.	Controlled by main power switch
D12	C12V	When it is lit, it indicates that the PCBA outputs a C12V power.	Controlled by analyzer power switch
D13	A5V	When it is lit, it indicates that the PCBA outputs an A5V power.	Controlled by analyzer power switch
D9	B24V	When it is lit, it indicates that the PCBA outputs a B24V power.	Controlled by analyzer power switch
D21	24VFAN	When it is lit, it indicates that the PCBA outputs a 24VFAN power.	Controlled by main power switch

Notes:

- Control by operating software means that the lamp voltage is controlled by the operating software while the main power switch and analyzer power switch are turned on.
- Control by analyzer power switch means that the power supply is controlled by the analyzer power switch with the prerequisite that the main power switch is turned on.
- Control by main power switch means that the power supply works normally when the analyzer power switch is turned on, and vice versa.

The table below shows the status of the indicators when the main power switch and analyzer power switch are turned on or off.

LED	Output Voltage	Control Description	Main Power ON Analyzer Power OFF	Main Power ON Analyzer Power ON	Main Power OFF
D11	B12V	When the main power	✓	✓	×

LED	Output Voltage	Control Description	Main Power ON Analyzer Power OFF	Main Power ON Analyzer Power ON	Main Power OFF
D15	A24V	switch is ON, indicators D11, D15 and D21 are lit and only controlled by the main power switch rather than the analyzer power switch.	✓	✓	×
D21	24VFAN		✓	✓	×
D12	C12V	The corresponding indicator is extinguished only when the main power switch is ON. When the analyzer power switch is ON, the corresponding indicator is lit and controlled by both switches.	×	✓	×
D13	A5V		×	✓	×
D14	-12V		×	✓	×
D20	D12V		×	✓	×
D9	B24V		×	✓	×
D10	A12V	Controlled by the main power switch and the operating software	×	×	×
		Turning on/off the lamp can be controlled through the operating software.			

Output connectors:

No.	Pins	Symbol	Output Description
J3	Pin1	ISE	A24V+
	Pin2		A24V-
J4	Pin1, 3, 5, 7	VALVE DRV	C12V positive output
	Pin9, 11, 12		B24V positive output
	Pin2, 4, 6, 8, 10		B24V and C12V output GND
J7	/	N	AC input N
J8	/	L	AC input L
J9	Pin1	24VFAN+	24VFAN positive output
	Pin2	24VFAN-	24VFAN negative output
	Pin3	C12VOUT	C12V positive output
	Pin4	/	C12V output GND
J12	Pin1	D12V	D12V output
	Pin2	-12V	-12V output
	Pin4, 5	/	D12V and -12V output GND
	Pin3	5VOUT	A5V positive output

No.	Pins	Symbol	Output Description
	Pin6	/	A5V output GND
J13	Pin1	AC PUMP	AC1 output, not used for BA80
	Pin3		AC2 output, not used for BA80
	Pin6, 8		AC input
J14	Pin1	5VOUT	A5V output
	Pin2, 3	B24VOUT	B24V output
	Pin4	C12VOUT	C12V output
	Pin5	-12V	-12V output
	Pin9	D12V	D12V output
	Pin6, 7, 8	/	A5V, B24V and C12V output GND
	Pin10	/	D12V and -12V output GND
J15	Pin1, 2, 3	B24VOUT	B24V output
	Pin4	5VOUT	A5V output
	Pin5	D12V	D12V output
	Pin6	-12V	-12V output
	Pin7, 8, 9, 10	GND2	A5V and B24V output GND
	Pin11, 12	DGND	D12V and -12V output GND
J16	Pin1	LAMP	A12V output for lamp
	Pin2		A12V output GND for lamp

Troubleshooting BA40 power supply assembly

Check the indicators on the power supply conversion board while the main power switch is turned on and the analyzer power switch is off. In normal conditions, indicators D11, D15 and D21 should be lit respectively indicating B12V, A24V and 24VFAN. If D11 is not lit, it indicates that B12V output is abnormal and the 12V power board may go wrong; if D15 and D21 are not lit, it indicates that 24VFAN output is abnormal and the 24V power board may go wrong; If the three indicators are not lit, please check whether the AC voltage of the board is normal or it is possible that both the 12V and 24V power board have gone wrong.

While the main power switch is turned on and the voltage input is normal, turn on the analyzer power switch. In normal conditions, all indicators on the conversion board except for the A12V indicator should be lit. If the B24V indicator is not lit, replug the load line; if the error remains, replace the 24V board and then check if the B24V indicator is lit. If the A5V indicator is not lit, perhaps it is short-circuited or the 12V board goes wrong. If the A5V indicator is lit but C12V or D12V/-12V indicator is extinguished, replug the load line; if the error remains, check the conversion board or replace the power supply board; if the error disappears when a new power supply board is installed, it indicates that the original power supply board needs to be repaired.

If the lamp A12V cannot be enabled through the operating software while the main power switch and analyzer power switch are turned on and working normally, replug the power cord of the lamp; if the error remains, troubleshoot the power supply conversion board by replacing it with a new one.

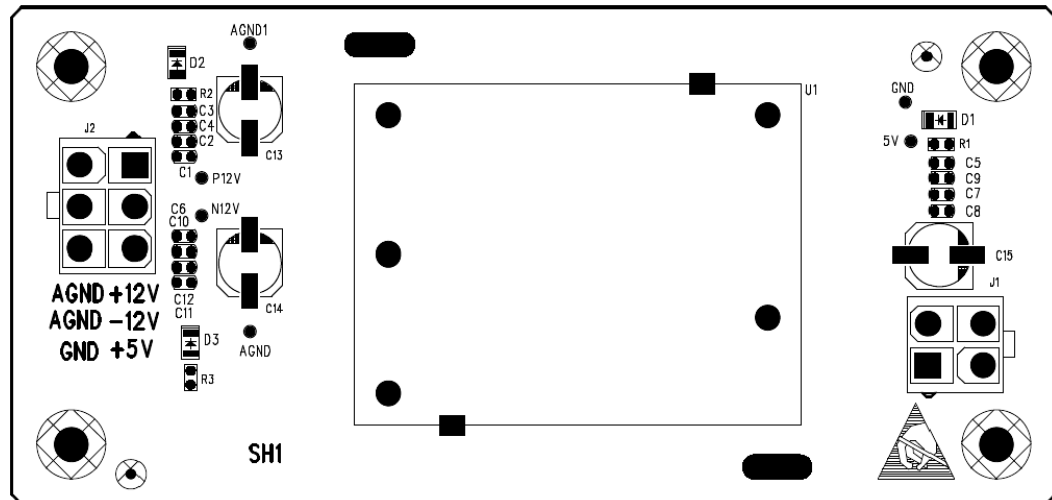
6.5.4 Analog power supply conversion board

Description

The instrument has one analog power supply conversion board, which is mainly used to: Convert the +5V digital into +/-12V in the way of DC/DC, and then provide it for the main control board to use as analog power for the AD collection board and preamplifier board.

Description

The PCB layout of the analog power supply conversion board is as shown below.



Connectors

The analog power supply conversion board contains the following connectors:

J1: connector for DC power supply conversion board input.

Pin No.	Signal
1	GND
2	A5V
3	NC
4	NC

J2: connector for main control board power supply output.

Pin No.	Signal
1	+12V analog
2	-12V analog
3	+5V digital
4, 5	Analog GND
6	Digital GND

Indicators

The analog power supply conversion board contains the following indicators:

- 12V power supply indicator (D3): green. When it is lit, it indicates that the -12V power supply has been connected.

- +12V power supply indicator (D2): green. When it is lit, it indicates that the +12V power supply has been connected.
- +5V power supply indicator (D1): green. When it is lit, it indicates that the +5V power supply has been connected.

Test points

In the following positions of the analog power supply conversion board can signal tests be performed.

- P12V: +12V power supply. Normal range: $12V \pm 5\%$, that is, 11.4 - 12.6V.
- N12V: -12V power supply. Normal range: $-12V \pm 5\%$, that is, -11.4 - -12.6V.
- +5V: +5V power supply. It is secondary power supply used to power the digital parts of the PCBA. Normal range: $5V \pm 5\%$, that is, 4.75 - 5.25V.
- GND: digital grounding terminal of the PCBA.
- AGND and AGND1: analog grounding terminals of the PCBA.

Installation methods and precautions



NOTE

Prior to removing the PCBA, disconnect the instrument from the power supply and wear a pair of anti-static gloves or wrist straps.

Make sure that the connectors are inserted properly into the end of the slots on the PCBA.

It requires great force to plug/unplug the connectors. Hold the PCBA by its edge while plugging/unplugging the connectors to prevent it from being deformed or damaged.

6.5.5 Maintenance of Power Supply System

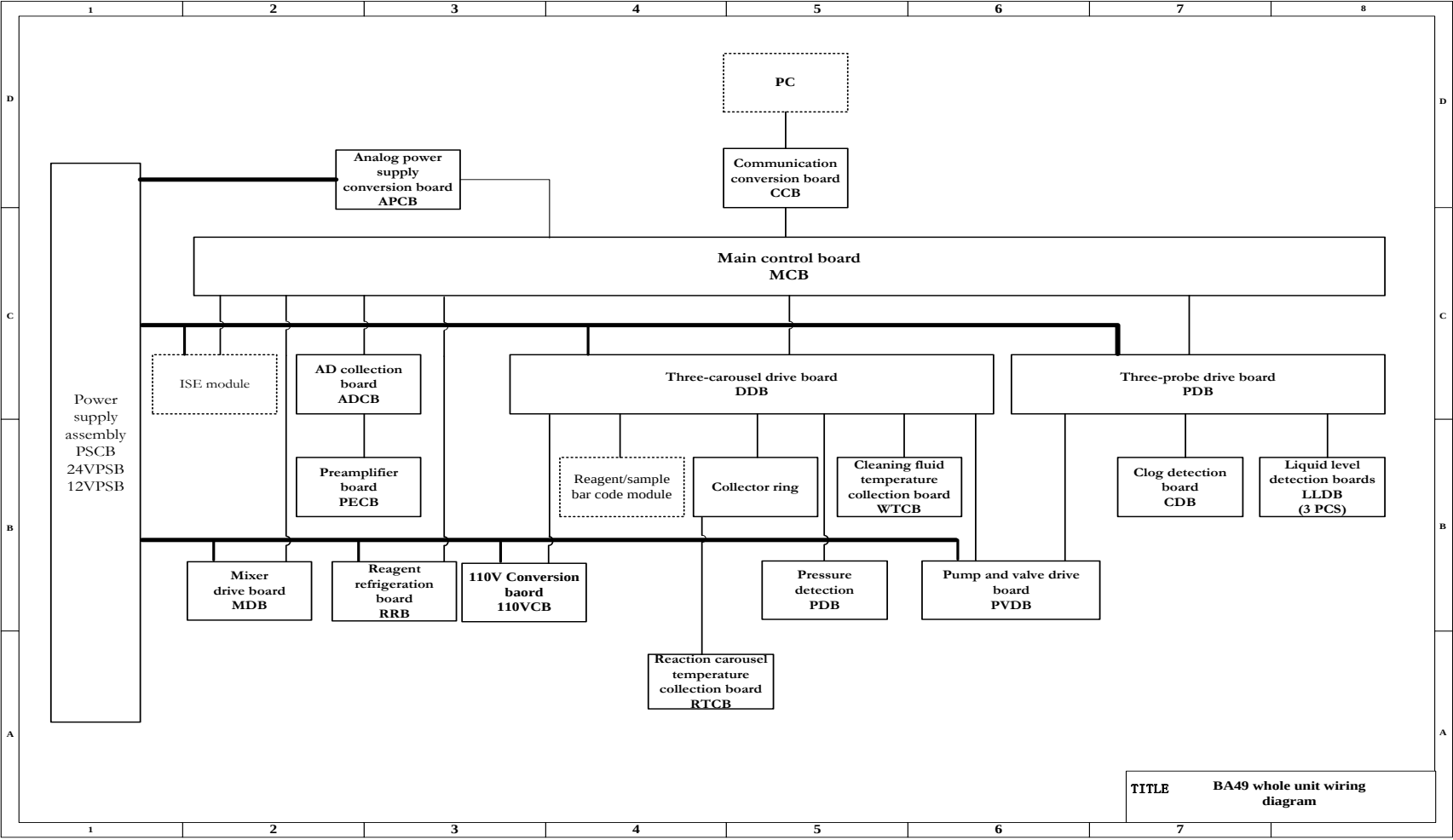
The following summary is about the control relations of the power supply system and provided to help service engineers to troubleshoot various failures.

For correspondence between the main power switch/analyzer power switch/hibernating status and the power supply working status, refer to relevant sections on the previous pages.

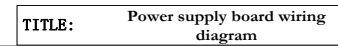
Voltage timing control:

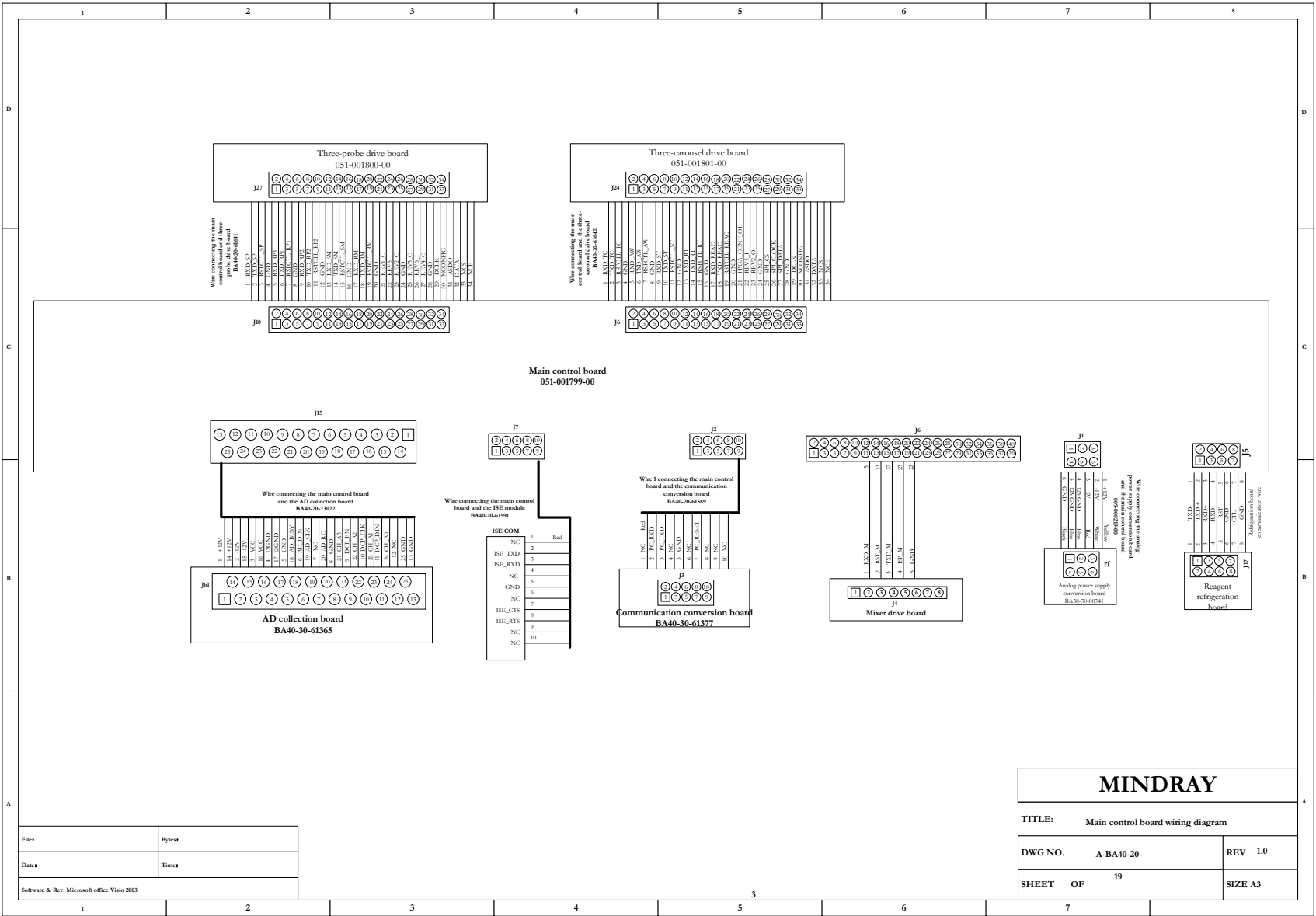
When the analyzer power switch is turned on, the A5V, C12V (D12V/E12V) and B24V are controlled successively in the order of A5V - C12V (D12V/-12V) - B24V. That is, only when the A5V is connected will the C12V (D12V/-12V) be provided, and this is the same for C12V and B24V; if the A5V is not output, the C12V (D12V/-12V) and B24V will not be provided.

6.6 Wiring Diagram of Analyzer

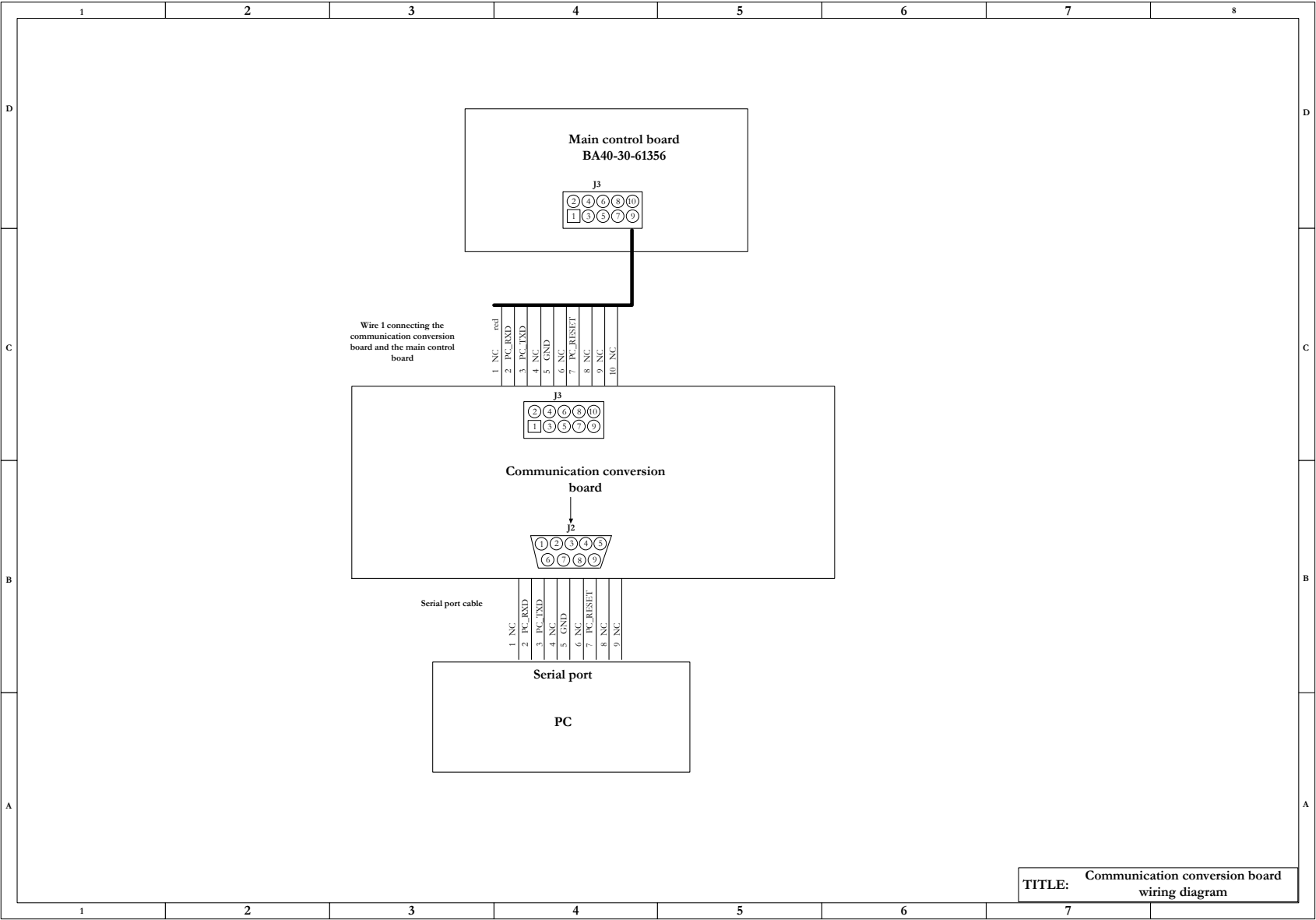


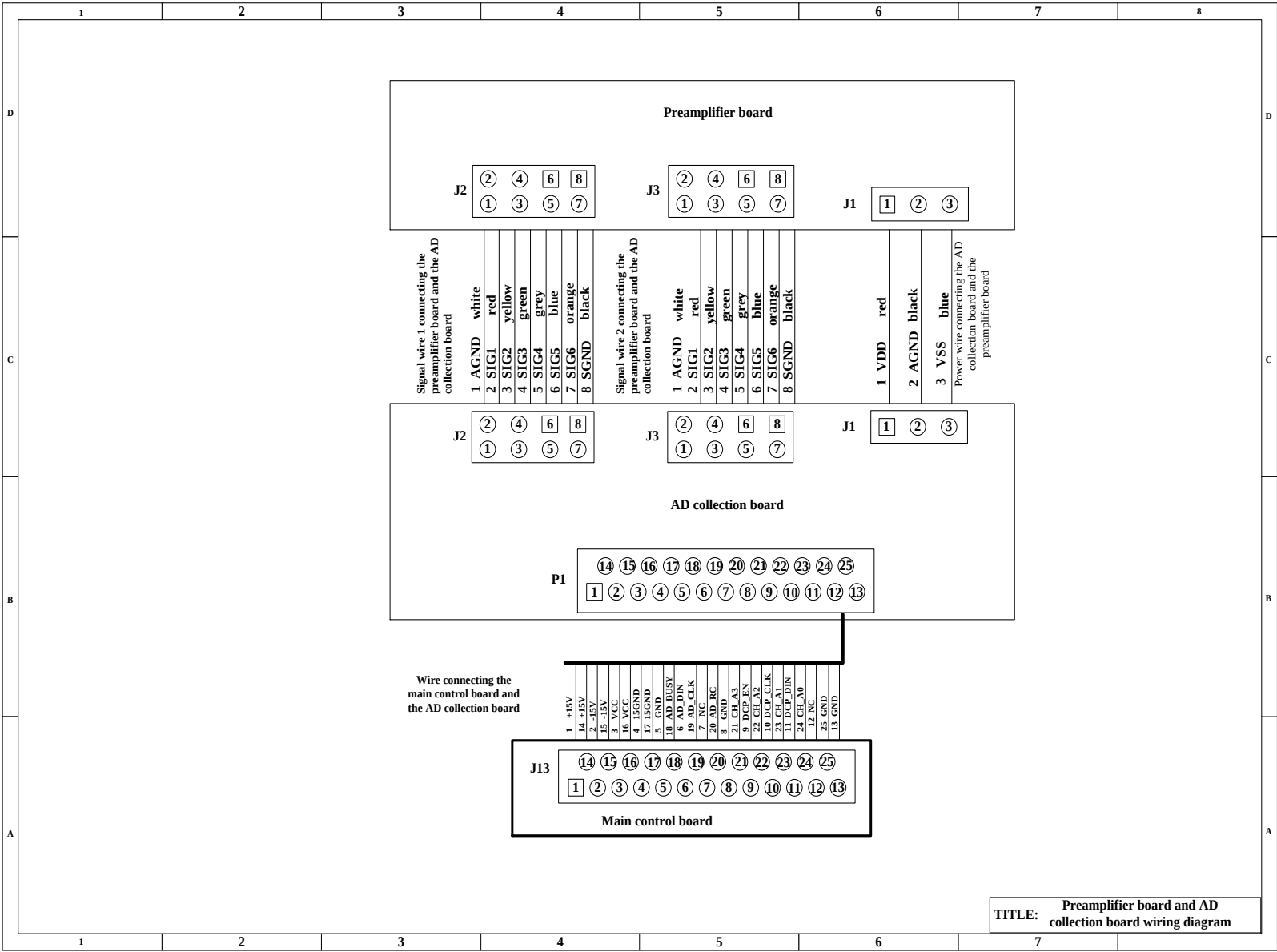
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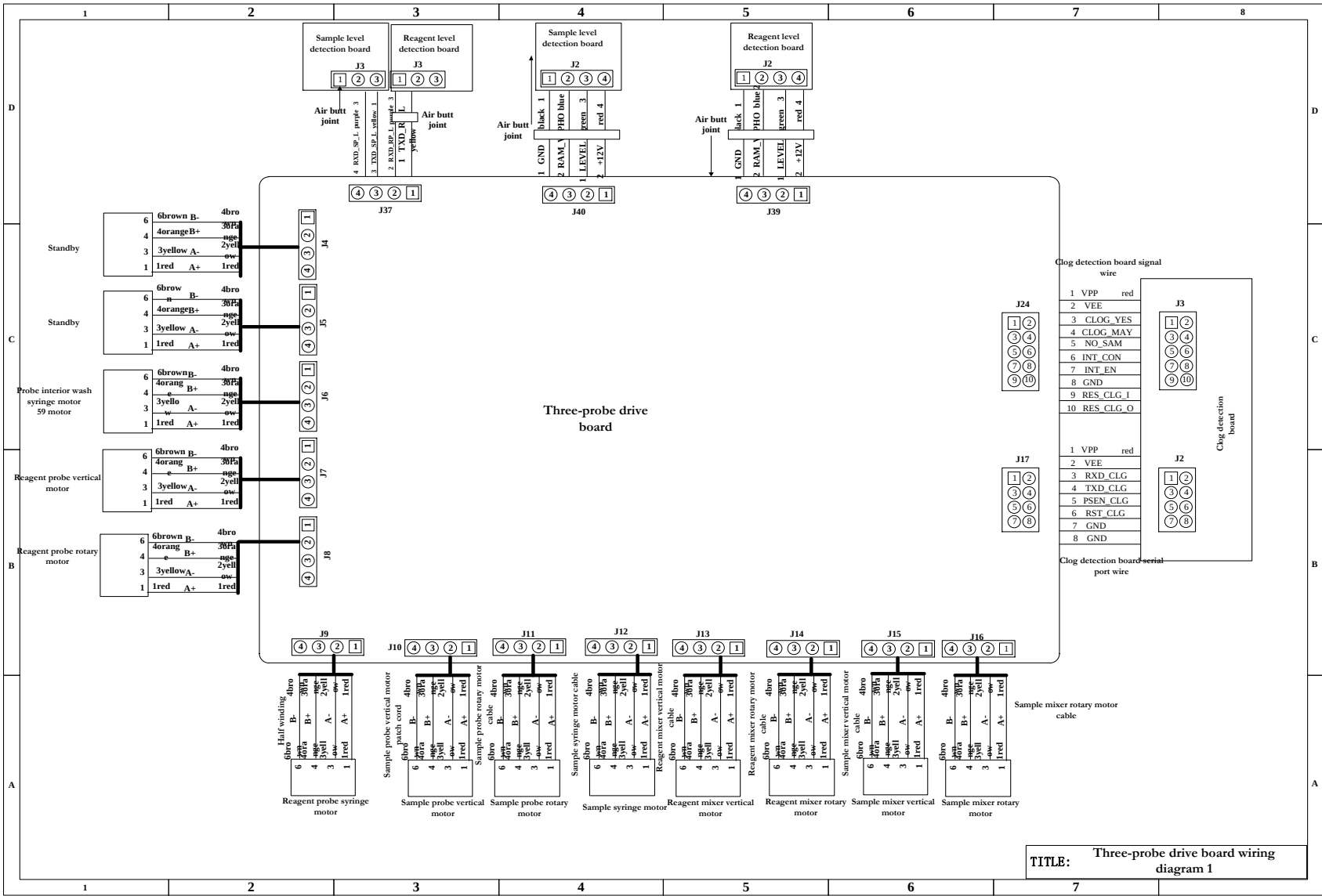


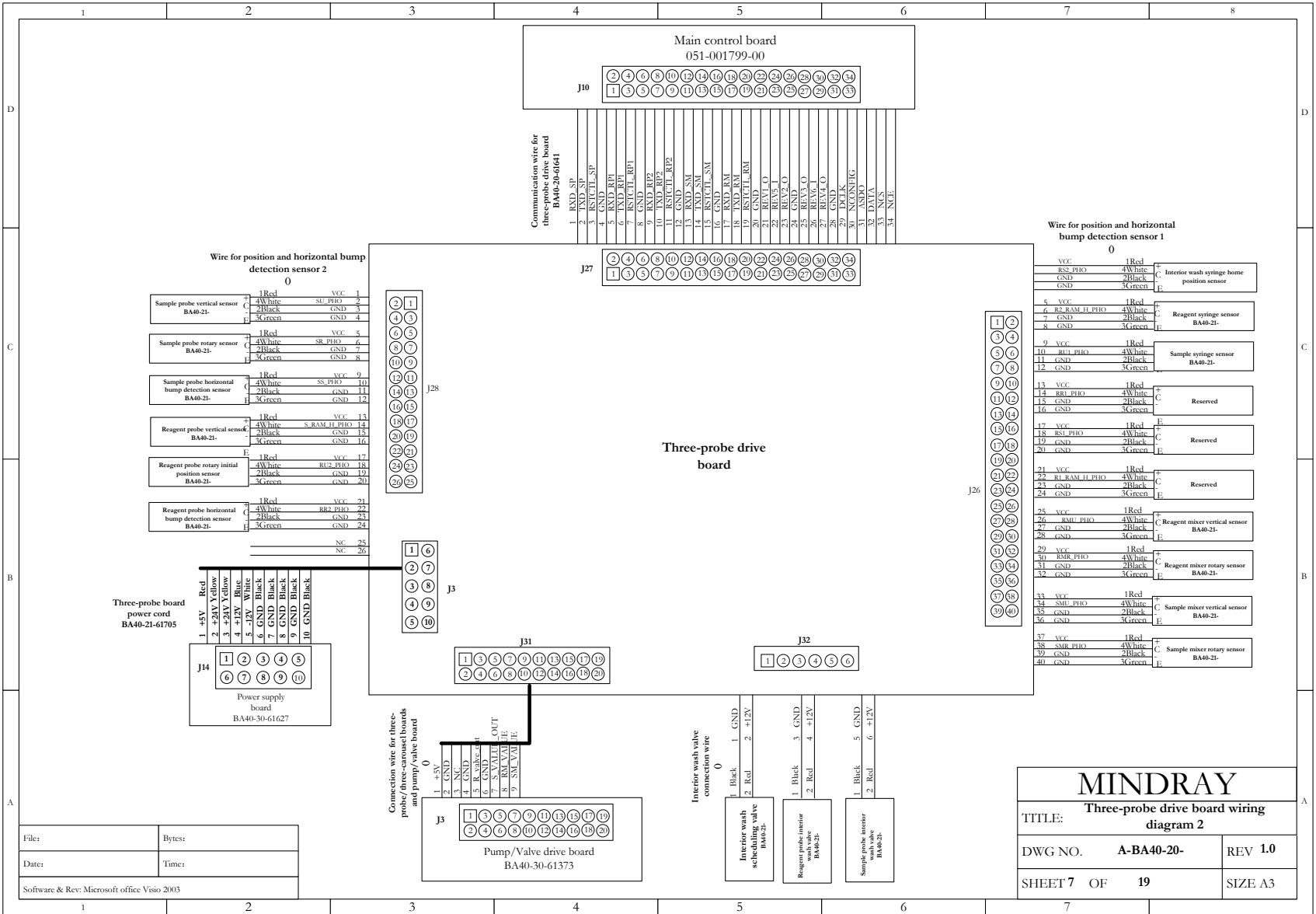
6 Hardware



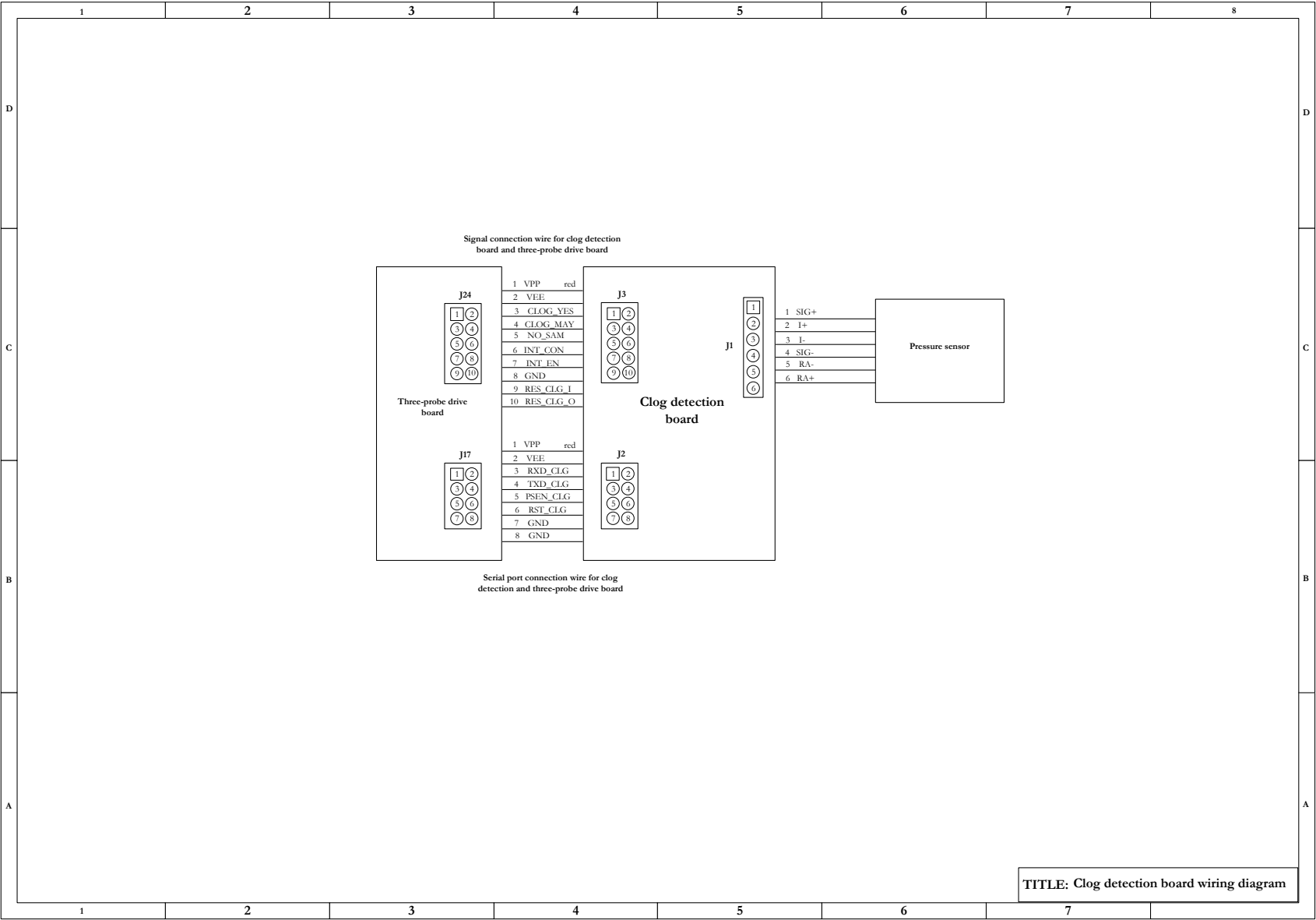


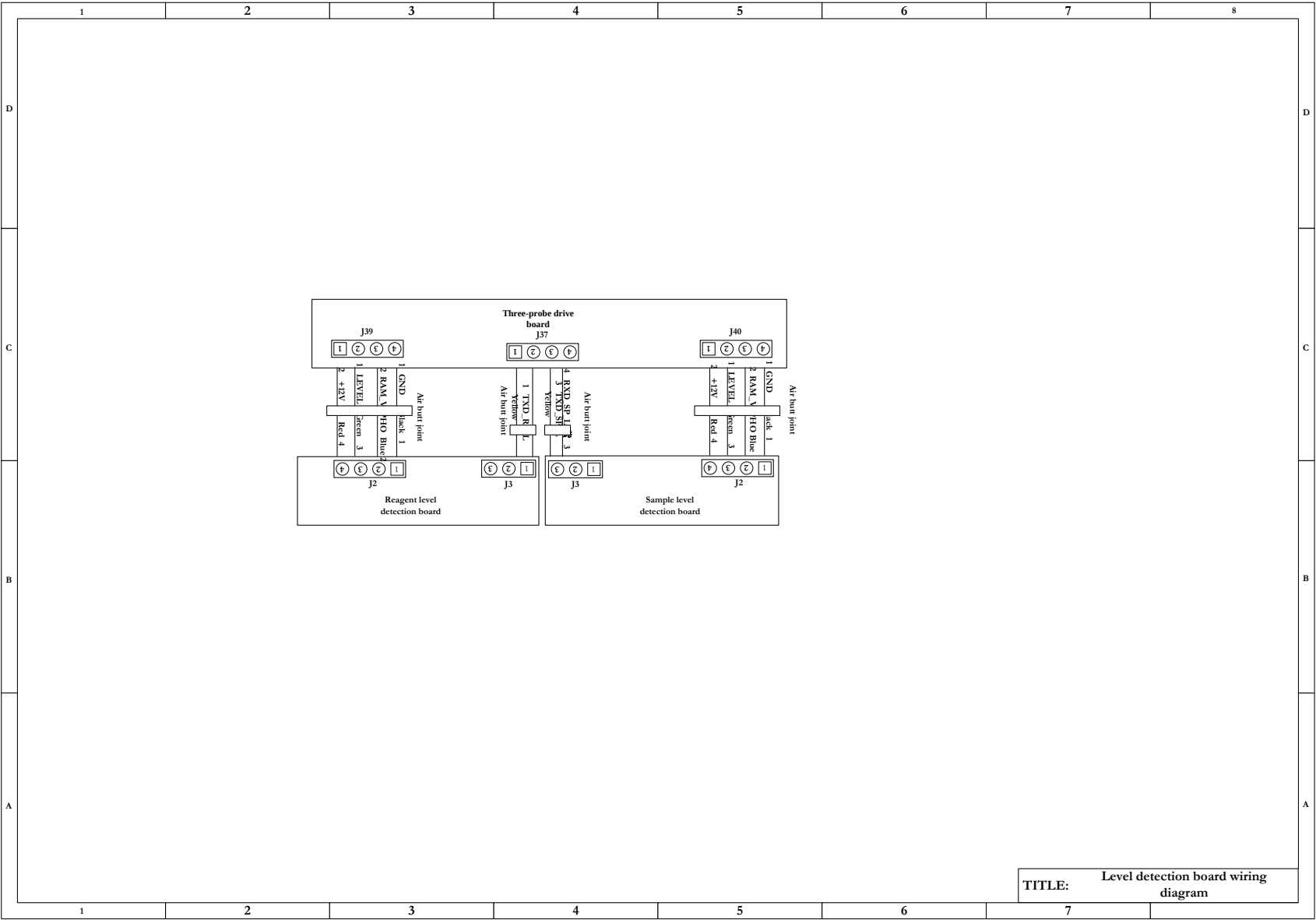
6 Hardware



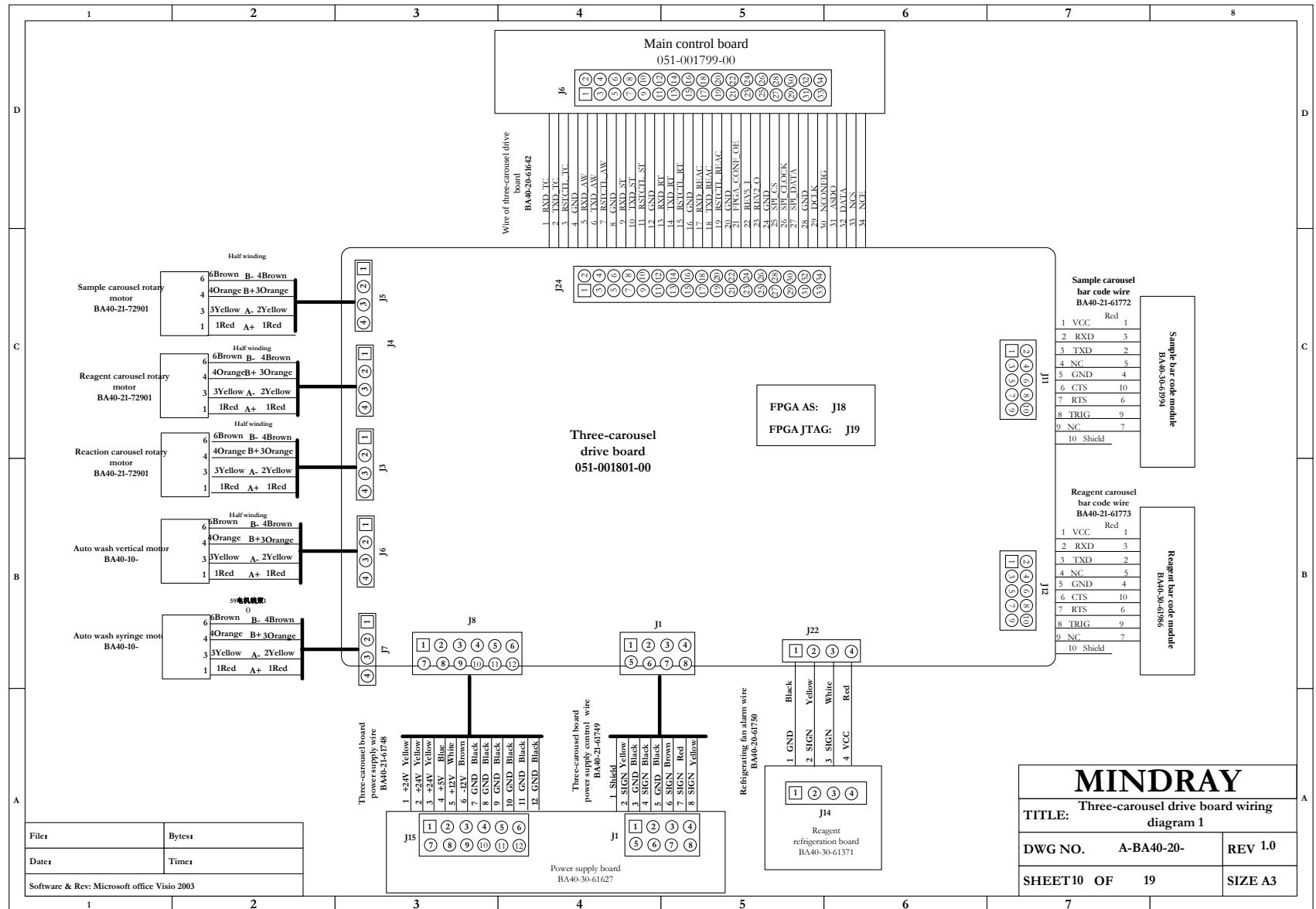


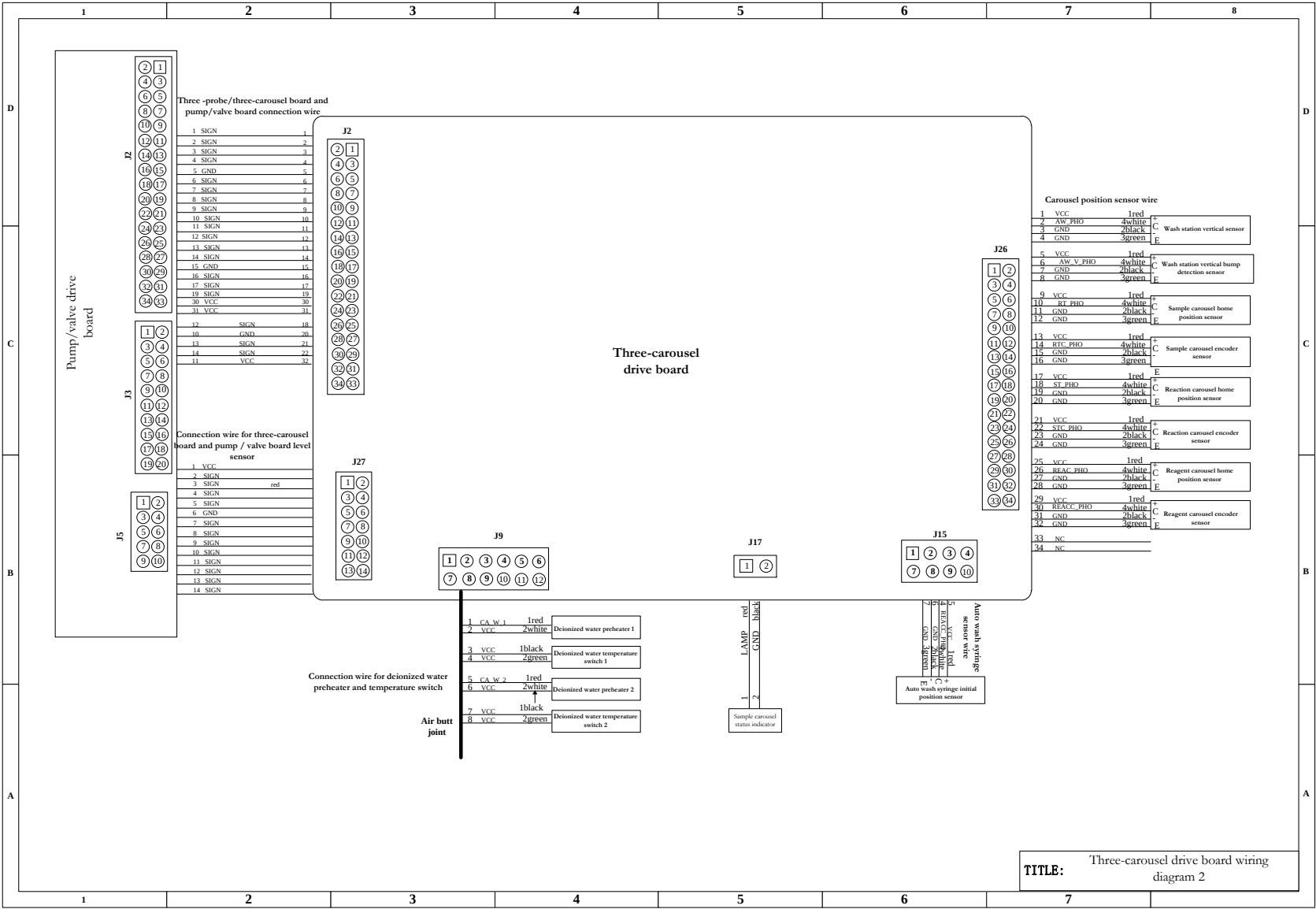
6 Hardware



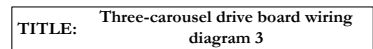


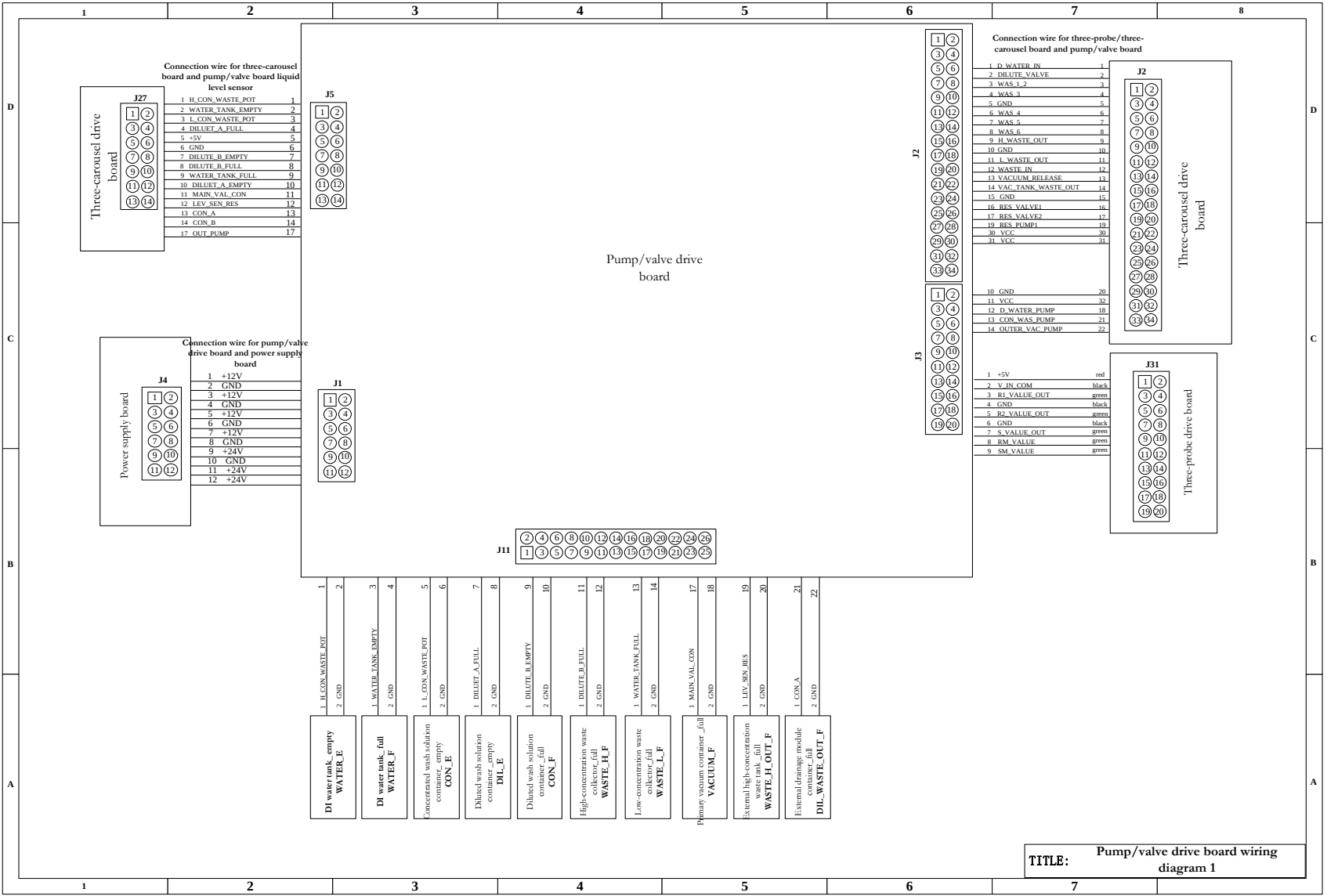
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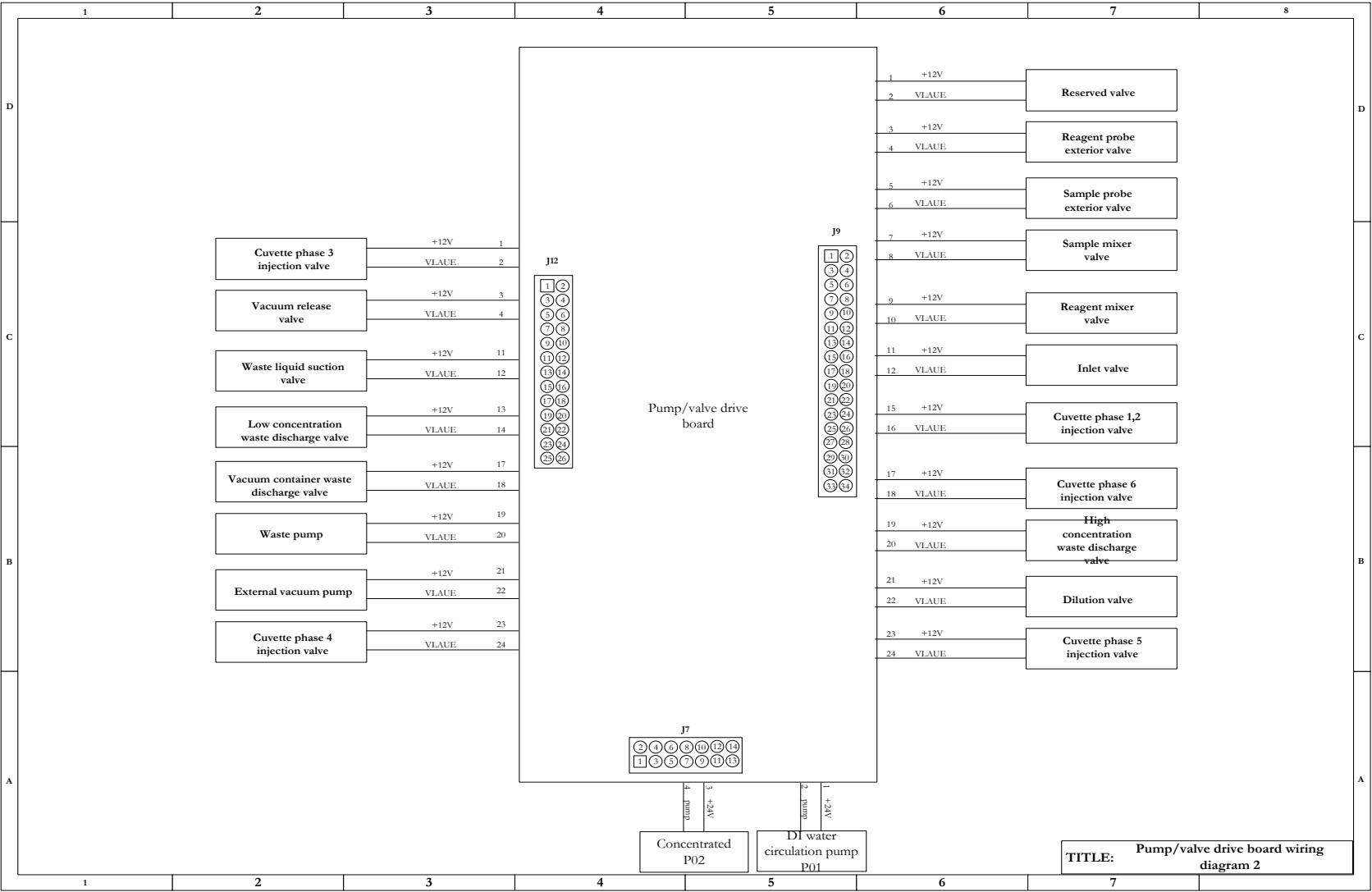


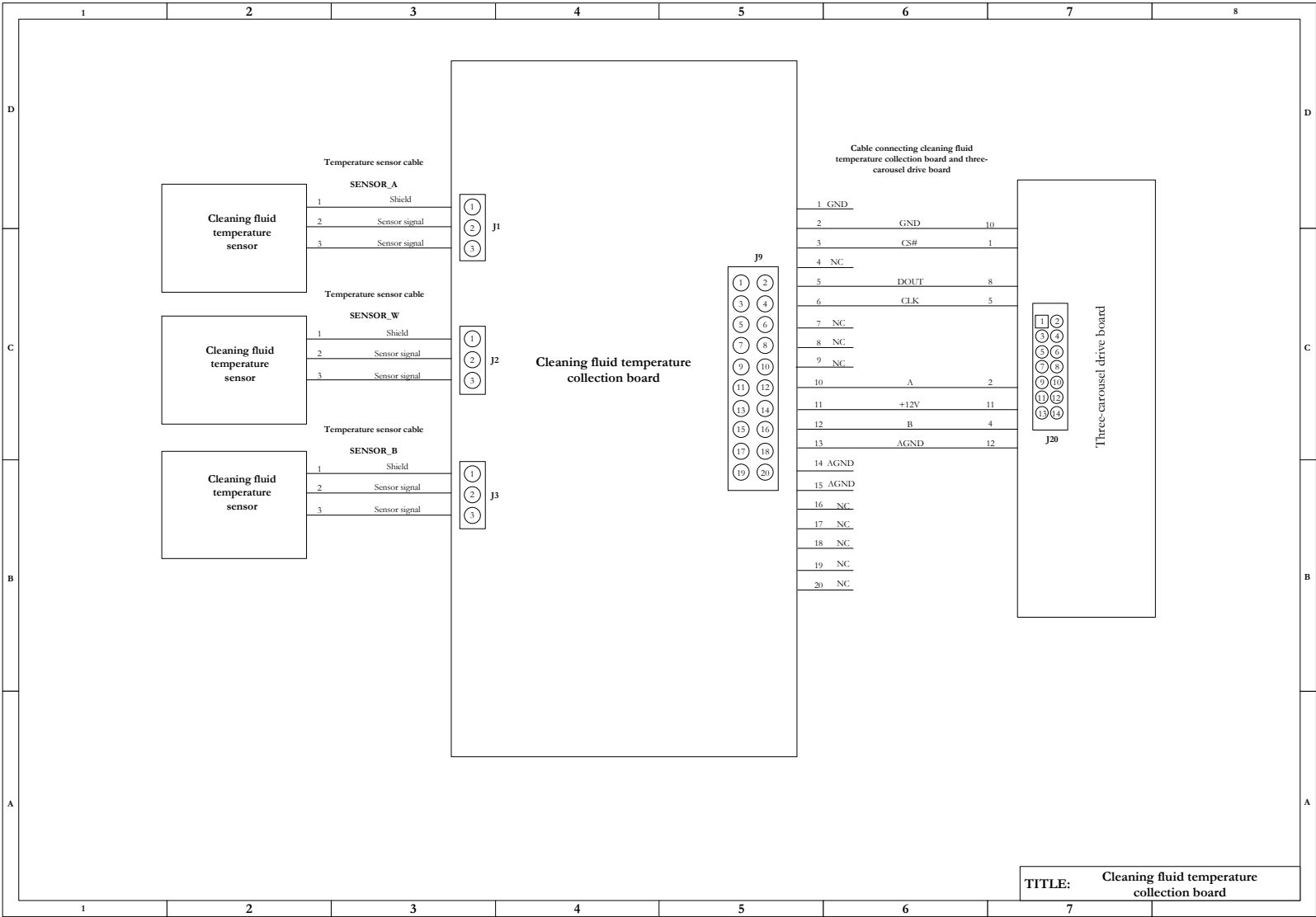
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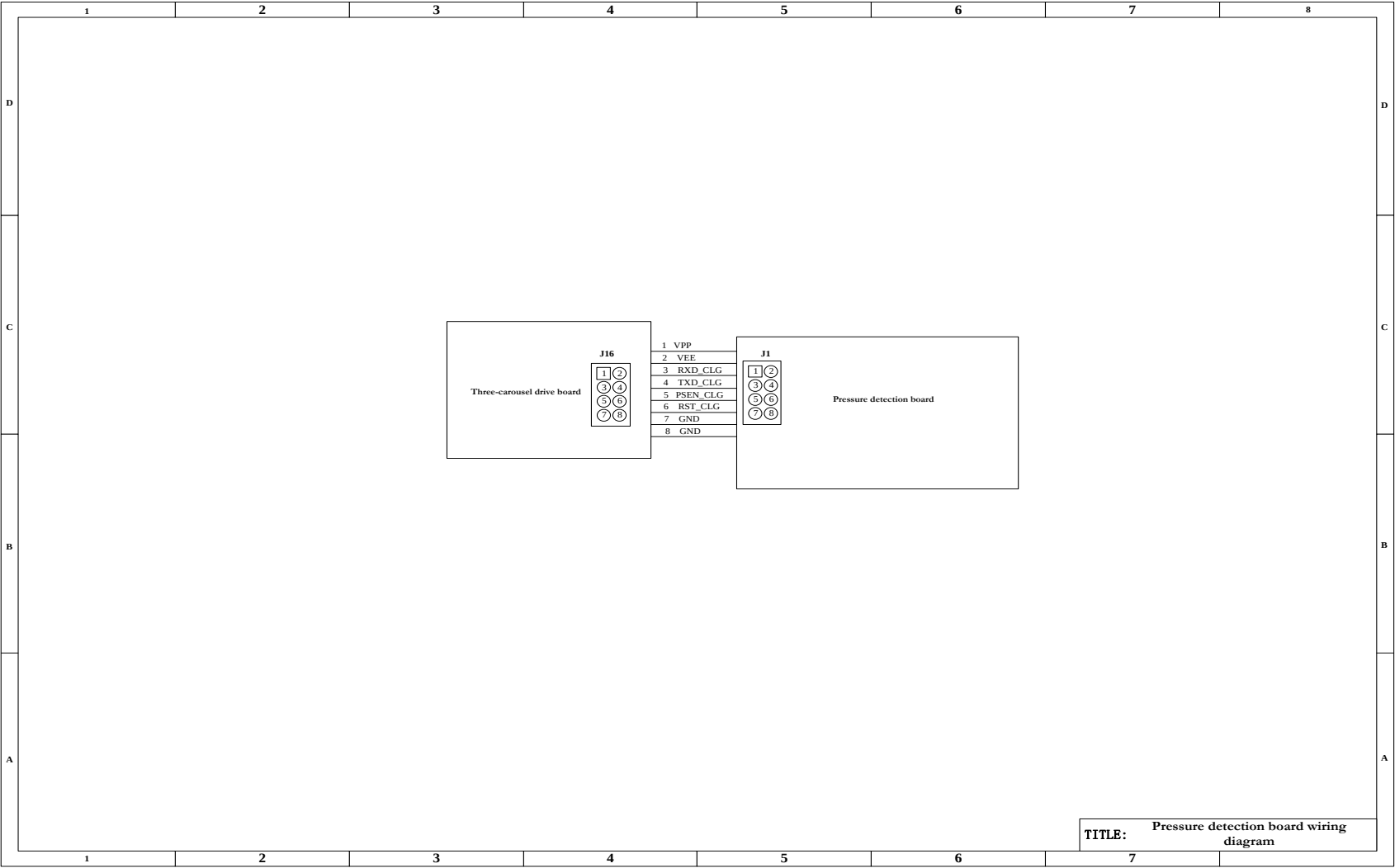


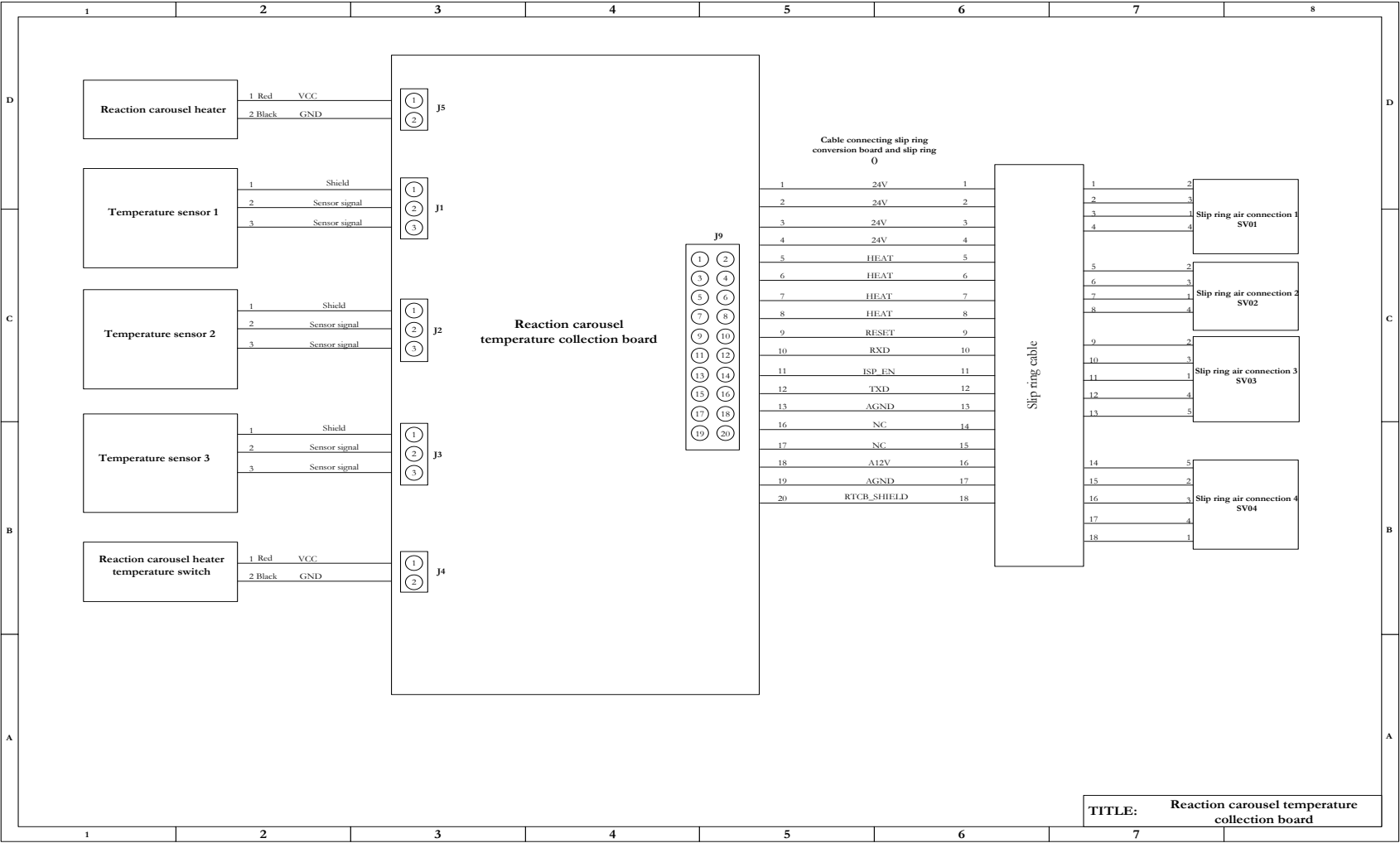
6 Hardware



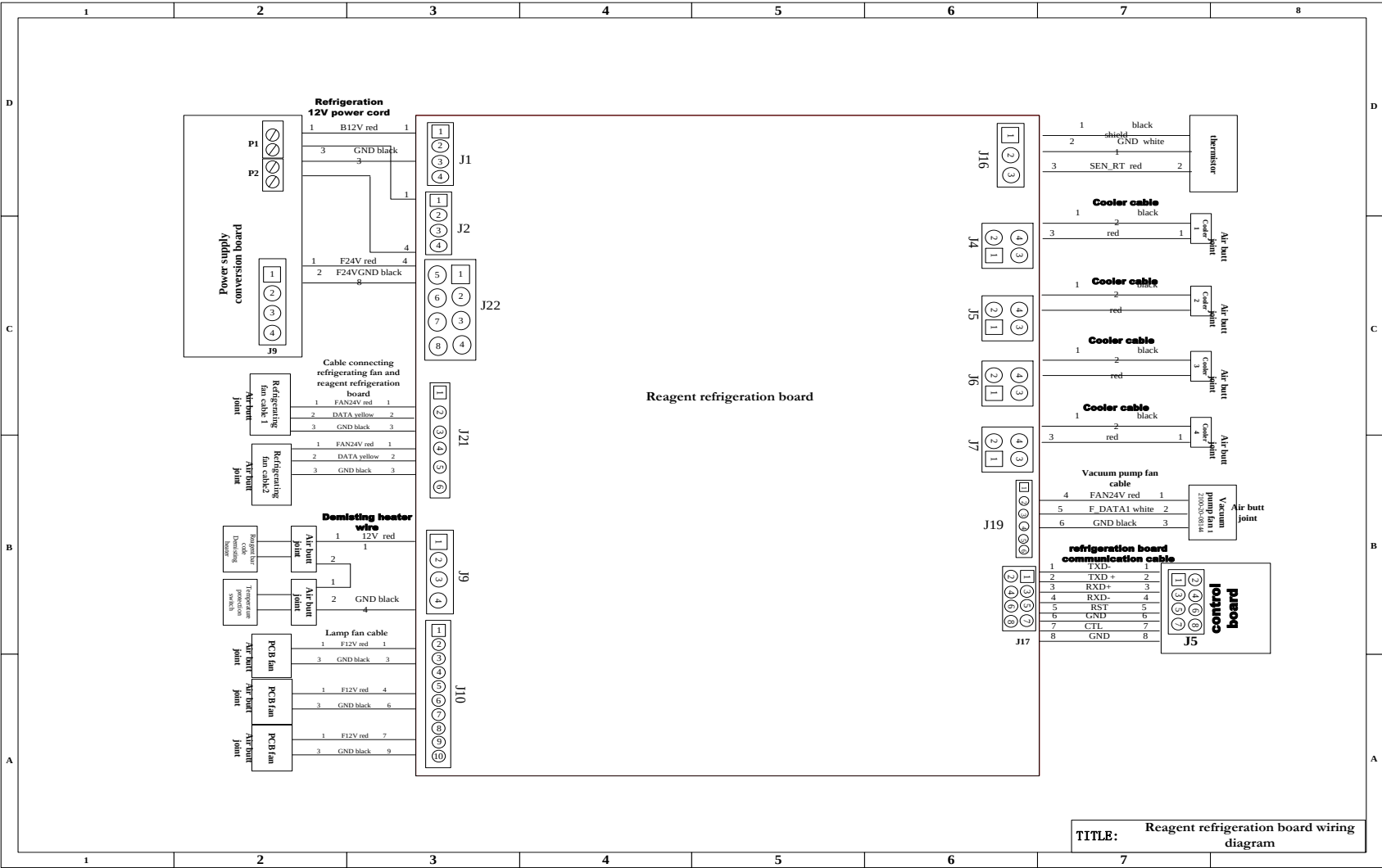


6 Hardware

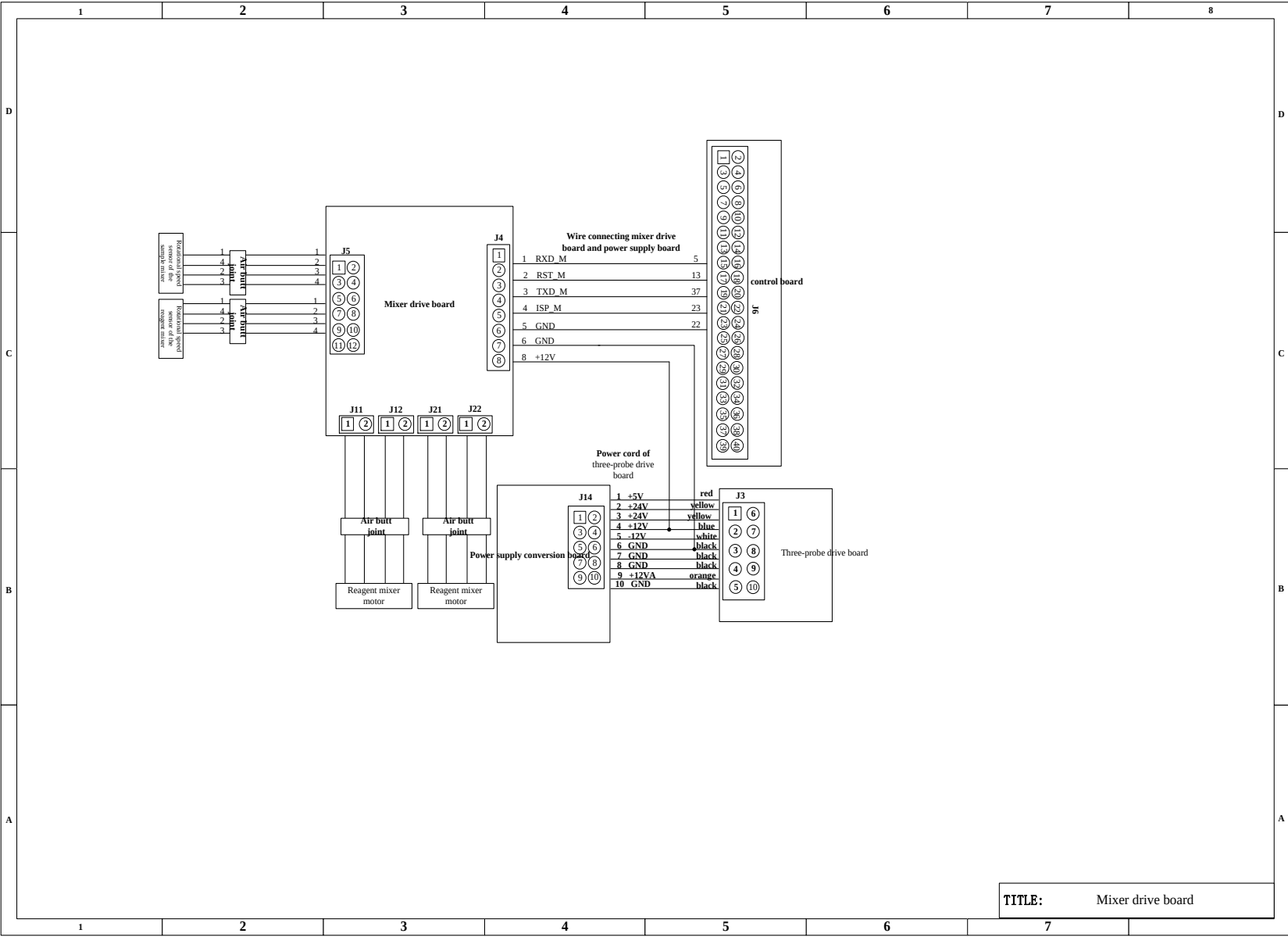




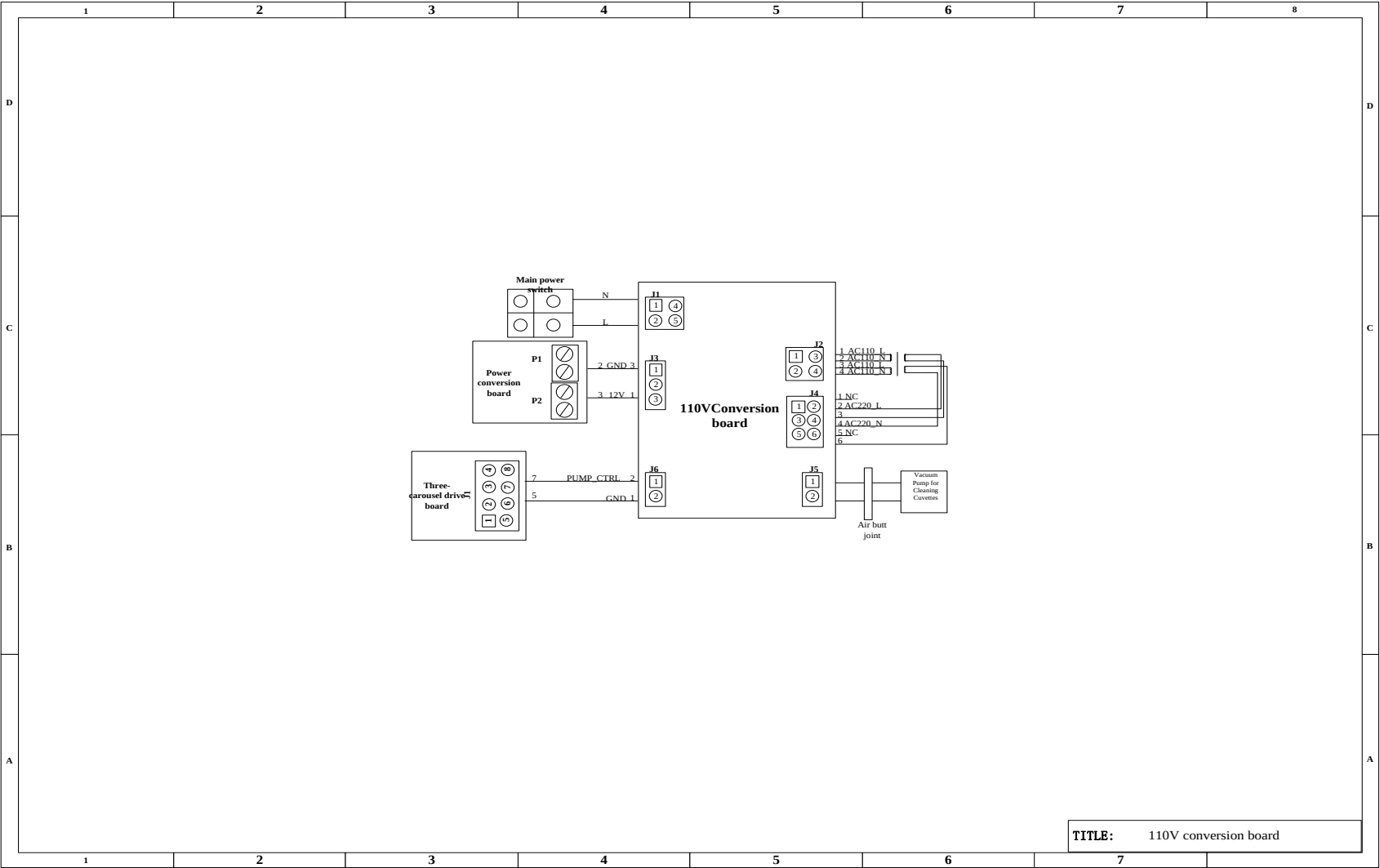
6 Hardware



TITLE: Reagent refrigeration board wiring diagram



6 Hardware



7

Hydropneumatic System

7.1 Overview

The Hydropneumatic system of the instrument includes the pneumatic part and hydraulic part.

The major function of the pneumatic part is:

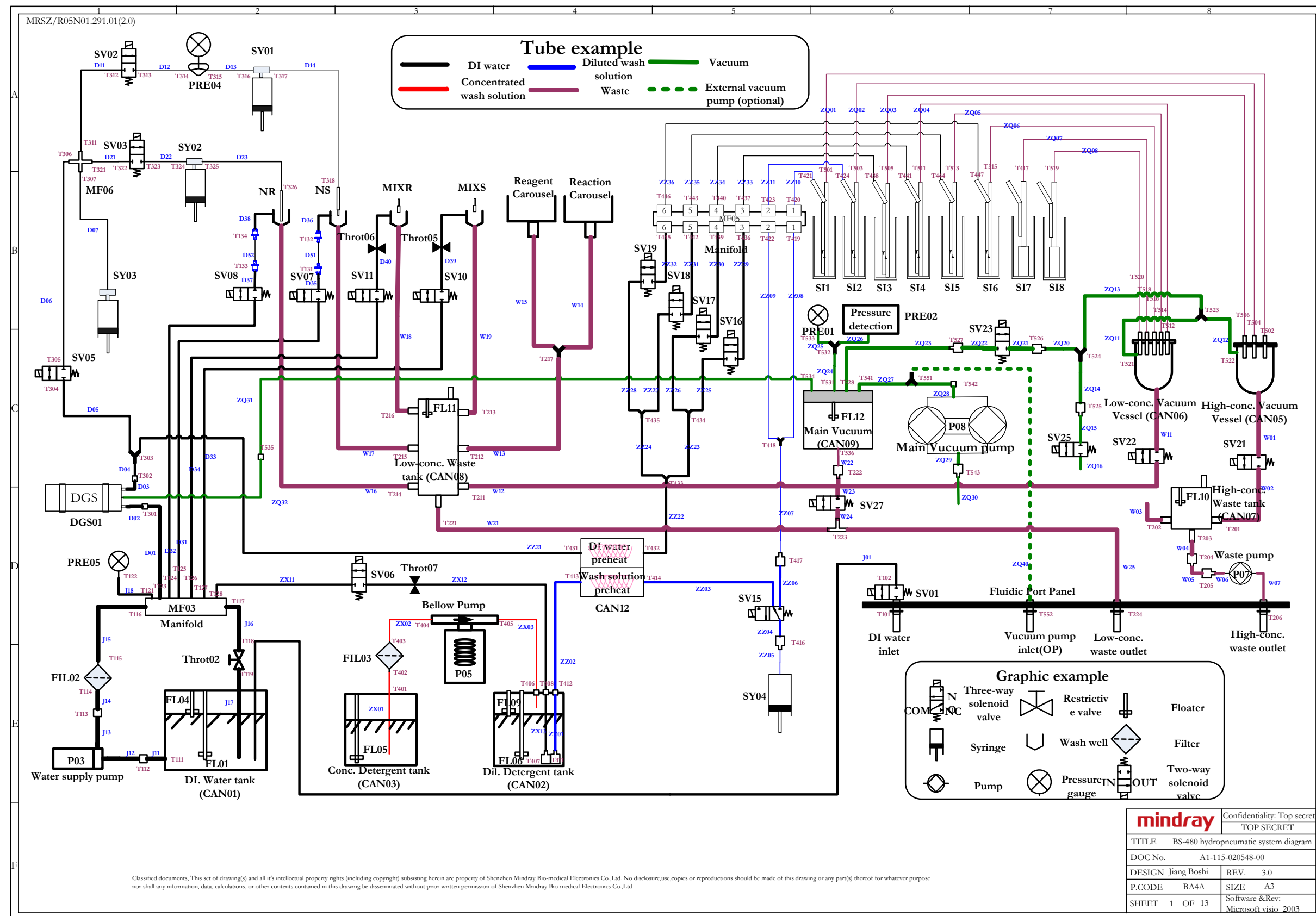
To provide vacuum for washing reaction cuvettes and probe interior.

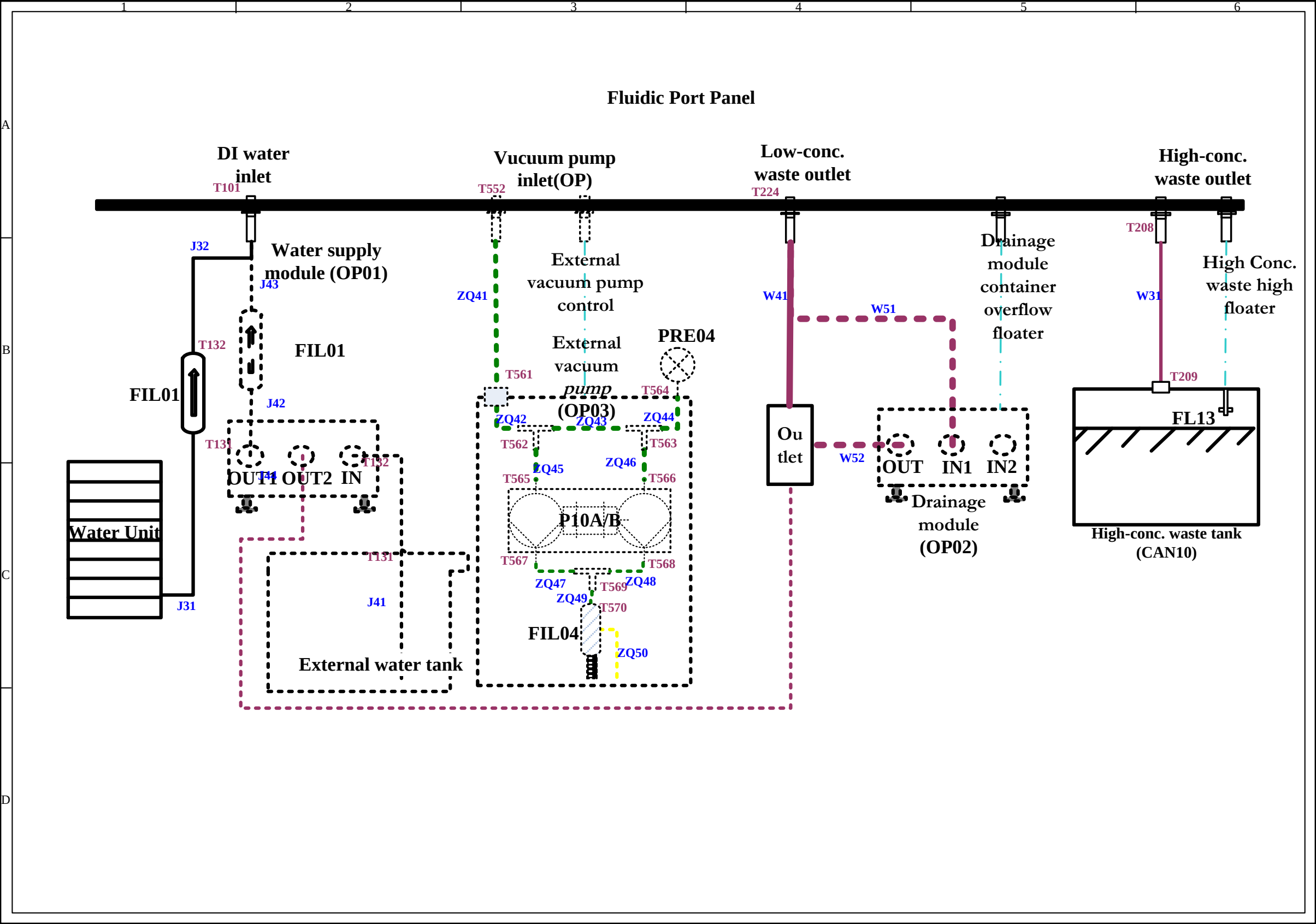
The major function of the hydraulic part is:

- To provide deionized water for washing interior/exterior of probes and mixers via the water supply unit.
- To provide deionized water and wash solution for washing cuvettes via the cuvette wash station.
- To provide the whole unit with deionized water via the water supply module and to discharge waste liquid via the drainage module.

This chapter describes the working principles and repairing methods of the Hydropneumatic system.

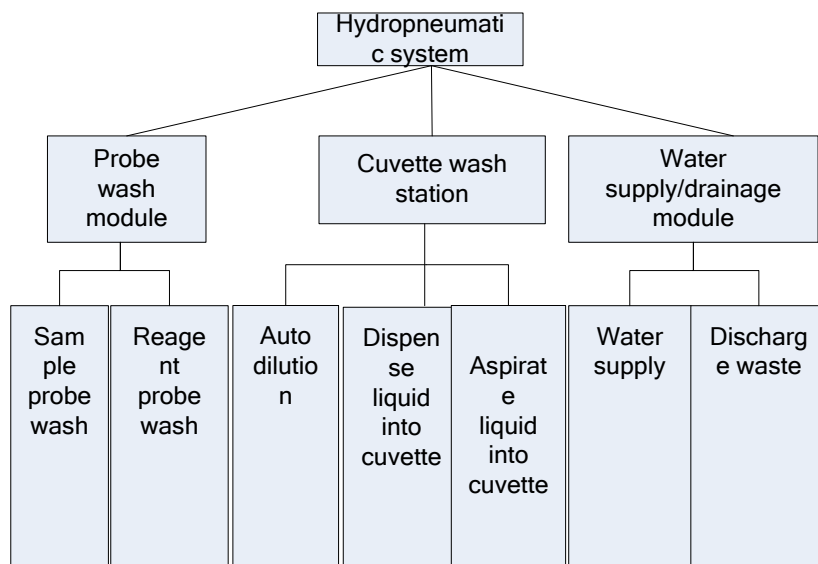
7.2 Fluidic Diagram





7.3 Principles of Hydropneumatic System

The hydropneumatic system of the BS-480 analyzer is composed of the probe wash module, cuvette wash station, and water supply/drainage module. See the figures below:



7.3.1 Probe wash module

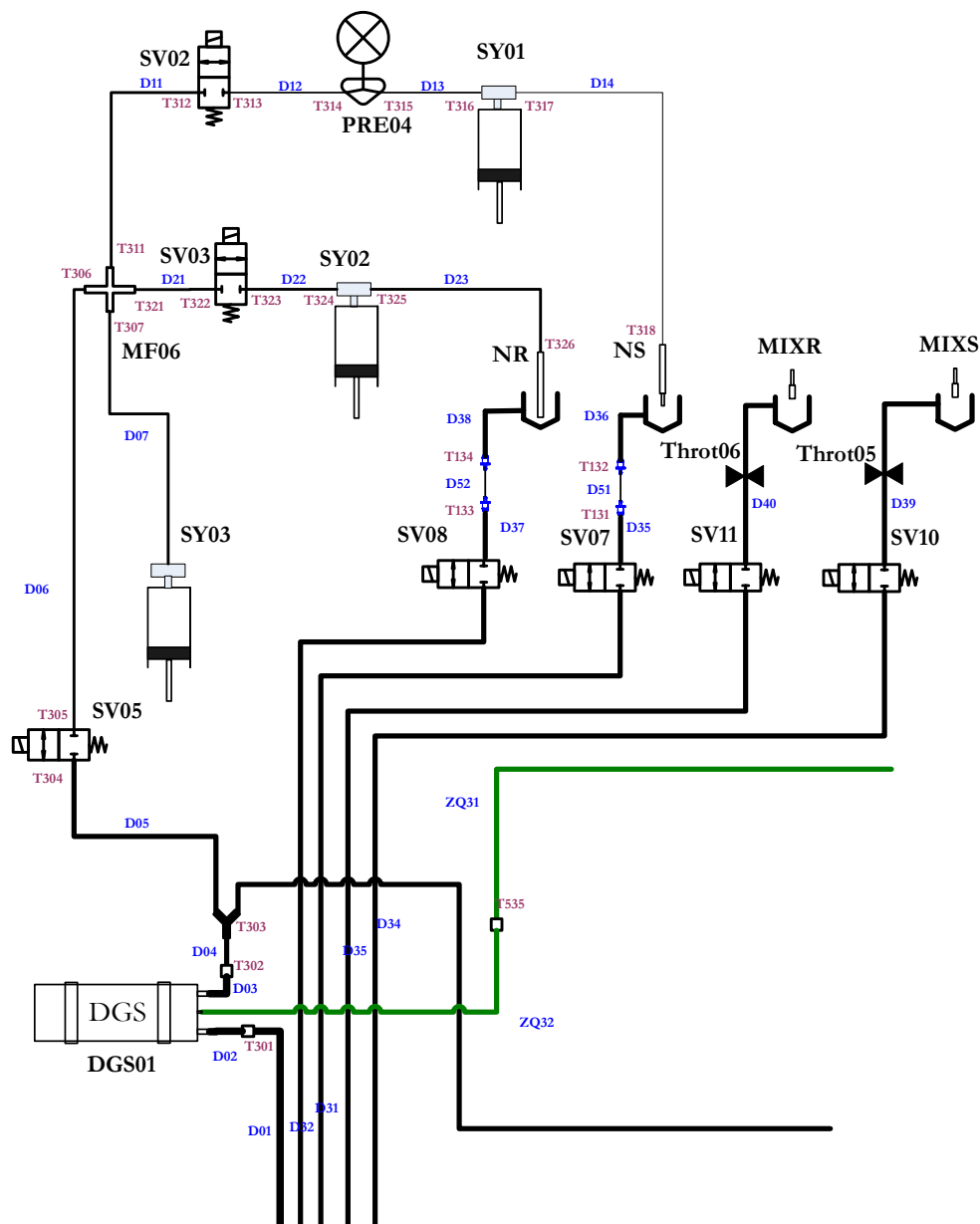
The probe wash module consists of one sample probe, one reagent probe, two mixers, and 3 syringes. The sample syringe is 100 μ l, the reagent syringe is 1ml, and the probe interior wash syringe is 10ml.

This module is used for fixed-quantity sampling and probe/mixer cleaning.

- Fixed-quantity sampling is realized through cooperation of the syringes, sampling valves and probes with the aim of delivering fixed-quantity reagent and sample.
- Interior washing of the sample probe and reagent probe is driven by the same syringe, and the probe/mixer exterior washing is driven by the deionized water circulating pump.
- To ensure accurate sampling, the deionized water used for cleaning needs to be degassed, that is, to be produced by passing through a degasser.

The schematic diagram of the probe wash module is as shown below.

Figure 7.1 Schematic diagram of probe wash module



7.3.2 Cuvette wash module

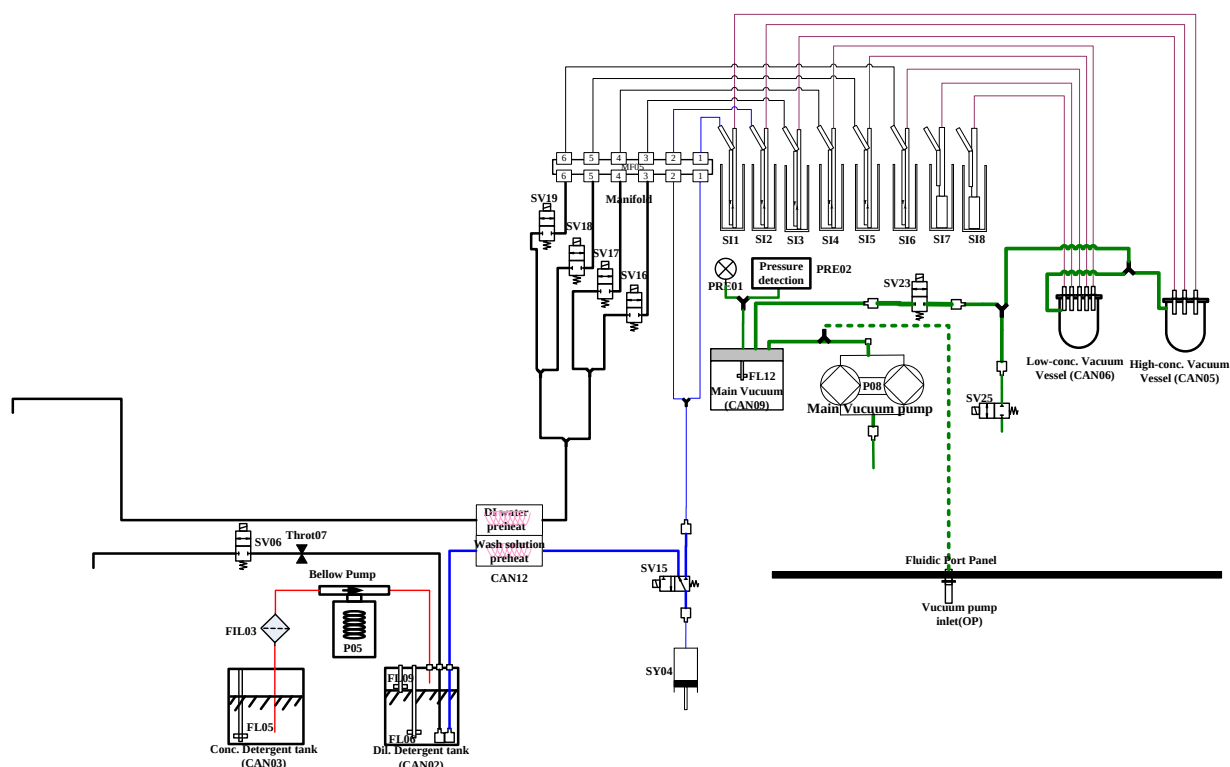
The cuvette wash unit is divided into dispense module and aspirate module, which cooperate with each other to clean the reaction cuvettes for 8 phases, making repeated use of cuvettes possible.

In phase 1 and 2, the cuvette wash syringe (10ml) provides diluted wash solution (alkaline); in phase 3 to 6, the deionized water circulating pump provides deionized water; in phase 7 and 8, no liquid is injected. The diluted wash solution provided for phase 1 and 2 is obtained by diluting the concentrated wash solution and deionized water at certain ratio; the concentrated wash solution is driven by the concentrated wash solution pump, and the diluting water by the deionized water pump.

A primary vacuum pump is used to discharge the waste fluid and wipe the cuvettes after cleaning. High-concentration waste is generated during phase 1-3 and low-concentration waste during phase 4-8. Wipe blocks are provided to absorb the remaining wash solution inside cuvettes during phase 7 and 8.

The schematic diagram of the module is as shown below.

Figure 7.2 Schematic diagram of cuvette wash unit



7.3.3 Water Supply/Drainage Module

The water supply/drainage module includes the water supply part and waste drainage part.

The water supply module delivers water from outside the instrument into the internal water tank, which is then transported by the deionized water pump to other modules. This module has the following features:

- Auto water supply and detection of fluid level are supported.
- The deionized water circulating pump is adopted to supply water for each wash unit, and the restrictive valve is used to adjust the water supply pressure.
- The water supply experiences two filtrations. The first filtration is executed before water is transported from the water supply equipment to the internal water tank, and the second filtration executed before the water is distributed by the deionized water pump to each wash unit.
- A water supply module, used for transporting deionized water to the internal water tank, is provided for optional configuration.

The drainage module allows discharging high-/low-concentration waste separately. The high-concentration waste is collected by gravity in the relevant container and then discharged by the waste pump to the external high-concentration waste tank; the low-concentration waste, also collected by gravity in the relevant container, is discharged by gravity or by an optional drainage module, out of the instrument. The high-/low-concentration waste containers and the external high-concentration waste tank are capable of detecting fluid level.

The high-concentration waste is produced during cuvette cleaning in phase 1-3.

Low-concentration waste comes from the following sources:

- Cleaning interior/exterior of sample probe and reagent probes

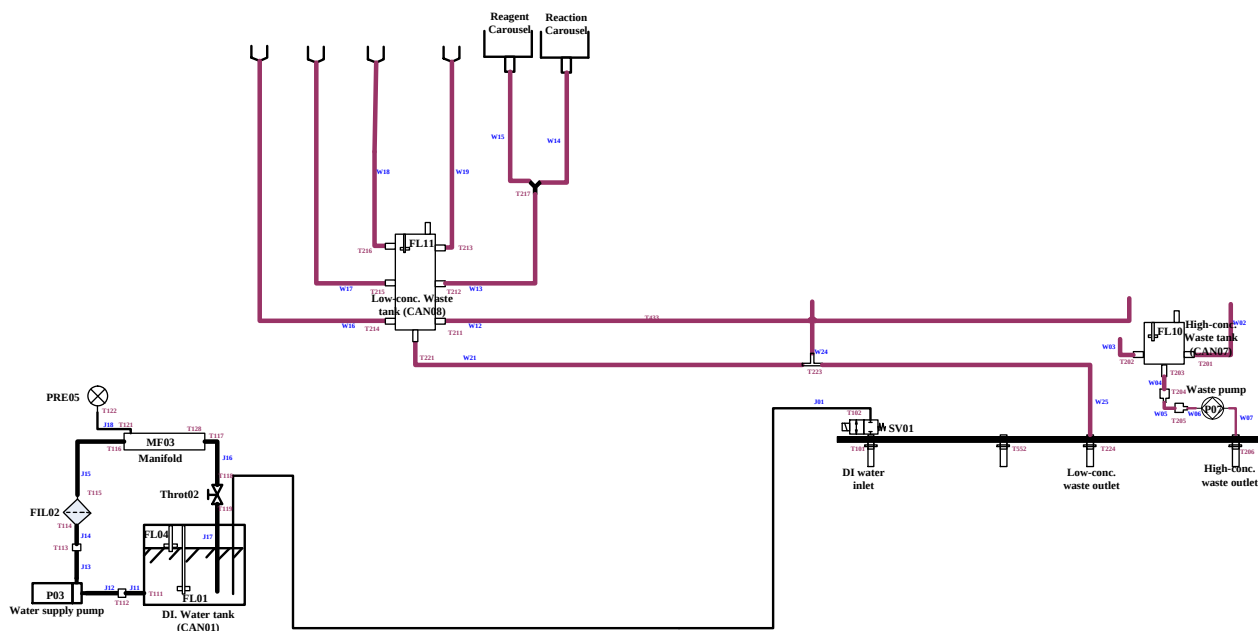
7 Hydropneumatic System

- Cleaning exterior of sample mixer and reagent mixer
- Auto cuvette cleaning in phase 4-8
- Condensate water of the reagent carousels
- Overflowed fluid of the reaction carousel

The first three are the major sources and account for over 99% of low-concentration waste.

The fluidic diagram of the module is as shown below.

Figure 7.3 Fluidic diagram of water supply/drainage module

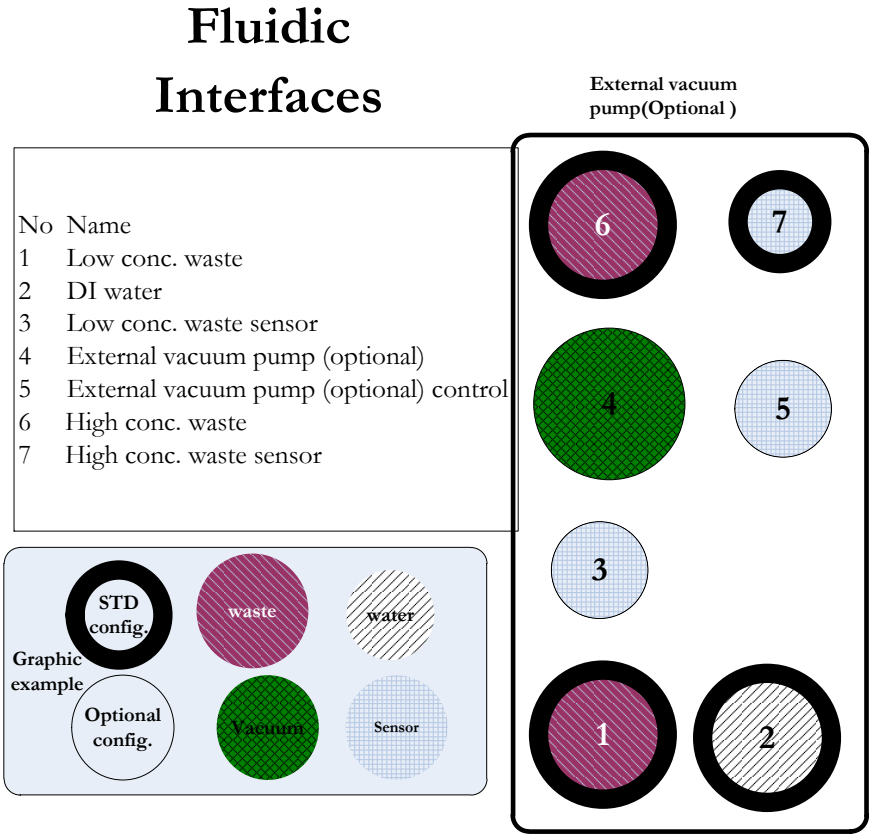


7.3.4 Other

The Hydropneumatic system of the instrument contains seven external interfaces, as shown in the figure below. Where,

- Two fluidic outlets: used for discharging high-/low-concentration waste.
- One fluidic inlet: used for supplying deionized water.
- One pneumatic outlet: used for connecting the external vacuum pump.
- Three control cable connectors: used for connecting the external vacuum pump, high-concentration waste level detection sensor, and low-concentration waste level detection sensor..
- .External fluidic interfaces of the instrument

Figure 7.4 External fluidic interfaces of the instrument



7.4 Introduction of Fluidic Actions

7.4.1 Fluidic initialization

The specific procedure is described below:

- 1 Power off all the pump and valve and they return to the home position.
- 2 Emptying primary vacuum container
- 3 The floater status of the water tank, diluted wash solution container, concentrated wash solution container, high concentration waste collector, low concentration waste collector, primary vacuum container, external high concentration waste tank, overflow container of the drainage module are checked. If related alarm is removed, the system moves to the next step. If not, the system gives out the alarm and the fluidic initialization fails.

Establish Vacuum

- 1 Cleaning Probes/Mixers Exterior is performed.
- 2 Pump P03 is turned on.
- 3 The interior wash syringe and phase1-2 wash syringe are reset.

Notes:

The fluidic initialization is one part of the startup, system recovery and system wakeup procedure.

7.4.2 Establish Vacuum

The specific procedure is described below:

- 1 The pressure of the current primary vacuum container is checked. If it is within the range and the system moves to the next step. If not, release vacuum instruction is performed.
- 2 SV23 and SV25 are turned on. After a short delay, Turning on P10 instruction is sent. P10
- 3 After a short delay, Powering on P18 instruction is sent.
- 4 After a short delay, SV23 and SV25 are turned off. Waiting for a short time.
- 5 The system checks the vacuum degree and if it meets the requirement, establishing vacuum succeeds.

7.4.3 Release Vacuum

The specific procedure is described below:

- 1 Primary Vacuum Pumps P08 is turned off.
- 2 SV23 and SV25 are turned on. After a short delay, SV23 and SV25 are turned off.
- 3 After turning off the valve, the system judges if the pressure of the current primary vacuum container is within the range and if releasing vacuum succeeds.

7.4.4 Emptying primary vacuum container

The specific procedure is described below:

7 Hydropneumatic system

- 1** The system judges if the pressure of the current primary vacuum container is 2 within the range; if not, releasing vacuum is performed
- 2** SV23, SV25 and SV27 are turned on for a while and then turned off.

7.5 Removing and Installing Hydropneumatic Components

This section describes removing and installation methods of the Hydropneumatic components, and provides schematic diagrams and pictures of components for service engineers to refer.

7.5.1 Overview

When an alarm is displayed on the operating software, service engineers can analyse the instrument status and finally locate the failure source, and if necessary, replace the failed part or component. Generally, service engineers are not recommended to remove the electric devices, such as pumps, solenoid valves, clog detection device, syringes, etc. Only when both hardware and software are confirmed normal but the Hydropneumatic alarm still remains are service engineers suggested to remove the relevant device and then analyse or replace it.

Prior to removing Hydropneumatic components, make sure that all pumps and solenoid valves of the Hydropneumatic system have been turned off, and both the analyzer power switch and main power switch have been placed to the OFF position.

Table 7.1 Necessary tools for removing/installing Hydropneumatic components

Name	Requirement	Quantity
Cross screwdriver	/	1
Hexagon screwdriver	/	1
Flathead screwdriver	/	1
Strap	/	Several
Diagonal pliers	/	1
Tube cutter	/	1

7.5.2 Probe wash module

Figure 7.5 Assembly drawing of whole unit -- Probe wash module-front view

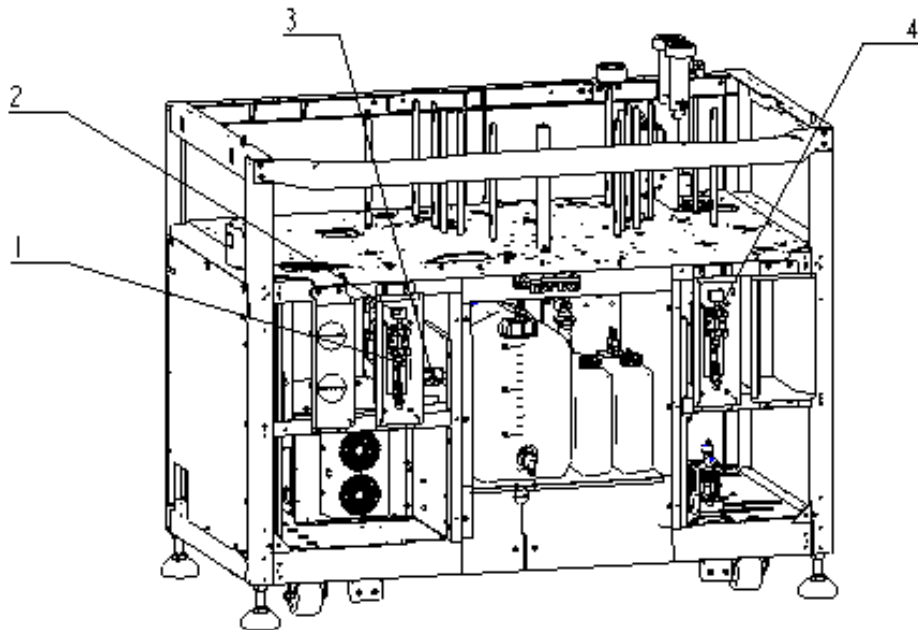


Figure 7.6 Assembly drawing of whole unit -- Probe wash module-rear view

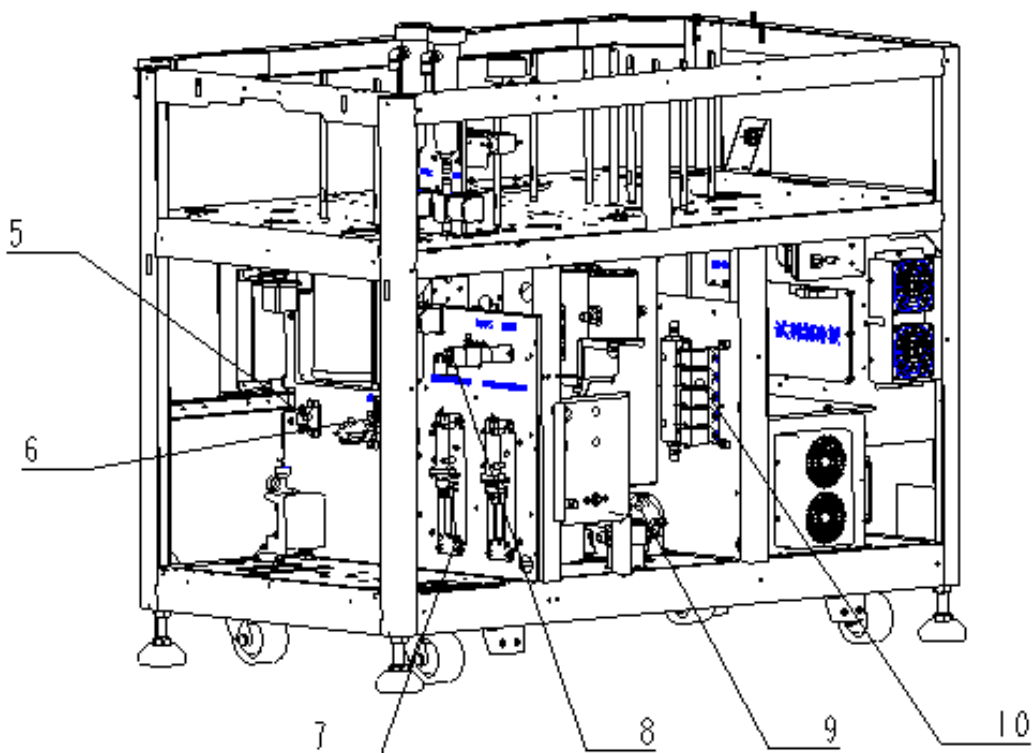


Table 7.2 List of materials

No.	FRU code and material code	Material Name	Remark
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7 Hydropneumatic System

No.	FRU code and material code	Material Name	Remark
1	801-BA80-00047-00	Syringe 1ml	FRU
2	801-BA38-00040-00	Syringe T piece	FRU
3	801-BA34-00146-00	Sample/reagent Probe Interior Wash	FRU
4	801-BA34-00041-00	Syringe 100ul	FRU
5	115-022008-00	Clog detection T-piece assembly	FRU
6	115-013114-00	Wash syringe assembly	FRU
7	801-BA80-00089-00	Probe interior wash return valve assembly	FRU
8	115-021041-00	Degasser assembly	FRU
9	801-BA40-00011-00	Solenoid valve WTA-2K-N4G cable	FRU
10	115-013149-00	Sample probe tubing assembly	FRU
11	115-013150-00	Reagent probe tubing assembly	FRU
12	041-000171-00	0.9mm restrictor ring	FRU
13	BA31-20-41660	0.6mm restrictor valve	FRU
14	115-034970-00	Restrictive tube assembly	FRU

Removing/Installing Wash Syringe

When to do

Analyze or replace the wash syringe when it has abnormal sound, worse precision, leaking, or other failures.

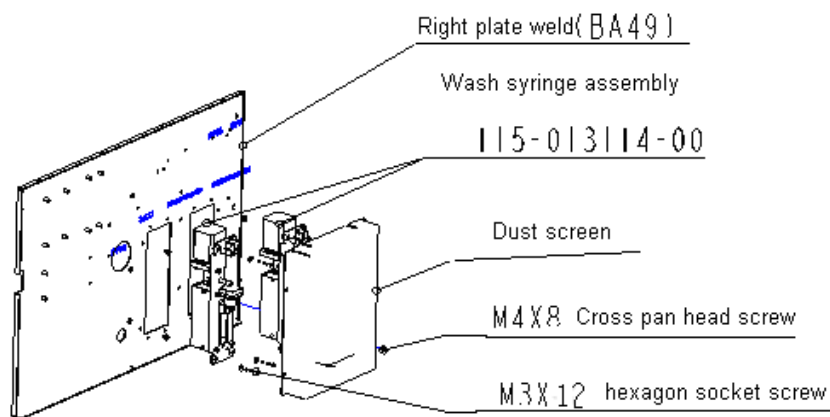
Steps

- 1 Remove the 4 M4*8 cross pan head screws used to fix the syringe shield and remove the shield.
- 2 Disconnect the syringe motor cable and sensor cable.
- 3 Remove the tube from the barbed connector of the syringe, and then apply strap on the tube opening to prevent liquid spraying.
- 4 Remove the four M3*12 hexagon socket cap head screws with spring washer and remove the syringe.

Installation Procedure

- 1 Install the four M3*12 hexagon socket cap head screws with spring washer and do not miss the shock pad.
- 2 Cut off a little from the tube's end and insert the tube to the barbed connector of the syringe. If a strap has been applied, restore it.
- 3 Connect the syringe motor cable and sensor cable.
- 4 Use four M4*8 cross pan head screws to fix the syringe shield on the right side plate.
- 5 Power on the analyzer and use the engineer account to log on the operating software. Select Utility-Maintenance-Maintenance-Engineer- Clear. If interior wash syringe has been replaced, tick "replace interior wash syringe". If wash syringe has been replaced, tick "replace wash syringe". Select OK to clear the using count of the syringes.

Figure 7.7 Details of wash syringe assembly



Verification steps

Refer to Section 8.12 Hydropneumatic Unit.

Removing/Installing Solenoid Valves

When to do

When a solenoid valve cannot be turned on or has leaks, it needs to be removed and then analyzed or replaced.

Steps

- 1 Disconnect the solenoid valve's power cord connector.
- 2 Mark the installation direction of the valve and the connection mode of the inlet and outlet tubes to prevent them from being confused.
- 3 Remove the tubes and apply strap on their openings to prevent liquid spraying.
- 4 Loosen the screws on the solenoid valve and then remove the solenoid valve.

Installation Procedure

- 1 Check the installation direction according to the marks, connect the inlet and outlet tubes, and then tighten the tube clamps or straps.
- 2 Tighten the retaining screws to fix the solenoid valve.
- 3 Make sure that the power cord is connected to the correct positive and negative ends.

Verification steps

Refer to Section 8.12 Hydropneumatic Unit.

Figure 7.8 Details of wash solenoid valve assembly

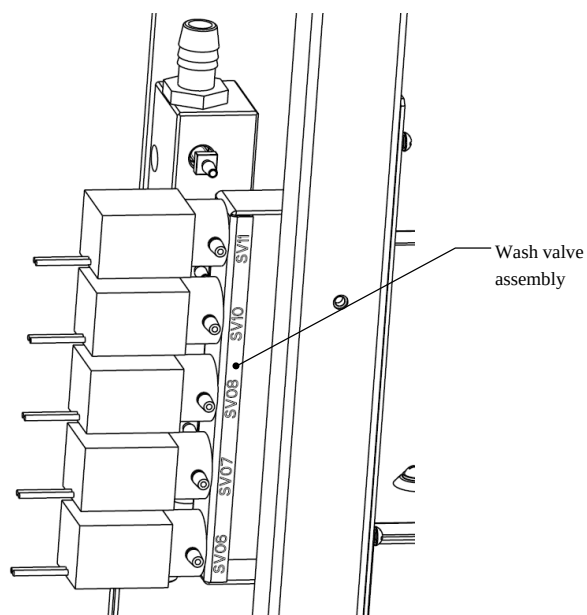
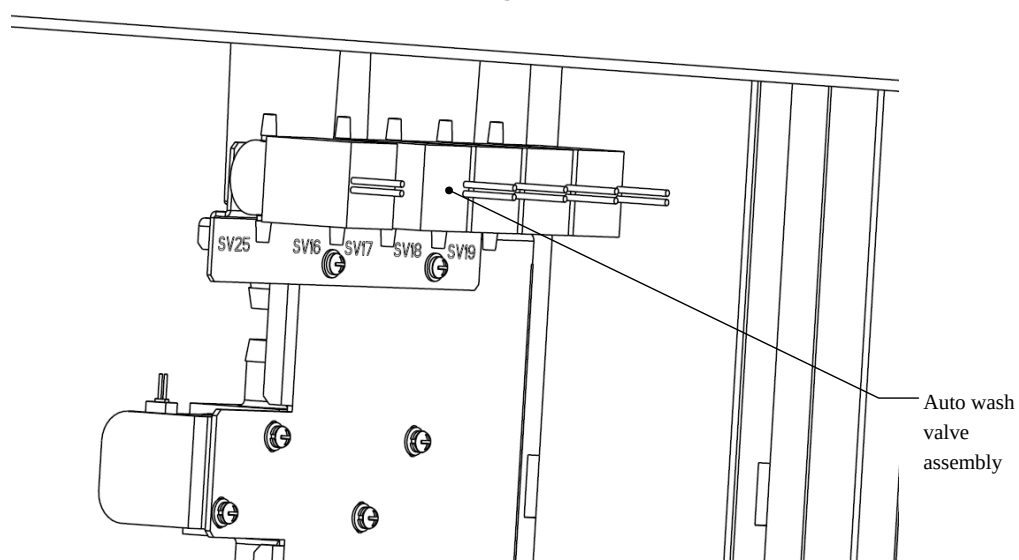


Figure 7.9 Details of cuvette wash solenoid valve assembly



Removing/Installing Restrictor Ring

When to do

Remove the restrictor ring for analysis or replacement if it is clogged with a result of inadequate water flow for mixer exterior wash or for auto dilution.

Steps

- 1 Remove the tube from the solenoid valve and pour the residual water of the tube into a container.
- 2 Use hands or sharp nose pliers to squeeze the tube to remove the restrictor ring.

Installation Procedure

- 1 Use hands or sharp nose pliers to squeeze the tube to restore the restrictor ring.
- 2 Connect the tube to the solenoid valve and then tighten the strap.

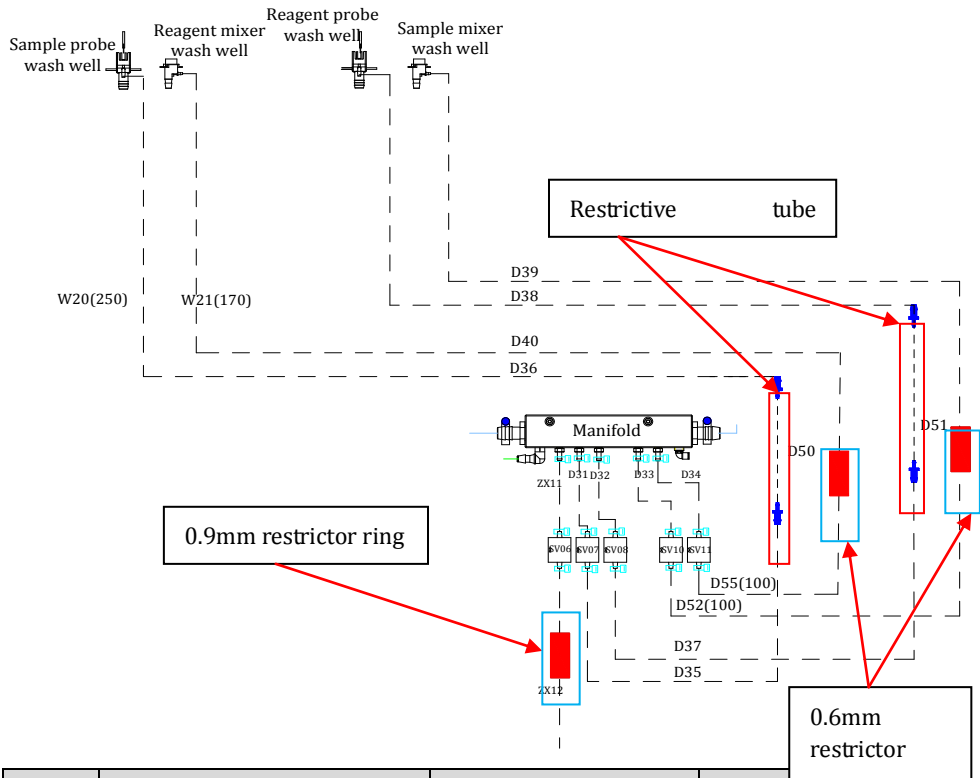
Verification steps

N/A



While removing or installing the restrictor, exercise caution to prevent the small hole on it from being blocked.

Figure 7.10 Tube connection diagram of restrictor ring



No.	Material name (FRU)	FRU Code	Remarks
12	0.9mm restrictor ring	041-000171-00	
13	0.6mm restrictor valve	BA31-20-41660	
14	Restrictive tube assembly	115-034970-00	

Removing and Reinstalling Restrictive Tube Assembly

When to do

When the sample probe or reagent probe exterior wash flow decreases, the restrictive tube may be clogged. Remove the restrictive tube for analysis or replacement.

Removing Steps

- 1 Check the connecting position of the restrictive tube assembly, and mark its inlet and outlet to avoid confusion.
- 2 Remove one end of the restrictive tube assembly and collect the residual water with a container.
- 3 Unplug the other end of the restrictive tube assembly.

Installation Procedure

Install a restrictive tube assembly with same length and specification in the original position.

Verification steps

See "Fluidic Prime" in 8.12 Hydropneumatic Unit.



NOTE

When removing or installing the restrictive tube, prevent misconnection and do not pull or bend it too much.

7.5.3 Cuvette wash module

Figure 7.11 Assembly drawing of whole unit -- Cuvette wash module(a) -right rear view

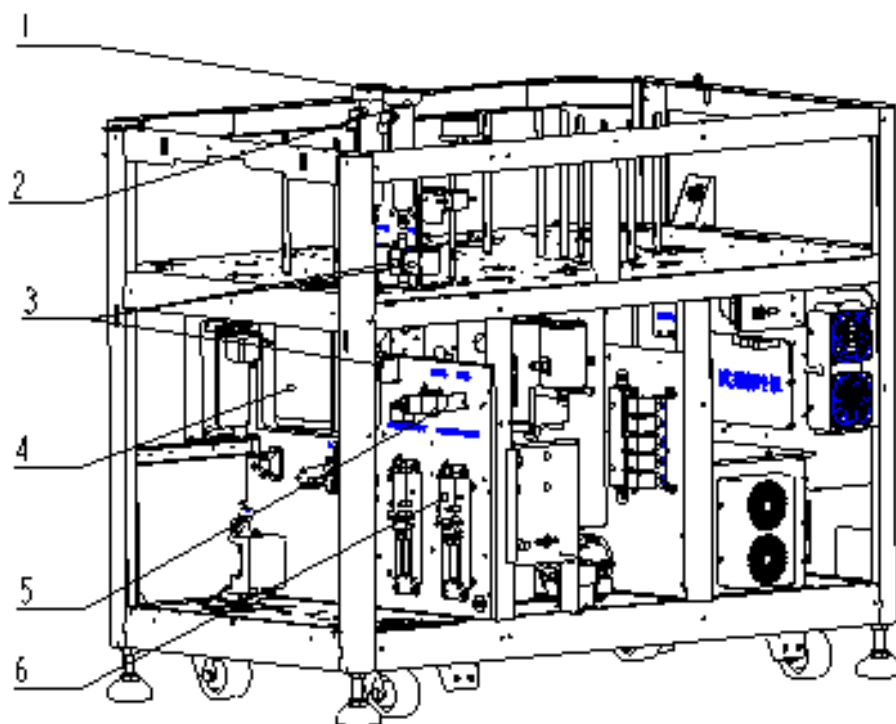


Figure 7.12 Assembly drawing of whole unit -- Cuvette wash module(b) -front view

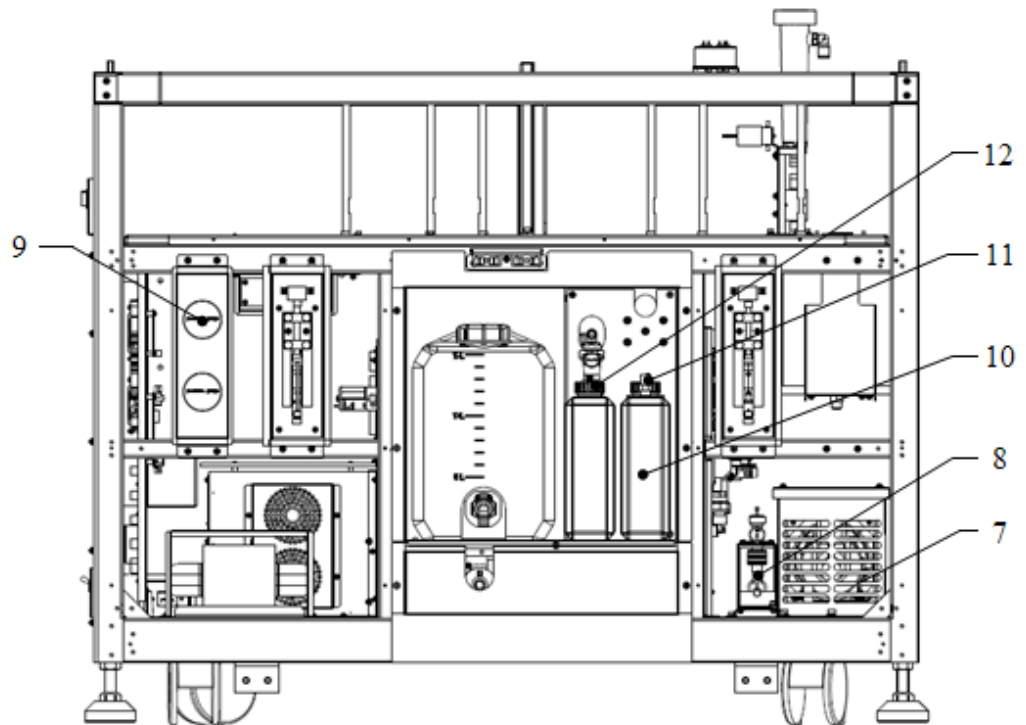


Figure 7.13 Assembly drawing of whole unit -- Cuvette wash module(c)-left rear view

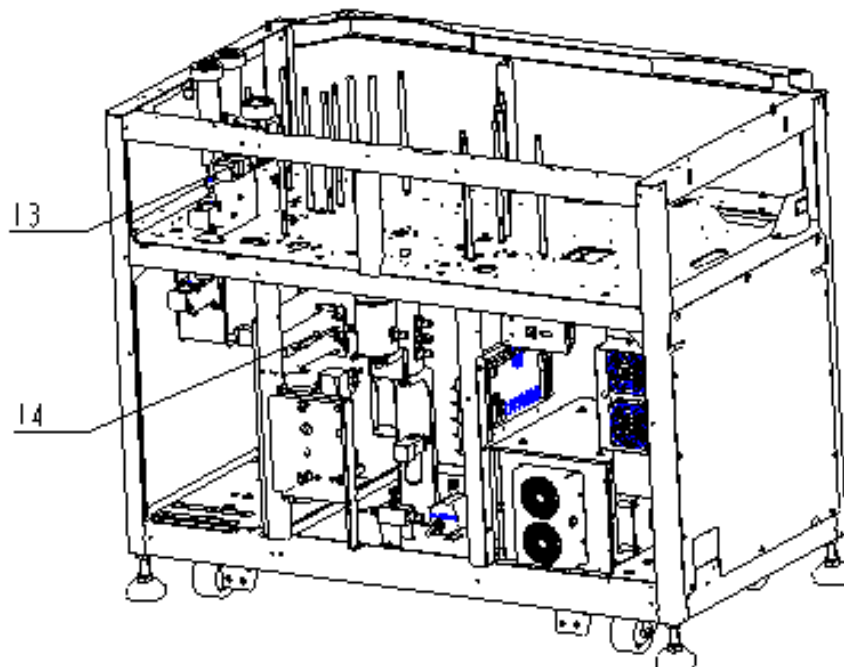


Table 7.3 List of materials

No.	FRU code and material code	Material Name	Remarks
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7 Hydropneumatic System

No.	FRU code and material code	Material Name	Remarks
1	115-013123-00	Low-concentration secondary vacuum container assembly	FRU
2	115-013122-00	High-concentration secondary vacuum container assembly	FRU
3	082-000333-00	valve	FRU
4	115-013125-00	Primary vacuum container assembly	FRU
5	BA38-21-88190	WTB-3R-N4E three-way solenoid valve cable	FRU
6	115-013114-00	Wash syringe assembly	FRU
7	082-001061-00	Vacuum Pump for Cleaning Cuvettes	FRU
8	115-005634-00	Concentrated wash solution pump	FRU
9	115-028135-00	Vacuum pressure gauge assembly (with connector)	FRU
10	041-004638-00	Diluted Wash Solution Tank	FRU
11	801-BA80-00017-01	Diluted wash solution level sensor assembly	FRU
12	801-BA80-00018-01	Concentrated wash solution sensor assembly	FRU
13	BA40-21-61680	Solenoid valve WTA-2K-N4G cable	FRU
14	115-020528-00	Preheating assembly	FRU
15	043-000422-00	Disk-like filter	FRU
16	049-000833-00	Silicone transit tube	FRU
17	115-013233-00	Wash probe assembly	FRU
18	041-006193-00	Wipe block	FRU

Removing/Installing fluidic pump

When to do

The pump does not work, that is, there is no flow or pressure.

The pump flow and pressure is low.

The pump has leaks.

Noise is produced when the pump is working.

Steps

- 1 Disconnect the pump's power cord connector.
- 2 Mark the connection mode of the inlet and outlet tubes and prevent them from being confused.
- 3 Remove the tubes and apply straps on their openings to prevent liquid from entering the instrument.
- 4 Loosen the screws fixing the pump and then remove the pump.

Installation Procedure

- 1 Connect the inlet and outlet tubes according to the marks and then tighten the tube clamps.
- 2 Tighten the retaining screws to fix the pump.
- 3 Make sure that the power cord is connected correctly.

Verification steps

Refer to Section 8.12 Hydropneumatic Unit.



NOTE

Exercise caution to avoid confusing the inlet tube with the outlet tube.

Make sure that the power cord of the DC pump is connected to the correct positive and negative ends.

Removing/Installing vacuum pump

When to do

The pump does not work, that is, there is no flow or pressure.

The pump flow and pressure is low.

Noise is produced when the pump is working.

Steps

- 1 Disconnect the pump's power cord and fan cable.
- 2 Unplug the quick connector of the pump hosing from the vacuum tank.
- 3 Remove the four M4*10 hexagon socket cap head screws with spring washer used to fix the vacuum pump and remove the vacuum pump assembly.
- 4 Remove the top cover and side cover of the vacuum pump.
- 5 Remove the hoop used to fix the power cord and wire of capacitor.
- 6 Mark the connection mode of the inlet and outlet tubes, and then remove the tubes. Exercise caution to prevent the tubes from being confused.
- 7 Remove the four screws used to fix the vacuum pump and remove the M6*16 hexagon socket cap head screws with spring washer and four hexagon nuts M6 to remove the vacuum pump.

Installation Procedure

- 1 Install the retaining screws of the vacuum pump.
- 2 Connect the inlet and outlet tubes according to the marks and then connect the tubes and tighten the pipe hoop.
- 3 Fix the tube and wire of the capacitor and the power cord of the vacuum pump.
- 4 Install the top cover and side cover of the vacuum pump.
- 5 Fix the vacuum pump assembly with screws on the frame.
- 6 Connect the inlet tube of the pump.
- 7 Connect the pump's power cord and fan cable.
- 8 Power on the analyzer and use the engineer account to log on the software. Select Utility-Maintenance-Maintenance-Engineer- Clear. Tick replace vacuum pump and click OK.

Verification steps

7 Hydropneumatic System

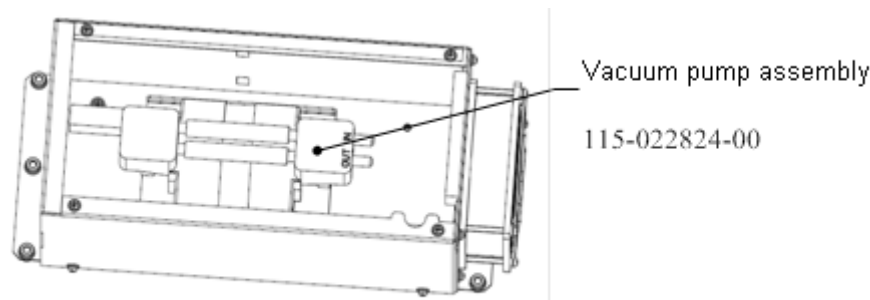
Refer to Section 8.12 Hydropneumatic Unit.



NOTE

Do not confuse the inlet pump with the outlet tube. (IN for vacuum tank assembly and OUT for atmosphere)

Figure 7.14 Structure of vacuum pump assembly



Removing/Installing Cuvette Wash Preheating Assembly

When to do

- The heater is damaged.
- The temperature sensor is damaged.
- The temperature protection switch is damaged.
- The canister has leaks.
- The connectors are leaking.

Steps

- 1 Switch off the power supply of the whole unit.
- 2 Open the rear panel.
- 3 Mark the connection modes of the following parts: inlet/outlet tubes, heaters, sensors and temperature protection switches for wash solution and deionized water.
- 4 Disconnect the connectors of the heaters, sensors and protection switches.
- 5 Loosen the two hexagon socket head screws fixing the cuvette wash preheating assembly.
- 6 Apply straps on the tube openings to prevent liquid from flowing out.
- 7 Disconnect the inlet and outlet tubes.

Installation Procedure

- 1 Prepare a new cuvette wash preheating assembly.
- 2 Connect the connectors according to the marks.
- 3 Connect the tubes according to the marks.
- 4 Use two M4*8 hexagon socket head screws to fix the preheating assembly.
- 5 Check if the installation is complete.
- 6 Restore the instrument.
- 7 Connect the power supply and perform fluidic prime before starting up the system.

Verification steps

Refer to Section 8.12 Hydropneumatic Unit.



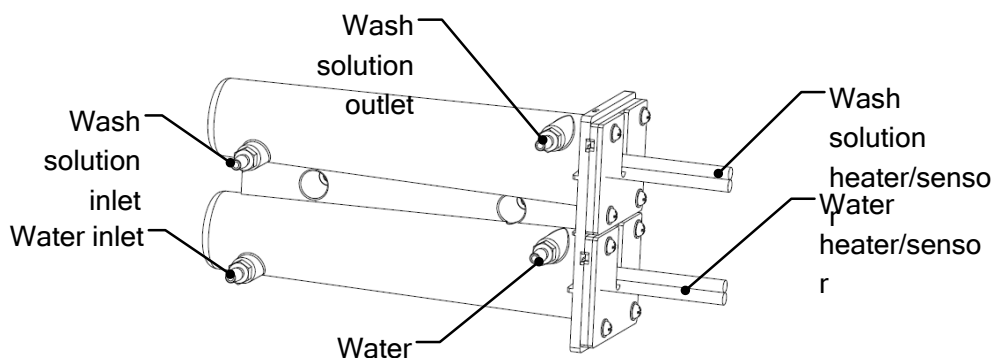
NOTE

Operate carefully to prevent liquid from entering the instrument.

Connect the tubes and connectors correctly.

Do not confuse the heater and sensor channel of wash solution with those of cleaning fluid.

Figure 7.15 Structure of cuvette wash preheating assembly



Removing/Installing vacuum and waste containers

When to do

When the containers have leaks or float alarms or need to be maintained regularly.

Steps

- 1 Confirm and record all tubes connected with the container, and distinguish the inlet tube from the outlet tube.
- 2 Remove the tubes, and then apply straps on the tube openings to prevent liquid inside the tubes from spraying out.
- 3 Use a screwdriver to remove the container.
- 4 Replace or maintain the container.

Installation Procedure

- 1 Connect the inlet and outlet tubes according to the recorded connection mode, and then tighten the tube clamps or straps.
- 2 Fix the container on the instrument.

Verification steps

N/A.

Removing/Installing acetabular filter

When to do

The deionized water filter should be maintained regularly.

Steps

- 1 Remove the Right plate assembly and you will see butterfly like filter beside the P05.
- 2 Remove the filter tube and the filter, and then apply straps on the tube opening to prevent liquid inside the tube from spraying out.

7 Hydropneumatic System

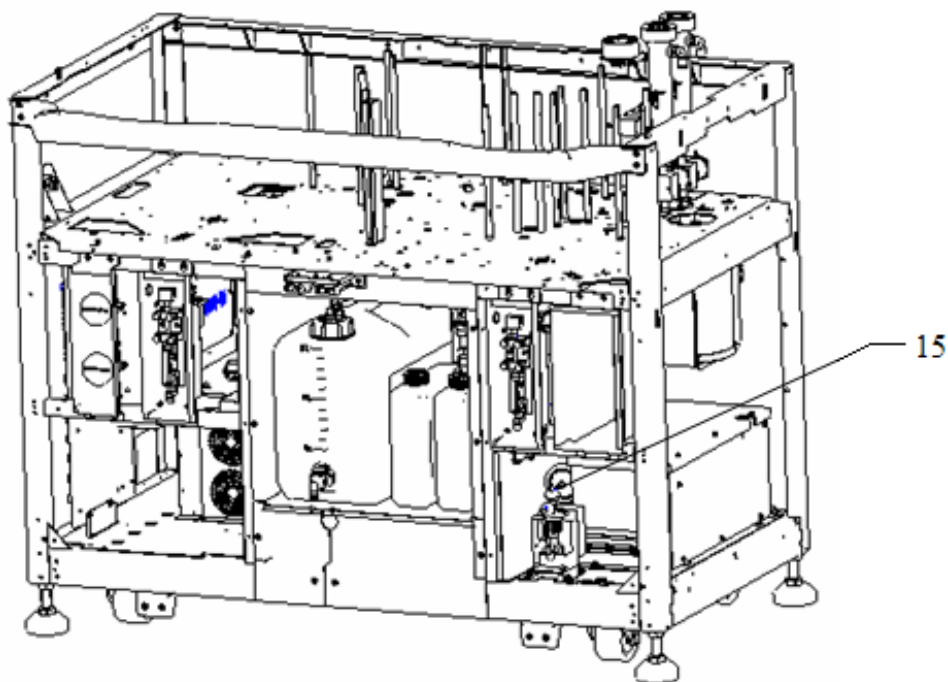
Installation Procedure

- 1 Install the filter tube and apply a strap.
- 2 Install back the Right plate assembly.

Verification steps

Refer to Section 8.12 Hydropneumatic Unit.

Figure 7.16 Assembly drawing of whole unit -- Acetabular filter



Removing/Installing Wash Probe Assembly

When to do

When fluid overflows the cuvettes or the dispensed cleaning fluid is less than the normal volume, probably a wash probe is clogged and needs to be removed and unclogged.

Steps

- 1 Manually loosen the retaining screws on the wash probe bracket.
- 2 Remove the wash probe assembly and place it in a container.
- 3 Turn on the dispense valves and vacuum valves, and then locate the clogged wash probe.
- 4 Use a needle to unclog the probe.

Installation Procedure

- 1 Align the locating holes on the wash probe assembly with the stop studs on the bracket, and then slightly tighten the retaining screws.
- 2 Lower the wash probe assembly, use an alignment tool or eyes to check the phase 7/8 wipe blocks corresponding to the cuvette wall, and then slightly adjust the wash probe assembly to prevent the wash probes from colliding with the cuvettes.
- 3 After finishing the second step, completely tighten the retaining screws on the wash probe assembly.

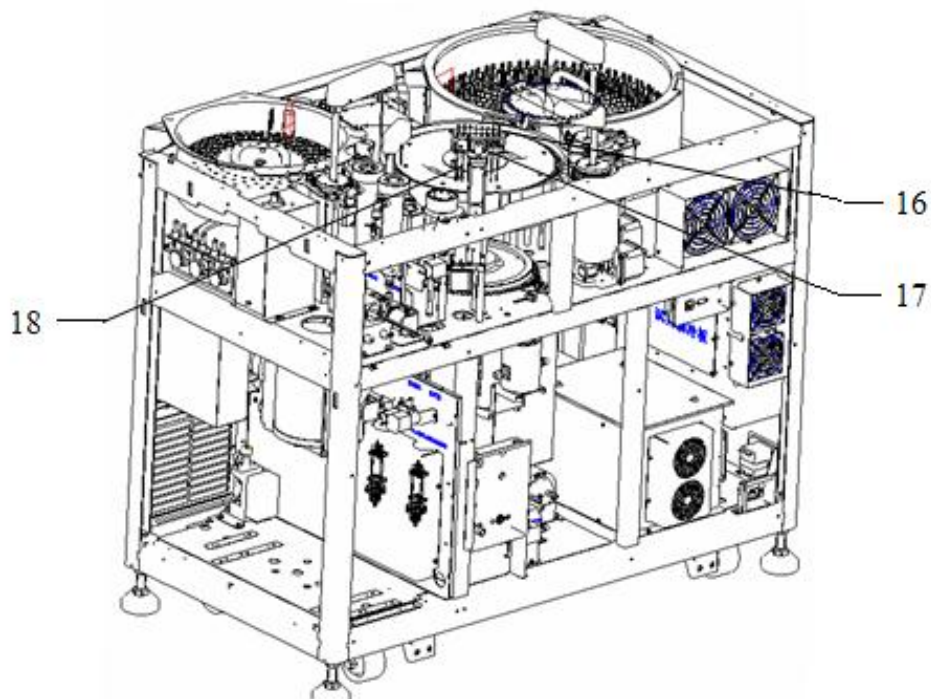
Verification steps

Refer to Section 8.12 Hydropneumatic Unit.

**NOTE**

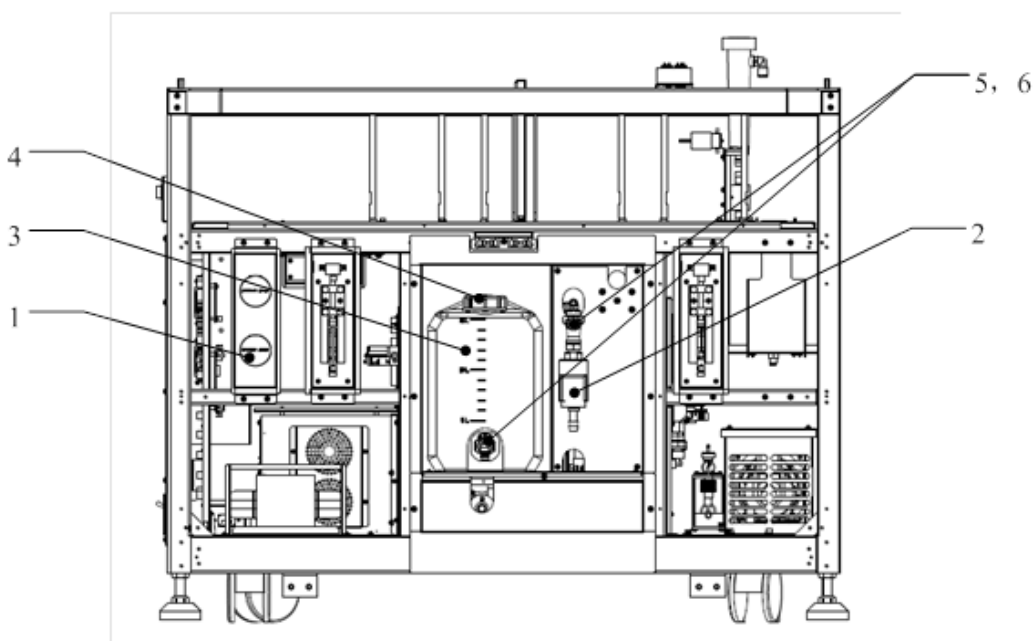
Operate carefully to prevent liquid from dropping into the reaction carousel and cuvettes.

Figure 7.17 Assembly drawing of whole unit -- Cuvette wash assembly



7.5.4 Water Supply/Drainage Module

Figure 7.18 Assembly drawing of whole unit -- Water supply/drainage module-front view



7 Hydropneumatic System

Figure 7.19 Assembly drawing of whole unit -- Water supply/drainage module-right rear view

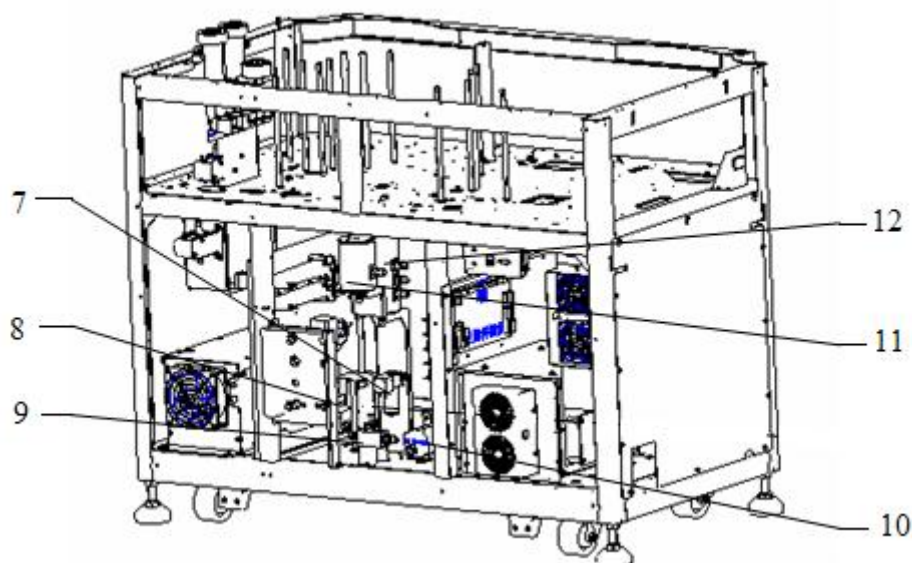


Table 7.4 List of materials

No.	FRU code and material code	Material Name	Remarks
1	115-028136-00	Deionized water pressure gauge assembly (with connector)	FRU
2	801-BA80-00020-00	Stainless steel filter core	FRU
3	801-BA80-00011-00	Water tank assembly	FRU
4	801-BA80-00012-01	Water tank fluid level sensor assembly	FRU
5	801-BA80-00172-00	Joint (quick female joint)	FRU
6	801-BA80-00173-00	Joint (quick joint)	FRU
7	801-BA34-00001-00	Rotational diaphragm pump assembly	FRU
8	115-013120-00	Inlet valve assembly (with bracket)	FRU
9	115-006999-00	Restrictive valve (self-made) assembly	FRU
10	009-002590-00	Deionized water pump cable	FRU
11	115-005644-00	High-concentration waste container assembly	FRU
12	115-005645-00	Low-concentration waste container assembly	FRU
13	801-BA80-00021-00	Sealing ring, O-shape	FRU

Removing/Installing water tank

When to do

The water tank should be maintained regularly.

Steps

- 1 Disconnect the CPC quick joint at the lower end of the water tank and drag out the water tank about 1/3 position. Take out the liquid level sensor assembly and do not drag the cable of the floater otherwise the cable may be damaged.
- 2 Drain about 3/4 water in the tank and remove the tube in the water tank.
- 2 Apply strap on the tube's opening or place the tube in a container to prevent liquid from entering the instrument.
- 3 Remove the water tank completely from the instrument and then clean it.

Installation Procedure

- 1 Insert the tube into the bottom of the water tank and adjust the tube to prevent tangling.
- 2 Place the liquid level sensor assembly into the water tank and sort the cable. Do not drag or pull the cable and the joint.
- 3 Connect the CPC quick connector with the water tank.

Verification steps

N/A.



NOTE

Do not drag or pull the cable of the liquid level sensor otherwise bad contact may occur.

Removing/Installing deionized water filter assembly

When to do

When the stainless steel filter core is regularly washed or replaced, or the deionized water filter assembly is replace.

Steps

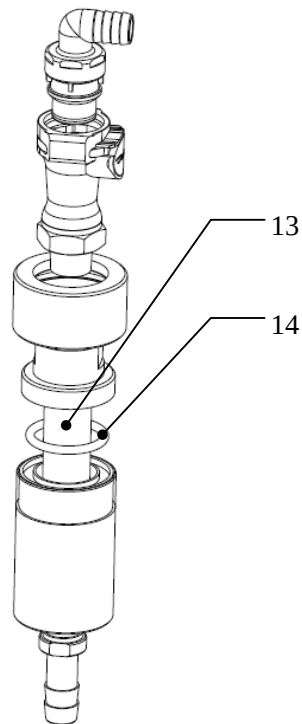
- 1 Unscrew the floaters and tubes on the concentrated wash solution tank and diluted wash solution tank, and then remove the tanks.
- 2 Disconnect the CPC connectors on the deionized water tank outlet and the water filter assembly, and then remove the filter from the mounting clamp.
- 3 Open the filter, and then clean it or replace the filter core.

Installation Procedure

- 1 Install the filter and fix it on the mounting bracket.
- 2 Install the CPC connectors on the deionized water tank outlet and the water filter assembly.
- 3 Install the floaters and tubes of the concentrated wash solution tank and diluted wash solution tank.

7 Hydropneumatic System

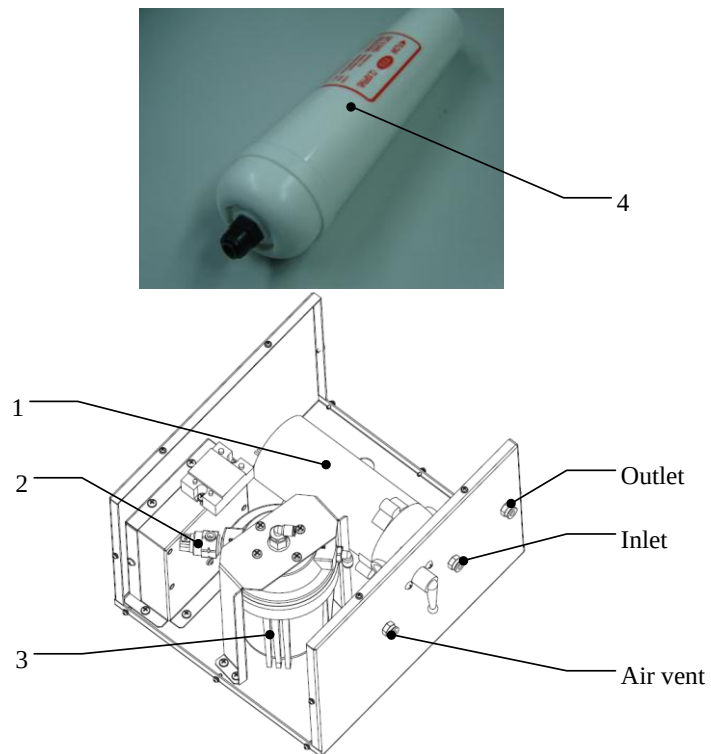
Figure 7.20 Exploded view of deionized water filter



7.5.5 External modules

Water supply module (optional)

Figure 7.21 Water supply module

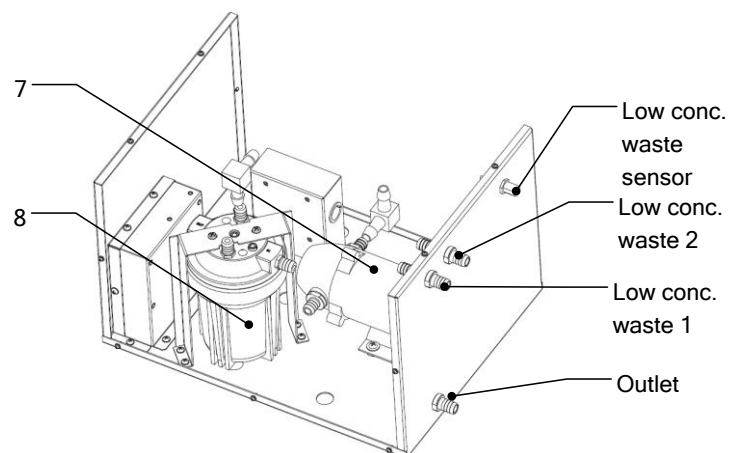


The external water supply module contains the following FRUs.

No.	FRU code and material code	Material Name	Remarks
1	801-BA40-00003-00	Diaphragm pump, 10W 24V 0.68MPa	FRU
2	801-BA40-00274-00	Pressure switch material pack	FRU
3	115-022610-00	5" tank assembly	FRU
4	115-021998-00	Water inlet filter assembly	FRU

Water drainage module (optional)

Figure 7.22 Water drainage module



The external water drainage module contains the following FRUs.

No.	FRU code and material code	Material Name	Remarks
5	801-BA40-00003-00	Diaphragm pump, 10W 24V 0.68MPa	FRU
6	115-022609-00	Waste buffer tank assembly	FRU

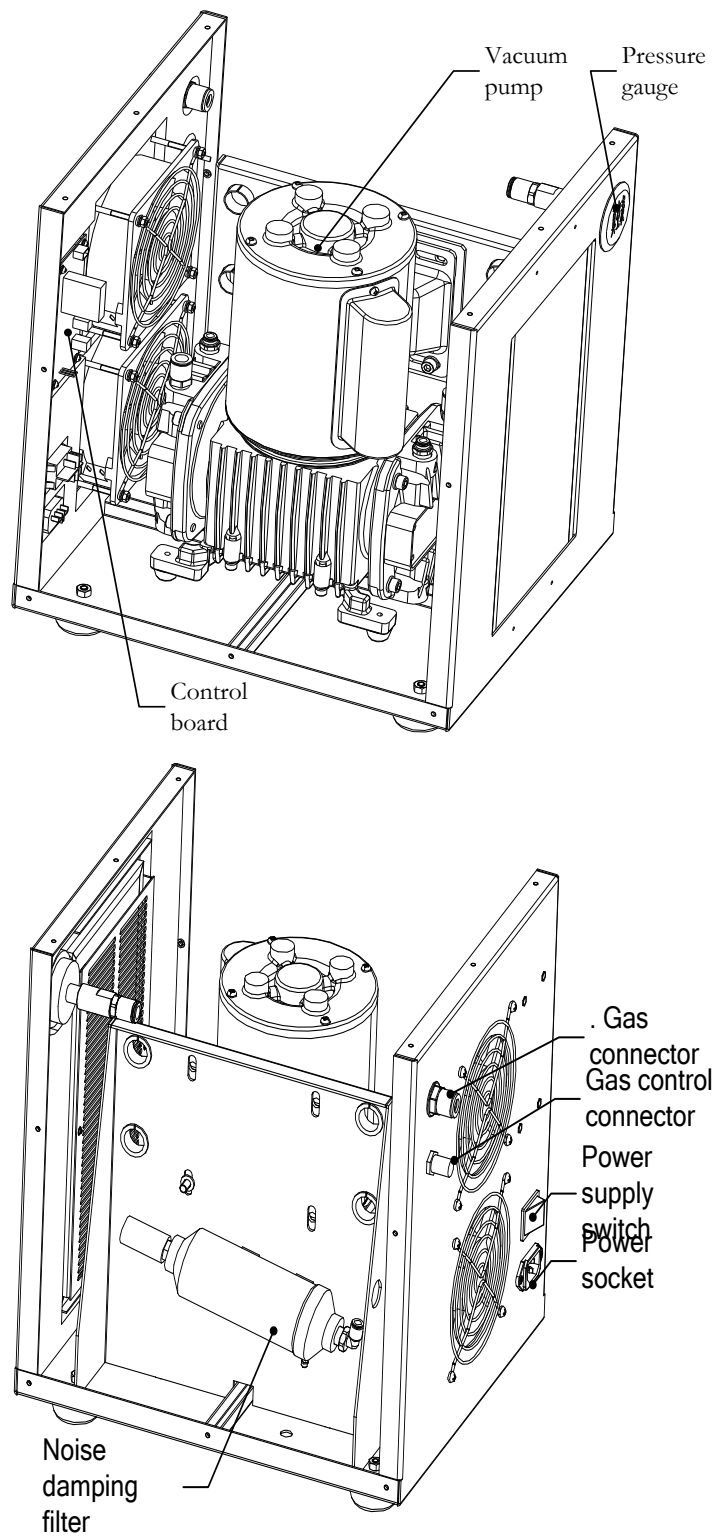
External high-concentration waste tank assembly

The external high-concentration waste tank assembly contains one FRU.

No.	FRU code and material code	Material Name	Remarks
7	801-BA38-00047-00	High-concentration waste tank cap assembly(FRU	FRU

External vacuum pump (optional)

Figure 7.23 External vacuum pump



8

Alignment

8.1 Basic Operation

8.1.1 Utility - Maintenance

No.	Description
1, 2	Maintenance and Diagnostics: used to maintain and diagnose the system at user end.
3	Alignment: used to align all mechanical positions and the hydropneumatic, pyrology units and ISE.
4	Parameters: used to inquire and configure parameters, and to export parameters of each unit.
5	SPT: provides basic performance tests for the system, including photometer test, precision test (dye), precision test (weight), cuvette residue test, and carryover test.

Figure 8.1 Utility - Maintenance



8.1.2 Alignment tool

No.	Code	Name	Precision	Quantity
1	BA80-J07	Cuvette alignment tool	/	3

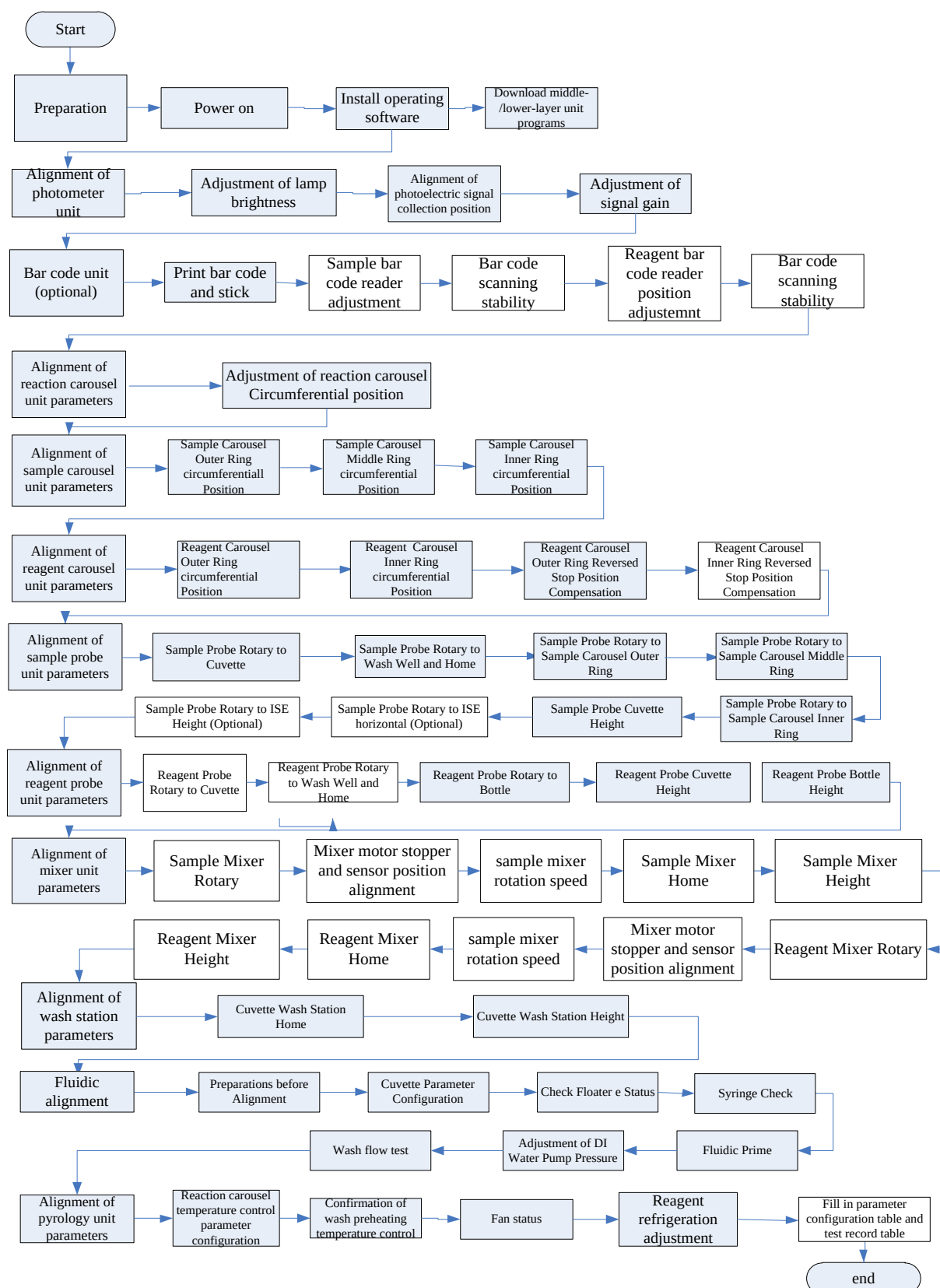
No.	Code	Name	Precision	Quantity
2	BA40-J24 (This tool will be replaced by BA49-J32 after EBJ249-F, because the carousel body will be molded as a whole.)	Reagent bottle alignment tool Reagent probe aspirate position alignment tool	/	6
3	BA48-J10	Sample position alignment tool	/	9
4	BA80-J08	Pseudo probe	/	2
5	BA49-J09	Mixer alignment tool	/	1
6	BA49-J05	Mixer alignment lever	/	2
7	BA48-J15	Mixer's cuvette horizontal position alignment tool	/	2
8	BA49-J20	Mixer motor stopper alignment tool	/	1
9	BA40-J76	Wash well alignment tool	/	1
10	BA80-J10	Wash station position alignment cylinderBA80-J10-001	/	2
11	BA80-J10	Wash station auxiliary alignment tool A80-J10-002	/	1
12	BA48-J12	Sample probe wash well alignment tool	/	2
13	BA48-J20 (This tool will be replaced by BA49-J31 after EBJ249-F, because the carousel body will be molded as a whole.)	Reagent carousel bar code reader alignment tool	/	1
14	BA48-J19	Reagent carousel alignment tool	/	1
15	BA49-J10	Sample carousel bar code reader alignment tool	/	1
16	BA48-J08	Temperature sensor fixture	/	1
17	/	Digital pressure gauges	No less than 1.5 Pa.	1
18	/	cylinder	5mL	1
19	/	feeler	0.05mm	1

8.1.3 Mechanical Reset

Alignment methods and steps:

- 1** Install the pseudo probe (BA80-J08), and block the collision sensor with folded thick paper.
- 2** Access the Alignment window and select Mech Reset to reset the mechanical parts of the system.

8.2 Alignment Procedure



8.3 Photometer Unit

Select Utility—>Maintenance—>Alignment—>Photometer Unit. The window is as shown below.

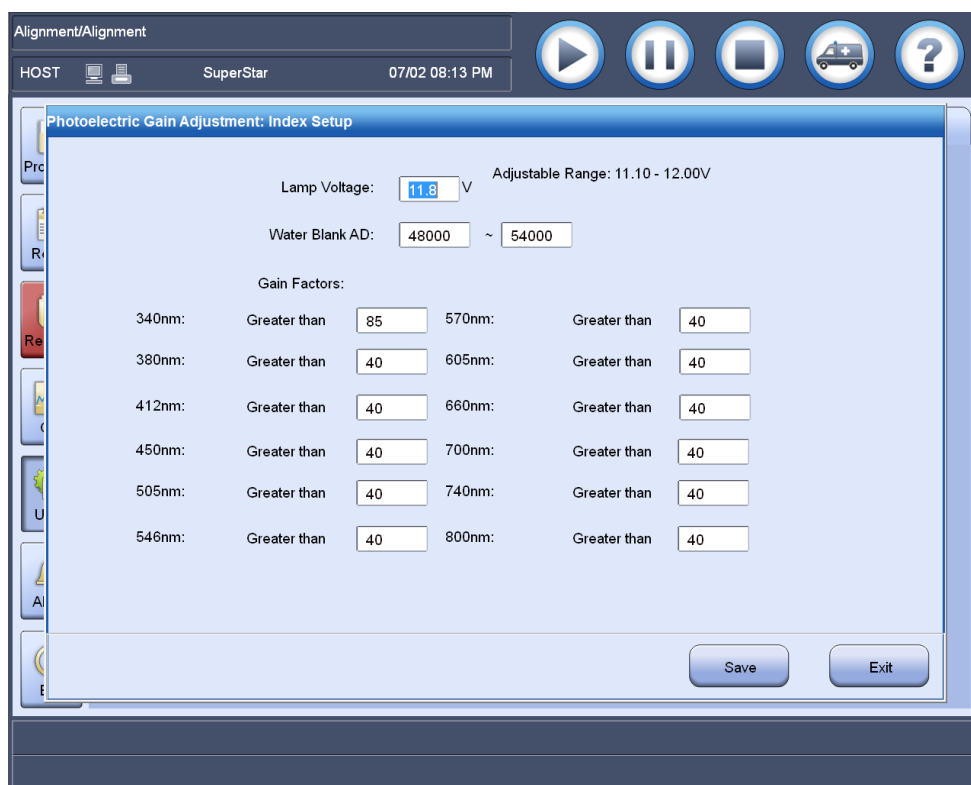


8.3.1 Lamp Brightness Adjustment

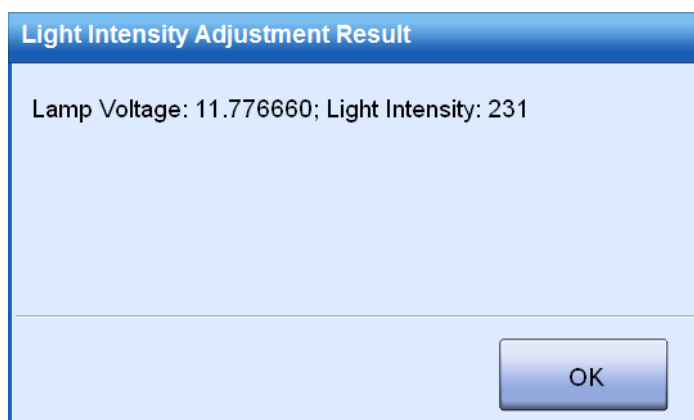
Alignment index: The lamp voltage is within the target voltage 11.10V~12.00 V.

Alignment methods and steps:

- 1 Select Index Setup to display the Photoelectric Gain Adjustment: Index Setup window. Set the Lamp Voltage as 11.80V. The indices are as shown below:



- 2 Select Save and then select Exit to close the window.
- 3 Select Lamp Brightness Adjustment, and then check if the current lamp voltage is within the target voltage range. If yes, select Cancel; otherwise, select Next to adjust the brightness.
- 4 After finishing the adjustment, a message pops up indicating adjustment succeeded. Select OK.



8.3.2 Signal Collecting Position Adjustment

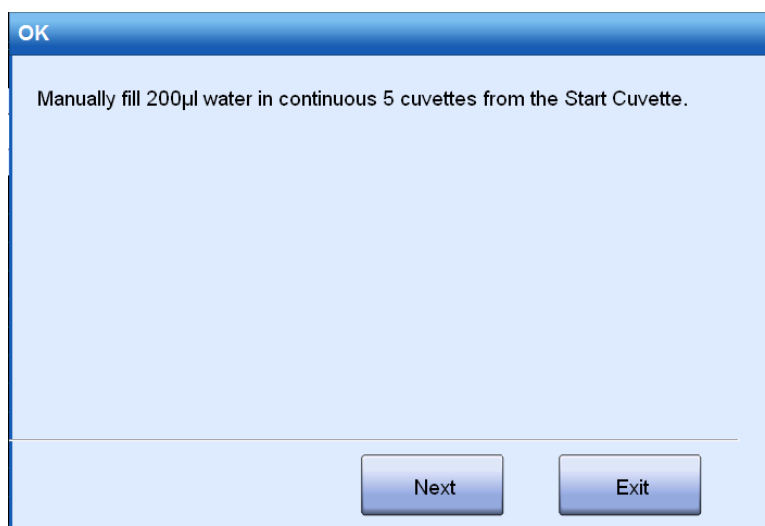
Alignment index: The photoelectric signal is in the center of the upper flat area, and the waveforms of each channel are consistent.

Alignment methods and steps:

- 1 Select Signal Collecting Position Adjustment.
- 2 Make sure the lamp has been turned on for over 1 minute. If not, wait for 1 minute, and then select Start to proceed to the next step.

8 Alignment

- 3 Find the position sensor behind the reaction carousel coder, and loose the 3 screws with an M3 hexagon spanner. Locate the sensor in the middle of adjustable range and then slightly tighten a screw.
- 4 Set up the Start Cuvette at the lower-left corner of the window. The scanning will start from it to the following 4 cuvettes. (The default is 51, that is, from 51-55.)
- 5 Select Start on the upper-right side of the window. A dialog box pops up reminding you to fill water. Use a pipettor to fill 200 μ l deionized water in cuvettes 51-55, and select Next.



- 6 Observe the waveforms in the upper part of the window and check if the photoelectric signal is in the center of the upper flat area as shown in the figure below. If the signal deviates from the center of the flat area, adjust the optical coupler.



To adjust the optical coupler, loosen the three M3 hexagon screws on the bracket, adjust the optical coupler clockwise to move the waveform leftwards or adjust it counterclockwise to move the waveform rightwards. Slightly tighten a screw and select Readjust. The software will test the photoelectric signal again.

- If the 340nm waveform is vertically beyond the display range, select Done, re-set the AD Range, and readjust the signal collecting position.

- If the optical coupler is in a proper position, select Done without any adjustment.
- 7 After confirming the signal collecting position, select Start at the bottom of the window to test photoelectric signal of all channels.
 - 8 Check if wave forms to the channels are uniform and tighten the retaining screws.

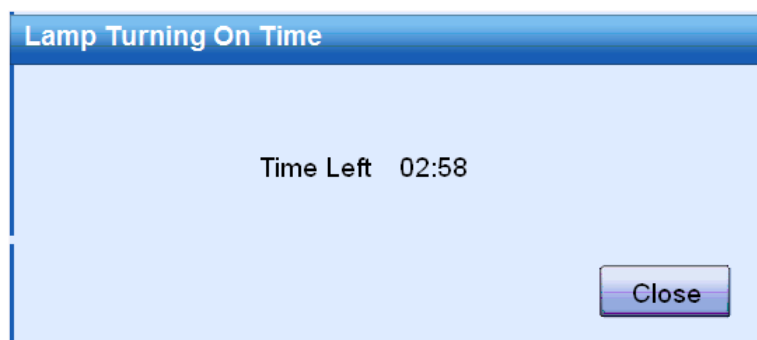
8.3.3 Photoelectric Gain Adjustment

Alignment index:

- The water blank AD value of each channel is within 48000-50000,
- the dark current of all channels is within 1-200;
- The photoelectric gain is no less than 85 for 340nm, and no less than 40 for other channels.

Alignment methods and steps:

- 1 Select Photoelectric Gain Adjustment.
- 2 Make sure the lamp has been turned on for over 3 minutes, and then select Done. If not, wait for 3 minutes.



- 3 Check cuvettes 51 to 55 for deionized water according to Step 4 on the screen. If there is, select Continue.
- 4 The photoelectric gain is adjusted automatically. When adjustment is complete, select Done.
- 5 Select Continue to test the dark current and the water blank AD of each channel.
- 6 If the water blank AD is greater than 54000, decrease the target voltage and readjust the lamp brightness. Decrease the target voltage by 0.1V for each excessive 2000 in the AD value. For instance, the water blank AD of 340nm is 56000, decrease the target voltage from 11.90V to 11.80V. Readjust the photoelectric gain till complying with the indices.
- 7 The photoelectric gain adjustment is complete.



NOTE

Alignment parameters can be saved only after each procedure is complete. Another alignment is required if a procedure is terminated. If the parameter is not successfully configured or out of range, the parameter range will be displayed by the alarm message and re-perform the alignment.

8.4 Reaction Carousel Unit

Only the Reaction Carousel Rotary procedure is performed based on the whole unit.

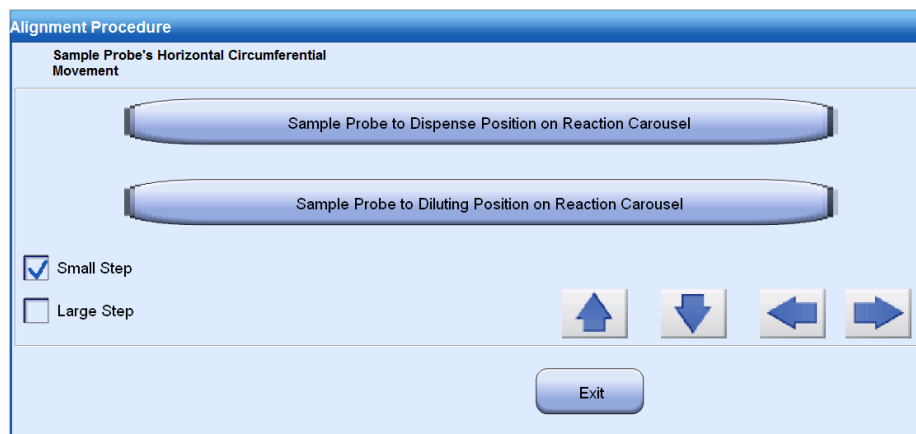
8.4.1 Reaction Carousel Rotary

Alignment index:

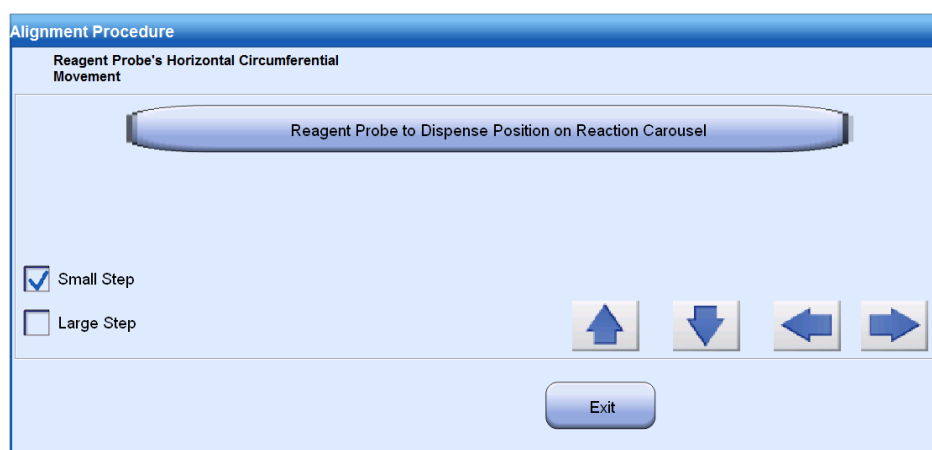
- Rotate the sample probe so that it can pass through the hole of the alignment tool placed in the sample dispense position and diluting position.
- The reagent probe can pass through the hole of the cuvette alignment tool placed in the reagent dispense position.

Operation methods and steps:

- 1 Select **Utility**—>**Maintenance**—>**Alignment**—>**Reaction Carousel Unit**.
- 2 Select Reaction Carousel Rotary.
- 3 Click Continue. The analyzer is reset mechanically.
- 4 According to the software prompts and the range displayed on the left corner, place a cuvette alignment tool (BA80-J07) in the reaction carousel optimal alignment position, diluting position, and reagent dispense position.
- 5 Move the sample probe to reaction carousel optimal alignment position, diluting position and select the up/down and left/right arrow buttons, and check if the probe tip can align with the center of the alignment tool. If it cannot, check if the sample probe is slightly deviating from cuvette in radial direction, and if necessary, adjust it with the alignment screw on the probe arm. If the radial length cannot meet the requirement, loosen the screw on the arm and the screws on the level sense board to adjust the position so as to make the anti-bumping stopper located in the middle of the slot of the optical coupler. Adjust the arm length until the requirement is met.



- 6 Select the command button to move the reagent probe to reagent probe dispense position, select the up/down arrow buttons, and then check if the probe tip can align with the dispense position. If it cannot, check if the reagent probe is slightly deviating from cuvette in radial direction, and if necessary, adjust it with the alignment screw on the probe arm. Select the left/right arrow buttons to adjust the reaction carousel and the up/down arrow buttons to adjust the reagent probe, till requests are met.



- 7 The alignment is complete. Select Exit to close the window.



NOTE

Alignment of each position is verified the next step.

The alignment tools will be used in following alignment. Leave them aside.

The left arrow button is equivalent to rotating clockwise and the right arrow button rotating counter clockwise.

When selecting Large Step, you can set up the concrete steps according to the reference range.

Once the probe arm length is determined, it must not be adjusted during alignment of probes and carousels.

Alignment parameters can be saved only after each procedure is complete. Another alignment is required if a procedure is terminated. If the parameter is not successfully configured or out of range, the parameter range will be displayed by the alarm message and re-perform the alignment.

8.5 Sample Carousel Unit

Alignment index: While rotated, the sample probe can align in each step with the hole on the sample position alignment tool.

Select Utility—>Maintenance—>Alignment—>Sample Carousel Unit. The window is as shown below.

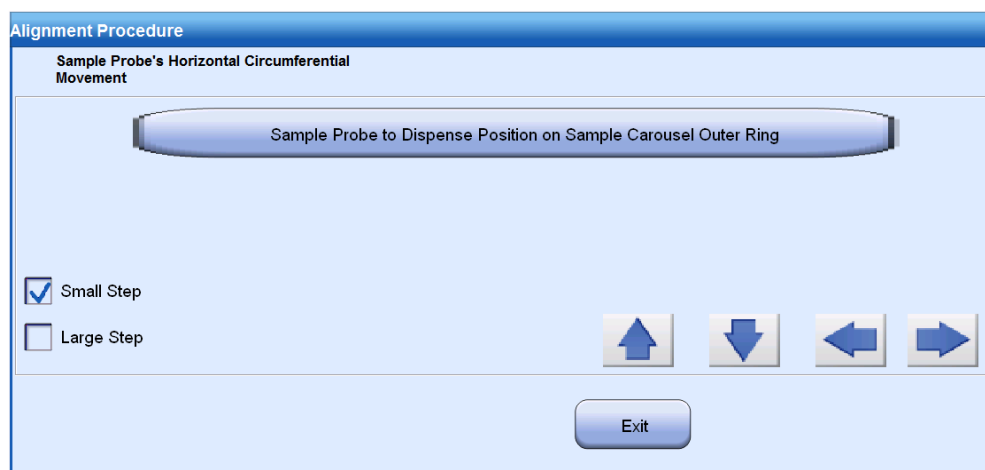


The sample carousel unit has three alignment items.

8.5.1 Sample Carousel Outer Ring Radial Position

Alignment methods and steps:

- 1 Select Sample Carousel Outer Ring Radial Position.
- 2 Place a sample position alignment tool BA48-J10 in position #1, #11 and #21 of the sample carousel, and then select Continue.
- 3 According to the requirements in Step 3-5, rotate the sample probe arm to the dispense position on the sample carousel outer ring, select the up/down and left/right arrow buttons, and check if the sample probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.



- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.5.2 Sample Carousel Middle Ring Radial Position

Alignment methods and steps:

- 1 Select Sample Carousel Middle Ring Radial Position.
- 2 Place a sample position alignment tool BA48-J10 in position #31, #41 and #51 of the sample carousel, and then select Continue.
- 3 According to the requirements in Step 3-5, rotate the sample probe arm to the dispense position on the sample carousel middle ring, select the up/down and left/right arrow buttons, and check if the sample probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.
- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.5.3 Sample Carousel Inner Ring Radial Position

Alignment methods and steps:

- 1 Select Sample Carousel Inner Ring Radial Position.
- 2 Place a sample position alignment tool BA48-J10 in position #E1, #S1 and #C1 of the sample carousel, and then select Continue.
- 3 According to the requirements in Step 3-5, rotate the sample probe arm to the dispense position on the sample carousel inner ring, select the up/down and left/right arrow buttons, and check if the sample probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.
- 4 Select Continue. The alignment is complete. Select Exit to close the window.



NOTE

When the alignment is complete, there is no need to take out the alignment tool.

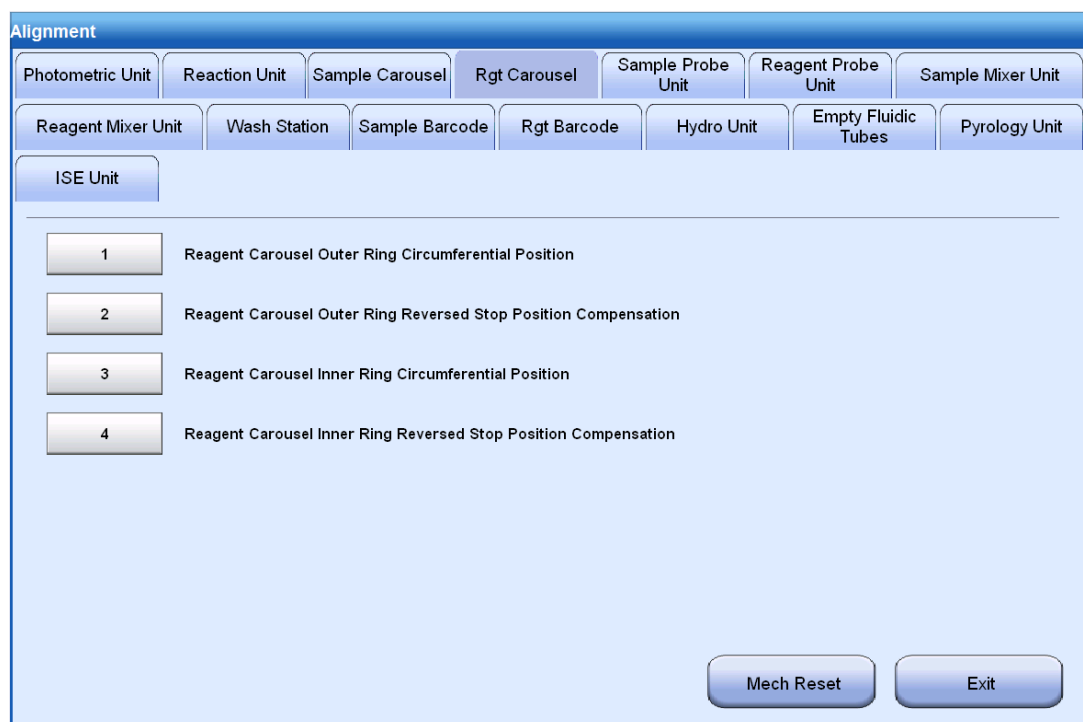
Alignment parameters can be saved only after each procedure is complete. Another alignment is required if a procedure is terminated. If the parameter is not successfully configured or out of range, the parameter range will be displayed by the alarm message and re-perform the alignment.

8.6 Reagent Carousel Unit

Alignment index:

While rotated, the reagent probe can align in each step with the hole on the reagent position alignment tool.

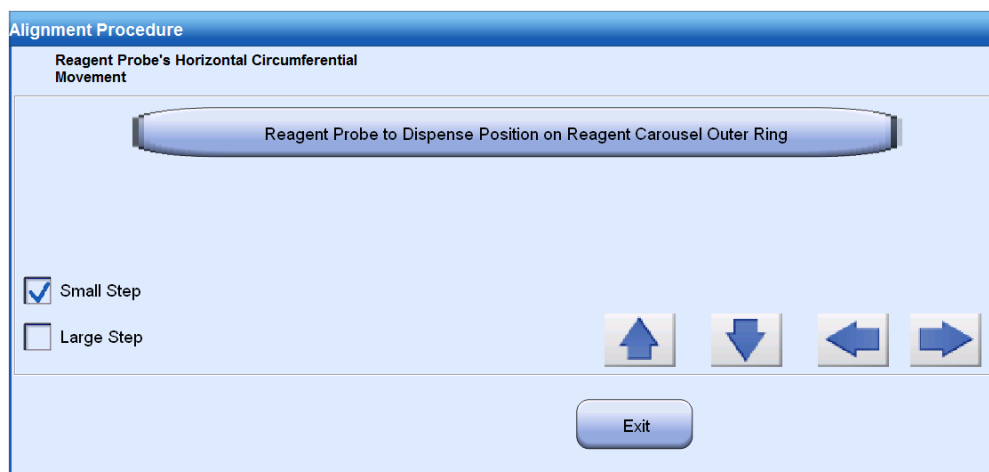
Select Utility—>Maintenance—>Alignment—>Reagent Carousel Unit. The window is as shown below.



8.6.1 Reagent Carousel Outer Ring Radial Position

Alignment methods and steps:

- 1 Select Reagent Carousel Outer Ring Radial Position.
- 2 Place the reagent probe aspirate position alignment tool (BA49-J32) in position #1, #14 and #27 of the reagent carousel, and then select Continue.
- 3 According to Step 3-5 on the screen, select the command button to rotate the reagent probe to the dispense position on the reagent carousel outer ring, select the up/down and left/right arrow buttons, and check if the reagent probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.



- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.6.2 Reagent Carousel Inner Ring Radial Position

Alignment methods and steps:

- 1 Select Reagent Carousel Inner Ring Radial Position.
- 2 Place the reagent probe aspirate position alignment tool (BA49-J32) in position #41, #54 and #67 of the reagent carousel, and then select Continue.
- 3 According to Step 3-5 on the screen, select the command button to rotate the reagent probe to the dispense position on the reagent carousel inner ring, select the up/down and left/right arrow buttons, and check if the reagent probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.
- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.6.3 Reagent Carousel Outer Ring Reversed Stop Position Compensation

Alignment methods and steps:

- 1 Select Reagent Carousel Outer Ring Reversed Stop Position Compensation.
- 2 Place a reagent position alignment tool (BA40-J24) in position #1, #14 and #27 of the reagent carousel, and then select Continue.
- 3 According to the requirements in Step 3-5, select the command button to rotate the reagent probe arm to the dispense position on the reagent carousel outer ring, select the up/down and left/right arrow buttons, and check if the reagent probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.
- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.6.4 Reagent Carousel Inner Ring Reversed Stop Position Compensation

Alignment methods and steps:

- 1 Select Reagent Carousel Inner Ring Reversed Stop Position Compensation.
- 2 Place a reagent position alignment tool (BA40-J24) in position #41, #54 and #67 of the reagent carousel, and then select Continue.

- 3 According to the requirements in Step 3-5, select the command button to rotate the reagent probe arm to the dispense position on the reagent carousel inner ring, select the up/down and left/right arrow buttons, and check if the reagent probe can pass through the alignment tool. If it cannot, adjust the steps with the left/right arrow buttons till requests are met.
 - 4 Select Continue. The alignment is complete. Select Exit to close the window.
-



NOTE

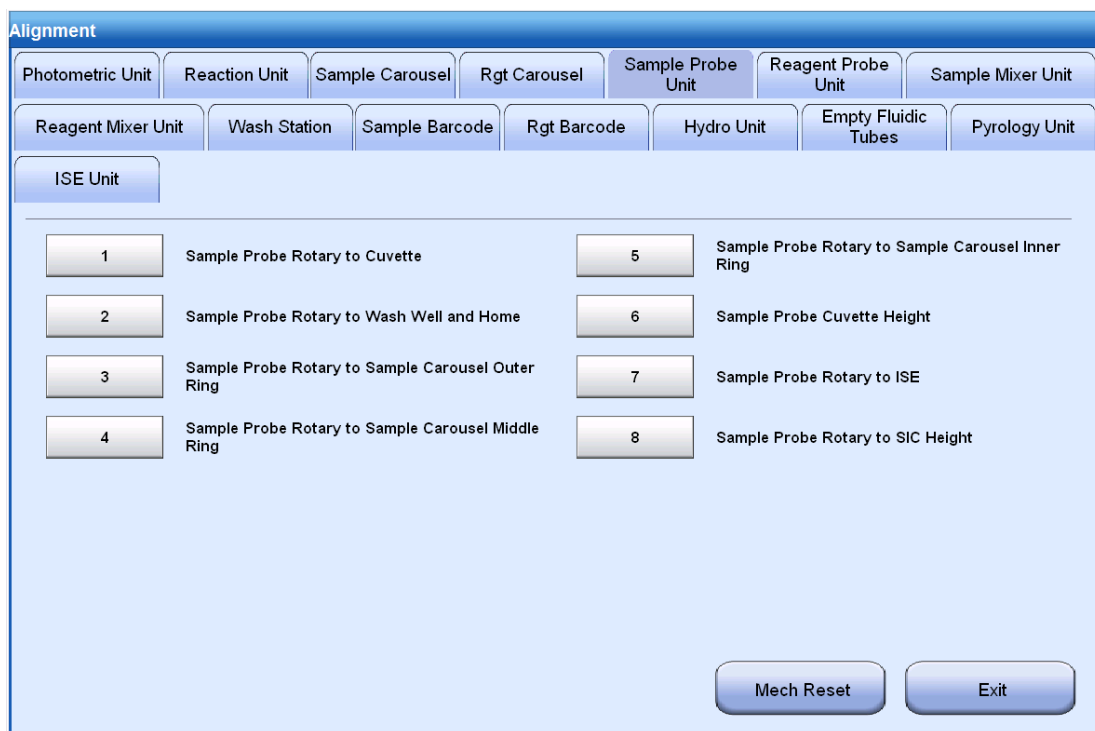
Alignment parameters can be saved only after each procedure is complete. Another alignment is required if a procedure is terminated.

If the parameter is not successfully configured or out of range, the parameter range will be displayed by the alarm message and re-perform the alignment. If the parameter is out of range, adjust the position of the sensor. Perform the alignment procedure again.

8.7 Sample Probe Unit

Select Utility—>Maintenance—>Alignment—>Sample Probe Unit.

The sample probe unit has 8 alignment items, as shown in the figure below:



8.7.1 Sample Probe Rotary to Cuvette

Alignment index: The sample probe tip can enter the central hole on the cuvette alignment tool.

Alignment methods and steps:

- 1 Select Sample Probe Rotary to Cuvette.
- 2 If no cuvette alignment tool (BA80-J07) has been placed, place one (BA80-J07) in reaction carousel optimal alignment position and diluting position and then select Continue.
- 3 According to Step 3 on the screen, select the down arrow button to lower the reagent probe to the top of the alignment tool so that it can align with the hole.
- 4 Use an M3 hexagon wrench to loosen the two retaining screws on the probe arm, adjust the arm radially till the probe can enter the hole on the alignment tool. Press the probe arm and tighten the two screws.
- 5 If there is deviation when the screws are tightened, select the left/right arrow buttons to adjust the parameter till requests are met.
- 6 Select Continue and verify the alignment with the up/down arrow buttons. If the index is not satisfied, select Previous to try again.
- 7 Select Continue to save parameters and exit the window.

8.7.2 Sample Probe Rotary to Wash Well and Home

Alignment index:

- Horizontal position: The sample probe tip can enter the central hole on the wash well alignment tool (BA48-J12).
- Vertical position: The sample probe enters the wash well alignment tool (BA48-J12), and is lifted for 0.15mm, through which a 0.15mm plug gauge can pass through.

Alignment methods and steps:

- 1 Select Sample Probe Rotary to Wash Well and Home.
- 2 According to Step 2 on the screen, place a wash well alignment tool (BA48-J12) in the sample probe wash well, and then select Continue.
- 3 According to Step 3 on the screen, select the up/down and left/right arrow buttons to move the sample probe so that it can lower into the center on the alignment tool. If the sample probe is greatly deviating from the alignment tool, use a big cross head screwdriver to loosen the two screws on the wash well, adjust the wash well till the sample probe can enter to the central hole on the alignment tool, and then tighten the screws.
- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 In Step 5, the sample probe is adjusted vertically above the wash well. Select up/down arrow buttons to adjust the lifting height till a 0.15mm clearance gauge can pass through.
- 6 The alignment is verified in Step 6. If requirements are not met, return to Step 5.
- 7 In Step 7, remove the alignment tool.
- 8 Select Continue to save parameters and exit the window.

8.7.3 Sample Probe Rotary to Sample Carousel Outer Ring

Alignment index: The sample probe tip can enter the central hole on the sample position alignment tool.

Alignment methods and steps:

- 1 Select Sample Probe Rotary to Sample Carousel Outer Ring.
- 2 Place a sample position alignment tool (BA48-J10) in position #1, #11 and #21 of the sample carousel, and then select Continue.
- 3 According to Step 3-5 on the screen, select the up/down and left/right arrow buttons to move the sample probe so that it can enter into the central hole on the alignment tool.
- 4 Select Continue to save parameters and exit the window.

8.7.4 Sample Probe Rotary to Sample Carousel Middle Ring

Alignment index: The sample probe tip can enter the central hole on the sample position alignment tool.

Alignment methods and steps:

- 1 Select Sample Probe Rotary to Sample Carousel Middle Ring.
- 2 Place a sample position alignment tool (BA48-J10) in position #31, #41 and #51 of the sample carousel, and then select Continue.
- 3 According to Step 3-5 on the screen, select the up/down and left/right arrow buttons to move the sample probe so that it can enter into the central hole on the alignment tool.
- 4 Select Continue to save parameters and exit the window.

8.7.5 Sample Probe Rotary to Sample Carousel Inner Ring

Alignment index: The sample probe tip can enter the central hole on the sample position alignment tool.

Alignment methods and steps:

- 1 Select Sample Probe Rotary to Sample Carousel Inner Ring.
- 2 Place a sample position alignment tool (BA48-J10) in position #1, #S1 and #C1 of the sample carousel, and then select Continue.
- 3 According to Step 3-5 on the screen, select the up/down and left/right arrow buttons to move the sample probe so that it can enter into the central hole on the alignment tool.
- 4 Select Continue to save parameters and exit the window.

8.7.6 Sample Probe Cuvette Height

Alignment index: The sample probe is lifted for 0.15mm from the bottom of a cuvette, and a No.15 clearance gauge can pass through.

Alignment methods and steps:

- 1 Select Sample Probe Cuvette Height. The mechanical parts of the system are reset.
- 2 Check the position #10, #40 and #70 for reaction cuvettes according to Step 2.
- 3 In Step 3, select the up/down arrow buttons to adjust the lifting height till a 0.15mm clearance gauge can pass through
- 4 In Step 4 and 5, adjust the vertical extreme position of the probe in the remaining two positions according to Step 3.
- 5 Select Continue to save parameters and exit the window.

8.7.7 Sample Probe Rotary to ISE (Optional)

Alignment index: The sample probe has its tip above the ISE sample injection port but deviating slightly from the central hole. The probe tip should not contact with the inwall of sample inject port.

Alignment methods and steps:

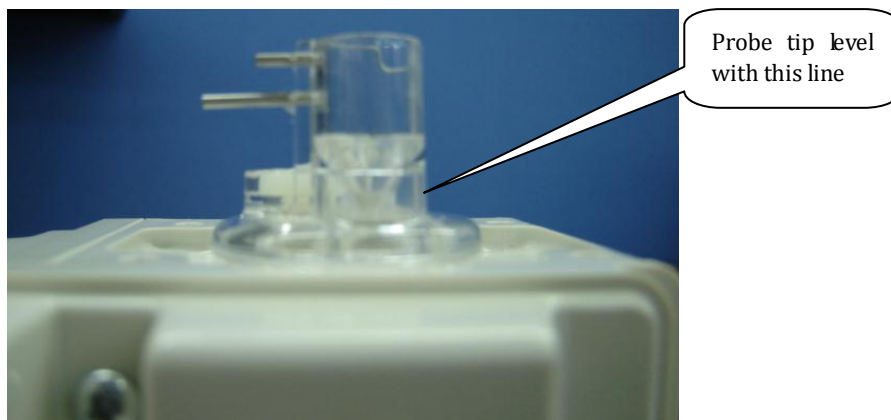
- 1 Ensure the ISE has been installed.
- 2 Select S Probe Rotary to ISE.
- 3 According to Step 3, select the down arrow button to lower the sample probe to above the ISE sample injection port. Select the left/right arrow buttons to adjust the sample probe radially till it is 1mm away from the center of the sample injection port or deviating slightly from the central hole. The probe tip should not contact with the inwall of sample inject port.
- 4 If the sample probe fails by adjusting parameters, loose the screws fixing the ISE module on the chassis, and then move it till requests are met.
- 5 If the problem still remains, use a cross head screwdriver to loosen the screws on the ISE module and move it till requests are met.
- 6 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 7 Select Continue to save parameters and exit the window.

8.7.8 Sample Probe Rotary to SIC Height (Optional)

Alignment index: The sample probe has its tip level with the third transition line in the sample injection cup.

Alignment methods and steps:

- 1 Check if the S Probe Rotary to ISE alignment has been completed.
- 2 Select S Probe Rotary to SIC Height
- 3 According to Step 3 on the screen select the up/down arrow buttons to make the probe tip level with the third transition line inside the sample injection cup. (You can see three transition lines, and the probe tip shall be level with the one at the bottom, as shown in the figure below.)
- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 Select Continue to save parameters and exit the window.



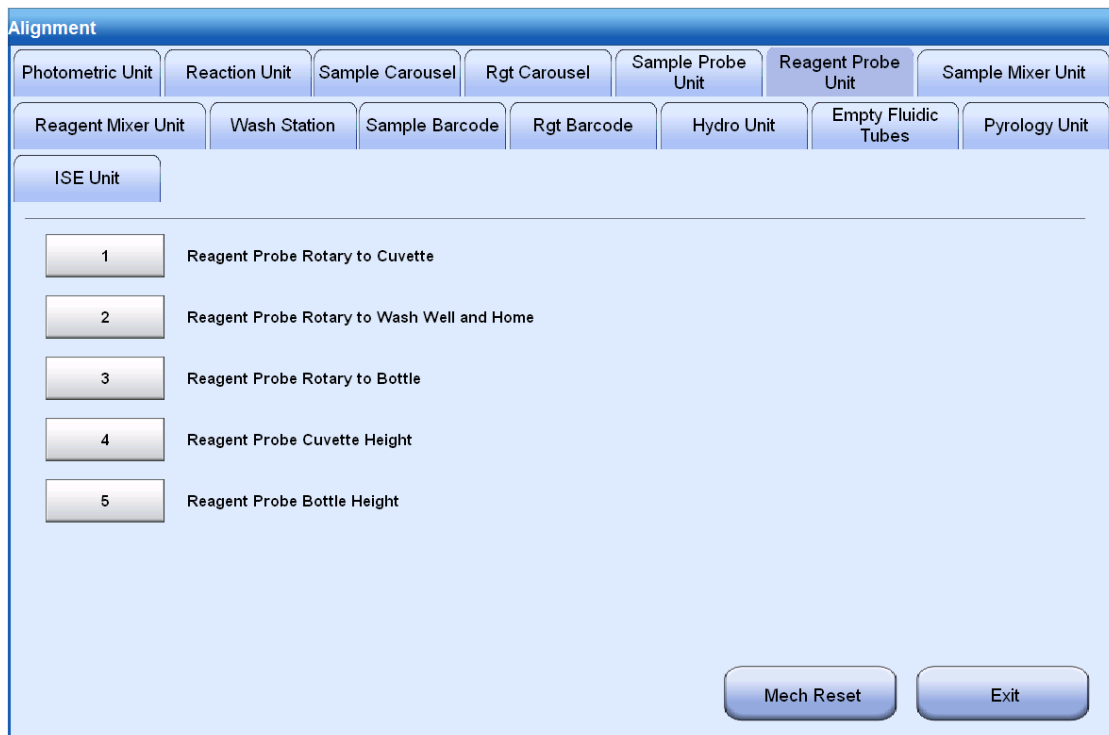
NOTE

Alignment parameters can be saved only after each procedure is complete. Another alignment is required if a procedure is terminated.

If the parameter is not successfully configured or out of range, the parameter range will be displayed by the alarm message and re-perform the alignment.

8.8 Reagent Probe Unit

Select Utility—>Maintenance—>Alignment—>Reagent Probe Unit. The window is as shown below.



8.8.1 Reagent Probe Rotary to Cuvette

Alignment index: The reagent probe tip can enter the central hole on the cuvette alignment tool (BA80-J07).

Alignment methods and steps:

- 1 Select Reagent Probe Rotary to Cuvette.
- 2 If no cuvette alignment tool (BA80-J07) has been placed, place one (BA80-J07) in reaction carousel optimal alignment position and then select Continue.
- 3 According to Step 3 on the screen select the down arrow button to lower the reagent probe to the top of the alignment tool so that it can align with the hole. Use an M3 hexagon wrench to loosen the two retaining screws on the probe arm, adjust the arm radially till the probe can enter the hole on the alignment tool. Press the probe arm and tighten the two screws. If there is deviation when the screws are tightened, select the left/right arrow buttons to adjust the parameter till requests are met.
- 4 The alignment is verified in Step 5. If requirements are not met, return to the previous step.
- 5 Select Continue to save parameters and exit the window.

8.8.2 Reagent Probe Rotary to Wash Well and Home

Alignment index:

- Horizontal position: The reagent probe tip can enter the central hole on the wash well alignment tool (BA48-J12).

- Vertical position: The reagent probe enters the wash well alignment tool (BA48-J12), and is lifted for 0.15mm, through which a 0.15mm plug gauge can pass through.

Alignment methods and steps:

- 1 Select Reagent Probe Rotary to Wash Well and Home.
- 2 According to Step 2 on the screen, place a wash well alignment tool (BA48-J12) in the reagent probe wash well, and then select Continue.
- 3 According to Step 3 on the screen, select the up/down and left/right arrow buttons to move the reagent probe so that it can lower into the center on the alignment tool. If the reagent probe is greatly deviating from the alignment tool, use a big cross head screwdriver to loosen the two screws on the wash well, adjust the wash well till the reagent probe can enter to the central hole on the alignment tool, and then tighten the screws.
- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 In Step 5, the reagent probe is adjusted vertically above the wash well. Select up/down arrow buttons to adjust the lifting height till a 0.15mm clearance gauge can pass through.
- 6 The alignment is verified in Step 6. If requirements are not met, return to Step 5.
- 7 Select Continue to save parameters and exit the window.

8.8.3 Reagent Probe Rotary to Bottle

Alignment index: The reagent probe tip can enter the central hole on the reagent bottle alignment tool (BA40-J24).

Alignment methods and steps:

- 1 Select Reagent Probe Rotary to Bottle.
- 2 Place a reagent bottle alignment tool (BA40-J24) in position #1, #14 and #27 of the reagent carousel outer ring and position #41, #54 and #67 of the inner ring, and then select Continue.
- 3 According to Step 3-8 on the screen, select the down arrow button to lower the reagent probe to the top of the alignment tool so that it can align with the hole. Select the left/right arrow buttons to adjust the reagent probe radially till it can enter the central hole on the alignment tool.
- 4 Select Continue. The alignment is complete. Select Exit to close the window.

8.8.4 Reagent Probe Cuvette Height

Alignment index: The reagent probe is lifted for 0.15mm from the bottom of a cuvette, and a No.15 clearance gauge can pass through.

Alignment methods and steps:

- 1 Select Reagent Probe Cuvette Height. The mechanical parts of the system are reset.
- 2 In Step 3, select the up/down arrow buttons to adjust the lifting height till a 0.15mm clearance gauge can pass through.
- 3 In Step 4 and 5, adjust the vertical extreme position of the probe in the remaining two positions according to Step 3.
- 4 Select Continue to save parameters and exit the window.

8.8.5 Reagent Probe Bottle Height

Alignment index: The reagent probe is lifted for 0.15mm from the bottom of a reagent bottle, and a No.15 clearance gauge can pass through.

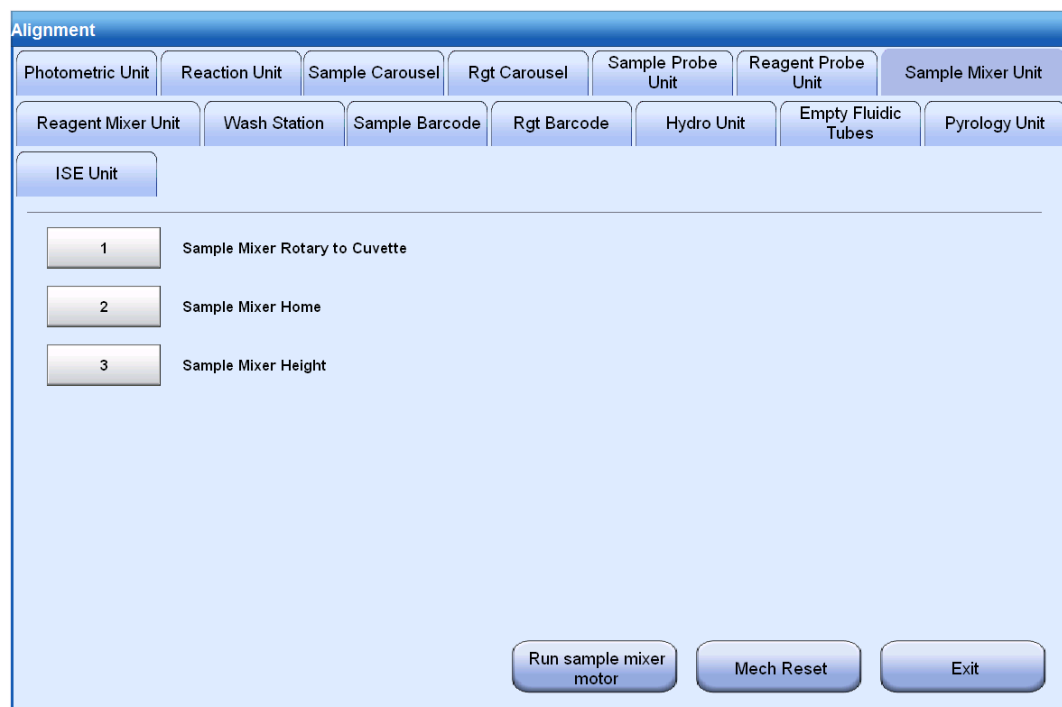
Alignment methods and steps:

- 1** Select Reagent Probe Bottle Height. The reagent probe resets mechanically.
- 2** According to Step 2, place empty reagent bottles in position #1, #14 and #27 of the reagent carousel outer ring.
- 3** In Step 3, select the up/down arrow buttons to adjust the lifting height till a 0.15mm clearance gauge can pass through.
- 4** In Step 4 and 5, adjust the vertical extreme position of the probe in the remaining two positions according to Step 3.
- 5** Select Continue to save parameters and exit the window.

8.9 Mixer Unit

The mixer unit includes the sample mixer and reagent mixer.

Select Utility—>Maintenance—>Alignment—>Sample Mixer Unit or Reagent Mixer Unit. The window is as shown below:



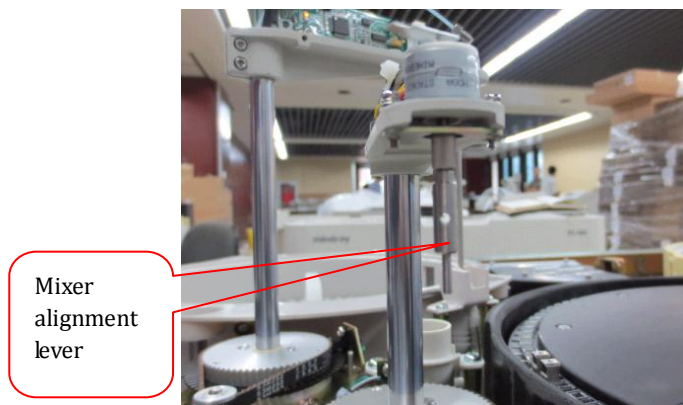
8.9.1 Sample Mixer Rotary

Alignment index: The mixer alignment lever (BA48-J05) can pass through the central hole on the horizontal position alignment tool (BA48-J15).

Alignment methods and steps:

- 1 Select Sample Mixer Rotary. The mechanical parts of the system are reset.
- 2 According to Step 2 on the screen, place a horizontal position alignment tool (BA48-J15) in reaction carousel optimal alignment position and diluting position. Move the sample mixer to the horizontal mixing position, and install an alignment lever (BA48-J16) on it.
- 3 In Step 3, select the up/down arrow buttons to lower the alignment lever (BA48-J16) to the top of the mixer horizontal position alignment tool (BA48-J15). Use an M3 hexagon wrench to loosen the retaining screw on the sample mixer arm, align the lever with the center of the alignment tool, and then slightly press the arm to tighten it. If the lever is not radially enough, use a cross head screwdriver to adjust the motor till the lever aligns with the alignment tool.
- 4 If the lever is not correct horizontally, select the left/right arrow buttons in Step 4 to adjust so that it can align with the alignment tool.
- 5 The operation in Step 4 is verified in Step 5. If requirements are not met, return to Step 4.
- 6 Select Continue. The sample mixer moves to the top of the alignment tool (BA48-J15) in dilution position. Use the up/down and left/right arrow buttons to adjust the mixer till the lever lies in the alignment tool.

- 7 The operation in Step 6 is verified in Step 7. If requirements are not met, return to Step 6.
- 8 Select Continue to save parameters and exit the window.
- 9 Remove the mixer horizontal position alignment tool (BA48-J15) and restore the cuvettes.



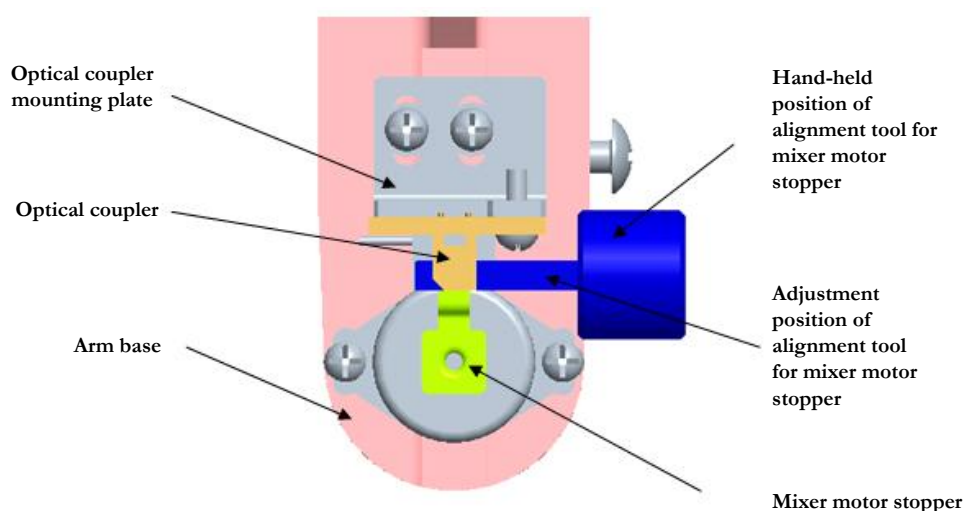
8.9.2 Mixer motor stopper and sensor position alignment

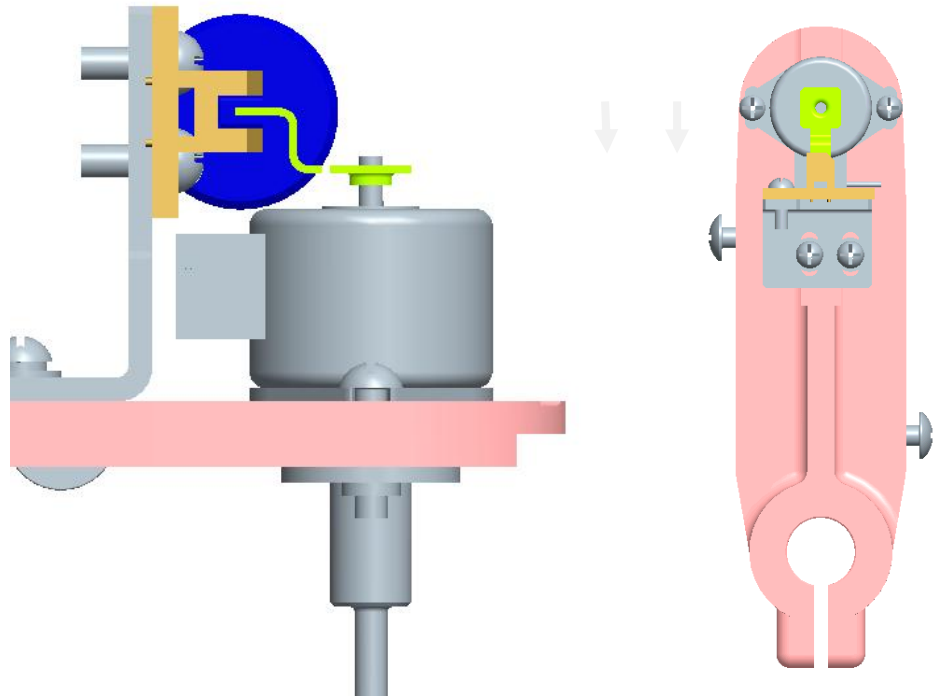
Alignment index: When the mixer motor stopper contacts with the inner side of its alignment tool, the motor stopper will not interfere with the alignment tool or be jammed as you rotate the stopper manually.

Alignment methods and steps:

Tip: This procedure should be performed after Mixer Rotary to Cuvette is done. If the position of the motor has been adjusted, perform this procedure.

- 1 Insert the alignment tool into the slot of the sensor and the slot of the tool faces outward.
- 2 Rotate the stopper to the slot and the stopper should contact with the slot of the tool. The motor stopper will not interfere with the alignment tool or be jammed as you rotate the stopper manually, otherwise adjust the screws on the sensor bracket until the requirement is met.
- 3 Remove the alignment tool and check the position of the stopper until the requirement is met.





8.9.3 Run Mixer motor

Alignment index: When the mixer motor is running, no abnormal rotation alarm is given and the stopper of the motor should not interfere with the optical coupler.

Alignment methods and steps:

- 1 Select Run sample mixer motor and check sample mixer rotation status. No abnormal rotation speed alarm is given out.

8.9.4 Sample Mixer Home

Alignment index: The lever can enter into the central hole on the wash well alignment tool (BA40-J76).

Alignment methods and steps:

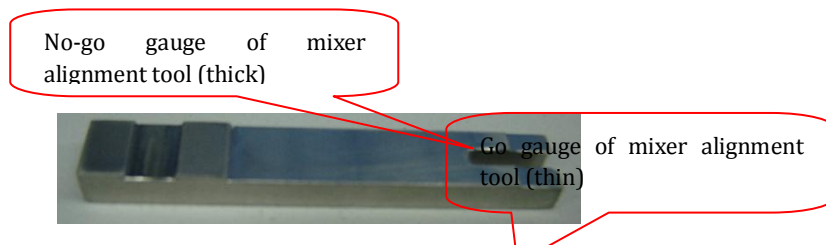
- 1 Select Sample Mixer Home. The sample mixer unit resets mechanically.
- 2 According to Step 2 on the screen, place a wash well alignment tool (BA40-J76) in the sample mixer wash well.
- 3 In Step 3, select the up/down and left/right arrow buttons so that the lever can enter into the central hole on the alignment tool.
- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 In Step 5, remove the wash well alignment tool (BA40-J76).
- 6 Select Continue to save parameters and exit the window.

8.9.5 Sample Mixer Height

Alignment index: The lever lowers into the cuvette, through the gap between them the go gauge of the mixer alignment tool (BA49-J09) can pass through and the no-go gauge cannot.

Alignment methods and steps:

- 1 Select Sample Mixer Height. Make sure the sample mixing position is holding a cuvette instead of alignment tool.
- 2 Select Continue. After the system resets mechanically, the sample mixer moves into the cuvette.
- 3 Select the up/down arrow buttons, and insert an alignment tool (BA49-J09) between the mixer's straight-knurled column and cuvette edge. Gently push the alignment tool (BA40-J06) to observe if the Go Gauge can easily pass while the No-go Gauge cannot.



- 4 The operation in Step 2 is verified in Step 3. If requirements are not met, return to Step 2.
- 5 Select Continue to save parameters and exit the window.

8.9.6 Reagent Mixer Rotary

Alignment index: The mixer alignment lever (BA48-J05) can pass through the central hole on the horizontal position alignment tool (BA48-J15).

Alignment methods and steps:

- 1 Select Reagent Mixer Rotary.
- 2 According to Step 2 on the screen, place a mixer alignment tool (BA48-J15) in reaction carousel optimal alignment position of the reaction carousel. Move the reagent mixer to the horizontal mixing position, and install an alignment lever (BA49-J05) on it.
- 3 In Step 3, select the up/down arrow buttons to lower the alignment lever to the top of the mixer horizontal position alignment tool (BA48-J15). Use an M3 hexagon wrench to loosen the retaining screw on the reagent mixer arm, and align the lever with the center of the alignment tool (BA48-J15). If the lever is not radially enough, use a cross head screwdriver to loosen the motor and adjust it till the lever aligns with the alignment tool.
- 4 If the lever is not correct horizontally, select the left/right arrow buttons in Step 4 to adjust so that it can align with the alignment tool.
- 5 The operation in Step 4 is verified in Step 5. If requirements are not met, return to Step 4.
- 6 Select Continue to save parameters and exit the window.
- 7 Remove the mixer horizontal position alignment tool (BA48-J15) and restore the cuvettes.

8.9.7 Mixer motor stopper and sensor position alignment

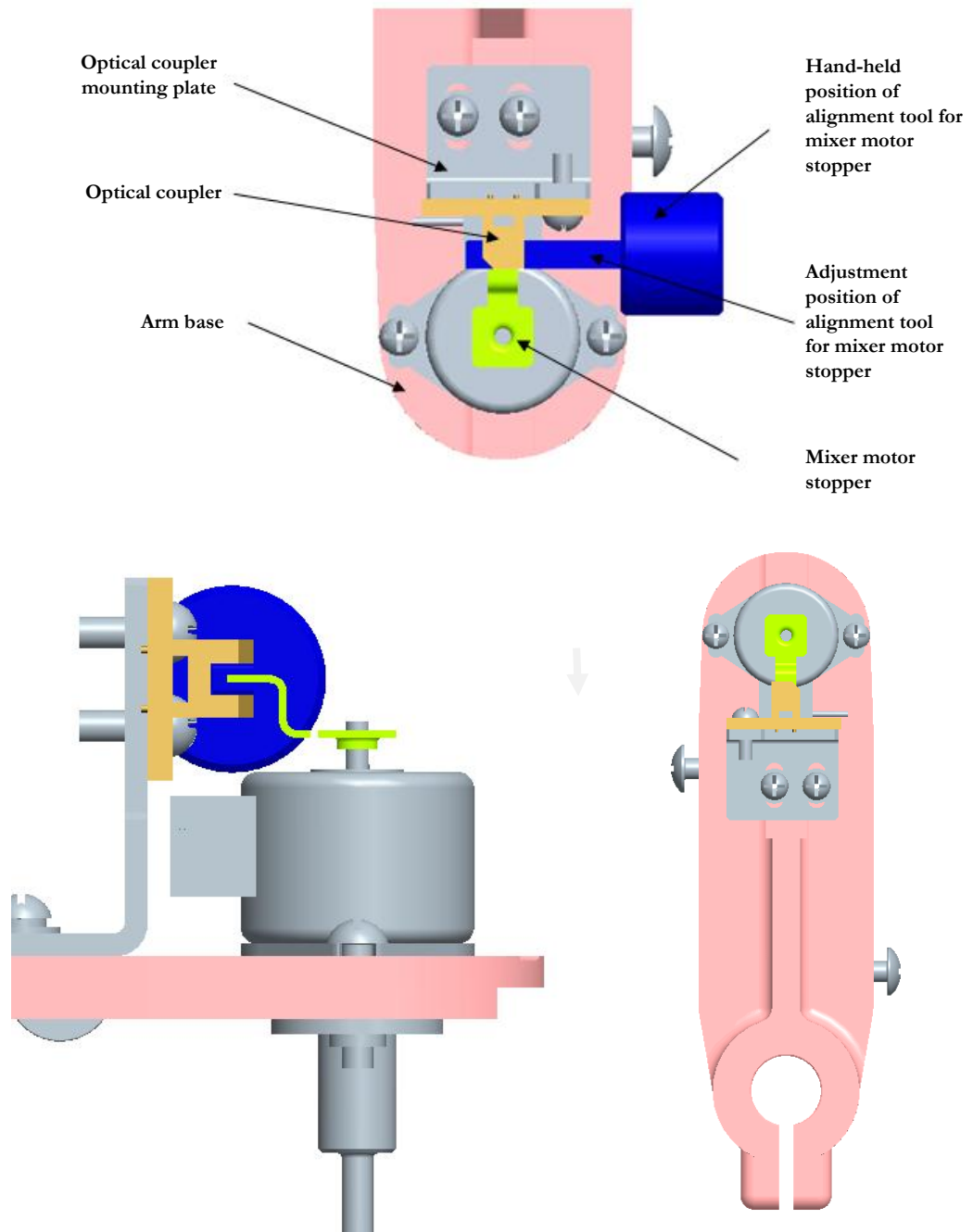
Alignment index: When the mixer motor stopper contacts with the inner side of its alignment tool, the motor stopper will not interfere with the alignment tool or be jammed as you rotate the stopper manually.

Alignment methods and steps:

Tip: This procedure should be performed after Mixer Rotary to Cuvette is done. If the position of the motor has been adjusted, perform this procedure.

8 Alignment

- 1 Insert the alignment tool into the slot of the sensor and the slot of the tool faces outward.
- 2 Rotate the stopper to the slot and the stopper should contact with the slot of the tool. The motor stopper will not interfere with the alignment tool or be jammed as you rotate the stopper manually, otherwise adjust the screws on the sensor bracket until the requirement is met.
- 3 Remove the alignment tool and check the position of the stopper until the requirement is met.



8.9.8 Run Mixer motor

Alignment index: When the mixer motor is running, no abnormal rotation alarm is given and the stopper of the motor should not interfere with the optical coupler.

Alignment methods and steps:

Select Run reagent mixer motor and check reagent mixer rotation status. No abnormal rotation speed alarm is given out.

8.9.9 Reagent Mixer Home

Alignment index: The lever can enter into the central hole on the wash well alignment tool (BA40-J76).

Alignment methods and steps:

- 1 Select Reagent Mixer Home.
- 2 According to Step 2 on the screen, place a wash well alignment tool in the reagent mixer wash well (BA40-J76).
- 3 In Step 3, select the up/down and left/right arrow buttons so that the lever can enter into the central hole on the alignment tool (BA40-J76).
- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 In Step 5, remove the alignment tool.
- 6 Select Continue to save parameters and exit the window.

8.9.10 Reagent Mixer Height

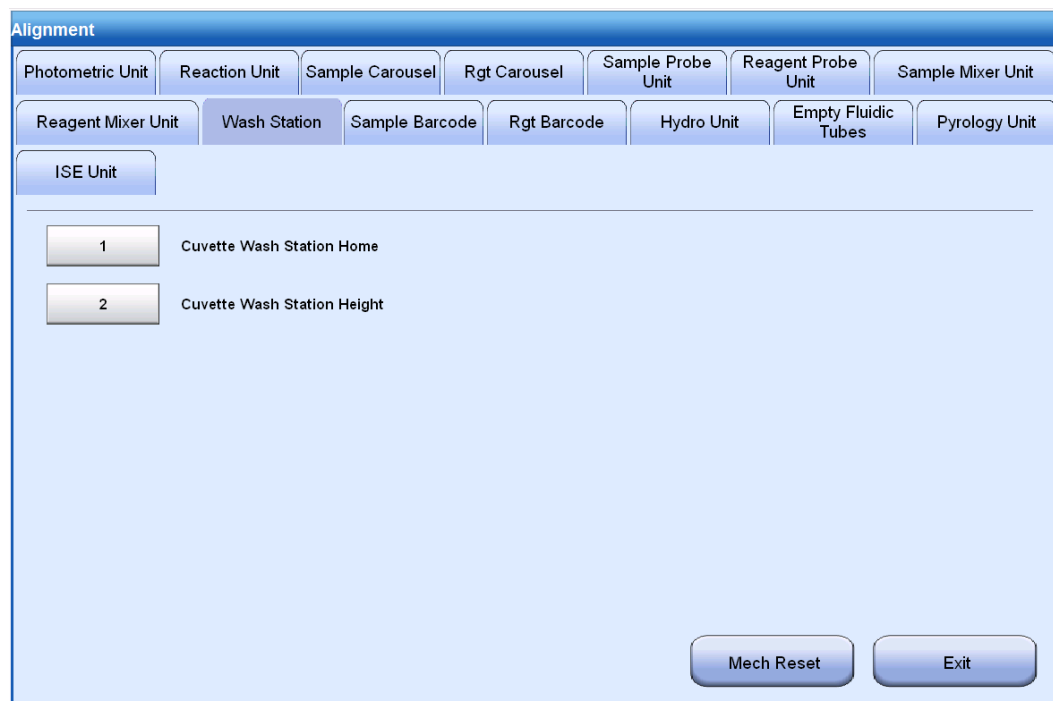
Alignment index: The lever lowers into the cuvette, through the gap between them the go gauge of the mixer alignment tool (BA49-J09) can pass through and the no-go gauge cannot.

Alignment methods and steps:

- 1 Select Reagent Mixer Height.
- 2 Select Continue. After the system resets mechanically, the reagent mixer moves into the cuvette.
- 3 Select the up/down arrow buttons, and insert an alignment tool (BA49-J09) between the mixer's straight-knurled column and cuvette edge. Gently push the alignment tool (BA49-J09) to observe if the Go Gauge can easily pass while the No-go Gauge cannot.
- 4 The operation in Step 2 is verified in Step 3. If requirements are not met, return to Step 2.
- 5 Select Continue to save parameters and exit the window.

8.10 Alignment of Cuvette Wash Unit

Select Utility—>Maintenance—>Alignment—>Cuvette Wash Station. Alignment of wash probes' horizontal position and vertical position is provided. The window is as shown below.



8.10.1 Cuvette Wash Station Home

Alignment index: The wipe blocks can move smoothly in the hole of the wash station position alignment rod (BA80-J10), phase-1 wash probe can enter into the hole on the wash station auxiliary alignment tool (BA80-J10), and wash probes of other phases are in the center of the cuvettes.

Alignment methods and steps:

- 1 Select Cuvette Wash Station Home.
- 2 Remove cuvettes under the wash probes of phase 1, 7 and 8, and place in alignment tools, including two alignment rods (BA80-J10) and one auxiliary alignment tool (having opening on one end). Make sure the alignment tool under phase-1 wash probe faces the probe with its opening, and the alignment rod under the wipe blocks has its long side in the radial direction. Select Continue.
- 3 According to Step 3 on the screen, use an M4 hexagon wrench to loosen the two screws on the wash probe conversion board, adjust the wash probes till the wipe blocks can move vertically in the hole of the alignment rod, and the wash probes of other phases are in the center of the cuvettes. When requests are met, tighten the two screws gradually. Exercise caution to prevent deforming the probes. Then remove the phase 1 probe and align it with the alignment tool. Place the cuvette properly and move the wash station with your hand. Ensure the wipe block of the wash station align with the center of the tool and other wipe blocks align with the center of the cuvettes. The wipe blocks can move vertically in the hole of the alignment tool and no interference is found. If the wipe block interferes with the alignment tool, realign till the requirement is met. Click Continue.

- 4 The operation in Step 3 is verified in Step 4. If requirements are not met, return to Step 3.
- 5 In Step 5, the reaction carousel rotates to the forward maximum deviating cuvette position. Manually move the wash station up and down so that the wipe blocks and wash probes are in the center of the cuvettes and can move smoothly.
- 6 In Step 6, the reaction carousel rotates to the reverse maximum deviating cuvette position. Manually move the wash station up and down so that the wipe blocks and wash probes are in the center of the cuvettes and can move smoothly.
- 7 Select Continue to save parameters and exit the window.
- 8 Remove the alignment tools and restore cuvettes.

8.10.2 Cuvette Wash Station Height

Alignment index: When the wash station lowers to the vertical washing position, the wash probes bounce for 0.7 to 1.0mm.

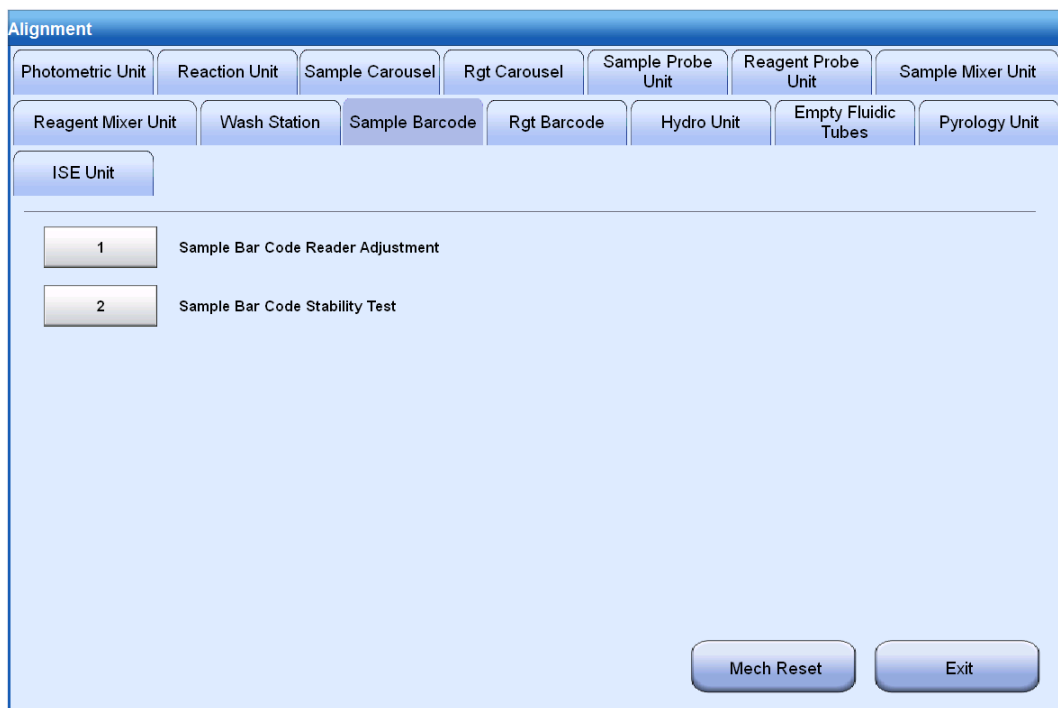
Alignment methods and steps:

- 1 Select Cuvette Wash Station Height.
- 2 Check if the wipe blocks can insert into the center of each cuvette. Select Continue.
- 3 When the wash probes move vertically to the bottom of the cuvettes, make them contact the cuvette bottom, and select Small Step or Large Step. Adjust the probes with the up/down arrow buttons while measuring the lifting height with 0.7mm and 1.0mm clearance gauge. Make sure the smallest gap between the anti-rotation blocks and the wash station bracket can let go a 0.7mm clearance gauge but 1.0mm gauge cannot pass the gap.
- 4 Verify Step 3. If requests are not met, return to Step 3, and select the up/down arrow buttons to adjust the mixer till requests are met.
- 5 Select Continue to save parameters and exit the window.

8.11 Alignment of Bar Code Unit (Optional)

8.11.1 Sample Barcode Unit

Select Utility—>Maintenance—>Alignment—>Sample Barcode.

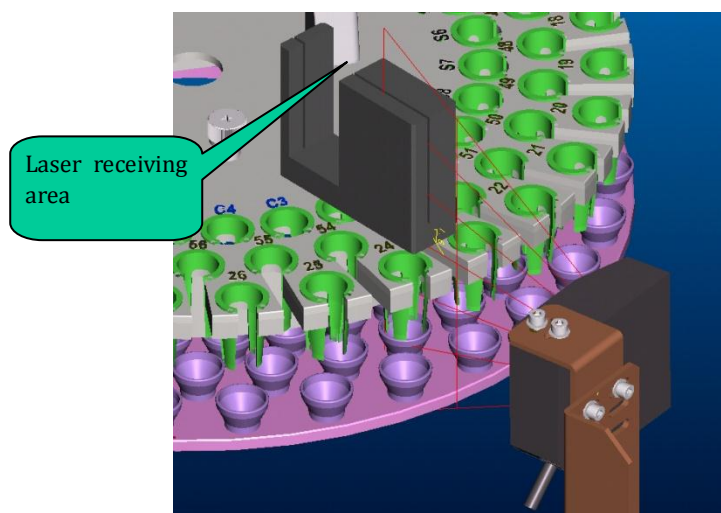


8.11.2 Sample Bar Code Reader Adjustment

Alignment index: The light beam transmitted from the sample bar code reader can pass through the bar code reader alignment tool (BA49-J10).

Alignment methods and steps:

- 1 Select Sample Bar Code Reader Adjustment.
- 2 Place a sample bar code reader alignment tool (BA49-J10) in position #31 of the sample carousel middle ring, and then select Continue.
- 3 If the light does not go through the fixture slot, loosen the two screws on the bar code reader to adjust the incident angle. If the light beam is wider than the fixture slot, adjust the bar code reader to allow the left of the light beam (counter-clockwise) to go through the slot. If the light beam is not vertical, adjust the two screws on the small bracket till requests are met. Tighten the screws.



- 4 In Step 3, if the light is slightly deviating in radial direction, use the left/right arrow buttons to rotate the sample carousel till requests are met.
- 5 Remove the alignment tool in position #31, and place bar-coded sample tubes in position 1-5 and 31-35. (The bar code label must be clean without dirt or scratches. It must be applied 13mm away from the tube bottom with inclination angle less than 5°.) When scanning is finished, check if the identified bar code is correct. If some bar code cannot be read, replace the relevant tube with another one having complete bar code. Repeat this step.
- 6 If all bar code cannot be read, check if the light direction meets the requirement and the bar code reader works normally.
- 7 Select Continue to save parameters and exit the window.

8.11.3 Bar Code Stability Test

Alignment index: When the sample carousel rotates for several circles, the bar code identified in the same position should be the same.

Alignment methods and steps:

- 1 Place bar-coded sample tubes in position #1-#60 of the sample carousel.
- 2 Select Bar Code Stability Test.
- 3 Select Start .After scanning, inconsistent sample bar code will appear in red.
- 4 If the stability test fails, find the reasons and take actions according to the following probabilities:

Check if the failed bar code meets the printing standard, the bar code label is not worn out or damaged and has been applied in correct position.

Check if the glass scanning window is clean without contamination.

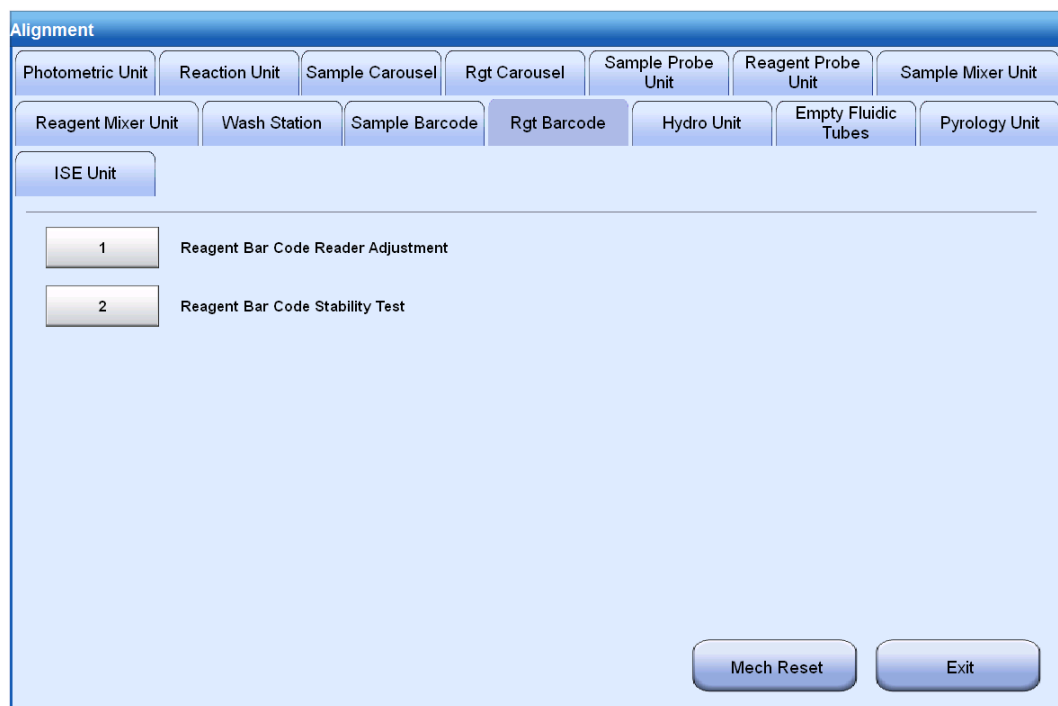
If none of the above happens, replace the bar code reader, adjust it properly and do the test again.

Generally, the three methods above are enough to troubleshoot the problem.

- 5 The test is successfully completed. Click Exit.

8.11.4 Reagent Barcode Unit

Select Utility—>Maintenance—>Alignment—>Reagent Barcode.

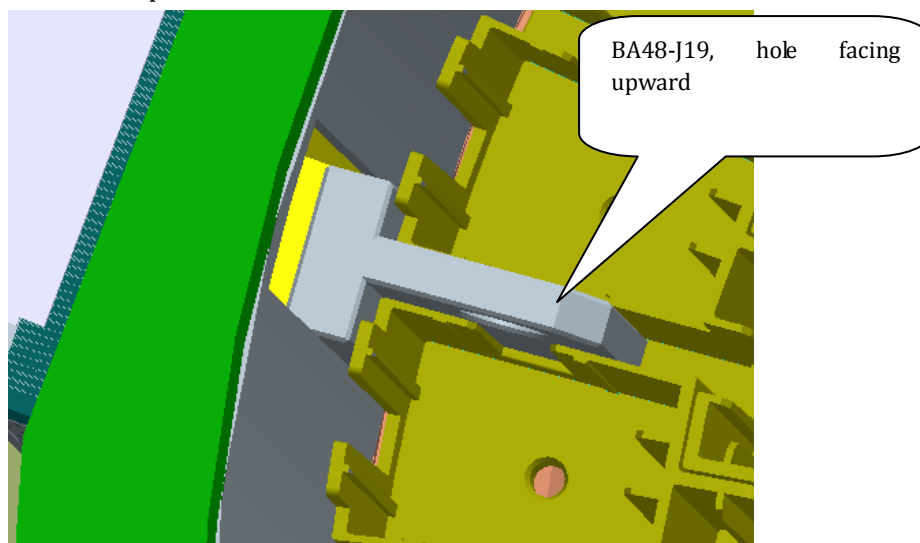


8.11.5 Reagent Bar Code Reader Adjustment

Alignment index: The light beam transmitted from the reagent bar code reader can pass through the bar code reader alignment tool (BA49-J31).

Alignment methods and steps:

- 1 Select Reagent Bar Code Reader Adjustment.
- 2 Click the left and right arrow to adjust reagent carousel position until the alignment tool (BA48-J19) can easily insert the groove between the 1# and 2# position on the outer ring of the reagent carousel. Before fine tuning the parameter, ensure the alignment tool has been removed, otherwise the reagent carousel may be damaged. Check if the reagent carousel alignment tool can easily insert into the slot and remove it. Note: if proper position cannot be adjusted to by the adjustment of parameters. Please re-fix the reagent carousel coder sensor and make it fixed in the middle position.



- 3 Place a reagent bar code alignment tool (BA49-J31) in position #43 of the reagent carousel inner ring. If the light does not go through the fixture slot, loosen the two screws on the bar code reader to adjust the incident angle. If the light beam is wider than the fixture slot, adjust the bar code reader to allow the left of the light beam (counter-clockwise) to go through the slot and fall in the laser receiving area. If the light beam is not vertical, adjust the two screws on the small bracket till requests are met. Tighten the screws.
- 4 In Step 3, if the light is slightly deviating in radial direction, use the left/right arrow buttons to rotate the reagent carousel till requests are met.
- 5 Remove the alignment tool in position #43, and place bar-coded reagent bottle in position 1-5 and 31-47. (The bar code label must be clean without dirt or scratches. It must be applied 13mm away from the tube bottom with inclination angle less than 5°.) When scanning is finished, check if the identified bar code is correct. If some bar code cannot be read, replace the relevant tube with another one having complete bar code. Repeat this step.
- 6 If all bar code cannot be read, check if the light direction meets the requirement and the bar code reader works normally.
- 7 Select Continue to save parameters and exit the window.

**NOTE**

Ensure the sponge around the scanning window can completely seal the bar code reader and it should not block the scanning light, otherwise readjust the position of the bar code reader.

8.11.6 Bar Code Stability Test

Alignment index: When the reagent carousel rotates for several circles, the bar code identified in the same position should be the same.

Alignment methods and steps:

- 1 Place bar-coded reagent bottles in position #1-#40 of the reagent carousel.
- 2 Select Bar Code Stability Test.
- 3 Select Start to start the bar code scanning.
- 4 After scanning, inconsistent sample bar code will appear in red. Take the same actions as sample bar code stability test.
- 5 The test is successfully completed. Click Exit.

Note: After alignment, check if the screws on the bar code reader and conversion board have been tightened.

8.12 Hydropneumatic Unit

Select Utility—>Maintenance—>Alignment—>Hydro Unit.



8.12.1 Preparations before Alignment

- 1 Check if all tubes have been connected correctly.
- 2 Check if the tubes of the wash station are properly connected. First the 8 phase aspirating tubes should be connected and then 6 phase dispensing tubes is connected. The 8 phase tubes should be inserted with same height and no obvious bend, twist or protrusion.
- 3 Check if the solenoid valves have connected to the correct cables.
- 4 Fill the concentrated wash solution tank with deionized water.
- 5 Fill the water tank with water.
- 6 Connect the water supply module to the water tank and analyzer, and connect the high-/low-concentration waste tanks.
- 7 Check if the inlet filter assembly has been installed.

8.12.2 Cuvette Parameter Configuration

Alignment methods and steps:

- 1 Select Cuvette Parameter Configuration.
- 2 Select search and check if the parameter matches that of the actual loaded cuvette. The result 1 indicates plastic cuvette and 0 indicates glass cuvette.

Alignment Procedure

Cuvette Parameter Configuration

Search Results Display

Parameter Definition

0 Glass Cuvette

1 Plastic Cuvette

- 3 Select Return to exit the window.

8.12.3 Check Floater and Valve Status

Valve Check

☐ SV01 Inlet	☐ SV16 Phase 3 Dispense
☐ SV02 S Probe Interior Wash	☐ SV17 Phase 4 Dispense
☐ SV03 R Probe Interior Wash	☐ SV18 Phase 5 Dispense
☐ SV05 Interior Wash Regulate	☐ SV19 Phase 6 Dispense
☐ SV06 Dilution	☐ SV21 Drain High
☐ SV07 S Probe Exterior Wash	☐ SV22 Drain Low
☐ SV08 R Probe Exterior Wash	☐ SV23 Suck Waste
☐ SV10 S Mixer Wash	☐ SV25 Release Vacuum
☐ SV11 R Mixer Wash	☐ SV27 Drain Vacuum Waste
☐ SV15 Phase 1/2 Dispense	

DI Water Tank
 Concentrated Wash Solution Tank
 Diluted Wash Solution Tank
 High-concentration Waste Tank
 Low-concentration Waste Tank
 Primary Vacuum Container
 High Concentration Waste Tank
 External Drain Canister

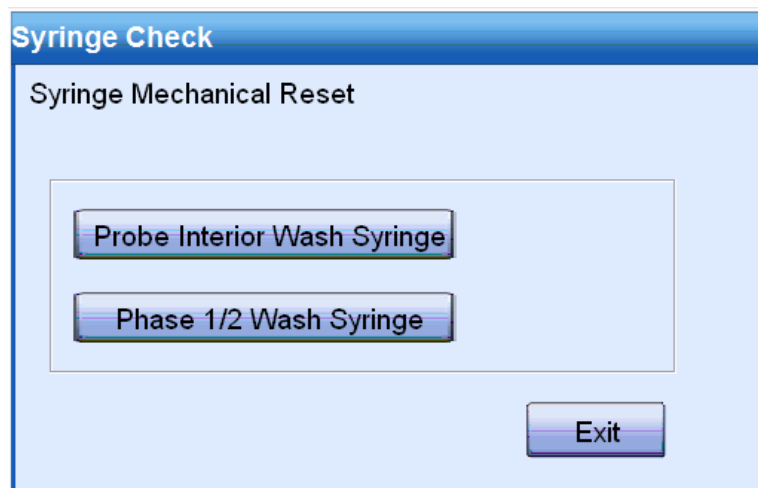
Shift the floater status according to the table below and then select query button to refresh the status of the floater. Check if the actual fluid level of each container is the same as the status displayed on the screen.

Tank Name	Floater Status	Screen description
Water tank and diluted wash solution tank	Low level floater(High) High level floater triggered(High)	Full
Water tank and diluted wash solution tank	Low level floater triggered(High) High level floater not triggered(Low)	Not Empty

Tank Name	Floater Status	Screen description
Water tank and diluted wash solution tank	Low level floater not triggered(Low) High level floater not triggered(Low)	Empty
Water tank and diluted wash solution tank	Low level floater not triggered(Low) High level floater not triggered(High)	Error
Concentrated Wash Solution Tank	Floater triggered(High)	Not Empty
Concentrated Wash Solution Tank	Floater not triggered(Low)	Empty

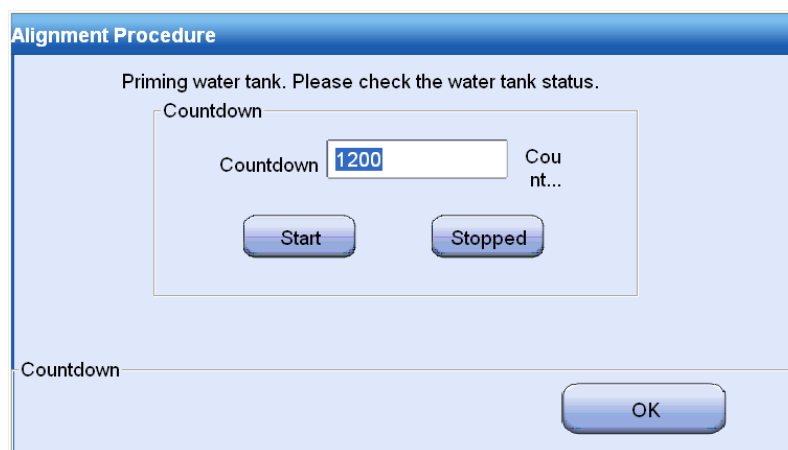
8.12.4 Syringe Check

- 1 Select Syringe Check.
- 2 Select Probe Interior Wash Syringe and Phase 1/2 Wash Syringe, and observe if the relevant syringe resets mechanically.

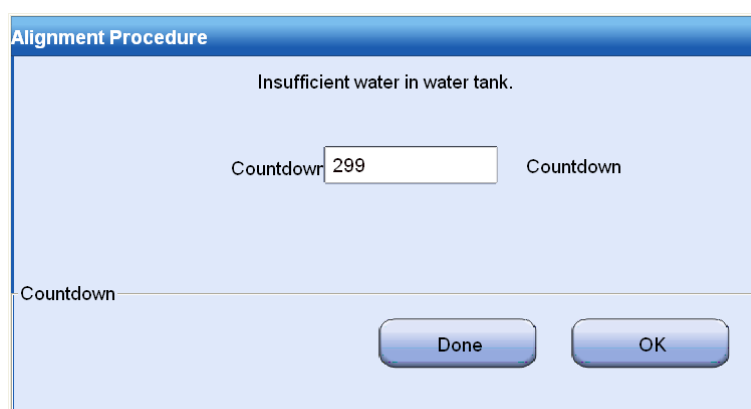


8.12.5 Fluidic Prime

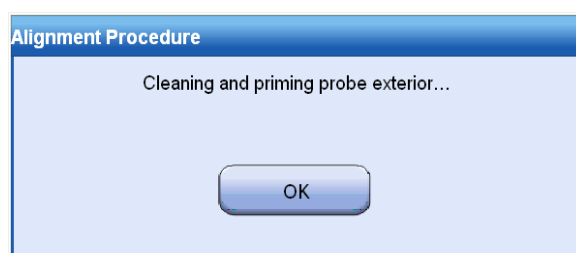
- 1 Fill the water tank with deionized water for half.
- 2 Select 5 Fluidic Prime.
- 3 In Step 2, select Continue to prime the water tank. Ensure the water tank is being primed, and select Stop to terminate the priming.



- 4 In Step 4, select Continue to prime the diluted wash solution tank. A 5 minute countdown is shown. When the time is up or the tank is full, select Done to exit.



- 5 In Step 5, select Start to prime the probes by their interior for 20 times. Check if cleaning fluid is sprayed out straightly from the sample probe and reagent probe. Loosen the screws on the syringe fixing block, loosen the syringes, pull and push the two syringe plungers until the syringes and T pieces are free of air bubbles. After priming, install the syringes back, making the V-shape slot level to scale 11 of the sample syringe and scale 15 of the reagent syringe. Select OK to exit.
- 6 In Step 6, the probes and mixers are cleaned and primed for their exteriors. Check if cleaning fluid can spray out continuously from the wash wells without air bubbles. Select OK and then select Continue to move to the next step.



- 7 In Step 7, remove the wash station and put it in an opening container, and then select Continue. In Step 8, select Start to start priming of phase 1 and 2. Make sure that the wash probes can spray continuously wash solution without air bubbles. Select Stop and then select Continue.
- 8 In Step 9, the wash probes are primed in phase 3-6. Make sure that the wash probes can spray wash solution continuously without air bubbles. Select Continue.
- 9 In Step 10, select Continue to lift the 8-phase wash probes off the fluid level.

8 Alignment

- 10 When vacuum is established in Step 11, check if the pressure of primary vacuum is less than 60% of the local atmospheric pressure. If it is, select Continue.
- 11 In Step 12, touch the wash probes to feel suction force.
- 12 In Step 13, install the wash station.
- 13 The cuvettes are cleaned for 10 times in step 14. Check that there is no liquid overflowing the cuvettes and the fluid level in cuvettes is 5-10mm from the cuvette opening.
- 14 In Step 15, the system is reset for 3 times and no alarms should occur.
- 15 Click Continue and exit the procedure.

8.12.6 Adjustment of DI Water Pressure

- 1 Make sure the deionized water pressure has been released. Unplug the pressure gauge tube and put it in an opening container off the analyzer. Deionized water will flow out from the tube. Lift the tube ascendingly and connect it to the digital pressure gauge, preventing liquid in it from flowing into the internal of the analyzer.
- 2 Select Adjustment of DI Water Pump Pressure. Select Continue. Spin the regulating valve knob in the back of the analyzer to make the reading on the digital pressure gauge within 33.5-42.5kPa. When the pressure is correct, tighten the knob and ensure the pressure remains unchanged during the process. Select OK. (Note: When performing the pressure test, try to adjust the pressure to the middle value of the pressure range.)
- 3 Connect the tube of the digital pressure gauge to the deionized water pump pressure gauge. When a dialog box pops up, check if the reading on the digital pressure gauge is within 33.5-42.5kPa. If it is not, replace the deionized water pressure gauge. Observe the reading and record the test result in the alignment record. Select OK.
- 4 Click Continue and exit.



NOTE

Once the DI water pressure valve is adjusted, flow test is required. Refer to the next section.

8.12.7 DI water flow test

Index: Sample/reagent probe exterior: 240-280ml

- 1 Use a knife to cut two 970mm tubes (082-000314-00) and two 800mm tubes (082-000305-00) and connect them to the wash wells to form a flow test tool. Note the tube error is ± 5 mm as the following figure shows:



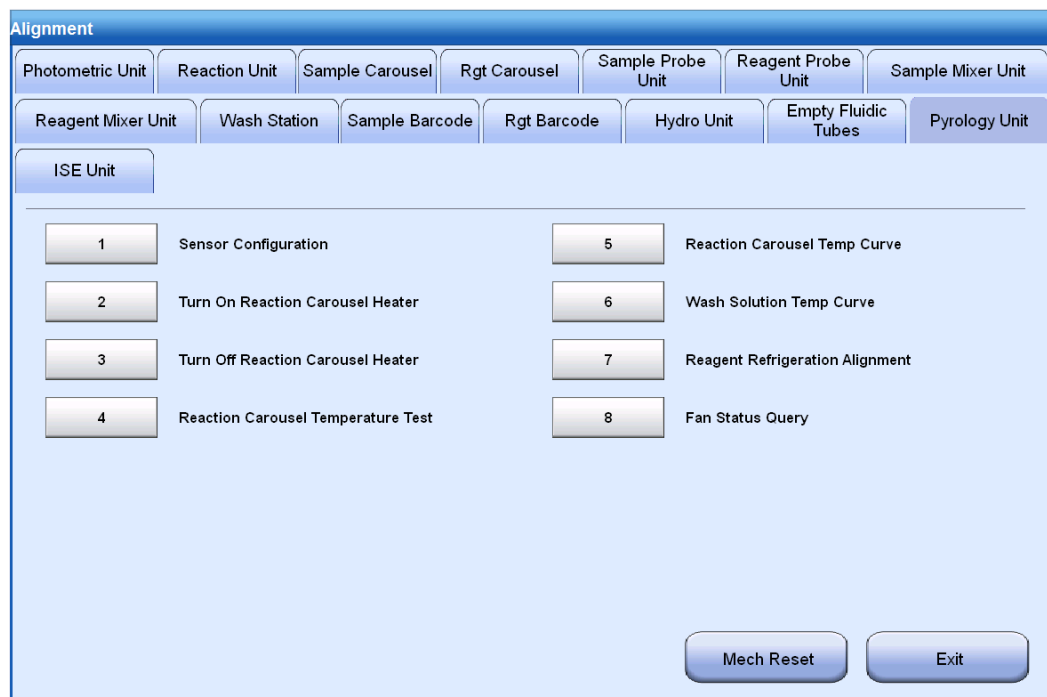
- 2** Select Wash Flow Test to enter alignment procedure. According to the prompt of step 1, there are two 1/8 connectors (M6Q-030063---) above the bracket of the left hydro assembly. Unplug the tubes of the sample well (D36) and reagent well (D38) and connect the thin tubes of the flow test with the two connectors.
- 3** Click Continue and check if the tubes are correctly connected. Select OK. The flow test tool will be primed. Click Continue.
- 4** According to the screen prompts, Insert the bigger tube of the alignment tool of the sample well flow test into the measuring cylinder and select OK.

Wait for 60 seconds. Observe the liquid volume inside the measuring cylinder and record the test result in the alignment record. (If the flow cannot meet the requirement, readjust the DI water pressure till the requirement is met.)
- 5** Similarly, insert the tube of the alignment tool of the reagent wash well flow test into the measuring cylinder and select OK.
- 6** Wait for 60 seconds. Observe the liquid volume inside the measuring cylinder and record the test result in the alignment record.
- 7** Remove the tubes and the connectors of probe exterior flow test tool. Cut off 15-20mm from the tubes on the wash wells of sample probe and reagent probe, and connect the tubes to the connectors (Note: cut the tube neatly and insert them properly).
- 8** Exit the alignment procedure.

8.13 Pyrology Unit

Select Utility—>Maintenance—>Alignment—>Pyrology Unit.

8.13.1 Reaction Carousel Temperature Control



Alignment index: Configure the parameters correctly according to the rated values of the reaction carousel sensors, and configure the ΔAD value of temperature sensors.

8.13.2 Sensor Configuration window

Alignment methods and steps:

- 1 Select Sensor Configuration.
- 2 According to the sequence number on the temperature sensor, find the resistance of rated temperature points (0, 37 and 100), select sensor 1 and input the resistance for each temperature point, and then select Calculate.
- 3 When a dialog box pops up indicating qualified sensor, select Configure to configure the parameters to the relevant sensor. Configure sensor 2 and 3 in the same way as sensor 1.
- 4 Observe the ΔAD value of PCB, and input it in the edit boxes at lower-right corner of the window for sensor 1, 2 and 3. Select Configure, and then select Search to verify them.
- 6 When configuration is complete, select Exit.

Sensor Configuration

Reaction Crsl Temp Sensors

Sensor 1				Sensor 2			
0	1000	R0	999.94	0	1000	R0	1001.21
37	1140	A	3.822e-003	37	1140	A	3.776e-003
100	1370	B	-5.91e-007	100	1370	B	-3.92e-007

Sensor 3				delta AD	
0	1000	R0	1000.01	Sensor 1	61
37	1140	A	3.829e-003	Sensor 2	65
100	1370	B	-6.25e-007	Sensor 3	64

Exit

8.13.3 Reaction Carousel Temperature Test

- 1 Select Reaction Carousel Temperature Test.
- 2 In Step 2, wait until the countdown is over. If the reaction carousel heater has been turned on for 30 minutes, select OK.
- 3 Install the reaction carousel cover; use a pipettor to fill 300 μ l deionized water in position #13 (sample dispense position) of the reaction carousel, and then select Continue.
- 4 When the reaction carousel rotates continuously for 30 circles, place a temperature sensor fixture (BA48-J08) in position #13, insert a thermometer into the fixture, wait for about 2 minutes, and record the temperature for every 30 seconds. Record temperature for 20 times for the following calculation. Calculate the average value $\bar{T}$, if $\bar{T}=37.0\pm0.3^{\circ}\text{C}$, and the fluctuation $(=(T_{\text{max}}-T_{\text{min}})/2(T_{\text{max}}, T_{\text{min}}))$ is within $0-0.1^{\circ}\text{C}$, no configuration is needed. Select Exit.
- 5 $\bar{T}$ If the ranges are exceeded, use the formula $DTh=37^{\circ}\text{C}-0x1+DTh1$ ($DTh1$ is the previous value and should be 0 for first alignment.)

Dth Setup

Dth Value(-0.6°C-0.6°C):

Configure Exit

- 6 Select Continue to exit the test.

8.13.4 Observe Temperature Curve

Alignment methods and steps:

- 1 Select Reaction Carousel Temperature Curve.
- 2 Select Turn On Heater.
- 3 Select Sensor Temp Curve, and then select Start. Observe the reaction carousel temperature curve and check if the curve of each sensor is straight and stable with average value within $37.0 \pm 0.3^\circ\text{C}$.
- 4 Select Stop. Select Exit to close the window.

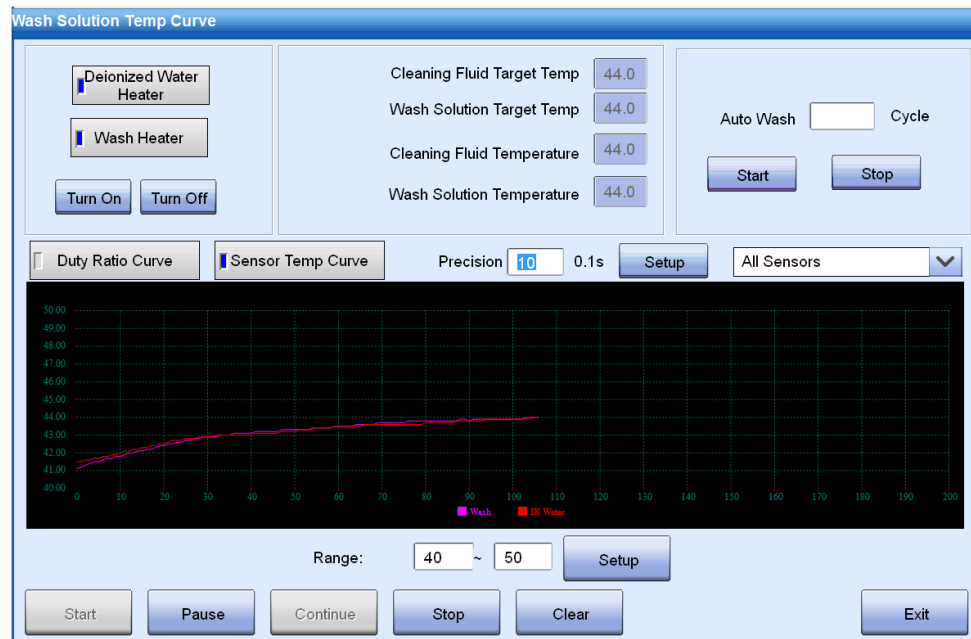


8.13.5 Wash Solution Temperature Control

Alignment index: The temperature of cleaning fluid and wash solution is within 43°C - 45°C .

Alignment methods and steps:

- 1 Select Wash Solution Temperature Curve.
- 2 Select Cuvette Wash Station on the Parameter Configuration window, and check if the prime cycle is 1.
- 3 Select Deionized Water Heater and Wash Solution Heater, and then select Start.
- 4 Select Start. Observe the temperature curves of deionized water and wash solution, and check if they are straight and stable with average value within 43°C - 45°C .
- 5 Select Stop. Select Exit to close the window. The window is closed.



8.13.6 Fan Status Query

The Fan Status Query option is used to inquire the current status of each fan.

Fan Status Query

Lamp Housing Fan	OK
Refrigeration Radiating Fan 1	OK
Refrigeration Radiating Fan 2	OK
Vacuum pump Fan	OK

9

Proactive Maintenance

9.1 Overview

Proactive maintenance is a preventive measure taken by service personnel of Mindray or authorized by Mindray with the aim of eliminating hidden troubles to ensure system reliability and achieve best performance during operation. The proactive maintenance cycle for the chemistry analyzer is one year and for ISE module, half year. The maintenance includes changing, cleaning or checking the parts as shown in the table below.

Category	Procedure	When to do
Replacement Item	Replace Cuvette Wash Connection Tube	Every 1 year
	Replace Online Filter	Every 1 year
	Replace Filter Core	Every 3 months
	Replace Sample Syringe	over 100,000 times
	Replace Reagent Syringe	over 300,000 times
	Replace Calibrator Tube(ISE configured)	Every 1 year
	Replace Pump Tube(ISE configured)	Every half year
	Replace Probe Washer	Every 1 year
	Replace Vacuum Pump	Insufficient vacuum pressure
	Replace external vacuum pump components(Optional)	Insufficient vacuum pressure
Clean Item	Photometer Lens Maintenance	As needed
	Clean Dust Screens	As needed
	Clean the dust of the radiator fan	As needed
	Clean Wash Wells	As needed
	Clean Wash Station	As needed
	Clean ISE injection port(ISE configured)	As needed
	Clean DI Water Tank	As needed
	Clean Sample/Reagent Compartment	As needed
	Clean Analyzer Panels	As needed
	Special wash	As needed
	Clean sample/reagent probes exterior	As needed
	Clean Mixers	As needed
Check Item	Pump Calibration(ISE configured)	As needed
	Air Bubble Detector Calibration(ISE configured)	As needed
	Cuvette Check	As needed
	Photometer Check	As needed
	Check Vacuum Pump	As needed

9.2 Maintenance tools

Too Name	Applicable Maintenance
Philips-head screwdriver $\phi 3.3 \times 100$	Removing the system enclosure and the cooling fans
Philips-head screwdriver $\phi 4.7 \times 100$	Installing and removing the probes and lamp
Slot-head screwdriver $\phi 4.7 \times 100$	Removing probe and pipe hoop
Round-head needle, 0.25+/- 0.01mm*125mm round tip	Unclogging the probes
Hair brush	Cleaning filter core and dust screens
Gauze	clean probe/mixer exterior and sample/reagent compartment
Cotton swabs	Clean Wash Wells
Suction cleaner	Cleaning fans and dust screens
Tweezers	Removing/Installing probes and syringe washers
Beaker	Maintaining wash station
Pipette	clean ISE sample injection port
Ethanol	Cleaning photometer, probes, mixers and wash station
NaClO (0.5% sodium hypochlorite solution)	Clean Wash Wells

9.3 Maintenance Procedure

Among the proactive maintenance items, some of them are performed with the main power switched off, while the other are performed through the maintenance window on the operating software, where maintenance logs are recorded. To improve the efficiency and shorten the field maintenance time, you are recommended to follow the procedure below:

No.	Procedure	Maintenance tools	Operations
1	Clean Analyzer Panels	Ethanol and gauze	Wipe
2	Clean Dust Screens	hair brush, Suction cleaner	Remove dust
3	Clean the dust of the radiator fan	hair brush, Suction cleaner	Remove dust
4	Photometer Lens Maintenance	Ethanol and tissue for wiping lens	Wipe
5	Replace Online Filter	Filter	Replace
6	Replace Filter Core	Filter core	Replace
7	Clean DI Water Tank	/	Washing
8	Replace Sample/Reagent Syringes	Sample/reagent syringes	Replace
9	Replace Probe Washer	Washer	Replace
10	Replace Cuvette Wash Connection Tube	Tube	Replace
11	Replace Pump Tube(ISE configured)	Pump Tube	Replace
12	Replace Calibrator Tube(ISE configured)	Calibrator Tube	Replace
13	Clean sample/reagent probes exterior/	Ethanol and gauze	Wipe
14	Clean Mixers	Ethanol and gauze	Wipe
15	Clean Wash Wells	NaClO solution and cotton swabs	Wipe
16	Clean Wash Station	ethanol,cotton swab	Wipe
17	Clean Sample/Reagent Compartment	Ethanol, gauze,cotton swab	Disassemble and wipe
18	Clean ISE injection port(ISE configured)	ethanol,cotton swab	Wipe
19	Check Vacuum Pump	/	Execute
20	Replace primary vacuum pump	Primary vacuum pump	Replace
21	Replace external vacuum pump components(Optional)	External vacuum pump	Replace
22	Pump Calibration(ISE configured)	/	Execute
23	Air Bubble Detector Calibration(ISE configured)	/	Execute

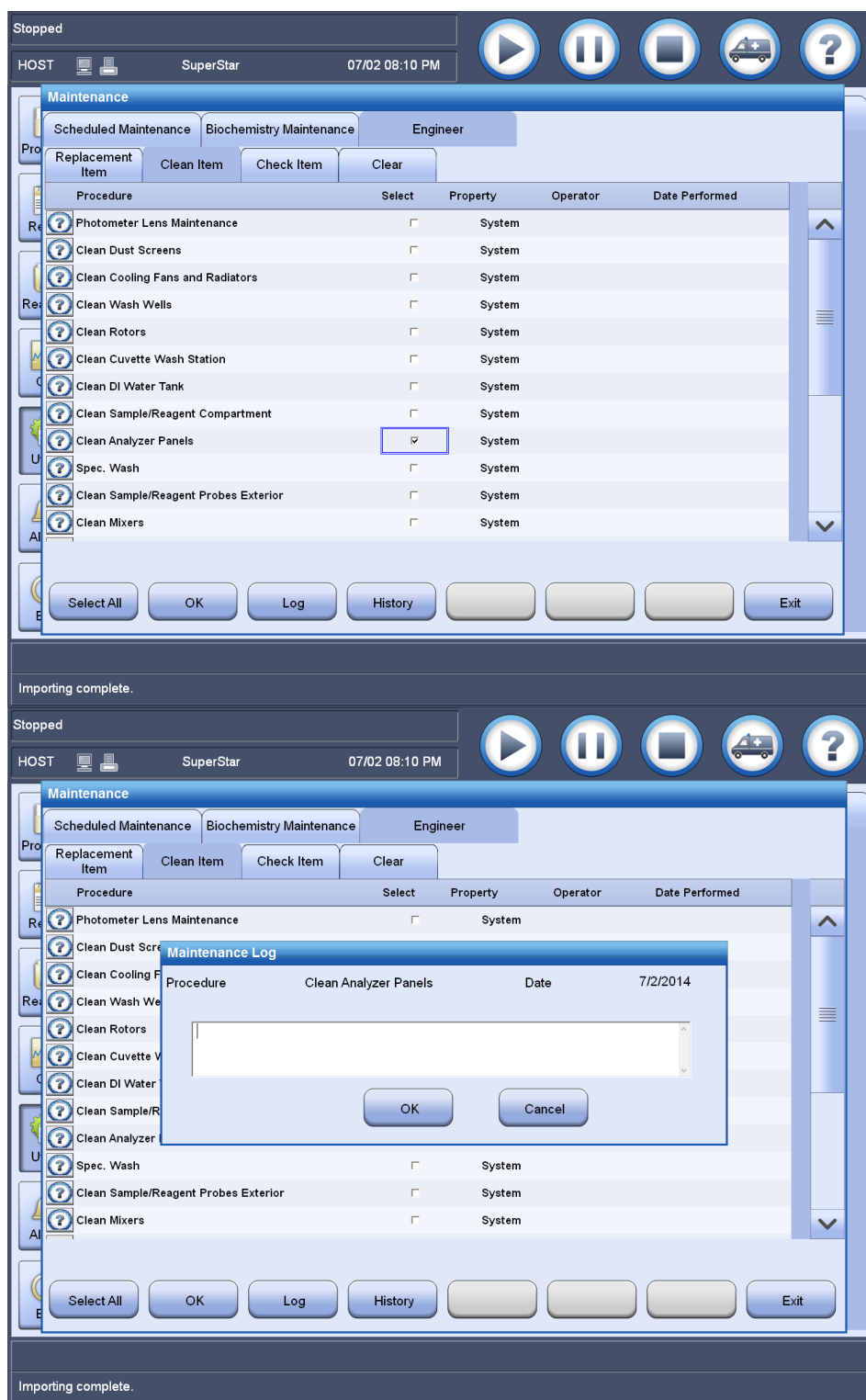
No.	Procedure	Maintenance tools	Operations
24	Special wash	Concentrated wash solution	Execute
25	Cuvette Check	/	Execute
26	Photometer Check	/	Execute

9.3.1 Steps

Clean Analyzer Panels

- 1** Open the upper protective shield of the analyzer.
- 2** Use clean gauze moistened with ethanol to clean the analyzer panels and carousel covers.
- 3** Use wash solution to clean the monitor screen and keyboard.
- 4** Restore the upper protective shield.
- 5** After maintenance, open the logs and record related information.

9 Proactive Maintenance



Clean Dust Screens

- 1 Switch off the analyzer power.
- 2 Open the front door of the analyzer.
- 3 Remove the dust screen. For the removing steps, refer to 5.2.4 Removing left/right dust screen.
- 4 Tidy and clean the dust screens.
- 5 Dry the dust screens in air.

- 6 Restore the dust screens.
- 7 Close the front door of the analyzer.
- 8 After maintenance, open the logs and record related information.

Clean the dust of the radiator fan

- 1 Refer to chapter 5 5.1.7 for cleaning dust of the fans.
- 2 After maintenance, open the logs and record related information.

Photometer Lens Maintenance

- 1 Refer to chapter 5 5.9.7 for Photometer Lens Maintenance.
- 2 After maintenance, open the logs and record related information.

Replace Online Filter

- 1 Check the maintenance log and when the online filter maintenance interval is more than 1 year, change it.
- 2 Refer to 7.5.3 Cuvette wash module in chapter 7 for the removing and installing steps.
- 3 After maintenance, open the logs and record related information.

Replace Filter Core

- 1 Check the maintenance log. It is recommended to replace the filter core for every 3 months.
- 2 Refer to 7.5.4 Removing/Installing deionized water filter assembly in chapter 7 for the removing and installing steps.
- 3 After maintenance, open the logs and record related information.

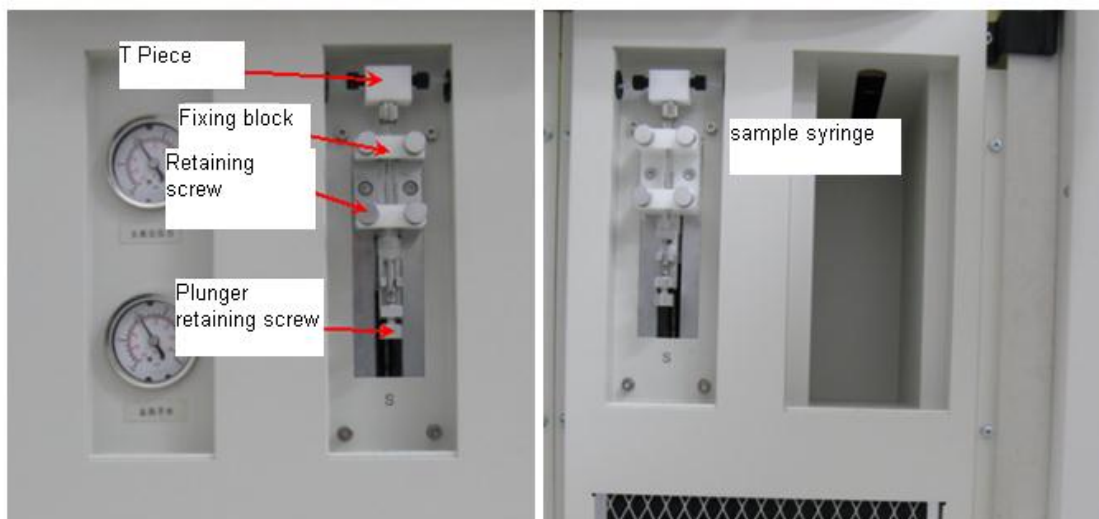
Clean DI Water Tank

- 1 Refer to 7.5.4 Removing/Installing water tank in chapter 7 for the removing and installing steps.
- 2 After maintenance, open the logs and record related information.

Replace Sample/Reagent Syringes

- 1 Check the using count of the syringe on the status screen. When the sample syringe has been used for 100,000times, change it. When the reagent syringe has been used for 300,000times, change it.
- 2 Prepare a new syringe plunger assembly and washer, put the plunger head in the deionized water beaker to remove air from the syringe, and then moisten the washer in the deionized water.
- 3 Place the analyzing unit power to the OFF position.
- 4 Open the front door of the analyzer. You will see two syringes. The left one is reagent syringe and the right one is sample syringe as shown in the figure below.

9 Proactive Maintenance



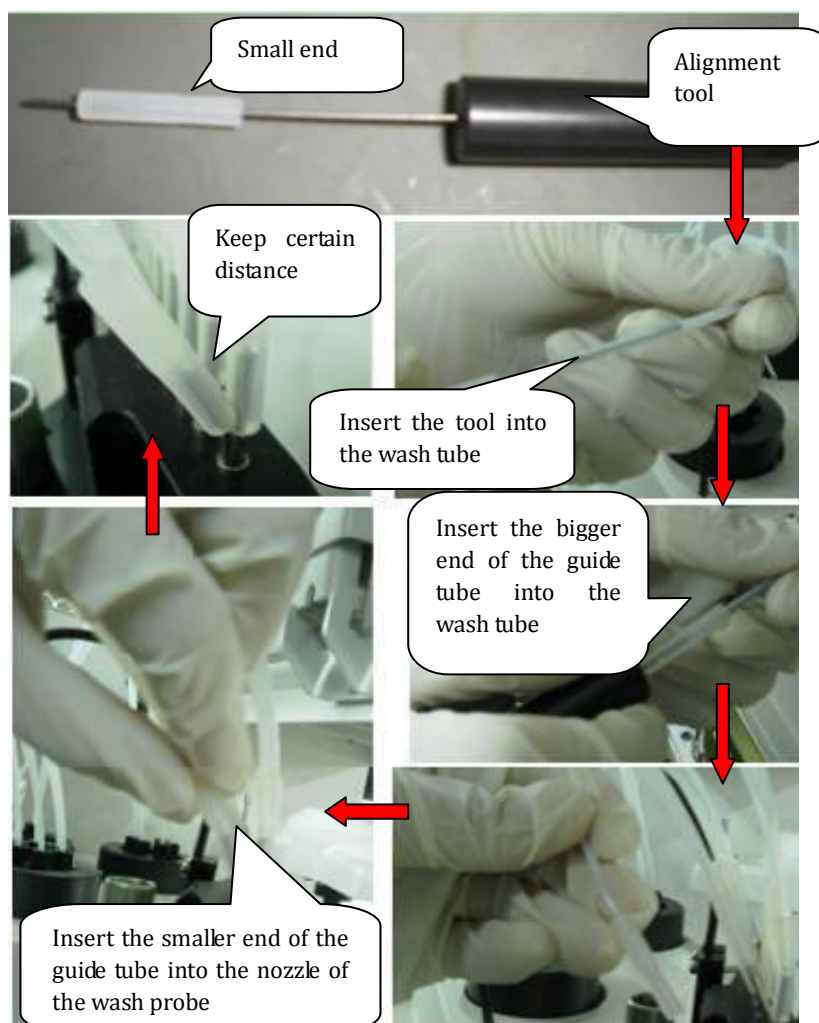
- 5 Loosen counterclockwise the four retaining screws on top of the syringe, and then remove the screws and the fixing blocks.
- 6 Loosen counterclockwise the retaining screw at the bottom of the syringe and then remove it.
- 7 Hold the T piece with one hand and the syringe connector with the other hand. Loosen the syringe counterclockwise and then remove the washer.
- 8 Loosen the plunger guide cap counterclockwise, hold the plunger head and pull it slightly to remove the plunger assembly from the syringe.
- 9 Insert the plunger head of the new plunger assembly into the bottom of the syringe, and then tighten the retaining screw to fix the plunger head. Tighten the plunger nut clockwise.
- 10 Soak the new syringe connector in the deionized water beak, pull the plunger head to aspirate half syringe of deionized water, and then push the plunger head to remove the air.
- 11 If no washer is found in the T piece, put the new washer in the T piece. Hold the T piece with one hand and the syringe connector with the other hand, and then screw the T piece clockwise. Install the syringe on the bracket.
- 12 Install the fixing blocks and 4 retaining screws while having the retaining screws not tightened.
- 13 Align the plunger head to the retaining screw at the bottom of the syringe, and then tighten clockwise the retaining screw.
- 14 Pinch the plunger guide cap to adjust the syringe height. For the sample syringe, make the syringe head over the upper fixing block for 11 scales; for the reagent syringes, make the syringe head over the upper fixing block for 15 scales.
- 15 Tighten the four retaining screws on the fixing blocks.
- 16 After finishing replacement, switch on the analyzer power.
- 17 Perform the Home maintenance procedure. Check the new syringe for leak and bubbles. If leak occurs, check the syringe and the connector.
- 18 Close the front door of the analyzer.
- 19 The procedure is completed. Write the maintenance log and input the maintenance information and clear the using count of the syringe.

Replace Probe Washer

- 1 Check the maintenance log. When the sample/reagent probe washer has been used for 1 year and change it.
- 2 Remove the sample/reagent probe and remove the washer.
- 3 Install the washer and install back the sample/reagent probe.
- 4 Run the operating software, and select Utility-Maintenance-Maintenance-Biochemistry Maintenance and then select Probe interior clean and prime.
- 5 After maintenance, open the logs and record related information.

Replace Cuvette Wash Connection Tube

- 1 Check the maintenance log. When Cuvette wash connection tube has been used for 1 year and change it.
- 2 The analyzing unit should be powered off, or in failure mode or in standby status.
- 3 Open the upper protective shield of the analyzer. Loosen the installation button of the wash station, and then remove the wash station.
- 4 Remove the connecting tube that needs replacing.
- 5 Insert the thin end of the connecting tube into the fixture, insert the fixture into the wash tube, insert the thick end of the connecting tube into the wash tube about 9~11 mm, and then insert the thin end into the nozzle of the wash probe. Make the ends of the tube and the nozzle remain in the middle of the connecting tube and keep a distance between the two ends. The installation steps as shown in the figure below.



9 Proactive Maintenance

- 6 Reinstall the wash station.
- 7 Execute Prime Wash Station on the Biochemistry Maintenance window. Deionized water flows through the tube, removing air bubbles from it.
- 8 Restore the upper protective shield of the analyzer.
- 9 After maintenance, open the logs and record related information.

Replace Pump Tube (ISE configured)

- 1 Check the maintenance log. When Pump Tube has been used for half year, change it.
- 2 Refer to 5.10.4 in chapter 5 for replace Pump Tube.
- 3 After maintenance, open the logs and record related information.

Replace Calibrator Tube (ISE configured)

- 1 View the maintenance log. When the Calibrator Tube has been used for more than 1 year, change it.
- 2 Refer to 5.10.5 in chapter 5 for Replace Calibrator Tube.
- 3 After maintenance, open the logs and record related information.

Clean Sample/Reagent Probes Exterior

- 1 Place the analyzing unit power to the OFF position.
- 2 Lift gently the probe to the utmost height and move it to the position convenient for maintenance.。
- 3 Use gauze soaked with ethanol to gently wipe the probe exterior. Clean the probe tip until it becomes clear without stain. Use gauze moistened with deionized water to clear the ethanol on the probe. Do not pull the probe vertically to prevent probe damage.
- 4 After the procedure, restore the upper protective shield of the analyzer. Power on the analyzing unit and select Home.
- 5 After maintenance, open the logs and record related information.

Clean Mixers

- 1 Place the analyzing unit power to the OFF position.
- 2 Lift gently the mixer to the utmost height and move it to the position convenient for maintenance.。
- 3 Use gauze soaked with ethanol to gently wipe the mixer exterior. Clean it until it becomes clear without stain. Use gauze moistened with deionized water to clear the ethanol on the mixer. Do not pull the mixer vertically to prevent probe damage.
- 4 After the procedure, restore the upper protective shield of the analyzer. Power on the analyzing unit and select Home.
- 5 After maintenance, open the logs and record related information.

Clean Wash Wells

- 1 Place the analyzing unit power to the OFF position.
- 2 Move the probe and mixer away from the wash well for the convenience of maintenance.
- 3 Use clean cotton swabs moistened with NaClO to clean the wash wells.
- 4 After the procedure switch on the analyzer power.

- 5 Execute Home command or Check Probes/Mixers/Wash Wells maintenance command. Check if the wash wells have a normal water flow.
- 6 After maintenance, open the logs and record related information.

Clean Wash Station

- 1 The analyzing unit should be powered off, or in failure mode or in standby or incubation status.
- 2 Open the upper protective shield of the analyzer. Remove the wash station, and use gauze moistened with ethanol to clean the nozzles and wipe blocks.
- 3 Use the gauze soaked with DI water to clear the ethanol left on the nozzles.
- 4 Install back the wash station and close the upper shielding cover of the analyzer.
- 5 Execute the Prime Wash Station maintenance command.
- 6 After maintenance, open the logs and record related information.

Clean Sample/Reagent Compartment

- 1 The analyzing unit should be powered off, or in failure mode or in standby or incubation status.
- 2 Remove the carousel cover and carousel, and then store them properly.
- 3 Use clean gauze soaked with deionized water or ethanol to clean the interior of the compartment. If necessary, you can use gauze moistened with neutral wash solution.
- 4 Use clean gauze soaked with deionized water or ethanol to clean the carousel body, and then use cotton swabs dipped with ethanol to clean the sample or reagent positions.
- 5 Install the carousel and the carousel cover.
- 6 After maintenance, open the logs and record related information.

Clean ISE injection port (ISE configured)

- 1 Make sure that the system status is Incubation or Standby.
- 2 Enter ISE maintenance screen and select Clean ISE injection port.
- 3 Open the upper protective shield of the analyzer.
- 4 Open the cover of the ISE module on the analyzer's front panel.
- 5 Use clean cotton swab soaked with ethanol to wipe the sample injection port (interior of the sample injection cup of the ISE module) until it is clean; then use a clean cotton swab soaked with DI water to wipe the interior and periphery of the sample injection port .
- 6 Enter the ISE Maintenance window and execute purge A and purge B for three times each.
- 7 Restore the cover of the ISE module on the analyzer's front panel. Restore the upper protective shield of the analyzer.
- 8 After maintenance, open the logs and record related information.

Check Vacuum Pump

- 1 Select Maintenance-Alignment-Hydro Unit and select pump check screen. Select Primary vacuum pump check and select Start to start the vacuum pump check. If "Checking cuvette wash vacuum pump pressure is succeeded" is given out, it indicates the vacuum pump meets the requirement. If insufficient vacuum pressure is prompted, record the total working hour of the vacuum and recommend the customer replace the vacuum pump.
- 2 If the external pump is configured, it should be checked as well. Select Maintenance-Alignment-Hydro Unit and select pump check screen. Select External vacuum pump (optional) check and select Start to start the vacuum pump check. If "Checking cuvette wash vacuum pump pressure is succeeded" is given out, it indicates the external vacuum pump meets the requirement. If insufficient vacuum pressure is prompted, record the total working hour of the external vacuum and recommend the customer replace the vacuum pump.
- 3 After maintenance, open the logs and record related information.

Replace Vacuum Pump

- 1 When the vacuum pump pressure is insufficient, change it.
- 2 Refer to 7.5.3 Removing/Installing vacuum pump in chapter 7.
- 3 Select Maintenance-Alignment-Hydro Unit and select pump check screen. Select Primary vacuum pump check and select Start to start the vacuum pump check. If "Checking cuvette wash vacuum pump pressure is succeeded" is given out, it indicates the primary vacuum pump meets the requirement. If insufficient vacuum pressure is prompted, it indicates the primary vacuum pump does not meet the requirement and check its installation status again
- 4 The procedure is completed. Write the maintenance log and input the maintenance information and clear the using count of the primary vacuum pump.

Replace external vacuum pump components (Optional)

- 1 When the external vacuum pump pressure is insufficient, change it.
- 2 Power off the external vacuum pump and open its pump box to remove it.
- 3 After replacement, install the new pump and power it on.
- 4 Select Maintenance-Alignment-Hydro Unit and select pump check screen. Select External vacuum pump (optional) check and select Start to start the vacuum pump check. If "Checking cuvette wash vacuum pump pressure is succeeded" is given out, it indicates the primary vacuum pump meets the requirement. If insufficient vacuum pressure is prompted, it indicates the external vacuum pump does not meet the requirement and check its installation status again.
- 5 After maintenance, open the logs and record related information.

Pump Calibration (ISE configured)

- 1 Make sure that the system status is Incubation or Standby.
- 2 Enter ISE maintenance screen and click Pump Calibration.
- 3 Fill sample cup with 500 μ l DI water, place it in the W position on the sample carousel.
- 4 Continue the procedure until it is completed.
- 5 After the procedure is completed, the calibration results are displayed. If calibration is successful, the procedure is completed. If not, ISE enters error status.
- 6 After maintenance, open the logs and record related information.

Air Bubble Detector Calibration (ISE configured)

- 1 Ensure ISE is standby.
- 2 Enter ISE maintenance screen, and select Air Bubble Detector Calibration. The procedure is performed automatically.
- 3 After the procedure is completed, the calibration results are displayed. If calibration is successful, the procedure is completed. If not, ISE enters error status.
- 4 After maintenance, open the logs and record related information.

Special wash

- 1 Open the upper protective shield of the analyzer.
- 2 Load over 40ml concentrated wash solution in position D on the reagent carousel and over 5ml concentrated wash solution in position D3 on the sample carousel.
- 3 Select Utility-Maintenance-Maintenance- Biochemistry Maintenance and select special wash.
- 4 Confirm if cuvette check is needed after the concentrated wash. If it is, mark the checkbox in front of Check Cuvettes.
- 5 Select Continue. The system starts cleaning the sample probe, reagent probes, mixers, cuvettes and wash station. When special wash is finished, the system performs cuvette check automatically.
- 6 Select Done.
- 7 Restore the upper protective shield of the analyzer.
- 8 After maintenance, open the logs and record related information.

Cuvette Check

- 1 Select Utility-Maintenance-Maintenance- Biochemistry Maintenance and then select Cuvette check.
- 2 After check, check the cuvette check results for yellow indication. Record the cuvettes highlighted in yellow and perform the Clean Cuvettes or Replace Cuvettes procedure.
- 3 After maintenance, open the logs and record related information.

Photometer Check

- 1 Select Utility-Maintenance-Maintenance- Biochemistry Maintenance and then select Photometer Check
- 2 If the result is Normal, the system will give no alarms; otherwise the following alarm will be given.
 - If the alarm indicates the lamp is off, check if the lamp has been turn on.
 - If the alarm light too strong is given, please perform photometric unit alignment or check if the photometric unit is normal.
 - If the alarm indicates light intensity too weak, replace the lamp.

10

Troubleshooting

10 Troubleshooting

Table 10.1 Error messages

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A00006	Command execution	Error	Main control unit E2PROM reading or writing error.	/	E2PROM read/write error	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A01006	Sample probe unit	Error	Sample probe vertical movement error 1. Sensor status is incorrect. 2. Home position is not found. 3. Collision occurs during operation other than aspirating. 4. Collision error remains. 5. Vertical movement is not allowed in current horizontal position. Sample probe horizontal movement error. 1. Sensor status is incorrect. 2. Home position is not found. 3. Collision occurs during horizontal movement. 4. Horizontal movement is not allowed in current vertical position. Sample syringe movement error. 1. Sensor status is incorrect. 2. Home position is not found.	/	Sample probe vertical movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driving circuit error; 3) Vertical motor error; 4) Vertical sensor error; 5) Belt pulley is loose; 3. Collision occurs during operation other than aspirating: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the sample probe; 3) Incorrect sample probe positioning; 4) Incorrect sample carousel positioning; 4. Collision error remains: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the sample probe; 3) Collision sensor is not blocked due to deformation of blocking plate's driving spring; 5. Vertical movement is not allowed in current horizontal position: Software error. Sample probe horizontal movement error. 1. Sensor status error: Belt or belt pulley is loose.	Sample probe vertical movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly; 3. Collision occurs during operation other than aspirating: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Rotate positioning sensor to remove dust or adjust probe to be vertical; if problem remains, adjust positioning parameters; 4) Adjust sample carousel positioning parameters; 4. Collision error remains: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Mount the spring of blocking plate again; 5. Vertical movement is not allowed in current horizontal position: Home instrument or sample probe unit. Sample probe horizontal movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					2. Failed to find the zero position: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 3. Collision occurs during horizontal movement: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 4. Horizontal movement is not allowed in current vertical position: Software error. Sample syringe movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driving circuit error; 3) Motor error; 4) Sensor error; 5) Belt pulley is loose.	if so; 2. Failed to find the zero position: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 3. Collision occurs during horizontal movement: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 4. Horizontal movement is not allowed in current vertical position: Home instrument or sample probe unit. Sample syringe movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A01007	Sample probe unit	Warning	Sample probe collides with an obstacle when aspirating Sample position: %s Sample ID/bar code: %s Specific position: %s	/	Sample probe vertical movement error 1. Collision occurs during aspirating: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the sample probe; 3) Incorrect sample probe positioning; 4) Incorrect sample carousel positioning;	Sample probe vertical movement error 1. Collision occurs during aspirating: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Rotate positioning sensor to remove dust or adjust probe to be vertical; if problem remains, adjust positioning parameters; 4) Adjust sample carousel positioning parameters.
A01021	Sample probe unit	Error	Clog detection board communication error	/	1. Connection error between clog detection board and wash control drive board; 2. Clog detection board malfunction	1. Check cable for any signs of fray or loose cable, or disconnection, or short circuit; 2. Check if the power indication light for clog detection board is on properly; check if clog detection board is working correctly; 3. Replace clog detection board if malfunction has been identified.
A01022	Sample probe unit	Warning	Sample syringe aspirates too much	/	The aspirate volume is beyond the range of the syringe.	Define the aspirate volume correctly.
A01023	Sample probe unit	Warning	Sample syringe dispenses too much	/	The dispense volume is beyond the range of the syringe.	Define the dispense volume correctly.
A01024	Sample probe unit	Warning	Insufficient sample Sample position: %s Sample ID/bar code: %s	/	1. No sample is loaded on the designated position; 2. Insufficient sample; 3. Sample probe level detection board error.	1. Check if there is sufficient sample, and run the test again; 2. If there are level detection error, check sample probe, level detection board and connecting cable; replace if any abnormality is identified.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A01026	Sample probe unit	Warning	Sample probe dispenses insufficient sample	/	1. Sample probe is clogged when aspirating; 2. Sample probe aspirates nothing; 3. Syringe assembly error.	1. Check if the sample meets requirement and is sufficient in volume, and then run the test again; 2. Check if there is any abnormality of syringe assembly.
A01027	Sample probe unit	Error	Sample probe aspirates nothing. 1. Sample probe aspirates nothing on the sample carousel. 2. Sample probe aspirates nothing on the reaction carousel.	/	1. No sample is loaded on the designated position; 2. Insufficient sample; 3. Sample probe level detection error; 4. Clog detection error.	1. Check if there is sufficient sample, and run the test again; 2. If level detection error is identified, check sample probe, level detection board and connecting cable; replace if any abnormality is identified; 3. If clog detection error is identified, check if there is loose connection or liquid leakage in sample probe hydro system.
A01028	Sample probe unit	Error	Sample probe fails to detect liquid level during cleaning. 1. Sample probe does not detect liquid level in wash well when reset. 2. Sample probe does not detect liquid level in wash well during cleaning.	/	1. No DI water or hydro error; 2. Sample probe level detection error.	1. Check if hydro is normal; troubleshoot if there is any abnormality; 2. If level detection error is identified, check sample probe, level detection board and connecting cable; replace if any abnormality is identified.
A01029	Sample probe unit	Warning	Sample probe is clogged when aspirating	/	1. Sample contains clots or insufficient sample; 2. Sample probe hydro error; 3. Clog detection system error.	1. Check if sample has been pre-processed properly; check if sample contains foreign matters such as clot; check if there is sufficient sample; 2. Clean the sample probe with diluted wash solution. If the problem remains, remove the sample probe and unclog it. 3. Check if hydro is normal, especially if there is any air bubbles

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						in clog detection pressure sensor; 4. Check clog detection board and clog detection pressure sensor, if previous two possible causes have been eliminated.
A01030	Sample probe unit	Error	Sample probe is clogged during cleaning	/	1. Sample probe is clogged; 2. Sample probe hydro error.	1. Clean the sample probe with diluted wash solution. Remove the sample probe and unclog it. 2. Check if sample probe tubing, pump and valve for interior wash is normal; 3. Check clog detection board and clog detection pressure sensor.
A01033	Sample probe unit	Warning	Sample probe fails to detect liquid level on reaction carousel when dispensing	/	1. There is no reagent or insufficient reagent in the cuvette; 2. Incorrect vertical extreme position parameter for sample probe on reaction carousel; 3. Sample probe level detection error.	1. Check if R1 volume is sufficient and the reagent bottle is free of air bubbles, and then try again; 2. Check if the vertical extreme position parameter for sample probe on reaction carousel is correct; 3. If level detection error is identified, check sample probe, level detection board and connecting cable; replace if any abnormality is identified.
A01036	Sample probe unit	Error	Sample probe level detection board data receiving error	/	Cable error for communication between level detection board and smart modules	Check relevant cable for any signs of fray or loose cable, or disconnection, or short circuit.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A01037	Sample probe unit	Error	Sample probe level detection board self-calibrating failed	/	1. Error of communication cable between level detection board and smart modules; 2. Sample probe error, or connection error between sample probe and level detection board; 3. Sample probe level detection board error.	1. Check relevant cable for any signs of fray or loose cable, or disconnection, or short circuit; 2. Check if sample probe is installed correctly; check if sample probe is bent, cracked or loose; 3. Check if level detection board is normal; replace it if not.
A01038	Sample Probe Unit	Error	Sample probe interior wash is abnormal.	/	Sample probe interior wash is abnormal.	1. Check if the cable of the valve is disconnected; the voltage of the valve is normal. If not, change it. 2. Check leak tightness of the syringe. If leak occurs, change it. 3. Check the tube connection. If necessary, change it.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A02006	Reagent probe unit	Error	Reagent probe vertical movement error 1. Sensor status is incorrect. 2. Home position is not found. 3. Collision occurs during operation other than aspirating. 4. Collision error remains. 5. Vertical movement is not allowed in current horizontal position. Reagent probe horizontal movement error. 1. Sensor status is incorrect. 2. Home position is not found. 3. Collision occurs during horizontal movement. 4. Horizontal movement is not allowed in current vertical position. Reagent syringe movement error. 1. Sensor status is incorrect. 2. Home position is not found.	/	Reagent probe vertical movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driving circuit error; 3) Vertical motor error; 4) Vertical sensor error; 5) Belt pulley is loose; 3. Collision occurs during operation other than aspirating: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the reagent probe; 3) Incorrect reagent probe positioning; 4) Incorrect reagent carousel positioning; 4. Collision error remains: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the reagent probe; 3) Collision sensor is not blocked due to deformation of blocking plate's driving spring; 5. Vertical movement is not allowed in current horizontal position: Software error. Reagent probe horizontal movement error1. Sensor status error: Belt or belt pulley is loose.	Reagent probe vertical movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly; 3. Collision occurs during operation other than aspirating: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Rotate positioning sensor to remove dust or adjust probe to be vertical; if problem remains, adjust positioning parameters; 4) Adjust reagent carousel positioning parameters; 4. Collision error remains: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Mount the spring of blocking plate again; 5. Vertical movement is not allowed in current horizontal position: Home instrument or reagent probe unit. Reagent probe horizontal movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					2. Failed to find the zero position: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 3. Collision occurs during horizontal movement: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 4. Horizontal movement is not allowed in current vertical position: Software error. Reagent syringe movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driving circuit error; 3) Motor error; 4) Sensor error; 5) Belt pulley is loose.	if so; 2. Failed to find the zero position: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 3. Collision occurs during horizontal movement: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 4. Horizontal movement is not allowed in current vertical position: Home instrument or reagent probe unit. Reagent syringe movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A02007	Reagent probe unit	Warning	Reagent probe collides with an obstacle when aspirating Reagent position: %s Specific position: %s	/	Reagent probe vertical movement error 1. Collision occurs during aspirating: 1) Collision sensor or circuit or cable failure; 2) Collision sensor is not blocked when assembling the reagent probe; 3) Incorrect reagent probe positioning; 4) Incorrect reagent carousel positioning;	Reagent probe vertical movement error 1. Collision occurs during aspirating: 1) Replace sensor or PCB or cable; 2) Reassemble; 3) Rotate positioning sensor to remove dust or adjust probe to be vertical; if problem remains, adjust positioning parameters; 4) Adjust reagent carousel positioning parameters.
A02021	Reagent probe unit	Warning	Reagent syringe aspirates too much	/	The aspirate volume is beyond the range of the syringe.	Define the aspirate volume correctly.
A02022	Reagent probe unit	Warning	Reagent syringe dispenses too much.	/	The dispense volume is beyond the range of the syringe.	Define the dispense volume correctly.
A02023	Reagent probe unit	Warning	Reagent probe fails to detect liquid level on reagent carousel	/	1. There is no reagent or insufficient reagent on the designated position; 2. Probe R1 level detection error.	1. Check if the reagent is sufficient, and then try again. 2. If there is level detection error, check probe R1, level detection board and connecting cable; replace if any abnormality is identified.
A02025	Reagent probe unit	Warning	Probe R1 dispenses insufficient reagent	/	1. The reagent probe aspirates nothing. 2. Syringe assembly error.	1. Check if the reagent meets the requirement and is sufficient in volume, and then try again. 2. Check if there is any abnormality of syringe assembly.
A02026	Reagent probe unit	Error	Reagent probe fails to detect liquid level during cleaning. 1. Reagent probe does not detect liquid level in wash well when reset. 2. Reagent probe does not detect liquid level in wash	/	1. No DI water or hydro error; 2. Probe R1 level detection error.	1. Check if tubing, pump and valve for exterior wash of probe R1 is normal; 2. If level detection error is identified, check probe R1, level detection board and connecting cable; replace if any abnormality is identified.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			well during cleaning.			
A02027	Reagent probe unit	Warning	Reagent probe detects liquid level in an empty cuvette	/	DI water residual exists in cuvette	Check if aspirating tubing and solenoid valves of wash station are normal; check if system vacuum is normal.
A02030	Reagent probe unit	Error	Reagent probe level detection board data receiving error	/	Cable error for communication between level detection board and smart modules	Check relevant cable for any signs of fray or loose cable, or disconnection, or short circuit.
A02031	Reagent probe unit	Error	Reagent probe level detection board self-calibrating failed	/	1. Cable error for communication between level detection board and smart modules; 2. Probe R1 error or level detection board connection error; 3. Probe R1 level detection error.	1. Check relevant cable for any signs of fray or loose cable, or disconnection, or short circuit; 2. Check if probe R1 is installed correctly; check if probe R1 is bent, cracked or loose; 3. Check if level detection board is normal; replace it if not.
A02032	Reagent probe unit	Warning	Reagent probe aspirates no reagent on the reagent carousel.	/	Air bubbles exist in the reagent bottle.	1. Check if the reagent bottle contains air bubbles, and then try again. 2. Check if the reagent bottle meets the requirements.
A02033	Reagent probe unit	Warning	Reagent probe fails to detect liquid level after dispensing	/	1. Insufficient reagent, or air bubbles exist in the reagent bottle. 2. Probe R1 level detection error.	1. Check if the reagent is sufficient and the reagent bottle contains air bubbles, and then try again. 2. If level detection error is identified, check probe R1, level detection board and connecting cable; replace if any abnormality is identified.
A04006	Sample	Error	Sample mixer vertical	/	Sample mixer vertical movement	Sample mixer vertical movement

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
	mixer		movement error 1. Sensor status is incorrect. 2. Home position is not found. 3. Vertical movement is not allowed in current horizontal position. Sample mixer horizontal movement error. 1. Sensor status is incorrect. 2. Home position is not found. 3. Horizontal movement is not allowed in current vertical position.		error 1. Sensor status error. Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driver circuit error; 3) Vertical motor error; 4) Vertical sensor error; 5) Belt pulley is loose; 3. Vertical movement is not allowed in current horizontal position. Software error. Sample mixer horizontal movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 3. Horizontal movement is not allowed in current vertical position Software error	error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly; 3. Vertical movement is not allowed in current horizontal position Home instrument or unit. Sample mixer horizontal movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 3. Horizontal movement is not allowed in current vertical position Home instrument or unit.
A4015	Sample Mixer	Error	Sample mixer rotation error	/	The mixer is obstructed by other object or interfered by the reaction cuvette.	Replace motor; Adjust gear or replace assembly; Replace sensor, or relevant cable, or circuit

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A05006	Reagent mixer unit	Error	Reagent mixer vertical movement error 1. Sensor status is incorrect. 2. Home position is not found. 3. Vertical movement is not allowed in current horizontal position. Reagent mixer horizontal movement error. 1. Sensor status is incorrect. 2. Home position is not found. 3. Horizontal movement is not allowed in current vertical position.	/	Reagent mixer vertical movement error 1. Sensor status error. Belt or belt pulley is loose. 2. Failed to find the zero position: 1) The assembly is jammed; 2) Driver circuit error; 3) Vertical motor error; 4) Vertical sensor error; 5) Belt pulley is loose; 3. Vertical movement is not allowed in current horizontal position. Software error. Reagent mixer horizontal movement error Reagent mixer horizontal movement error 1. Sensor status error: Belt or belt pulley is loose. 2. Failed to find the zero position: 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 3. Horizontal movement is not allowed in current vertical position Software error	Reagent mixer vertical movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor; 5) Tighten properly; 3. Vertical movement is not allowed in current horizontal position Home instrument or unit. Reagent mixer horizontal movement error Reagent mixer horizontal movement error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the zero position: 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 3. Horizontal movement is not allowed in current vertical position Home instrument or unit.
A05016	Reagent mixer unit	Error	Reagent mixer rotation error	/	The mixer is obstructed by other object or interfered by the reaction cuvette.	Replace motor; Adjust gear or replace assembly; Replace sensor, or relevant cable, or circuit

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A06006	Reaction carousel unit	Error	Reaction carousel movement error Error: %s	/	Reaction carousel movement error 1. Failed to find the home position Home position sensor error 2. The coder missed steps 1) Assembly is jammed; 2) Driving circuit error; 3) Motor error; 4) Sensor error; 5) Belt pulley is loose; 3. Steps missed when moving to the home position. Home position sensor error	Reaction carousel movement error 1. Failed to find the home position Replace sensor or cable 2. The coder missed steps 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor or cable; 5) Replace belt pulley and motor assembly; 3. Steps missed when moving to the home position. Replace sensor or cable
A07006	Sample Carousel Unit	Error	Sample carousel movement error. 1. Home position is not found. 2. Step missed during coder movement. 3. Step missed when moving to home position.	/	Sample carousel outer ring movement error 1. Failed to find the home position. Home position sensor error; 2. The coder missed steps 1) Assembly is jammed; 2) Driving circuit error; 3) Motor error; 4) Sensor error; 5) Belt pulley is loose; 3. The sample carousel missed steps when moving to the home position. Home position sensor error	Sample carousel outer ring movement error 1. Failed to find the home position. Replace sensor or cable; 2. The coder missed steps 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor or cable; 5) Replace belt pulley and motor assembly; 3. The reaction carousel missed steps when moving to the home position. Replace sensor or cable
A07009	Sample Carousel Unit	Error	Sample bar code reader does not work normally	/	Due to the system error, the sample bar code reader cannot work normally. If the error repeats and cannot be removed, then the bar	Recover the failure. If the error remains, initialize the sample bar code reader. If initialization cannot be performed, check if bar code

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					code reader has not performed the initialization normally or the communication cables are disconnected. If bar code reader is not available, please try again.	reader is connected properly. If the error remains, then replace the bar code reader or the control drive board.
A07010	Sample Carousel Unit	Warning	Received bar code contains no ending character.	/	Sample bar code reader does not work normally due to communication error.	Initialize the sample bar code reader and try again. If the problem still remains, check the cable of the bar code reader or replace the bar code reader or the control drive board.
A07011	Sample Carousel Unit	Error	Bar code sending buffer is full.	/	Sample bar code sending buffer is full due to communication error.	Recover the failure or reboot the analyzing unit.
A09006	Reagent Carousel Unit	Error	Reagent carousel cannot find home position. Reagent carousel movement error. 1. Home position is not found. 2. Step missed during coder movement. 3. Step missed when moving to home position.	/	Reagent carousel outer ring movement error 1. Failed to find the home position. Home position sensor error; 2. The coder missed steps 1) Assembly is jammed; 2) Driving circuit error; 3) Motor error; 4) Sensor error; 5) Belt pulley is loose; 3. The reagent carousel missed steps when moving to the home position. Home position sensor error	Reagent carousel outer ring movement error 1. Failed to find the home position. Replace sensor or cable; 2. The coder missed steps 1) Remove obstacle or replace with a new assembly; 2) Replace PCB; 3) Replace motor; 4) Replace sensor or cable; 5) Replace belt pulley and motor assembly; 3. The reagent carousel missed steps when moving to the home position. Replace sensor or cable
A09011	Reagent Carousel Unit	Error	Bar code reader does not work normally.	/	Reagent bar code reader connection error due to system failure. If the error repeats and cannot be removed, then the bar code reader has not performed the initialization	Recover the failure. If the error remains, initialize the sample bar code reader. If initialization cannot be performed, check if bar code reader is connected properly. If the

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					normally or the communication cables are disconnected. If bar code reader is not available, please try again.	error remains, then replace the bar code reader or the control drive board.
A09012	Reagent Carousel Unit	Warning	Received bar code contains no ending character.	/	Communication error. Reagent bar code instruction buffer is full.	Initialize the sample bar code reader and try again. If the problem still remains, check the cable of the bar code reader or replace the bar code reader or the control drive board.
A09014	Reagent Carousel Unit	Error	Bar code sending buffer is full.	/	Reagent bar coder reader does not work normally due to communication error.	Recover the failure or reboot the analyzing unit.
A11005	Wash unit	Error	Wash station movement error. 0. Home position is not found. 1. Sensor status is incorrect. 2. Collision occurs during movement. 3. Movement is forbidden in current status.	/	Wash station movement error 1. Sensor status error. Belt pulley is loose; 2. Failed to find the home position 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Belt pulley is loose; 3. The wash station collides with an obstacle when moving 1) Wash probes are not vertical; 2) Reaction carousel positioning is deviated; 3) Collision sensor or circuit error	Wash station movement error1. Sensor status error 1. Sensor status error: Check if belt or belt wheel is loose, tighten them if so; 2. Failed to find the home position 1) Replace PCB; 2) Replace motor; 3) Replace sensor; 4) Tighten properly; 3. The wash station collides with an obstacle when moving 1) Align wash probes; 2) Align reaction carousel positioning; 3) Replace sensor, cable or PCB
A11010	Wash unit	Error	Releasing vacuum failed.	/	1. Solenoid valves V23-V27 go wrong. 2. The vacuum pump goes wrong. 3. The vacuum sensor goes wrong.	1. Check if solenoid valves V23, V24, V25, V26, or V27 can be turned on and off normally; 2. Check if vacuum pump can be turned off;

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						3. Check if vacuum sensor is normal.
A11012	Wash unit	Warning	Water supplying is too slow	/	1. Water unit goes wrong. 2. Water inlet valve goes wrong. 3. Water inlet valve is not turned on. (When the value is turned on, the handle is horizontal.) 4. Water inlet ball valve goes wrong. 5. Low-level floater of water tank goes wrong. 6. Inlet tube is bent. 7. Filter of the inlet tube is blocked.	1. Check the water unit. 2. Check the handle status of the valve. 3. Check if the water inlet ball valve is blocked. 4. Check if the inlet valve can be turned on and off normally. 5. Check if the low-level floater signal is correct. 6. Check if the water supply tube is smooth. 7. Check if the water inlet filter is blocked.
A11013	Wash unit	Error	Water tank is empty	/	1. Water unit goes wrong. 2. Water inlet valve goes wrong. 3. Water inlet valve is not turned on. (When the value is turned on, the handle is horizontal.) 4. Water inlet ball valve goes wrong. 5. Low-level floater of water tank goes wrong. 6. Inlet tube is bent. 7. Filter of the inlet tube is blocked.	1. Check the water unit. 2. Check the handle status of the valve. 3. Check if the water inlet ball valve is blocked. 4. Check if the inlet valve can be turned on and off normally. 5. Check if the low-level floater signal is correct. 6. Check if the water supply tube is smooth. 7. Check if the water inlet filter is blocked.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A11014	Wash unit	Warning	Filling diluted wash solution tank is too slow	/	1. The solenoid valve V06 goes wrong. 2. The restrictor ring is clogged. 3. The inlet filter at the front panel is clogged. 4. The DI water circulating pump goes wrong. 5. The water tank is empty. 6. The concentrated wash solution tank is empty. 7. The concentrated wash solution pump goes wrong. 8. The low-level floater of the diluted wash solution tank goes wrong.	1. Check solenoid valve V06. 2. Check restrictor ring. 3. Check if inlet filter at the front panel is clogged. 4. Check if DI water pump is on and the pressure gauge reads between 40kPa-50kPa. 5. Check the floater of the deionized water tank. 6. Check the floater of the concentrated wash solution tank. 7. Check concentrated wash solution pump. 8. Check diluted wash solution low-level floater.
A11015	Wash unit	Error	Insufficient diluted wash solution	/	1. The solenoid valve V06 goes wrong. 2. The restrictor ring is clogged. 3. The inlet filter at the front panel is clogged. 4. The DI water circulating pump P03 goes wrong. 5. The water tank is empty. 6. The concentrated wash solution tank is empty. 7. The concentrated wash solution pump p05 goes wrong. 8. The low-level floater of the diluted wash solution tank goes wrong.	1. Check solenoid valve V06. 2. Check restrictor ring. 3. Check if inlet filter at the front panel is clogged. 4. Check if DI water pump is on and the pressure gauge reads between 40kPa-50kPa. 5. Check the floater of the deionized water tank. 6. Check the floater of the concentrated wash solution tank. 7. Check concentrated wash solution pump. 8. Check diluted wash solution low-level floater.
A11016	Wash unit	Warning	Concentrated wash solution tank is empty	/	1. Concentrated wash solution is exhausted and the floater status is Empty. 2. The low-level floater of the	1. Check if the concentrated wash solution is exhausted and the floater status is Empty. If yes, fill more concentrated wash solution.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					concentrated wash solution tank goes wrong.	2. Check if concentrated wash solution tank floater goes wrong.
A11017	Wash unit	Error	Liquid accumulates in primary vacuum container	/	1. Solenoid valve V27 goes wrong. 2. Waste pump P07 goes wrong. 3. The low-concentration waste tube is bent.	1. Check the solenoid valve V27. 2. Check if the waste pump can be turned on and off normally. 3. Check if the low-concentration waste tube is smooth.
A11018	Wash unit	Error	High concentration waste collector is full	/	1. Waste pump P07 goes wrong. 2. The high-concentration waste tube is bent.	1. Check if the waste pump can be turned on and off normally. 2. Check if the high-concentration waste tube is smooth.
A11019	Wash unit	Error	Low concentration waste collector is full	/	1. Waste pump P07 goes wrong. 2. The low-concentration waste tube is bent.	1. Check if the waste pump can be turned on and off normally. 2. Check if the low-concentration waste tube is smooth.
A11020	Wash unit	Error	High concentration waste tank is full	/	1. The high concentration waste tank is full 2. The floater of the high concentration waste tank goes wrong.	Check the high-concentration waste tank. If the tank is full, replace it. Cover the full tank and dispose of the waste properly. Check if high-concentration waste goes wrong.
A11027	Wash unit	Error	Wash instruction cannot be executed due to insufficient vacuum.	/	1. The primary vacuum pump P08 goes wrong. 2. The primary vacuum sensor goes wrong. 3. Connectors or tubes go wrong. 4. The primary vacuum container goes wrong. 5. Solenoid valve V23, V24, V25 or V26 goes wrong.	1. Check if primary vacuum pump P08 can be turned on and off normally. 2. Check primary vacuum sensor. 3. Check if connectors and tubing related to primary vacuum are leaking air. 4. Check if primary vacuum container is leaking air. 5. Check if solenoid valves V23, V24, V25 or V26 can be turned on and off normally.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A11028	Wash unit	Error	Water tank floater logic error	/	1. The high-/low-level floaters of the water tank go wrong.	1. Check if the high-/low-level floater signal of the water tank is correct.
A11029	Wash unit	Error	Diluted wash solution tank floater logic error	/	1. The high-/low-level floaters of the diluted wash solution tank go wrong.	1. Check if the high-/low-level floater signal of the diluted wash solution tank is correct.
A11030	Wash unit	Error	External Drain Canister Overflow	/	1. The drainage module is not powered on. 2. The pump of the drainage module is damaged. 3. The high-/low-level floaters of the drainage module go wrong. 4. The PCBA of the drainage module goes wrong. 5. The waste tube is leaking or bent.	1. Check if the drainage module has been powered on. 2. Check if the pump is working normally. 3. Check if the high-/low-level floaters are normal. 4. Check if the PCBA is normal. 5. Check if the waste tube is leaking or bent.
A11031	Wash unit	Error	Insufficient vacuum degree of the primary vacuum container	/	1. The primary vacuum pump P08 goes wrong. 2. The primary vacuum sensor goes wrong. 3. Connectors or tubes go wrong. 4. The primary vacuum container goes wrong. 5. Solenoid valve V21, V22, V23, V25 or V27 goes wrong.	1. Check if primary vacuum pump P08 can be turned on and off normally, and vacuum has been established successfully. 2. Check primary vacuum sensor. 3. Check if connectors and tubing related to primary vacuum are leaking air. 4. Check if primary vacuum container is leaking. 5. Check if solenoid valves V21, V22, V23, V25 and V27 can be turned on and off normally and are not leaking.
A11034	Wash unit	Error	Cuvette wash syringe movement error. 1) Sensor status is incorrect. 2) Mechanical home position is not found.	/	Cuvette wash syringe movement error. 1. Sensor status error Sensor is not connected or is damaged.	Wash station movement error 1. Sensor status error If the problem remains after resetting, check the sensor connection or replace it.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					2. Home position is not found. 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Component jam.	2. Home position is not found. 1) Check if the component resistance is too high. If it is, replace the component. 2) Replace PCB; 3) Replace motor; 4) Replace sensor;
A11037	Wash unit	Error	Water supply quality is below the standards.	/	1. The water supply does not meet the requirements. 2. The water supply sensor goes wrong.	1. Use qualified deionized water. 2. Check if water quality sensor is normal.
A12005	Temperature control unit	Warning	Reaction carousel temperature is out of range	T1	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error, cable error) 3. The temperature protection switch goes wrong. (component error, cable error) 4. The heater goes wrong. (component error, cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Reproduce failure. 2. Observe ambient temperature through the Status screen, to check if temperature is out of range. 3. Temperature sensor error: check for any signs of short circuit or broken circuit; check if connections are secure. 4. Temperature protection switch error: check if protection switch is shut and connections are secure. 5. Heater error: check if resistance under normal temperature is correct, and connections are secure. 6. PCB failure. 7. Parameters are lost. Check if the temperature control parameters are correct. 8. Electromagnetic interference exists.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A12006	Temperature control unit	Warning	Temperature of wash solution for cleaning cuvettes is out of range	/	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error, cable error) 3. The temperature protection switch goes wrong. (component error, cable error) 4. The heater goes wrong. (component error, cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Reproduce failure. 2. Observe ambient temperature through the Status screen, to check if temperature is out of range. 3. Temperature sensor error: check for any signs of short circuit or broken circuit; check if connections are secure. 4. Temperature protection switch error: check if protection switch is shut and connections are secure. 5. Heater error: check if resistance under normal temperature is correct, and connections are secure. 6. PCB failure. 7. Parameters are lost. Check if the temperature control parameters are correct. 8. Electromagnetic interference exists.
A12007	Temperature control unit	Warning	Temperature of deionized water for cleaning cuvettes is out of range	/	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error, cable error) 3. The temperature protection switch goes wrong. (component error, cable error) 4. The heater goes wrong. (component error, cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Reproduce failure. 2. Observe ambient temperature through the Status screen, to check if temperature is out of range. 3. Temperature sensor error: check for any signs of short circuit or broken circuit; check if connections are secure. 4. Temperature protection switch error: check if protection switch is shut and connections are secure. 5. Heater error: check if resistance under normal temperature is correct, and connections are secure.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						6. PCB failure. 7. Parameters are lost. Check if the temperature control parameters are correct. 8. Electromagnetic interference exists.
A12009	Temperature Control Unit	Warning	Internal temperature of the whole unit is out of range	/	The ambient temperature is out of range. The fan goes wrong. The dust screen is blocked. The air vent is blocked in the specified range.	1. Reproduce failure. Observe ambient temperature through the Status screen, to check if temperature is out of range. 3. Check if cooling fan is OK. 4. Check if dust screen is blocked and needs cleaning. 5. Check if air vent is blocked within specified distance.
A14006	Reagent Refrigeration Unit	Warning	Reagent refrigeration temperature is out of range.	/	The ambient temperature is out of range. The temperature sensor goes wrong. (component error, cable error) The temperature protection switch goes wrong. (component error, cable error) The radiator goes wrong. (component error, cable error) The fan goes wrong. (component error, cable error) The circulating pump goes wrong.	1. Reproduce failure. Observe ambient temperature through the Status screen, to check if temperature is out of range. Temperature sensor error: check for any signs of short circuit or broken circuit; check if connections are secure. Temperature protection switch error: check if protection switch is shut and connections are secure. Radiator error: check if radiator is

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					(component error, cable error) The refrigerant goes wrong. PCB error Parameters are lost. Electromagnetic interference exists.	broken, and connections are secure. 6. Fan error: check if fan is working normally, and connections are secure. Circulating pump error: check if circulating pump is working normally, and connections are secure. Refrigerant error: check if there is sufficient refrigerant (check after power off). 9. PCB error 10. Parameters are lost: reconfigure parameters. 11. Electromagnetic interference exists.
A14011	Temperature Control Unit	Warning	Reagent refrigerating fan 1 is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Reproduce failure. 2. Pull off and plug in fan socket. 3. Check if power supply for fan is normal. 4. Clean dust off the fan. 5. Replace fan.
A14012	Temperature Control Unit	Warning	Reagent refrigerating fan 2 is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Reproduce failure. 2. Pull off and plug in fan socket. 3. Check if power supply for fan is normal. 4. Clean dust off the fan. 5. Replace fan.
A14013	Temperature Control Unit	Warning	Light source fan is abnormal	/	Jam damaged. The power supply goes wrong.	1. Reproduce failure. Pull off and plug in fan socket. 3. Check if power supply for fan is normal. 4. Clean dust off the fan. 5. Replace fan.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A14014	Temperature control unit	Warning	Vacuum pump fan is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Reproduce failure. 2. Pull off and plug in fan socket. 3. Check if power supply for fan is normal. 4. Clean dust off the fan. 5. Replace fan.
A21001	Probe Interior Wash Unit	Error	Probe interior wash syringe movement error. 1) Sensor status is incorrect. 2) Mechanical home position is not found.	/	Probe interior wash syringe movement error. 1. Sensor status error Sensor is not connected or is damaged. 2. Home position is not found. 1) Driving circuit error; 2) Motor error; 3) Home position sensor error; 4) Component jam.	Wash station movement error 1. Sensor status error If the problem remains after resetting, check the sensor connection or replace it. 2. Home position is not found. 1) Check if the component resistance is too high. If it is, replace the component. 2) Replace PCB; 3) Replace motor; 4) Replace sensor;
A22001	ISE module	Error	1. Slope of Na electrode is out of range. 2. Slope of K electrode is out of range. 3. Slope of Cl electrode is out of range. 4. Slope of Na and Li electrodes is out of range. 5. Slope of K and Li electrodes is out of range. 6. Slope of K and Na electrodes is out of range. 7. Slope of Cl and Li electrodes is out of range. 8. Slope of Cl and Na	SLEX	1. Electrode is not installed correctly. 2. Calibration is invalid. 3. Electrode is degenerated. 4. Bubbles exist inside reference electrode. 5. Reference electrode has been used for a long time. 6. Electrodes interfere with each other. 7. Module or fluidic temperature exceeds 32°C.	1. Remove and reinstall the electrode. 2. Replace the calibrator. 3. Replace the failed electrode and run tests again. 4. Remove the electrode, clap it to remove the bubbles, reinstall it, and then run calibration. 5. Replace the reference electrode and run tests again. 6. Replace electrode by different combinations to troubleshoot the failed one. 7. Monitor the temperature. If the ambient temperature is too high,

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			electrodes is out of range. 9. Slope of Cl and K electrodes is out of range. 10. Slope of K, Na and Li electrodes is out of range. 11. Slope of Cl, Na and Li electrodes is out of range. 12. Slope of Cl, K and Li electrodes is out of range. 13. Slope of Cl, K and Na electrodes is out of range. 14. Slope of Cl, K, Na and Li electrodes is out of range.			move the instrument to another position.
A22002	ISE module	Error	Sample probe cannot detect fluid level on the sample carousel due to insufficient sample, or sample is detected containing bubbles in the electrode channel.	SMPA	1. Sample is insufficient, or contains bubbles after being dispensed. 2. No or insufficient sample is dispensed to the injection port. 3. Leaks exist due to improper installation of electrode. 4. Waste pump tube is aging or broken.	1. and 2. Increase the sample volume. Place enough sample in the tube, at least 90µl for ISE measurement. 3. Electrode is not installed correctly. Reinstall it. 4. Check the waste tube, and if necessary, replace it.
A22003	ISE module	Warning	1. Result of Na electrode is out of range. 2. Result of K electrode is out of range. 3. Result of Cl electrode is out of range. 4. Result of Na and Li electrodes is out of range. 5. Result of K and Li electrodes is out of range.	/	1. ISE calibration result error. 2. Electrode is invalid. 3. Concentration of sample is beyond the measurement range.	1. Recalibrate. 2. Replace the failed electrode. 3. Re-process the sample and ensure it is within the measurement range.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			6. Result of K and Na electrodes is out of range. 7. Result of Cl and Li electrodes is out of range. 8. Result of Cl and Na electrodes is out of range. 9. Result of Cl and K electrodes is out of range. 10. Result of K, Na and Li electrodes is out of range. 11. Result of Cl, Na and Li electrodes is out of range. 12. Result of Cl, K and Li electrodes is out of range. 13. Result of Cl, K and Na electrodes is out of range. 14. Result of Cl, K, Na and Li electrodes is out of range.			
A22004	ISE module	Error	Communicating with the ISE module failed.	/	1. Power supply of ISE module goes wrong. 2. Communication cable between ISE module and middle-layer unit goes wrong. 3. Communication interface or pins go wrong. 4. Main control board of the ISE module goes wrong.	1. Replace the 24V power supply board of the ISE module. 2. Replace the communication cable. 3. Replace the interface or pins. 4. Replace the main control board of the ISE module.
A22005	ISE module	Error	ISE unit communication response error	/	1. Communication cable between ISE module and middle-layer unit goes wrong. 2. Communication interface or pins go wrong. 3. Main control board of the ISE module goes wrong.	1. Replace the communication cable. 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 4. Upgrade or reinstall the software.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					4. Software error.	
A22006	ISE module	Error	Pure A/B failed according to the returned data of last purge A and purge B.	/	1. Leaks exist due to improper installation of electrode or O ring's missing. 2. Sample injection port of electrode inside is blocked. 3. Calibrator in the reagent pack is exhausted. 4. Purge cycle is not enough. 5. Peristaltic pump tube is aging, blocked or broken. 6. Calibrator cannot be output due to the blocked wand tube.	1. Reinstall the electrode and check for the O ring. 2. Use warm water to rinse the sample injection port, and unclog the electrode tube. Check the inside of the reference electrode for salt buildup. 3. Replace the reagent pack. 4. Increase the prime cycle. 5. Replace the pump tube. 6. Use warm water to unclog the wand tube.
A22007	ISE module	Warning	Inventory of ISE reagent pack is 0.	/	Calibrator is exhausted.	Replace the reagent pack with a new one.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A22008	ISE module	Error	Voltage of Na electrode is out of range during B calibration or sample measurement. Voltage of K electrode is out of range during B calibration or sample measurement. Voltage of Cl electrode is out of range during B calibration or sample measurement. Voltage of Na and Li electrodes is out of range during B calibration or sample measurement. Voltage of K and Li electrodes is out of range during B calibration or sample measurement. Voltage of K and Na electrodes is out of range during B calibration or sample measurement. Voltage of Cl and Li electrodes is out of range during B calibration or sample measurement. Voltage of Cl and Na electrodes is out of range during B calibration or sample measurement. Voltage of Cl and K electrodes is out of range during B calibration or sample measurement. Voltage of K, Na and Li	VOUT	1. Electrode or reagent pack is invalid. 2. Electrode is unstable. 3. New reagent pack is unstable. 4. Reference electrode has been used for over 6 months. 5. Main control board of the ISE module goes wrong.	1. Replace the failed electrode or reagent pack. 2. New electrode can become stable only after 15 minutes since installation. 3. Multiple calibrations are required after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the main control board.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			electrodes is out of range during B calibration or sample measurement. Voltage of Cl, Na and Li electrodes is out of range during B calibration or sample measurement. Voltage of Cl, K and Li electrodes is out of range during B calibration or sample measurement. Voltage of Cl, K and Na electrodes is out of range during B calibration or sample measurement. Voltage of Cl, K, Na and Li electrodes is out of range during B calibration or sample measurement. Voltage of Na electrode is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K electrode is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl electrode is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Na and Li electrodes is out of range for			

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K and Li electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K and Na electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and Li electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and Na electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and K electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K, Na and Li electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, Na and Li electrodes is out of range for calibration A in two-point calibration or serum mode,			

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			or for calibration B in urine mode. Voltage of Cl, K and Li electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, K and Na electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, K, Na and Li electrodes is out of range for calibration A in two-point calibration or serum mode, or for calibration B in urine mode.			

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A22009	ISE module	Error	Na electrode's voltage drift for calibration A or slope drift for two-point calibration in urine mode. K electrode's voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl electrode's voltage drift for calibration A or slope drift for two-point calibration in urine mode. Na and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. K and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. K and Na electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl and Na electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl and K electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. K, Na and Li electrodes' voltage drift for calibration A or	VDRF /SLD R	1. Electrode or reagent pack is invalid. 2. Electrode is unstable. 3. New reagent pack is unstable. 4. Reference electrode has been used for over 6 months. 5. Main control board of the ISE module goes wrong. 6. The ambient temperature is fluctuating greatly.	1. Replace the failed electrode or reagent pack. 2. New electrode can become stable only after 15 minutes since installation. 3. Multiple calibrations are required after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the main control board. 6. Monitor the ambient temperature and ensure the fluctuation is within +/-4.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			slope drift for two-point calibration in urine mode. Cl, Na and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl, K and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl, K and Na electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode. Cl, K, Na and Li electrodes' voltage drift for calibration A or slope drift for two-point calibration in urine mode.			

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A22010	ISE module	Error	Voltage of Na electrode has noise during B calibration or sample measurement. Voltage of K electrode has noise during B calibration or sample measurement. Voltage of Cl electrode has noise during B calibration or sample measurement. Voltage of Na and Li electrodes has noise during B calibration or sample measurement. Voltage of K and Li electrodes has noise during B calibration or sample measurement. Voltage of K and Na electrodes has noise during B calibration or sample measurement. Voltage of Cl and Li electrodes has noise during B calibration or sample measurement. Voltage of Cl and Na electrodes has noise during B calibration or sample measurement. Voltage of Cl and K electrodes has noise during B calibration or sample measurement. Voltage of K, Na and Li electrodes has noise during B calibration or sample measurement. Voltage of Cl, Na and Li electrodes has noise during B calibration or sample measurement.	NOIS	1. Electrode is invalid. 2. Temperature interference. 3. Main control board of the ISE module goes wrong. 4. Fluidic leaks exist, resulting in salt buildup around the electrode or tube.	1. Replace the electrode. 2. Relocate the instrument. 3. Replace the main control board of the ISE module. 4. Clean the tube and electrode.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			Voltage of Cl, K and Li electrodes has noise during B calibration or sample measurement. Voltage of Cl, K and Na electrodes has noise during B calibration or sample measurement. Voltage of Cl, K, Na and Li electrodes has noise during B calibration or sample measurement. Voltage of Na electrode has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K electrode has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl electrode has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Na and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K and Na electrodes has noise for			

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and Na electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl and K electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of K, Na and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, Na and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, K and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode. Voltage of Cl, K and Na electrodes has noise for calibration A in two-point calibration or serum mode,			

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			or for calibration B in urine mode. Voltage of Cl, K, Na and Li electrodes has noise for calibration A in two-point calibration or serum mode, or for calibration B in urine mode.			
A22011	ISE module	Error	Calibrator B is detected containing bubbles in the electrode channel, or is exhausted.	PUGB	1. Calibrator B is exhausted. 2. Calibrator B tube contains many air bubbles. 3. Peristaltic pump B tube is aging, clogged and broken. 4. Waste pump tube is aging, clogged and broken. 5. The sample injection port and fluidic tube is blocked or leaking. 6. The air bubble detector is invalid.	1. Replace the reagent pack with a new one. 2. Perform purge B to remove air bubbles. 3. and 4. Replace the peristaltic pump tube. 5. Clean the sample injection port, or reinstall the electrode. 6. Replace the air bubble detector.
A22012	ISE module	Error	Calibrator A is detected containing bubbles in the	PUGA	1. Calibrator A is exhausted. 2. Calibrator A tube contains many	1. Replace the reagent pack with a new one.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			electrode channel, or is exhausted.		air bubbles. 3. Peristaltic pump A tube is aging, clogged and broken. 4. Waste pump tube is aging, clogged and broken. 5. The sample injection port and fluidic tube is blocked or leaking. 6. The air bubble detector is invalid.	2. Perform purge B to remove air bubbles. 3. and 4. Replace the peristaltic pump tube. 5. Clean the sample injection port, or reinstall the electrode. 6. Replace the air bubble detector.
A22013	ISE module	Error	Pump calibration result error	/	1. Peristaltic pump tube is aging. 2. Aspiration and dispensing of sample probe goes wrong.	1. Replace the pump tube. 2. Replace the sample probe.
A22014	ISE module	Error	Air bubble detector failure	/	1. Air bubble detector board is eroded due to the leaks at the joint of sample injection port and bubble detector. 2. The air bubble detector is invalid.	Replace the air bubble detector.
A22015	ISE module	Error	Reading reagent pack chip error	/	1. Reagent pack is not installed. 2. Reagent pack wand is failed.	1. Install the reagent pack. 2. Install the reagent pack wand.
A22016	ISE module	Error	Writing reagent pack chip error	/	1. Reagent pack is not installed. 2. Reagent pack wand is failed.	1. Install the reagent pack. 2. Install the reagent pack wand.
A22017	ISE module	Error	Wash solution is detected containing bubbles in the electrode channel, or has not been loaded to correct position on the sample carousel.	/	1. ISE wash solution is insufficient. 2. Waste pump tube is aging, clogged and broken. 3. The sample injection port and fluidic tube is blocked or leaking. 4. The air bubble detector is invalid.	1. Place sufficient ISE wash solution. 2. Replace the peristaltic pump tube. 3. Clean the sample injection port, or reinstall the electrode. 4. Replace the air bubble detector.
A22018	ISE module	Error	The ISE tube has no liquid or has leaks.	CALF	1. Waste pump tube is aging, clogged and broken. 2. The sample injection port and fluidic tube is blocked or leaking. 3. The air bubble detector is invalid.	1. Replace the peristaltic pump tube. 2. Clean the sample injection port, or reinstall the electrode. 3. Replace the air bubble detector.
A22019	ISE module	Error	Stored calibration value	DEP	1. Communication cable between ISE	1. Replace the communication cable.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			error.		module and middle-layer unit goes wrong. 2. Communication interface or pins go wrong. 3. Main control board of the ISE module goes wrong. 4. Software error.	2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 4. Upgrade or reinstall the software.
A22021	ISE module	Error	Command format or execution error	RESP	1. Communication cable between ISE module and middle-layer unit goes wrong. 2. Communication interface or pins go wrong. 3. Main control board of the ISE module goes wrong. 5. Software error.	1. Replace the communication cable. 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 5. Upgrade or reinstall the software.
A22022	ISE module	Error	Air in segment	/	1. Waste pump tube is aging, clogged and broken. 2. The sample injection port and fluidic tube is blocked or leaking. 3. The air bubble detector is invalid.	1. Waste pump tube is aging, clogged and broken. 2. The sample injection port and fluidic tube is blocked or leaking. 3. The air bubble detector is invalid.
A22023	ISE module	Error	No reagent module has been loaded.	/	1. Reagent pack is not installed. 2. Reagent pack wand is failed.	1. Install the reagent pack. 2. Install the reagent pack wand.
A22024	ISE module	Error	Check digit in error code returned by ISE module is incorrect.	/	1. Communication cable between ISE module and middle-layer unit goes wrong. 2. Communication interface or pins go wrong. 3. Main control board of the ISE module goes wrong. 5. Software error.	1. Replace the communication cable. 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 5. Upgrade or reinstall the software.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A22025	System communication	Error	Communication with middle-/lower-layer units failed in home procedure.	/	The serial cable is not connected; or the analyzing unit power is switched off.	Check the serial port connection. Replug the cable. Check if the analyzing unit and rack feeder system are powered on. Start the initialization again. Restart the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
A22026	System communication	Error	Key parameters are not configured during home procedure.	/	Key parameters are not configured.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22027	System communication	Error	Fluidic prime is not performed during home procedure.	/	Fluidic prime is not performed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22028	System communication	Error	Downloading key parameters failed during home procedure.	/	Downloading key parameters failed, or reading parameters from E2PROM failed, or configuring parameters of low-layer unit failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22029	System communication	Error	Collecting dark current failed during home procedure.	/	Collecting dark current failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22030	System communication	Error	Discharging primary vacuum waste failed during home procedure.	/	Executing instruction of discharging primary vacuum waste failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
A22031	System communication	Error	Establishing vacuum failed during fluidic initialization of home procedure.	/	Establishing vacuum failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22032	System communication	Error	Floater are found abnormal during fluidic initialization of home procedure.	/	Low-level floater of water tank is full. Floater of diluted wash solution tank is full. Floater of concentrated wash solution tank is full. Floater of low-concentration waste container is empty. Primary vacuum container is empty. Floater of external high-concentration waste tank is empty.	1. Check the water unit and inlet tube. 2. Check floaters of the water tank, diluted wash solution tank, concentrated wash solution tank, low-concentration waste container, and external high-concentration waste tank. 3. Check the primary vacuum container.
A22033	System communication	Error	Resetting probe interior wash syringe failed during fluidic initialization of home procedure.	/	Resetting probe interior wash syringe failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22034	System communication	Error	Resetting cuvette wash syringe failed during fluidic initialization of home procedure.	/	Resetting cuvette wash syringe failed.	3. Check the primary vacuum container.
A22035	System communication	Error	Releasing vacuum failed during home procedure.	/	1. Solenoid valves V23-V27 go wrong. 2. The vacuum pump goes wrong. 3. The vacuum sensor goes wrong.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22036	System communication	Error	Initializing sample bar code reader failed during home procedure.	/	Initializing sample bar code reader failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						your local distributor.
A22037	System communication	Error	Initializing reagent bar code reader failed during home procedure.	/	Initializing reagent bar code reader failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22038	System communication	Error	Executing reagent bar code scanning failed during home procedure.	/	Scanning reagent bar code failed.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
A22039	System communication	Error	Software version does not match during home procedure.	/	1. Executing version query instruction failed. 2. Returned control software version and that stored in the operating software does not match.	Switch off the analyzing unit power and switch on it again. If this message appears for 3 times, contact our customer service department or your local distributor.
C00001	Operating system	Error	Operating system error	/	The operating system is not Windows XP.	Reinstall Windows XP and the operating software.
C00003	Operating system	Error	Resolution error	/	The screen resolution is not 1280*1024.	Set the screen resolution to 1280*1024.
C00004	Operating system	Error	Color error	/	The color is below 16 bit.	Set the color to above 16 and then reboot the operating system.
C00007	Operating system	Error	CPU performance low	/	The CPU is too busy.	Reboot the computer and operating software. If this message appears for 3 times, contact our customer service department or your local distributor.
C00008	Operating system	Warning	Printer cannot be connected	/	The printer is not powered on; the printer cable is not connected; or no driver is installed.	Check the printer connection; check if the printer is powered on and if the driver and default printer have been installed.
C00009	Operating system	Warning	Printer failure	/	Paper jam. No paper. No ink	Check for paper jam. Check if printer is busy, and print tasks are

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						too many.
C00010	Operating system	Warning	Print paper running out	/	Paper jam. No paper. No ink	Check for paper jam. Check if printer is busy, and print tasks are too many.
C00011	Operating system	Error	The operating software has not been closed normally.	/	The operating software is abnormal or the analyzer is powered off accidentally.	After rebooting the software, perform the special wash procedure.
C00012	Operating system	Warning	Sound card failure	/	No sound card is installed. Sound card failure. Incorrect sound card driver.	Reinstall the sound card or the sound card driver.
C01001	System communication	Error	Equipment cannot be connected	/	The serial cable is not connected; or the analyzing unit power is switched off; or the rack feeder system power is switched off.	Check the serial port connection. Replug the cable. Check if the analyzing unit and rack feeder system are powered on. Start the initialization again. Restart the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
C02001	Database	Error	Database initializing failed	/	Database software is not installed; database is not established; database system or database file is damaged or lost; database is opened exclusively	Install database software; re-establish database; if problem remains, contact our R&D department
C02002	Database	Error	Database upgrading failed	/	Database is severely damaged; database does not exist; database is not private for BS-480; database is opened exclusively	Install database software; re-establish database; if problem remains, contact our R&D department
C02004	Database	Warning	Database backup failed	/	Database record is under use; database is locked	Back up database again; if problem remains, contact our R&D department
C02005	Database	Warning	Reading/Writing database failed	/	Cannot read or write necessary information from or into database	Check if database has been damaged; contact our R&D

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						department
C03001	Result calculation	Warning	Response calculation error	RCE	Absorbance data for calculation is incomplete, or the dividend is 0.	Investigate the cause of photoelectric data loss; contact our R&D department
C03002	Result calculation	Warning	Absorbance out of range	ABS	The absorbance measured at the primary and secondary wavelength is greater than 3.3A.	Check the sample for foreign matters or interferences; check if the reagent is qualified and placed in the correct position; check the cuvette is clean; check if the photometric system is working normally.
C03003	Result calculation	Warning	R1 blank absorbance out of range	RBK	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if cuvette is clean or overflowed; check if reagent is sufficient, without air bubbles; check if chemistry parameters are reasonable; if previous problems do not exist, replace reagent and rerun the test.
C03004	Result calculation	Warning	Substrate depletion	BOE	The sample concentration is too high, and substrate depletion occurs during fixed-time measurements.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
C03005	Result calculation	Warning	No calculation interval	ENC	The sample concentration is too high, and substrate depletion occurs within the lag time of rate check measurements.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
C03006	Result calculation	Warning	Non-linear	LIN	The measuring points for result calculation are nonlinear, because the sample concentration is too high, or the substrate depletion limit is not specified or unreasonable.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
C03007	Result calculation	Warning	Prozone check error	PRO	Antigen excess occurs due to too high sample concentration.	Check the reaction curve and the prozone check parameters. Rerun the test with diluted sample.
C03008	Result calculation	Warning	Response greater than that of the maximum-	RRN	The sample concentration exceeds the high limit of the calibrator	Rerun the test with diluted sample.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
			concentration calibrator		concentration.	
C03009	Result calculation	Warning	Mixed blank absorbance out of range	MBK	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if the cuvette is clear and not overflowed, the reagent is sufficient without air bubbles, and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test.
C03010	Result calculation	Warning	Blank response out of range	BLK	The reagent goes wrong; insufficient reagent is dispensed; the cuvette contains air bubbles; the light drifts; or the cuvette is overflowed.	Check if the cuvette is not overflowed, the reagent is sufficient without air bubbles, the light does not drift and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test.
C03011	Result calculation	Warning	Calibration repeatability error.	DUP	The difference between the maximum and minimum response of the calibrator exceeds the specified limit.	Check if the acceptance limit is reasonable, troubleshoot the error, and then recalibrate.
C03012	Result calculation	Warning	Calibration sensitivity error.	SEN	The difference of final response of the maximum and minimum concentration calibrators exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03013	Result calculation	Warning	Calibration curve standard deviation out of range.	CSD	The calculated standard deviation of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03014	Result calculation	Warning	Calibration determination coefficient out of range	DET	The calculated fit of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03015	Result calculation	Warning	Calibration coefficient differential limit exceeds limit.	FAC	The slope difference is applicable to linear calibration only and refers to the K factor (slope) difference between two adjacent calibrations.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					It exceeds the specified limit.	
C03016	Result calculation	Warning	Calibration curve not monotonic.	MON	The calibration data and calibration curve are not monotonic.	Check if the calibrator is defined and placed correctly, and then recalibrate.
C03017	Result calculation	Warning	Calibration curve is not convergent.	COV	For nonlinear calibration, a satisfying base cannot be calculated and no calibration curve is drawn.	Check that the reagent and calibrator are normal, and then recalibrate. If troubleshooting fails, choose other calibration rules and recalibrate.
C03018	Result calculation	Warning	12S	12S	The QC result is between ± 2 and ± 3 standard deviations from the assigned mean concentration.	No actions are required.
C03019	Result calculation	Warning	13s	13s	The current QC result is greater than ± 3 standard deviations from the assigned mean concentration.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
C03020	Result calculation	Warning	22s	22s	Results of two controls or two results of one control within a run are simultaneously greater than +2 or -2 standard deviations from the assigned mean.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
C03021	Result calculation	Warning	R4S	R4S	One result of a run is greater than +2 standard deviations from the assigned mean and the other greater than -2SDs.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
C03022	Result calculation	Warning	41s	41s	Results of two runs (4 results), or 4 continuous results of a control are on the same side and greater than ± 1 standard deviation from the assigned mean concentration.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
C03023	Result calculation	Warning	10x	10x	Results of five runs (10 results), or 10 continuous results of a control are on the same side.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
C03024	Result calculation	Error	Biochemistry test period time out. Cannot continue	/	1. Software error 2. Operating system error	Rerun the test, or reboot the operating software and analyzing unit. If the error remains, contact our customer service department or your local distributor.
C03026	Result calculation	Warning	Photoelectric data is lost	/	Communication error.	If the error persists, contact our R&D department.
C03030	Result calculation	Error	Photoelectric measurement period is out of range	/	Software error	1. Rerun the operating software. 2. Reboot the operation unit. 3. If problem remains, contact our R&D department.
C03031	Result calculation	Error	Multiple photoelectric measurements are time out	/	Software error	1. Rerun the operating software. 2. Reboot the operation unit. 3. If problem remains, contact our R&D department.
C04001	Sample bar code	Warning	Sample bar code already exists.	/	Duplicate bar code is used.	Replace the duplicate sample bar code label.
C04002	Sample bar code	Warning	Bar code has no corresponding programming.	/	The sample of the bar code has not been programmed.	Program the sample of the bar code.
C04006	Sample bar code	Warning	Sample is expired.	/	The sample is loaded after its shelf life is exceeded.	The sample is expired. Replace the sample and program it again. Reject the expired sample. If the sample shelf life is too short, change it to a reasonable one.
C04008	Sample bar code	Warning	Sample bar code too long.	/	The bar code length is greater than the maximum value of 27 digits.	Redefine the bar code with no more than 27 digits.
C04010	Sample bar code	Error	Sample bar code parameter error	/	Query result of configuration parameters does not match the defined ones. The probable causes are: bar code reader does not receive configuration instruction due to communication error between bar code reader and smart modules; or	Re-configure the bar code reader.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
					the received instruction cannot be configured correctly due to communication error; or other errors occurred in bar code reader module. Reconfigure the bar code reader.	
C04012	Sample bar code	Warning	Sample bar code analyzing failed during sample programming.	/	Bar code data does not accord with the defined format.	Reset the bar code format, or reprint and scan the bar code.
C05001	Reagent bar code	Warning	Reagent of this bar code has already been loaded.	/	Incorrect reagent or reagent bar code is being used, or an invalid reagent bar code is being used in a closed-reagent system. Closed-reagent bar code is aligned with closed reagents, and cannot be used again for new reagent when a reagent is exhausted.	Reprint the reagent bar code, or replace the reagent bottle with an invalid bar code.
C05002	Reagent bar code	Warning	Reagent bar code information error.	/	Incorrect reagent bar code is being used, or reagent bar code is not configured reasonably. The reagent bar code contains incomplete or incorrect reagent information, such as expiration date, reagent volume, etc. If the reagent is placed incorrectly, adjust them to the correct positions, that is, R1 to reagent carousel 1 and R2 to reagent carousel 2.	Print the new reagent bar code with correct settings and check the bar code against the settings. In the case of a closed-reagent system, replace the reagent bottle, or contact the reagent supplier. If the reagent is placed incorrectly, adjust them to the correct positions, that is, R1 to reagent carousel 1 and R2 to reagent carousel 2.
C05003	Reagent bar code	Warning	Reagent bar code is incorrect and does not meet the requirements.	/	Incorrect reagent bar code is being used, or reagent bar code settings are incorrect. Non-closed reagent bar code is being used in a closed-reagent system. The system is failed to extract reagent information from the bar code.	Check the reagent bar code settings, or reprint the reagent bar code against the settings. In the case of a closed-reagent system, contact the reagent supplier.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
C05006	Reagent bar code	Error	Other reagents are placed in the fixed reagent position.	/	Reagent rather than wash solution is placed in the fixed wash solution position on reagent carousel 1.	Reposition the reagent, or remove it from the fixed wash solution position.
C05008	Reagent bar code	Error	Other reagents are placed in the fixed reagent position.	/	Reagent rather than physiological saline is placed in the fixed physiological saline position on reagent carousel 1.	Reposition the reagent, or remove it from the fixed wash solution position.
C06001	Host communication	Error	LIS initialization error	/	Host file is damaged or does not exist.	Reboot operating software; initialize LIS module
C06002	Host communication	Error	LIS communication parameter error	/	Host parameters error	Check system configuration parameters; re-connect LIS
C06003	Host communication	Error	LIS communication error	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error occurs for several times, contact LIS vendor. If problem remain, contact our R&D department.
C06004	Host communication	Error	LIS host cannot be connected	/	Abnormal network connection, or the LIS host is not started.	Check if LIS host and LIS station is started normally.
C06005	Host communication	Warning	Sending sample results failed.	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error occurs for several times, contact LIS vendor. If problem remain, contact our R&D department.
C06006	Host communication	Warning	Sending sample information failed.	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error occurs for several times, contact LIS vendor. If problem remain, contact our R&D department.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
C06007	Host communication	Warning	Inquiring sample information failed	/	LIS host failure	If the error occurs accidentally, neglect it. Neglect the error. If the error occurs for several times, contact LIS vendor. If the problem remains, contact our customer service department or your local distributor.
C06008	Host communication	Warning	Downloading sample failed.	/	Incorrect channel settings; or insufficient or redundant chemistries on the LIS host.	Check and re-set the chemistry correspondence between the operating software and the LIS host.
C07003	Light Source	Error	Light intensity is too weak	/	1. The lamp is not installed correctly. 2. The cuvette is contaminated. 3. The lamp is aging. 4. The wash station dispenses liquid incorrectly.	1. Open the lamp replacement door, and then check if the lamp is installed correctly. 2. Perform the diluted wash procedure and then the lamp check procedure. 3. Replace the lamp. 4. Check if the wash station dispenses liquid with correct volume to reaction cuvettes. 5. If problem remains, contact our R&D department.
C07004	Light Source	Warning	Cuvette blank out of range	/	1. The cuvette is contaminated. 2. The lamp is aging. 3. The lamp is not installed correctly. 4. The wash station dispenses liquid incorrectly. 5. The photoelectric collection board goes wrong.	1. Open the reaction carousel and check if the lamp is turned on. 1. If it is not, rerun the operating software. 2. Open the lamp replacement door, and then check if the lamp is installed correctly. 3. Perform the diluted wash procedure and then the cuvette check procedure. 4. Replace or clean the abnormal cuvette.

Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
						6. Check if the wash station dispenses liquid with correct volume to reaction cuvettes. 7. If problem remains, contact our R&D department.
C07005	Light Source	Error	Lamp is not turned on	/	1. The lamp is damaged. 2. The lamp cable is not connected properly. 3. The power board of the lamp is not connected properly. 4. The power supply of the analyzing unit is disconnected. 5. The photoelectric collection board goes wrong.	1. Open the reaction carousel and check if the lamp is turned on. 1. If it is not, rerun the operating software. 2. Open the lamp replacement door, and then check if the lamp cable is tightened. 3. Replace the lamp. 4. Check if the connect of the lamp power board is loose, and if necessary, reinsert the connector. 5. Check if the PCBs work normally.
C07006	Light Source	Error	Light intensity is too strong	/	1. A cuvette position has no cuvette installed. 2. The circuit gain is too high and beyond the measurement range.	1. Check if all cuvette positions have cuvettes installed. 2. Readjust the gain.
C07007	Light Source	Error	Dark current is too high	/	1. The circuit gain is too high and beyond the measurement range. 2. The power board of the lamp is not connected properly. 3. The photoelectric collection board goes wrong.	1. Perform dark current fluctuation test to identify failure. 2. Check if the abnormal channel has something to do with the channel gain. If it does, adjust the channel gain. 3. Check if the connect of the lamp power board is loose, and if necessary, reinsert the connector.
C07008	Light Source	Warni	The lamp has been used over	/	1. The lamp has been used over	1. Replace the lamp.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
		ng	its lift span.		2,000 hours. 2. The lamp has been replaced incorrectly.	2. Perform the Replace Lamp procedure again.
C07009	Light Source	Error	Water blank out of range (10X).	/	1. The cuvette is overflowing. 2. The lamp has been replaced incorrectly. 3. Cuvette check is not performed after maintenance. 4. The cable connectors are not tightened. 5. The retaining screw is not tightened. 6. The lamp is aging. 7. The photometer goes wrong.	1. Check if the cuvette is overflowing. 2. Check if the Replace Lamp command is executed during lamp replacement. 3. Execute the Cuvette Check command again. 4. Check if the cleaning liquid inside the cuvette is less than half of the cuvette. 5. Check if the cable connectors and retaining screw of the lamp have been tightened. 6. Check if the reaction curve fluctuates irregularly. If yes, replace the lamp. 7. If the error remains, contact our customer service department.
C07012	Other	Warning	Storage device error. Cannot import data	/	No floppy disk or U disk is inserted. No file is found in the floppy disk or U disk, or file error, or file is damaged. The floppy disk or U disk is locked or damaged.	Check if a USB drive or floppy disk is inserted or full. Check if the storage device is damaged.
C07013	Other	Warning	Storage device error. Cannot export data	/	No floppy disk or U disk is inserted. Insufficient disk space. The floppy disk or U disk is locked or damaged.	Check if a USB drive or floppy disk is inserted or full. Check if the storage device is damaged.
C07014	Other	Warning	Biochemistry reagent exhausted	/	All reagents of the reagent type for the chemistry are less than the minimum limit. All reagents of the type are too little to be detected.	Refill or replace the reagent.

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
C07016	Other	Warning	Insufficient wash solution. Position: %s	/	Insufficient wash solution on the reagent carousel.	Refill the wash solution on the reagent carousel.
C07017	Other	Warning	Wash solution exhausted. Position: %s	/	The wash solution on the reagent carousel is exhausted.	Refill the wash solution on the reagent carousel.
C07022	Other	Warning	Less than %s tests are left in biochemistry reagent Chemistry: %s Position: %s	/	All reagents of the reagent type for the chemistry are less than the minimum limit. All reagents of the type are too little to be detected.	Refill or replace the reagent.
C07023	Other	Warning	Calibration of one or more chemistries will be expired	/	The calibration factors are about to be expired.	Recalibrate the chemistry.
C07027	Other	Warning	Calibrator %s has been expired	/	The calibrator is expired.	Replace the calibrator.
C07028	Other	Warning	Chemistry: %s, lot number: %s, position: %s, has been expired	/	The reagent is expired.	Replace the reagent.
C07029	Other	Warning	Chemistry: %s, lot number: %s, position: %s, has exceeded the uncapping time	/	The uncapping time of the reagent pack is too long.	Replace the reagent.
C07034	Other	Warning	Insufficient physiological saline on reagent carousel	/	Insufficient physiological saline on reagent carousel	Refill the physiological saline on the reagent carousel.
C07035	Other	Warning	Physiological saline on reagent carousel is exhausted	/	Physiological saline is exhausted	Refill the physiological saline on the reagent carousel.
C07036	Other	Warning	Chemistry: %s. Calibration factors are expired	/	The calibration factors have been expired.	Recalibrate the chemistry.
C07037	Other	Warning	Reagent bottle number of %s chemistry is changed. Please recalibrate	/	Serial number of the reagent is changed.	Reagent serial number of the chemistry is changed. Please recalibrate
C07038	Other	Warning	Reagent lot number of %s chemistry is changed. Please recalibrate	/	Lot number of the reagent is changed.	Reagent lot number of the chemistry is changed. Please recalibrate

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Error Code	Component	Event class	Description	Flag	Probable Causes	Corrective Actions
C07039	Other	Warning	The calibration factors are expired.	/	The calibration factors are expired.	The calibration factors are expired.
C07040	Other	Warning	Biochemistry reagent exhausted	/	Reagent is running out, or reagent level cannot be detected.	Refill or replace the reagent.
C07041	Other	Error	Less than X% tests are left in ISE chemistry reagent. (X can be defined)	/	Inventory of the ISE reagent is lower than the alarm limit.	Check the inventory, and refill ISE reagent if necessary.
C07042	Others	Warning	The lot number of the solution is expired.	/	The non-reagent solution or the reagent is expired.	Replace the expired solution or the reagent.

